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# 1 Template

## 1.1 Main

```

1 #include <bits/stdc++.h>
2 #include <ext/pb_ds/assoc_container.hpp>
3 #include <ext/pb_ds/tree_policy.hpp>
4
5 using namespace std;
6 using namespace __gnu_pbds;
7
8 #define El_Hefnawys \
9     ios_base::sync_with_stdio(false); \
10     cin.tie(nullptr); \
11     cout.tie(nullptr)
12
13 typedef long long ll;
14 typedef unsigned long long ull;
15
16 #define int ll
17
18 typedef vector<vector<int>>> GRAPH;
19
20 const int MOD = 1e9 + 7;
21 const int OO = 1e15;
22 const double PI = acos(-1.0);
23 const double EPS = 1e-9;
24
25 int dx4[4] = {0, 1, 0, -1};
26 int dy4[4] = {1, 0, -1, 0};
27
28 int dx8[8] = {0, 0, 1, -1, 1, 1, -1, -1};
29 int dy8[8] = {1, -1, 0, 0, 1, -1, 1, -1};
30
31 void solve()
32 {
33 }
34
35 signed main()
36 {
37     El_Hefnawys;
38     int t{1};
39     // cin >> t;
40     while (t--)
41     {
42         solve();
43     }
44     return 0;
45 }
```

# 2 Data Structures

## 2.1 Centroid decomposition

### 2.1.1 Centroid decomposition

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: CENTROID DECOMPOSITION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Paths of length K", "Distance between all pairs", "Update node
6  *   color, find closest red node".
7  * - Classic Scenarios: You need to answer heavy path queries on a massive tree
8  *   where standard DFS would Time Limit Exceed (TLE).
9  * - The Magic: We find the "center of gravity" (Centroid) of the tree, make it a
10  *   root, and "remove" it.
11  * This shatters the graph into a forest. We repeat this recursively. The
12  * resulting "Centroid Tree"
13  * connects these centers and mathematically guarantees a maximum depth of  $O(\log N)$ .
14  * You can safely brute-force climbing from any node to the root of the
15  * Centroid Tree in just ~17 steps!
16  *
17  * 2. HOW TO USE IT (THE BLACK BOX)
18  * - Initialization: Pass the size of the tree (1-indexed) and the adjacency list
19  *   .
20  *   CentroidDecomposition cd(N, adj);
21  *
22  * - Navigation: The `cd.centroid_parent[u]` array tells you the parent of node `
23  *   u` in the new tree.
24  * - Distance: Use `cd.get_dist(u, v)` to find the distance between nodes in the
25  *   ORIGINAL tree.
26  *
27  * - Complexity:
28  *   Time:  $O(N \log N)$  to build the tree and the LCA tables.
29  *   Space:  $O(N \log N)$  memory for binary lifting tables.
30  *
31  * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
32  * - How to solve a problem with it:
33  *   Instead of answering queries globally, answer them for "all paths passing
34  *   through the current centroid".
35  *   For point updates (e.g., painting a node red), start at the updated node and
36  *   climb up its
37  *   `centroid_parent` path, updating the answer at each ancestor using `get_dist
38  *   (u, ancestor)`.
39  */
40
41 #include <bits/stdc++.h>
42 using namespace std;
43
44 struct CentroidDecomposition
```

```

35 {
36     int n;
37     vector<vector<int>> adj;
38
39     // Variables for the Centroid Tree
40     vector<int> sz;
41     vector<bool> removed;
42     vector<int> centroid_parent;
43
44     // Variables for O(1) distance queries in the original tree (Binary Lifting)
45     const int LOG = 20;
46     vector<vector<int>> up;
47     vector<int> depth;
48
49     CentroidDecomposition(int nodes, const vector<vector<int>> &graph)
50     {
51         n = nodes;
52         adj = graph;
53
54         sz.assign(n + 1, 0);
55         removed.assign(n + 1, false);
56         centroid_parent.assign(n + 1, -1);
57
58         up.assign(n + 1, vector<int>(LOG, 0));
59         depth.assign(n + 1, 0);
60
61         // 1. Precompute LCA for fast distance queries
62         lca_dfs(1, 1);
63
64         // 2. Build the Centroid Tree
65         decompose(1, -1);
66     }
67
68     // Standard Binary Lifting DFS to compute depths and ancestors
69     void lca_dfs(int u, int p)
70     {
71         up[u][0] = p;
72         for (int i = 1; i < LOG; i++)
73         {
74             up[u][i] = up[up[u][i - 1]][i - 1];
75         }
76         for (int v : adj[u])
77         {
78             if (v != p)
79             {
80                 depth[v] = depth[u] + 1;
81                 lca_dfs(v, u);
82             }
83         }
84     }
85
86     int get_lca(int u, int v)

```

```

87     {
88         if (depth[u] < depth[v])
89             swap(u, v);
90         int diff = depth[u] - depth[v];
91         for (int i = 0; i < LOG; i++)
92         {
93             if ((diff >> i) & 1)
94                 u = up[u][i];
95         }
96         if (u == v)
97             return u;
98         for (int i = LOG - 1; i >= 0; i--)
99         {
100             if (up[u][i] != up[v][i])
101             {
102                 u = up[u][i];
103                 v = up[v][i];
104             }
105         }
106         return up[u][0];
107     }
108
109     // Get the shortest path distance between u and v in the ORIGINAL tree
110     int get_dist(int u, int v)
111     {
112         return depth[u] + depth[v] - 2 * depth[get_lca(u, v)];
113     }
114
115     // Your exact dfs function, slightly modified to ignore "removed" nodes!
116     void get_sz(int u, int p)
117     {
118         sz[u] = 1;
119         for (int v : adj[u])
120         {
121             if (v == p || removed[v])
122                 continue; // Ignore dead ends
123             get_sz(v, u);
124             sz[u] += sz[v];
125         }
126     }
127
128     // Your exact centroid function, slightly modified to ignore "removed" nodes!
129     int get_centroid(int u, int p, int total_nodes)
130     {
131         for (int v : adj[u])
132         {
133             if (v == p || removed[v])
134                 continue; // Ignore dead ends
135
136             if (sz[v] > total_nodes / 2)
137             {
138                 return get_centroid(v, u, total_nodes);

```

```

139     }
140 }
141 return u;
142 }
143
144 // The master function that shatters the graph and builds the new tree
145 void decompose(int u, int p)
146 {
147     // 1. Calculate the subtree sizes of the current shattered component
148     get_sz(u, -1);
149     int total_nodes = sz[u];
150
151     // 2. Find the exact center of gravity (Centroid) for this component
152     int c = get_centroid(u, -1, total_nodes);
153
154     // 3. Link it to the Centroid Tree!
155     centroid_parent[c] = p;
156
157     // 4. "Delete" this centroid to shatter the tree into smaller forests
158     removed[c] = true;
159
160     // 5. Recursively find the centroids of the remaining pieces
161     for (int v : adj[c])
162     {
163         if (!removed[v])
164         {
165             decompose(v, c);
166         }
167     }
168 }
169 };
170
171 void solve()
172 {
173     int n, q;
174     // Read N nodes and Q queries
175     if (!(cin >> n >> q))
176         return;
177
178     vector<vector<int>> adj(n + 1);
179     for (int i = 0; i < n - 1; i++)
180     {
181         int u, v;
182         cin >> u >> v;
183         adj[u].push_back(v);
184         adj[v].push_back(u);
185     }
186
187     // Build the magical O(log N) depth tree
188     CentroidDecomposition cd(n, adj);
189
190     // EXAMPLE QUERY: "Update node U's color, then find closest node with that

```

```

191         color"
192     // To solve this, you would start at U and literally climb the centroid tree!
193     /*
194     int u = 5;
195     int curr = u;
196     while (curr != -1) {
197         int distance_in_original_tree = cd.get_dist(u, curr);
198         // update_answer(curr, distance_in_original_tree);
199         curr = cd.centroid_parent[curr];
200     }
201     */
202 }
203
204 int main()
205 {
206     // Fast I/O
207     ios_base::sync_with_stdio(false);
208     cin.tie(NULL);
209     solve();
210     return 0;
211 }

```

## 2.2 DSU

### 2.2.1 DSU(Hassan)

```

1 class DSU {
2 public:
3     vector<int>parent,sizes,mn,mx;
4     DSU(int n) {
5         parent.resize(n+1);
6         sizes.resize(n+1);
7         mx.resize(n+1);
8         mn.resize(n+1);
9         for (int i = 1; i <= n; i++)
10             parent[i] = mn[i] = mx[i] = i,sizes[i] = 1 ;
11     }
12
13     int findRoot(int u ) {
14         return parent[u] == u ? u : parent[u] = findRoot(parent[u]);
15     }
16
17     bool merge(int u, int v) {
18         int root_u = findRoot(u), root_v = findRoot(v);
19         if (root_u == root_v)
20             return false;
21
22         if (sizes[root_u] > sizes[root_v])
23             swap(root_u, root_v);
24
25         mx[root_v] = max(mx[root_v], mx[root_u]);

```

```

26     mn[root_v] = min(mn[root_v], mn[root_u]);
27     parent[root_u] = root_v;
28     sizes[root_v] += sizes[root_u];
29     return true;
30 }
31 };
32
33 void Hassan() {
34     int n,q;cin >>n>>q;
35     DSU d(n);
36     while (q--){
37         string op;int u,v;cin >> op ;
38         if (op == "get") {
39             cin >>v;
40             int r = d.findRoot(v);
41             cout << d.mn[r] <<"_" << d.mx[r] <<"_"<< d.sizes[r] <<endl;
42         }else {
43             cin >>u>>v;
44             d.merge(u,v);
45         }
46     }
47 }

```

### 2.2.2 DSU

```

1 class DSU
2 {
3 private:
4     vector<int> parent, size;
5     int forests{};
6
7     void link(int x, int y)
8     {
9         if (size[x] > size[y])
10             swap(x, y);
11         parent[x] = y;
12         size[y] += size[x];
13     }
14
15 public:
16     DSU(int n)
17     {
18         parent = vector<int>(n), size = vector<int>(n);
19         forests = n;
20         for (int i = 0; i < n; i++)
21             parent[i] = i, size[i] = 1;
22     }
23
24     int find_set(int x)
25     {
26         if (x == parent[x])

```

```

27         return x;
28         return parent[x] = find_set(parent[x]);
29     }
30
31     int union_sets(int x, int y)
32     {
33         x = find_set(x), y = find_set(y);
34         if (x != y)
35         {
36             link(x, y);
37             forests--;
38         }
39         return x != y;
40     }
41
42     bool same_set(int x, int y)
43     {
44         x = find_set(x), y = find_set(y);
45         return x == y;
46     }
47
48     int get_size(int x)
49     {
50         return size[find_set(x)];
51     }
52
53     int get_forests() const
54     {
55         return forests;
56     }
57 };

```

### 2.2.3 DSUOnTree

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DSU ON TREE (SACK / SMALL TO LARGE)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Subtree queries", "Offline queries", "Distinct colors in subtree", "Frequencies at depth".
6  * - Classic Scenarios: You have a tree and Q offline queries asking about properties within the subtree of node U (e.g., "Can we form a palindrome using nodes at depth H in U's subtree?").
7  * A naive DFS for each query gives TLE  $O(N^2)$ .
8  * - The Magic: We use "Small to Large" merging. We find the "heavy" child (largest subtree).
9  * We process light children and ERASE their data, then process the heavy child and KEEP its data.
10  * Finally, we add light children back. Since a node is a "light child" at most  $O(\log N)$  times,

```



```

12 * the complexity drops to  $O(N \log N)$ .
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization: Pass the number of nodes, the string of characters, and
    number of queries.
16 *     DSUOnTree dsu_tree(N, s, Q);
17 *
18 * - Updates: Add edges to build the tree, and add queries offline.
19 *     dsu_tree.addEdge(u, v);
20 *     dsu_tree.addQuery(node, depth, query_index);
21 *
22 * - Execution: Call solve(root) to process everything and get the answers.
23 *     vector<string> answers = dsu_tree.process(1);
24 *
25 * - Complexity:
26 *     Time:  $O(N \log N)$  processing time.
27 *     Space:  $O(N)$  memory.
28 *
29 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
30 * - 0-indexed or 1-indexed?
31 * The template naturally handles 1-indexed nodes. If your string is 0-indexed,
    it automatically
32 * shifts it to 1-indexed internally for safety.
33 * - Changing the problem condition:
34 * Modify the `modifyState()` function to count frequencies, update maximums,
    or manage a `std::set`.
35 * Then modify the "Answer queries" section in `dfsDSU()` to check your
    specific condition.
36 */
37
38 #include <bits/stdc++.h>
39 using namespace std;
40
41 struct DSUOnTree
42 {
43 private:
44     int n;
45     string s;
46     vector<vector<int>>> adj;
47     vector<vector<pair<int, int>>> queries; // {depth, query_index}
48
49     vector<int> sz;
50     vector<int> lvl;
51     vector<int> mask; // Bitmask to track odd/even frequencies of characters
52     vector<bool> big; // Marks the heavy child
53     vector<string> ans;
54
55     // Step 1: Precalculate subtree sizes and depths
56     void dfsSize(int node, int parent, int level)
57     {
58         lvl[node] = level;
59         sz[node] = 1;

```

```

60     for (int child : adj[node])
61     {
62         if (child != parent)
63         {
64             dfsSize(child, node, level + 1);
65             sz[node] += sz[child];
66         }
67     }
68 }
69
70 // Step 2: Add or remove a node's contribution from our global state
71 void modifyState(int node, int parent, int x)
72 {
73     // ? THE MAGIC: We toggle the i-th bit to flip the parity of the
        character's frequency
74     mask[lvl[node]] ^= (1 << (s[node] - 'a'));
75
76     for (int child : adj[node])
77     {
78         // ? CRITICAL: Do NOT traverse the heavy child if it's already marked
        !
79         if (child != parent && !big[child])
80         {
81             modifyState(child, node, x);
82         }
83     }
84 }
85
86 // Step 3: The core DSU on Tree logic
87 void dfsDSU(int node, int parent, bool keep)
88 {
89     int maxSize = -1, bigChild = -1;
90
91     // Find the heavy child
92     for (int child : adj[node])
93     {
94         if (child != parent && sz[child] > maxSize)
95         {
96             maxSize = sz[child];
97             bigChild = child;
98         }
99     }
100
101     // Process all light children first, and ERASE their data (keep = 0)
102     for (int child : adj[node])
103     {
104         if (child != parent && child != bigChild)
105         {
106             dfsDSU(child, node, 0);
107         }
108     }
109

```

```

110 // Process the heavy child, and KEEP its data (keep = 1)
111 if (bigChild != -1)
112 {
113     dfsDSU(bigChild, node, 1);
114     big[bigChild] = true; // Mark as heavy so `modifyState` ignores it
115 }
116
117 // Add the current node and its light children back to the state
118 modifyState(node, parent, 1);
119
120 // --- ANSWER QUERIES FOR THIS SUBTREE ---
121 for (auto& q : queries[node])
122 {
123     int target_depth = q.first;
124     int query_id = q.second;
125
126     // ? THE MAGIC: If at most 1 bit is set, the characters can form a
127     // palindrome.
128     if (__builtin_popcount(mask[target_depth]) <= 1)
129     {
130         ans[query_id] = "Yes";
131     }
132     else
133     {
134         ans[query_id] = "No";
135     }
136 }
137
138 // Cleanup: Unmark the heavy child
139 if (bigChild != -1)
140 {
141     big[bigChild] = false;
142 }
143
144 // ? CRITICAL: If this whole subtree was a light child, erase it
145 // completely!
146 if (keep == 0)
147 {
148     modifyState(node, parent, -1);
149 }
150 public:
151 DSUOnTree(int n, string input_s, int q_count)
152 {
153     this->n = n;
154     // Pad the string so it becomes 1-indexed to match the nodes
155     this->s = " " + input_s;
156
157     adj.resize(n + 1);
158     queries.resize(n + 1);
159     sz.assign(n + 1, 0);
160
161     lvl.assign(n + 1, 0);
162     mask.assign(n + 1, 0);
163     big.assign(n + 1, false);
164     ans.resize(q_count);
165 }
166
167 void addEdge(int u, int v)
168 {
169     adj[u].push_back(v);
170     adj[v].push_back(u);
171 }
172
173 void addQuery(int node, int depth, int query_index)
174 {
175     queries[node].push_back({depth, query_index});
176 }
177
178 vector<string> process(int root = 1)
179 {
180     dfsSize(root, 0, 1);
181     dfsDSU(root, 0, 0);
182     return ans;
183 }
184
185 void solve()
186 {
187     int n, q;
188     // Example Input: N nodes, Q queries
189     if (!(cin >> n >> q)) return;
190
191     string s;
192     cin >> s;
193
194     DSUOnTree dsu_tree(n, s, q);
195
196     for (int i = 1; i < n; i++)
197     {
198         int u, v;
199         cin >> u >> v;
200         dsu_tree.addEdge(u, v);
201     }
202
203     for (int i = 0; i < q; i++)
204     {
205         int v, h;
206         cin >> v >> h;
207         // 0-indexed queries
208         dsu_tree.addQuery(v, h, i);
209     }
210
211     // Process all queries offline

```

```

212     vector<string> results = dsu_tree.process(1);
213
214     for (const string& res : results)
215     {
216         cout << res << "\n";
217     }
218 }
219
220 int main()
221 {
222     ios_base::sync_with_stdio(false);
223     cin.tie(NULL);
224     // solve();
225     return 0;
226 }

```

## 2.2.4 DSU move erase

```

1 struct DSU
2 {
3     vector<int> _t;
4
5     DSU(int N) : _t(2 * N, -1)
6     {
7         iota(_t.begin(), _t.begin() + N, N);
8     }
9
10    int find(int x)
11    {
12        return _t[x] < 0 ? x : _t[x] = find(_t[x]);
13    }
14
15    bool join(int x, int y)
16    {
17        auto fx = find(x), fy = find(y);
18        if (is_del(x) || is_del(y) || fx == fy)
19            return false;
20        if (_t[fx] > _t[fy])
21            swap(fx, fy);
22        _t[fx] += _t[fy];
23        _t[fy] = fx;
24        return true;
25    }
26
27    bool move(int x, int y)
28    {
29        auto fx = find(x), fy = find(y);
30        if (is_del(x) || is_del(y) || fx == fy)
31            return false;
32        _t[fx]++;
33        _t[fy]--;

```

```

34        _t[x] = fy;
35        return true;
36    }
37
38    bool erase(int x)
39    {
40        auto fx = find(x);
41        if (is_del(x))
42            return false;
43        _t[fx]++;
44        _t[x] = -1;
45        return true;
46    }
47
48    bool is_del(int x)
49    {
50        return _t[x] == -1;
51    }
52
53    int sz(int x)
54    {
55        return -_t[find(x)];
56    }
57 };

```

## 2.2.5 DSU with rollback

```

1 class DSU
2 {
3 private:
4     vector<int> parent, size;
5     stack<int> changes;
6     int forests{};
7
8     void link(int x, int y)
9     {
10        if (size[x] > size[y])
11            swap(x, y);
12        changes.push(x);
13        parent[x] = y;
14        size[y] += size[x];
15    }
16
17 public:
18     DSU(int n)
19     {
20        parent = vector<int>(n), size = vector<int>(n), changes = stack<int>();
21        forests = n;
22        for (int i = 0; i < n; i++)
23            parent[i] = i, size[i] = 1;
24    }

```

```

25
26 int find_set(int x)
27 {
28     while (x != parent[x])
29         x = parent[x];
30     return x;
31 }
32
33 int union_sets(int x, int y)
34 {
35     x = find_set(x), y = find_set(y);
36     if (x != y)
37     {
38         link(x, y);
39         forests--;
40     }
41     return x != y;
42 }
43
44 void rollback()
45 {
46     if (changes.empty())
47         return;
48     int x = changes.top();
49     changes.pop();
50     size[parent[x]] -= size[x];
51     parent[x] = x;
52     forests++;
53 }
54
55 bool same_set(int x, int y)
56 {
57     x = find_set(x), y = find_set(y);
58     return x == y;
59 }
60
61 int get_size(int x)
62 {
63     return size[find_set(x)];
64 }
65
66 int get_forests() const
67 {
68     return forests;
69 }
70 };

```

## 2.2.6 KRT

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: KRUSKAL RECONSTRUCTION TREE (KRT)

```

```

3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Minimax path", "Reachability under weight limit", "Max edge on
    path".
6  * - Classic Scenarios:
7  * A) You are asked to find a path between U and V that MINIMIZES the MAXIMUM
    edge weight along the way.
8  * B) You are at city U. You can only use roads with a weight <= W. How many
    cities can you reach?
9  * - The Magic: During Kruskal's MST algorithm, instead of just merging two
    components, we create a
10 * BRAND NEW artificial node to represent the edge! We make the roots of the
    two components children
11 * of this new artificial node. The result is a tree where the "leaves" are the
    original cities, and
12 * the "internal nodes" are the edges! The weight of the ancestors strictly
    increases as you go up.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization: Pass the number of nodes (1-indexed) and a vector of Edges.
16 *     vector<Edge> edges = {{1, 2, 10}, {2, 3, 20}, {1, 3, 5}};
17 *     KRT krt(3, edges);
18 *
19 * - Minimax Path Query:
20 *     // Find the maximum weight edge on the optimal path between U and V
21 *     long long max_edge = krt.getMaxEdgeOnPath(U, V);
22 *
23 * - Reachability Query:
24 *     // From U, how many cities can I reach using ONLY edges with weight <= W
    ?
25 *     int cities = krt.getReachableLeaves(U, W);
26 *
27 * - Complexity:
28 *     Time: O(E log E) to sort edges and build. O(log N) per query!
29 *     Space: O(N log N) memory for Binary Lifting tables.
30 *
31 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
32 * - Max Spanning Tree instead of Min Spanning Tree?
33 * If you want a path that MAXIMIZES the MINIMUM edge, change the `operator<`
    in the `Edge` struct
34 * to `return w > other.w;`. Then, in `getReachableLeaves`, change the jump
    condition to
35 * `val[p] >= min_w`.
36 * - Disconnected Graphs?
37 * The template natively supports disconnected graphs! `getMaxEdgeOnPath`
    returns `-1` if U and V
38 * are completely disconnected.
39 */
40
41 #include <bits/stdc++.h>
42 using namespace std;
43

```

```

44 struct Edge
45 {
46     int u, v;
47     long long w;
48     // Sort ascending for standard Min-MST KRT
49     bool operator<(const Edge &other) const
50     {
51         return w < other.w;
52     }
53 };
54
55 struct KRT
56 {
57 private:
58     int n, krt_nodes;
59     const int LOG = 21; // Safe for up to 10^6 nodes
60
61     // DSU parent array (crucial that we don't use size/rank here, we MUST link
62     // to the new node)
63     vector<int> dsu_parent;
64
65     // KRT structure
66     vector<long long> val; // The weight of the edge this node represents
67     vector<vector<int>> adj; // Tree edges
68     vector<int> depth; // Depth for LCA
69     vector<vector<int>> up; // Binary Lifting table
70     vector<int> leaves_count; // How many original cities are in this subtree?
71
72     int find(int i)
73     {
74         if (i == dsu_parent[i])
75             return i;
76         // Path compression is perfectly safe here because we only use DSU to
77         // find the current root
78         return dsu_parent[i] = find(dsu_parent[i]);
79     }
80
81     // Standard Binary Lifting DFS
82     void dfs(int u, int p)
83     {
84         up[u][0] = p;
85         for (int i = 1; i < LOG; i++)
86         {
87             up[u][i] = up[up[u][i - 1]][i - 1];
88         }
89
90         // Base case: If it has no children, it's an original city (Leaf)
91         if (adj[u].empty())
92         {
93             leaves_count[u] = 1;
94         }
95         else
96         {
97             for (int v : adj[u])
98             {
99                 if (v != p)
100                 {
101                     depth[v] = depth[u] + 1;
102                     dfs(v, u);
103                     // Bubble up the leaf count
104                     leaves_count[u] += leaves_count[v];
105                 }
106             }
107         }
108     }
109
110 public:
111     KRT(int n, vector<Edge> &edges)
112     {
113         this->n = n;
114         // The KRT will have exactly N original leaves, and at most N-1
115         // artificial edge-nodes
116         int max_nodes = 2 * n + 5;
117
118         dsu_parent.resize(max_nodes);
119         iota(dsu_parent.begin(), dsu_parent.end(), 0);
120
121         val.assign(max_nodes, 0);
122         adj.resize(max_nodes);
123         leaves_count.assign(max_nodes, 0);
124         depth.assign(max_nodes, 0);
125         up.assign(max_nodes, vector<int>(LOG, 0));
126
127         // 1. Sort the edges to process in Kruskal's order
128         sort(edges.begin(), edges.end());
129
130         krt_nodes = n; // Nodes 1 to N are the original cities
131
132         // 2. Build the Tree
133         for (auto &e : edges)
134         {
135             int ru = find(e.u);
136             int rv = find(e.v);
137
138             if (ru != rv)
139             {
140                 krt_nodes++; // Create a new artificial node for this edge
141                 val[krt_nodes] = e.w;
142
143                 // Connect the new node to the roots of the two components
144                 adj[krt_nodes].push_back(ru);
145                 adj[krt_nodes].push_back(rv);

```

```

145                                     196
146         // ? THE MAGIC: The new node becomes the ultimate DSU root for
147         both components!
148         dsu_parent[ru] = krt_nodes;
149         dsu_parent[rv] = krt_nodes;
150     }
151 }
152 // 3. Process the DFS for LCA and Subtree sizes
153 // We loop through all nodes in case the graph is a disconnected forest
154 for (int i = 1; i <= krt_nodes; i++)
155 {
156     if (find(i) == i)
157     {
158         // We found a supreme root! Run DFS downwards.
159         dfs(i, 0);
160     }
161 }
162 }
163
164 // Returns the max edge weight on the path between U and V
165 long long getMaxEdgeOnPath(int u, int v)
166 {
167     if (find(u) != find(v))
168         return -1; // They are in disconnected components
169     if (u == v)
170         return 0; // Same node
171
172     // Standard LCA Logic
173     if (depth[u] < depth[v])
174         swap(u, v);
175     int diff = depth[u] - depth[v];
176
177     for (int i = 0; i < LOG; i++)
178     {
179         if ((diff >> i) & 1)
180             u = up[u][i];
181     }
182
183     if (u == v)
184         return val[u]; // One is an ancestor of the other
185
186     for (int i = LOG - 1; i >= 0; i--)
187     {
188         if (up[u][i] != up[v][i])
189         {
190             u = up[u][i];
191             v = up[v][i];
192         }
193     }
194
195     // The LCA node IS the edge that connected them!
196
197     int lca = up[u][0];
198     return val[lca];
199 }
200
201 // Returns how many original cities can be reached from U using edges <=
202 // max_w
203 int getReachableLeaves(int u, long long max_w)
204 {
205     // Jump up the tree as long as the parent edge weight is valid
206     for (int i = LOG - 1; i >= 0; i--)
207     {
208         int p = up[u][i];
209         // Parent 0 means we hit the root boundaries.
210         if (p != 0 && val[p] <= max_w)
211         {
212             u = p; // Safe to jump!
213         }
214     }
215     // Once we reach the highest valid ancestor, its subtree contains ALL
216     // reachable cities!
217     return leaves_count[u];
218 }
219
220 void solve()
221 {
222     int n, m, q;
223     // Read N nodes, M edges, Q queries
224     if (!(cin >> n >> m >> q))
225         return;
226
227     vector<Edge> edges(m);
228     for (int i = 0; i < m; i++)
229     {
230         cin >> edges[i].u >> edges[i].v >> edges[i].w;
231     }
232
233     KRT krt(n, edges);
234
235     while (q--)
236     {
237         int type;
238         cin >> type;
239         if (type == 1)
240         {
241             int u, v;
242             cin >> u >> v;
243             long long ans = krt.getMaxEdgeOnPath(u, v);
244             if (ans == -1)
245                 cout << "Disconnected\n";
246             else
247                 cout << ans << "\n";
248         }
249     }
250 }

```

```

246     }
247     else
248     {
249         int u;
250         long long w;
251         cin >> u >> w;
252         cout << krt.getReachableLeaves(u, w) << "\n";
253     }
254 }
255 }
256
257 int main()
258 {
259     ios_base::sync_with_stdio(false);
260     cin.tie(NULL);
261     // solve();
262     return 0;
263 }

```

### 2.2.7 Offline Dynamic connectivity

```

1 class DSU
2 {
3 private:
4     vector<int> parent, size;
5     stack<int> changes;
6     int forests{};
7
8     void link(int x, int y)
9     {
10         if (size[x] > size[y])
11             swap(x, y);
12         changes.push(x);
13         parent[x] = y;
14         size[y] += size[x];
15     }
16
17 public:
18     DSU(int n)
19     {
20         parent = vector<int>(n), size = vector<int>(n), changes = stack<int>();
21         forests = n;
22         for (int i = 0; i < n; i++)
23             parent[i] = i, size[i] = 1;
24     }
25
26     int find_set(int x)
27     {
28         while (x != parent[x])
29             x = parent[x];
30         return x;

```

```

31     }
32
33     int union_sets(int x, int y)
34     {
35         x = find_set(x), y = find_set(y);
36         if (x != y)
37         {
38             link(x, y);
39             forests--;
40         }
41         return x != y;
42     }
43
44     void rollback()
45     {
46         if (changes.empty())
47             return;
48         int x = changes.top();
49         changes.pop();
50         size[parent[x]] -= size[x];
51         parent[x] = x;
52         forests++;
53     }
54
55     bool same_set(int x, int y)
56     {
57         x = find_set(x), y = find_set(y);
58         return x == y;
59     }
60
61     int get_size(int x)
62     {
63         return size[find_set(x)];
64     }
65
66     int get_forests() const
67     {
68         return forests;
69     }
70 };
71
72 struct edge
73 {
74     ll from, to;
75
76     edge()
77     {
78     }
79
80     edge(ll from, ll to) : from(from), to(to)
81     {
82     }

```

```

83
84     bool operator==(const edge &e)
85     {
86         return from == e.from && to == e.to;
87     }
88 };
89
90 struct Node
91 {
92     int val;
93
94     Node()
95     {
96         val = 0;
97     }
98
99     Node(int val) : val(val)
100    {
101    }
102
103    void change(int x)
104    {
105        val = x;
106    }
107 };
108
109 struct segmentTree
110 {
111     int sz;
112     vector<Node> seg;
113     vector<vector<edge>> edges;
114     DSU d;
115
116     segmentTree(int n) : d(n)
117     {
118         sz = 1;
119         while (sz < n)
120             sz *= 2;
121         seg.assign(2 * sz, Node());
122         edges.assign(2 * sz, {});
123     }
124
125     void set(int l, int r, int node, int lx, int rx, edge edge)
126     {
127         if (lx >= l && rx <= r)
128         {
129             edges[node].push_back(edge);
130             return;
131         }
132         if (rx <= l || lx >= r)
133             return;
134         int mid = (lx + rx) / 2;

```

```

135         set(l, r, 2 * node + 1, lx, mid, edge);
136         set(l, r, 2 * node + 2, mid, rx, edge);
137     }
138
139     void set(int l, int r, edge edge)
140     {
141         set(l, r, 0, 0, sz, edge);
142     }
143
144     void get(int node, int lx, int rx)
145     {
146         int cnt{};
147         for (auto &[x, y] : edges[node])
148             cnt += d.union_sets(x, y);
149         if (rx - lx == 1)
150         {
151             // answer queries
152         }
153         else
154         {
155             int mid = (lx + rx) / 2;
156             get(2 * node + 1, lx, mid);
157             get(2 * node + 2, mid, rx);
158         }
159         while (cnt--)
160             d.rollback();
161     }
162
163     void get()
164     {
165         return get(0, 0, sz);
166     }
167 };

```

## 2.2.8 Partially persistent DSU

```

1 /**
2  *  THE ULTIMATE CP TEMPLATE BLUEPRINT: PARTIALLY PERSISTENT DSU
3  *
4  *  1.  WHAT PROBLEM DOES THIS SOLVE?
5  *  - Keywords: "At what time did U and V connect?", "Were they connected after K
6  *  - Classic Scenarios: You have a graph where edges are slowly being built over
7  *  You get queries asking if two cities were connected at a specific year in
8  *  - The Magic: Because we ONLY use Union-by-Size (and strictly ban path
9  *  naturally forms a tree with a maximum depth of  $O(\log N)$ . By simply recording
   the `time`

```



```

10 * an edge was formed on the child node pointing to the parent, we can "time
    travel".
11 * To find the root at time T, we just climb the parent pointers, but we STOP
    climbing if
12 * the edge above us was built AFTER time T!
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization: Pass the number of nodes (1-indexed).
16 *   PartiallyPersistentDSU pp_dsu(N);
17 *
18 * - Updates: Merge U and V at time T. (T must be strictly increasing across
    calls).
19 *   pp_dsu.unite(U, V, T);
20 *
21 * - Queries:
22 *   // Were U and V connected at time T?
23 *   bool connected = pp_dsu.isConnected(U, V, T);
24 *
25 *   // What is the EXACT time U and V became connected?
26 *   int time_joined = pp_dsu.timeConnected(U, V);
27 *
28 * - Complexity:
29 *   Time:  $O(\log N)$  per merge and per query.
30 *   Space:  $O(N)$  memory.
31 *
32 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
33 * - 0-indexed nodes?
34 * The template naturally allocates `n + 1` space. It works flawlessly for both
    0-indexed
35 * and 1-indexed inputs without changing any code!
36 * - Getting the size of a component at time T?
37 * This requires a bit more work. You would need to make `sz` a vector of `
    vector<pair<int, int>>`
38 * to store the history of sizes: `sz[root].push_back({time, new_size})`, and
    then binary search it.
39 * If you just need standard connectivity, this template is perfectly
    lightweight.
40 */
41
42 #include <bits/stdc++.h>
43 using namespace std;
44
45 struct PartiallyPersistentDSU
46 {
47 private:
48     vector<int> parent;
49     vector<int> sz;
50     vector<int> time_joined; // The time this node became a child of its parent
51     const int INF = 2e9 + 7;
52
53 public:
54     PartiallyPersistentDSU(int n)
55     {
56         parent.resize(n + 1);
57         sz.assign(n + 1, 1);
58         time_joined.assign(n + 1, INF);
59         iota(parent.begin(), parent.end(), 0);
60     }
61
62     // ? CRITICAL: Notice there is NO path compression `parent[i] = find(...)`!
63     // Path compression would destroy the historical edges.
64     int find(int i, int t)
65     {
66         // We stop climbing if we hit a root, or if the edge above us hasn't been
67         // built yet at time `t`
68         if (i == parent[i] || time_joined[i] > t)
69         {
70             return i;
71         }
72         return find(parent[i], t);
73     }
74
75     // Merge u and v at time t. (Assumes t is strictly increasing as you feed
76     // edges)
77     bool unite(int u, int v, int t)
78     {
79         // When merging, we only care about the present moment, so we pass INF as
80         // time
81         int root_u = find(u, INF);
82         int root_v = find(v, INF);
83
84         if (root_u == root_v)
85         {
86             return false;
87         }
88
89         // Union by Size ensures the tree depth stays  $O(\log N)$ 
90         if (sz[root_u] < sz[root_v])
91         {
92             swap(root_u, root_v);
93         }
94
95         parent[root_v] = root_u;
96         sz[root_u] += sz[root_v];
97
98         // ? THE MAGIC: We record EXACTLY when root_v surrendered to root_u
99         time_joined[root_v] = t;
100
101         return true;
102     }
103
104     // Check if u and v were in the same component at time t
105     bool isConnected(int u, int v, int t)
106     {
107         return find(u, t) == find(v, t);
108     }
109 }

```

```

104
105 // Find the exact time u and v became connected. Returns INF if they never
    connect.
106 int timeConnected(int u, int v)
107 {
108     if (!isConnected(u, v, INF))
109         return INF; // Never connected
110
111     // Binary search the timeline to find the exact moment of connection!
112     int l = 0, r = INF - 1;
113     int ans = INF;
114
115     while (l <= r)
116     {
117         int mid = l + (r - l) / 2;
118         if (isConnected(u, v, mid))
119         {
120             ans = mid;
121             r = mid - 1; // Try to find an earlier time
122         }
123         else
124         {
125             l = mid + 1; // We need more time
126         }
127     }
128     return ans;
129 }
130 };
131
132 void solve()
133 {
134     int n, m, q;
135     // Example Input: N nodes, M edges added sequentially, Q queries
136     if (!(cin >> n >> m >> q))
137         return;
138
139     PartiallyPersistentDSU pp_dsu(n);
140
141     for (int i = 1; i <= m; i++)
142     {
143         int u, v;
144         cin >> u >> v;
145         // Assume edges are added at time `i`
146         pp_dsu.unite(u, v, i);
147     }
148
149     while (q--)
150     {
151         int type, u, v;
152         cin >> type >> u >> v;
153
154         if (type == 1)

```

```

155     {
156         // Were u and v connected at time T?
157         int t;
158         cin >> t;
159         cout << (pp_dsu.isConnected(u, v, t) ? "YES" : "NO") << "\n";
160     }
161     else
162     {
163         // At what time did u and v finally connect?
164         int t = pp_dsu.timeConnected(u, v);
165         if (t > m)
166             cout << "Not connected\n";
167         else
168             cout << "Connected at edge " << t << "\n";
169     }
170 }
171 }
172
173 int main()
174 {
175     ios_base::sync_with_stdio(false);
176     cin.tie(NULL);
177     // solve();
178     return 0;
179 }

```

## 2.2.9 Small to large

```

1 // Small to large merging
2
3 vector<int> colors;
4 GRAPH graph;
5
6 set<int> dfs(int u, int p)
7 {
8     set<int> st{colors[u]};
9     for (auto &ch : graph[u])
10     {
11         if (ch == p)
12             continue;
13         set<int> child_st = dfs(ch, u);
14         if (int(child_st.size()) > int(st.size()))
15             swap(st, child_st);
16         st.merge(child_st);
17     }
18     return st;
19 }

```

## 2.3 Fenwick Tree

### 2.3.1 2D bit

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: 2D FENWICK TREE (POINT UPDATE & RANGE
   QUERY)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "2D Grid", "Update a single cell", "Sum of subgrid", "Rectangle
   Queries".
6  * - Classic Scenarios: You have an N x M grid. You need to repeatedly change the
   value of
7  * a specific cell (X, Y). Then, you need to query the sum of all elements
   inside a
8  * subgrid bounded by top-left (X1, Y1) and bottom-right (X2, Y2).
9  * - The Magic: Replaces a bulky, complex 2D Segment Tree. It computes 2D prefix
   sums dynamically
   in O(log N * log M) time using ridiculously short and fast bitwise loops.
10 *
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Initialization: Pass the dimensions (rows, cols) to the constructor.
14 * (Note: Fenwick Trees MUST be 1-indexed. The bounds are [1, N] and [1, M]).
15 * FenwickTree2D fenwick(N, M);
16 *
17 * - Point Updates (1-indexed (x, y)):
18 * // Add 'val' to the cell at row X, col Y
19 * fenwick.update(X, Y, val);
20 *
21 * - Range Queries (Inclusive 1-indexed [x1, x2] and [y1, y2]):
22 * // Get the sum of all cells in the rectangle
23 * long long total = fenwick.query(x1, y1, x2, y2);
24 *
25 * - Complexity:
26 * Time: O(log N * log M) per update and query.
27 * Space: O(N * M) memory.
28 *
29 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
30 * - "Point Set" instead of "Point Add"?
31 * Fenwick Trees can ONLY add values. If the problem asks you to REPLACE the
   value at (X, Y)
32 * with `V`, you must maintain a 2D array `grid` of the current values.
33 * Then, do:
34 * long long diff = V - grid[X][Y];
35 * fenwick.update(X, Y, diff);
36 * grid[X][Y] = V;
37 *
38 * - 0-Indexed Input?
39 * Simply add +1 to all coordinates before passing them into `update` and `
   query`.
40 * NEVER change the internal 1-based indexing of the BIT.
41 */
42
43 #include <bits/stdc++.h>
44 using namespace std;
45
46 struct FenwickTree2D
47 {
48 private:
49     int n, m;
50     vector<vector<long long>> bit;
51
52     // Get the sum of the rectangle from (1, 1) to (x, y)
53     long long queryPrefix(int x, int y)
54     {
55         long long sum = 0;
56         for (int i = x; i > 0; i -= i & -i)
57         {
58             for (int j = y; j > 0; j -= j & -j)
59             {
60                 sum += bit[i][j];
61             }
62         }
63         return sum;
64     }
65
66 public:
67     FenwickTree2D(int rows, int cols)
68     {
69         n = rows;
70         m = cols;
71         // 1-indexed, so allocate size + 1
72         bit.assign(n + 1, vector<long long>(m + 1, 0));
73     }
74
75     // Add `val` to the cell at (x, y)
76     void update(int x, int y, long long val)
77     {
78         for (int i = x; i <= n; i += i & -i)
79         {
80             for (int j = y; j <= m; j += j & -j)
81             {
82                 bit[i][j] += val;
83             }
84         }
85     }
86
87     // Get the sum of the rectangle from (x1, y1) to (x2, y2)
88     long long query(int x1, int y1, int x2, int y2)
89     {
90         // Standard 2D Inclusion-Exclusion Principle
91         return queryPrefix(x2, y2) - queryPrefix(x1 - 1, y2) - queryPrefix(x2, y1
           - 1) + queryPrefix(x1 - 1, y1 - 1);
92     }

```

```

93 };
94
95 void solve()
96 {
97     int n, m, q;
98     // Example Input Format: N rows, M cols, Q queries
99     if (!(cin >> n >> m >> q))
100         return;
101
102     FenwickTree2D fenwick(n, m);
103
104     // Optional: Maintain original grid if "Point Set" operations are required
105     // vector<vector<long long>> grid(n + 1, vector<long long>(m + 1, 0));
106
107     while (q--)
108     {
109         int type;
110         cin >> type;
111
112         if (type == 1)
113         { // Point Add
114             int x, y;
115             long long val;
116             cin >> x >> y >> val;
117             fenwick.update(x, y, val);
118         }
119         else if (type == 2)
120         { // Rectangle Query
121             int x1, y1, x2, y2;
122             cin >> x1 >> y1 >> x2 >> y2;
123             cout << fenwick.query(x1, y1, x2, y2) << "\n";
124         }
125     }
126 }
127
128 int main()
129 {
130     // Fast I/O
131     ios_base::sync_with_stdio(false);
132     cin.tie(NULL);
133     solve();
134     return 0;
135 }

```

### 2.3.2 2D bit rangeUpdate rangeQuery

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: 2D FENWICK TREE (RANGE UPDATE & QUERY)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?

```

```

5  * - Keywords: "2D Grid", "Add X to subgrid", "Sum of subgrid", "Rectangle
6  * - Classic Scenarios: You are given an N x M grid (like a terrain map or matrix
7  * You need to repeatedly add a value V to all cells in a rectangle from (X1,
8  * Then, you need to query the sum of all cells in a rectangle from (X1, Y1) to
9  * - The Magic: A 2D Segment tree with Lazy Propagation is notoriously difficult
10 * gigantic amounts of memory. This 2D Fenwick Tree uses 4 parallel grids to
11 * track the inclusion-exclusion bounds of the updates. It uses a fraction of
12 * and runs blazingly fast in O(log N * log M) time.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization: Pass the dimensions (rows, cols) to the constructor.
16 * (Note: Fenwick Trees MUST be 1-indexed. The bounds are [1, N] and [1, M]).
17 *     FenwickTree2DRange fenwick(N, M);
18 *
19 * - Range Updates (Inclusive 1-indexed [x1, x2] and [y1, y2]):
20 *     // Add 'val' to every cell in the rectangle bounded by top-left (x1, y1)
21 *     fenwick.update(x1, y1, x2, y2, val);
22 *
23 * - Range Queries (Inclusive 1-indexed [x1, x2] and [y1, y2]):
24 *     // Get the sum of all cells in the rectangle
25 *     long long total = fenwick.query(x1, y1, x2, y2);
26 *
27 * - Complexity:
28 *     Time: O(log N * log M) per update and query.
29 *     Space: O(N * M) memory.
30 *
31 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
32 * - 0-Indexed Input?
33 * If your problem gives coordinates as 0-indexed (e.g., (0, 0) to (N-1, M-1)),
34 * simply
35 * add +1 to all coordinates before passing them into `update` and `query`.
36 * NEVER change the internal 1-based indexing of the BIT, as `index & -index`
37 * fails on 0!
38 * - Modulo Arithmetic?
39 * If the problem asks for the answer modulo 10^9+7, wrap every `+=` and `-=`
40 * inside the
41 * `add` and `queryPrefix` functions with modulo logic: `res = (res % MOD + MOD
42 * ) % MOD;`
43 */
44 #include <bits/stdc++.h>
45 using namespace std;
46
47 struct FenwickTree2DRange

```

```

45 {
46 private:
47     int n, m;
48
49     // We need 4 parallel 2D BITs to mathematically calculate the 2D Prefix Sum
50     //  $D[i][j], D[i][j] * i, D[i][j] * j, D[i][j] * i * j$ 
51     vector<vector<long long>> bit1, bit2, bit3, bit4;
52
53     // Add value to coordinate (x, y) across all 4 mathematical states
54     void add(int x, int y, long long val)
55     {
56         for (int i = x; i <= n; i += i & -i)
57         {
58             for (int j = y; j <= m; j += j & -j)
59             {
60                 bit1[i][j] += val;
61                 bit2[i][j] += val * x;
62                 bit3[i][j] += val * y;
63                 bit4[i][j] += val * x * y;
64             }
65         }
66     }
67
68     // Get the sum of the rectangle from (1, 1) to (x, y)
69     long long queryPrefix(int x, int y)
70     {
71         long long res = 0;
72         for (int i = x; i > 0; i -= i & -i)
73         {
74             for (int j = y; j > 0; j -= j & -j)
75             {
76                 // The mathematical expansion of 2D Difference Array Prefix Sum
77                 //  $Sum = D * (x+1)(y+1) - (D*x) * (y+1) - (D*y) * (x+1) + (D*x*y)$ 
78                 res += bit1[i][j] * (x + 1) * (y + 1) - bit2[i][j] * (y + 1) -
79                     bit3[i][j] * (x + 1) + bit4[i][j];
80             }
81         }
82         return res;
83     }
84 public:
85     FenwickTree2DRange(int rows, int cols)
86     {
87         n = rows;
88         m = cols;
89         // 1-indexed, so allocate size + 1
90         bit1.assign(n + 1, vector<long long>(m + 1, 0));
91         bit2.assign(n + 1, vector<long long>(m + 1, 0));
92         bit3.assign(n + 1, vector<long long>(m + 1, 0));
93         bit4.assign(n + 1, vector<long long>(m + 1, 0));
94     }
95
96     // Add `val` to the rectangle from (x1, y1) to (x2, y2)
97     void update(int x1, int y1, int x2, int y2, long long val)
98     {
99         // 2D Difference Array Inclusion-Exclusion
100         add(x1, y1, val);
101         add(x2 + 1, y1, -val);
102         add(x1, y2 + 1, -val);
103         add(x2 + 1, y2 + 1, val);
104     }
105
106     // Get the sum of the rectangle from (x1, y1) to (x2, y2)
107     long long query(int x1, int y1, int x2, int y2)
108     {
109         // 2D Prefix Sum Inclusion-Exclusion
110         return queryPrefix(x2, y2) - queryPrefix(x1 - 1, y2) - queryPrefix(x2, y1
111             - 1) + queryPrefix(x1 - 1, y1 - 1);
112     }
113
114     void solve()
115     {
116         int n, m, q;
117         // Example Input Format: N rows, M cols, Q queries
118         if (!(cin >> n >> m >> q))
119             return;
120
121         FenwickTree2DRange fenwick(n, m);
122
123         while (q--)
124         {
125             int type;
126             cin >> type;
127
128             if (type == 1)
129             { // Range Add
130                 int x1, y1, x2, y2;
131                 long long val;
132                 cin >> x1 >> y1 >> x2 >> y2 >> val;
133                 fenwick.update(x1, y1, x2, y2, val);
134             }
135             else if (type == 2)
136             { // Range Query
137                 int x1, y1, x2, y2;
138                 cin >> x1 >> y1 >> x2 >> y2;
139                 cout << fenwick.query(x1, y1, x2, y2) << "\n";
140             }
141         }
142     }
143
144     int main()
145     {
146         // Crucial fast I/O setup for dense CP test cases

```

```

147 ios_base::sync_with_stdio(false);
148 cin.tie(NULL);
149 solve();
150 return 0;
151 }

```

### 2.3.3 Bit rangeUpdate rangeQuery

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: 1D FENWICK TREE (RANGE UPDATE & QUERY)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Range addition", "Subarray sum", "Add X to range [L, R]".
6  * - Classic Scenarios: You are given an array. You need to repeatedly add a
   value V
7  * to all elements from index L to R. Then, you need to query the sum of all
   elements
8  * from index L to R.
9  * - The Magic: Normally, you need a Lazy Segment Tree for this. However, by
   using
10 * two parallel Fenwick Trees, we can track the difference array and its index
   multiplier
11 * to mathematically simulate Lazy Propagation. It is ridiculously fast and
   takes zero
12 * extra memory.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization: Pass the size of the array to the constructor.
16 * (Note: Fenwick Trees MUST be 1-indexed. The bounds are [1, N]).
17 * FenwickTreeRange fenwick(N);
18 *
19 * - Range Updates (Inclusive 1-indexed [L, R]):
20 * // Add 'val' to every cell from index L to R
21 * fenwick.update(L, R, val);
22 *
23 * - Range Queries (Inclusive 1-indexed [L, R]):
24 * // Get the sum of all cells from index L to R
25 * long long total = fenwick.query(L, R);
26 *
27 * - Complexity:
28 * Time: O(log N) per update and query.
29 * Space: O(N) memory.
30 *
31 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
32 * - Initial Array Values?
33 * If the problem gives you an initial array `A`, you don't need a special
   build function.
34 * Simply loop through the array and call `fenwick.update(i, i, A[i]);` for
   each element.
35 * - 0-Indexed Input?

```

```

36 * If the problem uses 0-based indices, add +1 to L and R before passing them
   to
37 * `update` or `query`. NEVER change the internal 1-based logic, as `idx & -idx`
   breaks on 0.
38 * - Modulo Arithmetic?
39 * If the problem asks for the sum modulo  $10^9+7$ , wrap the `+=` and `-=`
   assignments inside
40 * `add` and `queryPrefix` with modulo logic: `res = (res % MOD + MOD) % MOD;`.
41 */
42
43 #include <bits/stdc++.h>
44 using namespace std;
45
46 struct FenwickTreeRange
47 {
48 private:
49     int n;
50
51     // bit1 tracks the standard difference array: D[i]
52     // bit2 tracks the difference array multiplied by the index: D[i] * i
53     vector<long long> bit1, bit2;
54
55     // Point add to both parallel BITS
56 void add(int idx, long long val)
57 {
58     // We use the actual `idx` where the value was inserted for the bit2
   multiplier
59     for (int i = idx; i <= n; i += i & -i)
60     {
61         bit1[i] += val;
62         bit2[i] += val * idx;
63     }
64 }
65
66 // Get the sum from index 1 to idx
67 long long queryPrefix(int idx)
68 {
69     long long res = 0;
70     for (int i = idx; i > 0; i -= i & -i)
71     {
72         // The magic algebraic expansion: Sum = (idx + 1) * D[i] - (D[i] * i)
73         res += bit1[i] * (idx + 1) - bit2[i];
74     }
75     return res;
76 }
77
78 public:
79     FenwickTreeRange(int size)
80     {
81         n = size;
82         bit1.assign(n + 1, 0);
83         bit2.assign(n + 1, 0);

```

```

84     }
85
86     // Add `val` to the inclusive range [l, r]
87     void update(int l, int r, long long val)
88     {
89         // Standard difference array inclusion-exclusion logic
90         add(l, val);
91         add(r + 1, -val);
92     }
93
94     // Get the sum of the inclusive range [l, r]
95     long long query(int l, int r)
96     {
97         return queryPrefix(r) - queryPrefix(l - 1);
98     }
99 };
100
101 void solve()
102 {
103     int n, q;
104     // Example Input Format: N elements, Q queries
105     if (!(cin >> n >> q))
106         return;
107
108     FenwickTreeRange fenwick(n);
109
110     while (q--)
111     {
112         int type;
113         cin >> type;
114
115         if (type == 1)
116         { // Range Add
117             int l, r;
118             long long val;
119             cin >> l >> r >> val;
120             fenwick.update(l, r, val);
121         }
122         else if (type == 2)
123         { // Range Query
124             int l, r;
125             cin >> l >> r;
126             cout << fenwick.query(l, r) << "\n";
127         }
128     }
129 }
130
131 int main()
132 {
133     // Fast I/O for competitive programming
134     ios_base::sync_with_stdio(false);
135     cin.tie(NULL);

```

```

136     solve();
137     return 0;
138 }

```

### 2.3.4 Fenwick tree

```

1 template <typename T>
2 struct FenwickTree
3 {
4     int n;
5     vector<T> tree;
6
7     // 1. Initialize an empty tree of size n (all zeros)
8     FenwickTree(int n)
9     {
10         this->n = n;
11         tree.assign(n + 1, 0);
12     }
13
14     // 2. Fast O(N) initialization from an existing array
15     FenwickTree(const vector<T> &a)
16     {
17         n = a.size();
18         tree.assign(n + 1, 0);
19
20         // Step 1: Copy values in (shifting to 1-based indexing)
21         for (int i = 0; i < n; i++)
22         {
23             tree[i + 1] = a[i];
24         }
25
26         // Step 2: Propagate the sums forward to the immediate parent
27         for (int i = 1; i <= n; i++)
28         {
29             int parent = i + (i & -i);
30             if (parent <= n)
31             {
32                 tree[parent] += tree[i];
33             }
34         }
35     }
36
37     void add(int i, T delta)
38     {
39         for (; i <= n; i += (i & -i))
40         {
41             tree[i] += delta;
42         }
43     }
44
45     T query(int i)

```

```

46 {
47     T sum = 0;
48     for (; i > 0; i -= (i & -i))
49     {
50         sum += tree[i];
51     }
52     return sum;
53 }
54
55 T query(int l, int r)
56 {
57     return query(r) - query(l - 1);
58 }
59 };

```

### 2.3.5 Lower bound on bit

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: LOWER BOUND ON BIT (BINARY LIFTING)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Find the K-th smallest element", "First index with prefix sum >= X".
6  * - Classic Scenarios: You have an array of frequencies (how many times a number appears).
7  * You need to dynamically add/remove numbers and frequently ask: "What is the K-th
8  * smallest number currently in my set?"
9  * - The Magic: Normally, you'd use a C++ PBDS (Policy Based Data Structure) for this.
10 * If PBDS is banned, or you need raw execution speed, this algorithm does a
11 * search directly ON the Fenwick tree's internal structure. It takes exactly
12 * steps instead of  $O(\log^2 N)$ .
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization: Pass the size of the array to the constructor.
16 * FenwickTreeLB fenwick(N);
17 *
18 * - Standard Point Updates (1-indexed):
19 * // Add 'val' to index 'idx'
20 * fenwick.update(idx, val);
21 *
22 * - The K-th Element Query (Lower Bound):
23 * // Find the smallest index where the prefix sum is >= V
24 * int ans_idx = fenwick.lower_bound(V);
25 *
26 * - Complexity:
27 * Time:  $O(\log N)$  per update and lower_bound query.
28 * Space:  $O(N)$  memory.

```

```

29 *
30 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
31 * - CRITICAL RULE: MONOTONICITY
32 * This function ONLY works if the original array contains ONLY non-negative
33 * numbers!
34 * If you add negative numbers, the prefix sums go up and down randomly. Binary
35 * search
36 * (and therefore this binary lifting trick) will completely fail.
37 * - Out of Bounds Check:
38 * If you search for a value `V` that is larger than the total sum of the
39 * entire array,
40 * the function will return `N + 1`. Always check `if (ans_idx > N)` if you
41 * aren't sure
42 * the sum exists!
43 */
44
45 #include <bits/stdc++.h>
46 using namespace std;
47
48 struct FenwickTreeLB
49 {
50 private:
51     int n;
52     int LOGN;
53     vector<long long> bit;
54
55 public:
56     FenwickTreeLB(int size)
57     {
58         n = size;
59         bit.assign(n + 1, 0);
60
61         // Find the highest power of 2 that is <= n
62         LOGN = 0;
63         while ((1 << (LOGN + 1)) <= n)
64         {
65             LOGN++;
66         }
67     }
68
69     // Add `val` to the cell at `idx`
70     void update(int idx, long long val)
71     {
72         for (; idx <= n; idx += idx & -idx)
73         {
74             bit[idx] += val;
75         }
76     }
77
78     // Get standard prefix sum up to `idx`
79     long long query(int idx)
80     {
81

```



```

77     long long sum = 0;
78     for (; idx > 0; idx -= idx & -idx)
79     {
80         sum += bit[idx];
81     }
82     return sum;
83 }
84
85 // ? THE MAGIC: Binary Lifting directly on the Fenwick Tree
86 // Returns the 1-based index of the FIRST element whose prefix sum is >= v
87 int lower_bound(long long v)
88 {
89     if (v <= 0)
90         return 1; // Edge case: Target is already reached at the start
91
92     int idx = 0;
93     long long current_sum = 0;
94
95     // Jump down from the highest power of 2 to 1
96     for (int i = LOGN; i >= 0; i--)
97     {
98         int next_idx = idx + (1 << i);
99
100        // If taking this jump keeps us inside the array bounds,
101        // AND the sum AT this jump is still STRICTLY LESS than our target
102
103        if (next_idx <= n && current_sum + bit[next_idx] < v)
104        {
105            // We commit to the jump!
106            idx = next_idx;
107            current_sum += bit[next_idx];
108        }
109
110        // `idx` is now the largest index where the sum is STRICTLY LESS than `v`
111
112        // Therefore, `idx + 1` is the first index where the sum is >= `v`!
113        return idx + 1;
114 }
115
116 void solve()
117 {
118     int n, q;
119     // Example Input Format: Array of size N, Q queries
120     if (!(cin >> n >> q))
121         return;
122
123     FenwickTreeLB fenwick(n);
124
125     while (q--)
126     {

```

```

127         int type;
128         cin >> type;
129
130         if (type == 1)
131         { // Point Add
132             int idx;
133             long long val;
134             cin >> idx >> val;
135             fenwick.update(idx, val);
136         }
137         else if (type == 2)
138         { // Lower Bound Query
139             long long target_sum;
140             cin >> target_sum;
141
142             int ans = fenwick.lower_bound(target_sum);
143
144             if (ans > n)
145             {
146                 cout << "Target sum not reached!\n";
147             }
148             else
149             {
150                 cout << "Index: " << ans << "\n";
151             }
152         }
153     }
154 }
155
156 int main()
157 {
158     // Fast I/O
159     ios_base::sync_with_stdio(false);
160     cin.tie(NULL);
161     solve();
162     return 0;
163 }

```

## 2.4 HLD

### 2.4.1 HLD

```

1 struct Node
2 {
3     int val;
4
5     Node()
6     {
7         val = -00;
8     }
9

```

```

10     Node(int val) : val(val)
11     {
12     }
13
14     void change(int x)
15     {
16         val = x;
17     }
18 };
19
20 struct segmentTree
21 {
22     int sz;
23     vector<Node> seg;
24
25     segmentTree(int n)
26     {
27         sz = 1;
28         while (sz < n)
29             sz *= 2;
30         seg.assign(2 * sz, Node());
31     }
32
33     void init(vector<int> &arr, int node, int lx, int rx)
34     {
35         if (rx - lx == 1)
36         {
37             if (lx < int(arr.size()))
38                 seg[node] = Node(arr[lx]);
39             return;
40         }
41         int mid = (lx + rx) / 2;
42         init(arr, 2 * node + 1, lx, mid);
43         init(arr, 2 * node + 2, mid, rx);
44         seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
45     }
46
47     void init(vector<int> &arr)
48     {
49         init(arr, 0, 0, sz);
50     }
51
52     Node merge(Node &lf, Node &rt)
53     {
54         Node ans = Node();
55         ans.val = max(lf.val, rt.val);
56         return ans;
57     }
58
59     void set(int idx, int val, int node, int lx, int rx)
60     {
61         if (rx - lx == 1)

```

```

62         {
63             seg[node].change(val);
64             return;
65         }
66         int mid = (lx + rx) / 2;
67         if (idx < mid)
68             set(idx, val, 2 * node + 1, lx, mid);
69         else
70             set(idx, val, 2 * node + 2, mid, rx);
71         seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
72     }
73
74     void set(int idx, int val)
75     {
76         set(idx, val, 0, 0, sz);
77     }
78
79     Node get(int l, int r, int node, int lx, int rx)
80     {
81         if (lx >= l && rx <= r)
82             return seg[node];
83         if (rx <= l || lx >= r)
84             return Node();
85         int mid = (lx + rx) / 2;
86         Node lf = get(l, r, 2 * node + 1, lx, mid);
87         Node ri = get(l, r, 2 * node + 2, mid, rx);
88         return merge(lf, ri);
89     }
90
91     int get(int l, int r)
92     {
93         return get(l, r, 0, 0, sz).val;
94     }
95 };
96
97 struct HLD
98 {
99 private:
100     vector<int> sz, par, dep, tin, tout, head;
101     vector<vector<int>>> graph;
102     vector<int> a;
103     int timer{1};
104     segmentTree seg;
105
106 public:
107     HLD(int n, GRAPH graph, vector<int> &v) : graph(std::move(graph)), seg(n + 1)
108     {
109         sz = vector<int>(n + 1), par = vector<int>(n + 1), dep = vector<int>(n + 1), tin = vector<int>(n + 1), tout = vector<int>(n + 1), head = vector<int>(n + 1), a = vector<int>(n + 1);
110         dfs1(1, 0);
111         head[1] = 1;

```

```

112     dfs2(1);
113     for (int i = 1; i <= n; i++)
114         a[tin[i]] = v[i];
115     seg.init(a);
116 }
117
118 void dfs1(int u, int p)
119 {
120     dep[u] = dep[p] + 1;
121     par[u] = p;
122     sz[u] = 1;
123     if (p != 0)
124         graph[u].erase(find(graph[u].begin(), graph[u].end(), p));
125     for (auto &x : graph[u])
126     {
127         dfs1(x, u);
128         sz[u] += sz[x];
129         if (sz[x] > sz[graph[u][0]])
130             swap(x, graph[u][0]);
131     }
132 }
133
134 void dfs2(int u)
135 {
136     tin[u] = timer++;
137     for (auto &x : graph[u])
138     {
139         head[x] = x == graph[u][0] ? head[u] : x;
140         dfs2(x);
141     }
142     tout[u] = timer;
143 }
144
145 int lca(int u, int v)
146 {
147     while (head[u] != head[v])
148     {
149         if (dep[head[u]] > dep[head[v]])
150             u = par[head[u]];
151         else
152             v = par[head[v]];
153     }
154     if (dep[u] < dep[v])
155         return u;
156     return v;
157 }
158
159 void point_update(int u, int val)
160 {
161     seg.set(tin[u], val);
162 }
163

```

```

164 int query_subtree(int u)
165 {
166     return seg.get(tin[u], tout[u]);
167 }
168
169 // now querying for maximum
170 int query_path(int u, int v)
171 {
172     int res{-00}; // TO
173     while (head[u] != head[v])
174     {
175         if (dep[head[u]] < dep[head[v]])
176             swap(u, v);
177         res = max(res, seg.get(tin[head[u]], tin[u] + 1)); // TO
178         u = par[head[u]];
179     }
180     if (dep[u] < dep[v])
181         swap(u, v);
182     res = max(res, seg.get(tin[v], tin[u] + 1)); // TO
183     return res;
184 }
185 };

```

## 2.5 Misc

### 2.5.1 CDQ

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: CDQ DIVIDE AND CONQUER
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "3D Partial Order", "Dynamic 2D Range Queries", "Offline range
   counting".
9  * - Classic Scenarios: You need to count how many points i have (x_i <= X, y_i
   <= Y, z_i <= Z).
10 * - The Magic: CDQ DYC reduces dimensionality.
11 * - Dimension 1 (X): Handled by sorting the array.
12 * - Dimension 2 (Y): Handled by the Divide and Conquer recursion.
13 * - Dimension 3 (Z): Handled by a 1D Fenwick Tree (BIT).
14 *
15 * 2. HOW TO USE IT (THE BLACK BOX)
16 * - Define a struct Element {x, y, z, id}.
17 * - Sort the elements initially by x.
18 * - Call cdq(0, n-1) to process.
19 *
20 * - Complexity:
21 *     Time: O(N log^2 N) where N is the number of elements.
22 *     Space: O(N) memory.

```

```

23 */
24
25 struct Element
26 {
27     int x, y, z, id;
28 };
29
30 // 1D Fenwick Tree for the 3rd dimension
31 struct BIT
32 {
33     int n;
34     vector<int> tree;
35     BIT(int n) : n(n), tree(n + 1, 0) {}
36
37     void update(int i, int delta)
38     {
39         for (; i <= n; i += i & -i)
40             tree[i] += delta;
41     }
42
43     int query(int i)
44     {
45         int sum = 0;
46         for (; i > 0; i -= i & -i)
47             sum += tree[i];
48         return sum;
49     }
50 };
51
52 int n;
53 vector<Element> arr;
54 vector<int> result;
55 BIT bit(1000005); // Adjust size to coordinate range
56
57 void cdq(int l, int r)
58 {
59     if (l >= r)
60         return;
61     int mid = l + (r - l) / 2;
62
63     cdq(l, mid);
64     cdq(mid + 1, r);
65
66     // Sort both halves by Y (Dimension 2)
67     vector<Element> temp;
68     int i = l, j = mid + 1;
69
70     while (i <= mid && j <= r)
71     {
72         if (arr[i].y <= arr[j].y)
73             bit.update(arr[i].z, 1);
74

```

```

75         temp.push_back(arr[i++]);
76     }
77     else
78     {
79         result[arr[j].id] += bit.query(arr[j].z);
80         temp.push_back(arr[j++]);
81     }
82 }
83
84 // Process remaining
85 while (j <= r)
86 {
87     result[arr[j].id] += bit.query(arr[j].z);
88     temp.push_back(arr[j++]);
89 }
90
91 // Clear BIT before moving on
92 for (int k = l; k < i; k++)
93     bit.update(arr[k].z, -1);
94
95 while (i <= mid)
96     temp.push_back(arr[i++]);
97 for (int k = 0; k < temp.size(); k++)
98     arr[l + k] = temp[k];
99 }
100
101 int main()
102 {
103     ios_base::sync_with_stdio(false);
104     cin.tie(NULL);
105
106     // Example: Read N elements with (x, y, z) coordinates
107     // Ensure all coordinates are mapped to 1-indexed values for BIT
108
109     // cdq(0, n - 1);
110
111     return 0;
112 }

```

## 2.5.2 Leftist heap

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: LEFTIST HEAP (MELDABLE)
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Meldable Heaps", "Priority Queue on Trees", "Greedy Tree DP".
9  * - Classic Scenarios: You need to maintain a set of values for each node in a
   tree

```

```

10 * and frequently merge them with children. std::priority_queue cannot merge. 60
11 * - The Magic: It keeps the tree "left-heavy" to guarantee that the right spine 61 Node *push(Node *root, int val)
    is 62 {
12 * always short ( $O(\log N)$ ). Merging two heaps is just merging their right 63     return merge(root, new Node(val));
    spines! 64 }
13 * 65
14 * 2. HOW TO USE IT 66 Node *pop(Node *root)
15 * - Initialization: Root = nullptr. 67 {
16 * - Merge: root = merge(root1, root2); 68     Node *left = root->left;
17 * - Complexity:  $O(\log N)$  per merge, push, and pop. 69     Node *right = root->right;
18 */ 70     delete root; // Clean up memory
19 71     return merge(left, right);
20 struct Node 72 }
21 { 73
22     int val, npl; 74 int top(Node *root)
23     Node *left, *right; 75 {
24 76     return root->val;
25     Node(int v) : val(v), npl(0), left(nullptr), right(nullptr) {} 77 }
26 }; 78
27 79 int main()
28 int getNpl(Node *n) 80 {
29 { 81     // Example: Standard Usage
30     return n ? n->npl : -1; 82     Node *root = nullptr;
31 } 83     root = push(root, 10);
32 84     root = push(root, 5);
33 // The core MELD operation 85
34 Node *merge(Node *a, Node *b) 86     cout << "Top:_" << top(root) << endl; // 5
35 { 87     root = pop(root);
36     if (!a) 88     cout << "Top_after_pop:_" << top(root) << endl; // 10
37         return b; 89
38     if (!b) 90     return 0;
39         return a; 91 }
40
41 // To maintain a Min-Heap, ensure 'a' has the smaller root
42 if (a->val > b->val)
43     swap(a, b);
44
45 // ! PERSISTENCE NOTE: If you need a fully persistent heap,
46 // uncomment the line below to clone the node before modifying.
47 // a = new Node(*a);
48
49 a->right = merge(a->right, b);
50
51 // Maintain Leftist Property: Left NPL >= Right NPL
52 if (getNpl(a->left) < getNpl(a->right))
53 {
54     swap(a->left, a->right);
55 }
56
57 a->npl = getNpl(a->right) + 1;
58 return a;
59 }

```

### 2.5.3 Monotonic queue

```

1 class Monotonic_queue
2 {
3     deque<pair<int, int>> dq;
4     int k;
5
6 public:
7     Monotonic_queue(int k) : k(k) {}
8
9     void push(int idx, int val)
10    {
11        if (!dq.empty() && dq.front().second <= idx - k)
12            dq.pop_front();
13        while (!dq.empty() && dq.back().second <= val)
14            dq.pop_back();
15        dq.push_back({idx, val});
16    }

```

```

17
18     int max() const
19     {
20         return dq.front().second;
21     }
22 };

```

## 2.5.4 Monotonic stack

```

1 class MonotonicStack
2 {
3 public:
4     static vector<int> getNextGreater(const vector<int> &nums)
5     {
6         int n = nums.size();
7         vector<int> nextGreater(n, n);
8         stack<int> s;
9         for (int i = 0; i < n; ++i)
10        {
11            while (!s.empty() && nums[i] > nums[s.top()])
12            {
13                nextGreater[s.top()] = i;
14                s.pop();
15            }
16            s.push(i);
17        }
18        return nextGreater;
19    }
20 };

```

## 2.5.5 ODT

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: OLD DRIVER TREE (ODT / CHTHOLLY TREE)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Color blocks", "Range assign", "Randomized data", "Interval set"
6  * - Classic Scenarios: You have an array that is mostly composed of long
   segments of identical
7  * values. You need to perform Range Assignments ( $A[i] = V$  for  $L$  to  $R$ ) and
   other Range Queries.
8  * - The Magic: An ODT stores the array as a set of intervals  $\{l, r, v\}$ .
9  * The "Assign" operation is the key: it erases all intervals in the set
   between  $[L, R]$ 
10 * and inserts one single new interval. This keeps the number of intervals very
   small.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Initialization: Insert the initial  $[1, N]$  interval into the `odt` set.

```

```

14 * - Split: You MUST call `split(l)` and `split(r+1)` before ANY operation to
   ensure
15 * the boundaries are exact.
16 * - Complexity:
17 *     Time: Amortized  $O(N \log \log N)$  on random data.  $O(N^2)$  on worst-case non-
   random data.
18 *     Space:  $O(N)$  memory.
19 */
20
21 #include <bits/stdc++.h>
22 using namespace std;
23
24 struct Node
25 {
26     int l, r;
27     mutable long long v;
28
29     Node(int l, int r = -1, long long v = 0) : l(l), r(r), v(v) {}
30
31     // Set must be sorted by left endpoint
32     bool operator<(const Node &o) const
33     {
34         return l < o.l;
35     }
36 };
37
38 set<Node> odt;
39
40 // Splits the interval containing 'pos' into [l, pos-1] and [pos, r]
41 // Returns an iterator to the interval starting at 'pos'
42 auto split(int pos)
43 {
44     auto it = odt.lower_bound({pos});
45     if (it != odt.end() && it->l == pos)
46         return it;
47
48     --it;
49     int l = it->l, r = it->r;
50     long long v = it->v;
51
52     odt.erase(it);
53     odt.insert({l, pos - 1, v});
54     return odt.insert({pos, r, v}).first;
55 }
56
57 // HOW TO ADAPT IT (THE DIALS & SWITCHES)
58 // - If the problem asks for Range Sum: loop through the set from itl to itr and
   sum (r-l+1)*v.
59 // - If the problem asks for K-th smallest or Count: you must keep a secondary
   data
60 // structure (like a Segment Tree) updated alongside the ODT.
61 // The core efficiency of ODT: replaces range [l, r] with one single interval

```

```

62 void assign(int l, int r, long long v)
63 {
64     auto itr = split(r + 1), itl = split(l);
65     odt.erase(itl, itr);
66     odt.insert({l, r, v});
67 }
68
69 void solve()
70 {
71     int n, m;
72     cin >> n >> m;
73
74     // Initially add [1, n] with value 0
75     odt.insert({1, n, 0});
76 }
77
78 int main()
79 {
80     ios_base::sync_with_stdio(false);
81     cin.tie(NULL);
82     solve();
83     return 0;
84 }

```

## 2.6 More Data structures

### 2.6.1 Wavelet tree

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: WAVELET TREE
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "K-th smallest in range", "Range frequency count", "Count elements
   < K".
6  * - Classic Scenarios: You have an array and you need to answer queries like:
7  * "Find the k-th smallest element in subarray [L, R]" or "Count occurrences
8  * of values in [A, B] within index range [L, R]".
9  * - The Magic: It is essentially a Merge Sort Tree, but instead of storing
10 * sorted vectors (which consume  $O(N \log N)$  memory), it stores a bit-vector
11 * representation of the partitioning. This allows queries in  $O(\log(\text{MaxVal}))$ 
   time.
12 *
13 * 2. HOW TO USE IT (THE BLACK BOX)
14 * - Initialization:
15 *     // Pass the range of values [min_val, max_val] and the input vector
16 *     WaveletTree wt(1, 1000000, array);
17 *
18 * - Queries:
19 *     // How many numbers < K in subarray [L, R]
20 *     int count = wt.count_less(L, R, K);
21 *     // Find the k-th smallest number in subarray [L, R] (1-indexed k)

```

```

22 *     int val = wt.kth_smallest(L, R, k);
23 *
24 * - Complexity:
25 *     Time:  $O(N \log(\text{MaxVal}))$  to build.  $O(\log(\text{MaxVal}))$  per query.
26 *     Space:  $O(N \log(\text{MaxVal}))$  memory.
27 */
28
29 #include <bits/stdc++.h>
30 using namespace std;
31
32 struct WaveletTree
33 {
34     int low, high;
35     WaveletTree *left, *right;
36     vector<int> pref; // Prefix sums of elements going to the left child
37
38     WaveletTree(int l, int r, const vector<int> &a) : low(l), high(r), left(
        nullptr), right(nullptr)
39     {
40         if (low == high || a.empty())
41             return;
42
43         int mid = low + (high - low) / 2;
44
45         // Define function to check if value goes to left child
46         auto is_left = [&](int x)
47         { return x <= mid; };
48
49         pref.reserve(a.size() + 1);
50         pref.push_back(0);
51
52         vector<int> l_vec, r_vec;
53         for (int x : a)
54         {
55             if (is_left(x))
56             {
57                 l_vec.push_back(x);
58                 pref.push_back(pref.back() + 1);
59             }
60             else
61             {
62                 r_vec.push_back(x);
63                 pref.push_back(pref.back());
64             }
65         }
66
67         left = new WaveletTree(low, mid, l_vec);
68         right = new WaveletTree(mid + 1, high, r_vec);
69     }
70
71     // Count numbers <= k in range [ql, qr] (0-indexed)
72     int count_less_equal(int ql, int qr, int k)

```

```

73 {
74     if (ql > qr || low > k)
75         return 0;
76     if (high <= k)
77         return qr - ql + 1;
78
79     int l_count_before = pref[ql];
80     int r_count_before = ql - l_count_before;
81     int l_count_all = pref[qr + 1];
82     int r_count_all = (qr + 1) - l_count_all;
83
84     return left->count_less_equal(l_count_before, l_count_all - 1, k) +
85         right->count_less_equal(r_count_before, r_count_all - 1, k);
86 }
87
88 // Find k-th smallest number in range [ql, qr]
89 int kth_smallest(int ql, int qr, int k)
90 {
91     if (low == high)
92         return low;
93
94     int l_count_before = pref[ql];
95     int r_count_before = ql - l_count_before;
96     int l_count_all = pref[qr + 1];
97     int r_count_all = (qr + 1) - l_count_all;
98
99     int left_size = l_count_all - l_count_before;
100
101     if (k <= left_size)
102         return left->kth_smallest(l_count_before, l_count_all - 1, k);
103     else
104         return right->kth_smallest(r_count_before, r_count_all - 1, k -
105             left_size);
106 };

```

## 2.7 SQRT decomposition MO

### 2.7.1 MO

```

1
2 // how many distinct values are there in a range [l,r]
3
4 int SQ = 400;
5 struct Query
6 {
7     int l, r, idx;
8     bool operator<(const Query &q) const
9     {
10         if (l / SQ != q.l / SQ)
11             {

```

```

12         return l / SQ < q.l / SQ;
13     }
14     return (l / SQ & 1) ? r < q.r : r > q.r;
15 }
16 };
17
18 void solve()
19 {
20     int n, q;
21     cin >> n >> q;
22     vector<int> a(n), frq(n + 10);
23     map<int, int> mp;
24     int id{};
25     for (int i = 0; i < n; i++)
26     {
27         cin >> a[i];
28         if (!mp.count(a[i]))
29             {
30                 mp[a[i]] = id++;
31             }
32     }
33     for (int i = 0; i < n; i++)
34         a[i] = mp[a[i]];
35
36     vector<Query> query(q);
37     for (int i = 0; i < q; i++)
38     {
39         cin >> query[i].l >> query[i].r;
40         query[i].l--, query[i].r--;
41         query[i].idx = i;
42     }
43     sort(query.begin(), query.end());
44     int ans = 0;
45     auto add = [&](int i)
46     {
47         // if (frq[a[i]] == 0)
48         // {
49         //     ans++;
50         // }
51         // frq[a[i]]++;
52     };
53     auto remove = [&](int i)
54     {
55         // frq[a[i]]--;
56         // if (frq[a[i]] == 0)
57         // {
58         //     ans--;
59         // }
60     };
61
62     vector<int> res(q);
63     int l = 0, r = -1;

```



```

64   for (auto [ql, qr, idx] : query)
65   {
66       while (l > ql)
67       {
68           --l;
69           add(l);
70       }
71       while (r < qr)
72       {
73           ++r;
74           add(r);
75       }
76       while (l < ql)
77       {
78           remove(l);
79           ++l;
80       }
81       while (r > qr)
82       {
83           remove(r);
84           --r;
85       }
86       res[idx] = ans;
87   }
88   for (int i = 0; i < q; i++)
89       cout << res[i] << "\n";
90 }

```

## 2.7.2 MO on tree

```

1  /**
2  *  THE ULTIMATE CP TEMPLATE BLUEPRINT: MO'S ALGORITHM ON TREES
3  *
4  *  1.  WHAT PROBLEM DOES THIS SOLVE?
5  *  - Keywords: "Path queries", "Distinct values on path", "Offline tree queries"
6  *  - Classic Scenarios: You are given a tree where each node has a color. You are
7  *    asked Q queries:
8  *    "How many distinct colors exist on the simple path between node U and node V?"
9  *  - The Magic: We flatten the tree using an Euler Tour. Every node appears twice
10 *    in the array:
11 *    once when we enter it (tin), and once when we leave it (tout).
12 *    If we want the path between U and V (where tin[U] < tin[V]), we query the
13 *    flat array:
14 *    A) If U is an ancestor of V: Query range [tin[U], tin[V]].
15 *    B) If U is NOT an ancestor: Query range [tout[U], tin[V]] PLUS the LCA(U, V)
16 *    .
17 *    Because dead-end branches will appear twice in this range (entry and exit),
18 *    we simply
19 *    TOGGLE their state. 1st appearance = Add. 2nd appearance = Remove.
20 *

```

```

16 * 2.  HOW TO USE IT (THE BLACK BOX)
17 * - Initialization: Pass your tree edges and node values, then run `dfs(1, 0)`.
18 * - L and R Logic: We maintain an `in_path` boolean array. The `toggle()`
19 *   function handles
20 *   adding OR removing automatically based on the boolean!
21 * - Complexity:
22 *   Time:  $O(N \log N)$  to setup LCA +  $O((N + Q) * \sqrt{2N})$  for Mo's
23 *   processing.
24 *   Space:  $O(N \log N)$  for LCA lifting table +  $O(N)$  arrays.
25 * 3.  HOW TO ADAPT IT (THE DIALS & SWITCHES)
26 * - Changing the logic:
27 *   Right now, the template counts "Distinct Colors" using a frequency array.
28 *   Modify the `add()` and `remove()` functions inside `toggle()` to track
29 *   whatever
30 *   property your specific problem requires.
31 */
32 #include <bits/stdc++.h>
33 using namespace std;
34
35 const int MAXN = 100005;
36 const int LOG = 20;
37
38 int n, q;
39 vector<int> adj[MAXN];
40 long long val[MAXN]; // The color or value of each node
41
42 // Euler Tour Variables
43 int timer_tour = 0;
44 int tin[MAXN], tout[MAXN];
45 int tour[2 * MAXN]; // The flattened 2N array
46
47 // LCA Variables
48 int up[MAXN][LOG];
49 int depth[MAXN];
50
51 void dfs(int u, int p)
52 {
53     // 1. Record Entry Time
54     tin[u] = ++timer_tour;
55     tour[timer_tour] = u;
56
57     // 2. Build Binary Lifting Table for LCA
58     up[u][0] = p;
59     for (int i = 1; i < LOG; i++)
60     {
61         up[u][i] = up[up[u][i - 1]][i - 1];
62     }
63
64     // 3. Traverse Children

```

```

65     for (int v : adj[u])
66     {
67         if (v != p)
68         {
69             depth[v] = depth[u] + 1;
70             dfs(v, u);
71         }
72     }
73
74     // 4. Record Exit Time
75     tout[u] = ++timer_tour;
76     tour[timer_tour] = u;
77 }
78
79 int get_lca(int u, int v)
80 {
81     if (depth[u] < depth[v])
82         swap(u, v);
83
84     int diff = depth[u] - depth[v];
85     for (int i = 0; i < LOG; i++)
86     {
87         if ((diff >> i) & 1)
88             u = up[u][i];
89     }
90
91     if (u == v)
92         return u;
93
94     for (int i = LOG - 1; i >= 0; i--)
95     {
96         if (up[u][i] != up[v][i])
97         {
98             u = up[u][i];
99             v = up[v][i];
100         }
101     }
102     return up[u][0];
103 }
104
105 int BLOCK_SIZE;
106
107 struct Query
108 {
109     int l, r, id, lca;
110
111     bool operator<(const Query &other) const
112     {
113         int block_own = l / BLOCK_SIZE;
114         int block_other = other.l / BLOCK_SIZE;
115         if (block_own != block_other)
116             return block_own < block_other;
117     }
118 }
119
120 // Parity optimization for slightly faster R pointer movement
121 return (block_own & 1) ? r < other.r : r > other.r;
122 };
123
124 bool in_path[MAXN];
125 long long freq[MAXN]; // Example: Frequency array for colors
126 long long distinct_colors = 0;
127
128 // ? THE MAGIC: Toggling a node.
129 // If it's entering our path, we add it. If we hit its exit time, it cancels out
130 // and we remove it!
131 void toggle(int node)
132 {
133     long long c = val[node];
134     if (in_path[node])
135     {
136         // It was in the path, so we are removing it
137         freq[c]--;
138         if (freq[c] == 0)
139             distinct_colors--;
140     }
141     else
142     {
143         // It wasn't in the path, so we are adding it
144         if (freq[c] == 0)
145             distinct_colors++;
146         freq[c]++;
147     }
148     in_path[node] ^= 1; // Flip the boolean state
149 }
150
151 void solve()
152 {
153     if (!(cin >> n >> q))
154         return;
155
156     // Reset globals for multiple test cases (if needed)
157     timer_tour = 0;
158     distinct_colors = 0;
159     memset(in_path, 0, sizeof(in_path));
160     memset(freq, 0, sizeof(freq));
161
162     // Coordinate compression for colors might be required here if values > 10^5!
163     for (int i = 1; i <= n; i++)
164         cin >> val[i];
165
166     for (int i = 0; i < n - 1; i++)

```

```

168 {
169     int u, v;
170     cin >> u >> v;
171     adj[u].push_back(v);
172     adj[v].push_back(u);
173 }
174
175 // 1. Flatten the tree!
176 depth[1] = 0;
177 dfs(1, 1);
178
179 // Block size for an array of size 2N
180 BLOCK_SIZE = max(1, (int)sqrt(2 * n));
181
182 vector<Query> queries(q);
183 for (int i = 0; i < q; i++)
184 {
185     int u, v;
186     cin >> u >> v;
187
188     // Ensure u is visited before v in the Euler Tour
189     if (tin[u] > tin[v])
190         swap(u, v);
191
192     int lca = get_lca(u, v);
193
194     if (lca == u)
195     {
196         // SCENARIO A: U is an ancestor of V.
197         // The path is strictly the sequence of entries from U down to V.
198         queries[i].l = tin[u];
199         queries[i].r = tin[v];
200         queries[i].lca = -1; // No extra LCA needed
201     }
202     else
203     {
204         // SCENARIO B: U and V are in different branches.
205         // We take the path from the exit of U, up to the LCA, and down to
206         // the entry of V.
207         queries[i].l = tout[u];
208         queries[i].r = tin[v];
209         queries[i].lca = lca; // We MUST manually include the LCA!
210     }
211     queries[i].id = i;
212 }
213
214 // 2. Sort Queries
215 sort(queries.begin(), queries.end());
216
217 vector<long long> ans(q);
218
219 // 3. Run Mo's Pointers on the 2N Tour Array

```

```

219 int L = 1, R = 0;
220
221 for (const Query &qry : queries)
222 {
223     while (L > qry.l)
224     {
225         L--;
226         toggle(tour[L]);
227     }
228     while (R < qry.r)
229     {
230         R++;
231         toggle(tour[R]);
232     }
233     while (L < qry.l)
234     {
235         toggle(tour[L]);
236         L++;
237     }
238     while (R > qry.r)
239     {
240         toggle(tour[R]);
241         R--;
242     }
243
244     // If U and V were on different branches, the LCA itself was NOT included
245     // in the [tout[U], tin[V]] range. We must temporarily add it!
246     if (qry.lca != -1)
247         toggle(qry.lca);
248
249     // Record the answer
250     ans[qry.id] = distinct_colors;
251
252     // Revert the temporary LCA addition to keep the L/R pointers clean
253     if (qry.lca != -1)
254         toggle(qry.lca);
255 }
256
257 for (int i = 0; i < q; i++)
258 {
259     cout << ans[i] << "\n";
260 }
261 }
262
263 int main()
264 {
265     ios_base::sync_with_stdio(false);
266     cin.tie(NULL);
267     solve();
268     return 0;
269 }

```

## 2.7.3 Mo's with updates

```

1  /**
2  *  THE ULTIMATE CP TEMPLATE BLUEPRINT: 3D MO'S ALGORITHM (WITH UPDATES)
3  *
4  *  1.  WHAT PROBLEM DOES THIS SOLVE?
5  *  - Keywords: "Point Updates", "Range Queries", "Offline", "Distinct elements
      with updates".
6  *  - Classic Scenarios: You are given an array of size N. You have two types of
      queries:
7  *    Type 1: Query the number of distinct elements in range [L, R].
8  *    Type 2: Change the element at index X to V.
9  *  - The Magic: A standard Segment Tree cannot count "Distinct Elements".
      Standard Mo's
10 *    Algorithm cannot handle "Updates". 3D Mo combines them by adding a "Time"
      pointer (T).
11 *    Queries are sorted into 3D blocks: [L_block, R_block, Time].
12 *
13 *  2.  HOW TO USE IT (THE BLACK BOX)
14 *  - Initialization:
15 *    As you read the input, split it! If it's an update, push to the `updates`
      array.
16 *    If it's a query, push to the `queries` array and record the CURRENT size of
      the
17 *    `updates` array as its "Time" stamp!
18 *
19 *  - Complexity:
20 *    Time:  $O(N^{\frac{5}{3}})$  optimally.
21 *    Space:  $O(N + Q)$  memory.
22 *
23 *  3.  HOW TO ADAPT IT (THE DIALS & SWITCHES)
24 *  - The Block Size (B):
25 *    Standard Mo uses  $B = N^{\frac{1}{2}}$ . 3D Mo MUST use  $B = N^{\frac{2}{3}}$ .
26 *    If  $N = 100,000$ ,  $N^{\frac{2}{3}}$  is roughly 2154. The template calculates this
      dynamically!
27 *  - Coordinate Compression:
28 *    This template assumes array values are  $\leq 10^6$ . If values are up to  $10^9$ ,
      you MUST
29 *    coordinate compress the initial array AND all the update values before
      running Mo!
30 */
31
32 #include <bits/stdc++.h>
33 using namespace std;
34
35 const int MAXN = 200005;
36 const int MAX_VAL = 1000005; // Maximum possible value in the array
37
38 int n, q;
39 int a[MAXN];           // The active array
40 int freq[MAX_VAL];     // Frequency array for distinct element counting
41 int current_ans = 0;   // Tracks the current number of distinct elements
42 int BLOCK_SIZE;
43
44 struct Update
45 {
46     int idx;
47     int val; // When an update is applied, this holds the NEW value.
48             // When reverted, it automatically swaps to hold the OLD value!
49 };
50
51 struct Query
52 {
53     int l, r, t, id;
54     int l_block, r_block;
55
56     // Sort by L block, then R block, then Time.
57     // Includes a 3D parity optimization for faster pointer movement!
58     bool operator<(const Query &other) const
59     {
60         if (l_block != other.l_block)
61             return l_block < other.l_block;
62
63         if (r_block != other.r_block)
64         {
65             return (l_block & 1) ? r_block < other.r_block : r_block > other.
                r_block;
66         }
67
68         return (r_block & 1) ? t < other.t : t > other.t;
69     }
70 };
71
72 vector<Update> updates;
73 vector<Query> queries;
74 vector<int> ans;
75
76 // Add an element at index 'idx' to our active window
77 inline void add(int idx)
78 {
79     if (freq[a[idx]] == 0)
80     {
81         current_ans++;
82     }
83     freq[a[idx]]++;
84 }
85
86 // Remove an element at index 'idx' from our active window
87 inline void remove(int idx)
88 {
89     freq[a[idx]]--;
90     if (freq[a[idx]] == 0)
91     {
92         current_ans--;

```

```

93     }
94 }
95
96 // ? THE MAGIC: The Swap Trick
97 // If we move forward in time, we swap the current array value with the update's
    value.
98 // Because we swap, the `updates[t_idx].val` now holds the OLD value!
99 // This means if we need to travel backward in time, we just call the exact same
    function!
100 inline void apply_time(int t_idx, int L, int R)
101 {
102     int idx = updates[t_idx].idx;
103
104     // If the index being updated is currently INSIDE our active [L, R] window,
105     // we must remove its contribution using the old value.
106     if (L <= idx && idx <= R)
107     {
108         remove(idx);
109     }
110
111     // Swap the array value and the update cache!
112     swap(a[idx], updates[t_idx].val);
113
114     // If the index is in our active window, add its contribution using the new
115     // value!
116     if (L <= idx && idx <= R)
117     {
118         add(idx);
119     }
120 }
121 void solve()
122 {
123     if (!(cin >> n >> q))
124         return;
125
126     for (int i = 1; i <= n; i++)
127     {
128         cin >> a[i];
129     }
130
131     // 1-based indexing for updates is safer for the Time pointer
132     updates.push_back({0, 0}); // Dummy update at T = 0
133
134     // Parse queries and updates
135     for (int i = 0; i < q; i++)
136     {
137         int type;
138         cin >> type;
139         if (type == 1)
140         {
141             // Type 1: Range Query (L, R)
142
143             int l, r;
144             cin >> l >> r;
145             queries.push_back({l, r, (int)updates.size() - 1, (int)queries.size()
146                 , 0, 0});
147         }
148         else
149         {
150             // Type 2: Point Update (Index X becomes Value V)
151             int idx, val;
152             cin >> idx >> val;
153             updates.push_back({idx, val});
154         }
155     }
156
157     // ? THE MATH: Optimal block size for 3D Mo is  $N^{(2/3)}$ 
158     BLOCK_SIZE = max(1, (int)pow(n, 2.0 / 3.0));
159
160     for (auto &qry : queries)
161     {
162         qry.l_block = qry.l / BLOCK_SIZE;
163         qry.r_block = qry.r / BLOCK_SIZE;
164     }
165
166     sort(queries.begin(), queries.end());
167     ans.resize(queries.size());
168
169     int L = 1, R = 0, T = 0;
170
171     for (const auto &qry : queries)
172     {
173         // Step 1: Expand the spatial window [L, R]
174         while (L > qry.l)
175         {
176             L--;
177             add(L);
178         }
179         while (R < qry.r)
180         {
181             R++;
182             add(R);
183         }
184
185         // Step 2: Shrink the spatial window [L, R]
186         while (L < qry.l)
187         {
188             remove(L);
189             L++;
190         }
191         while (R > qry.r)
192         {
193             remove(R);
194             R--;
195         }
196     }

```

```

193     }
194
195     // Step 3: Fast-forward or Rewind the Time window (T)
196     // We do Time AFTER Space to ensure the active window bounds are correct
197     // for the `apply_time` intersection check!
198     while (T < qry.t)
199     {
200         T++;
201         apply_time(T, L, R);
202     }
203     while (T > qry.t)
204     {
205         apply_time(T, L, R);
206         T--;
207     }
208
209     // Record the answer for this specific snapshot in space and time
210     ans[qry.id] = current_ans;
211 }
212
213 for (int i = 0; i < ans.size(); i++)
214 {
215     cout << ans[i] << "\n";
216 }
217 }
218
219 int main()
220 {
221     // Fast I/O is mandatory due to heavy pointer manipulation
222     ios_base::sync_with_stdio(false);
223     cin.tie(NULL);
224     solve();
225     return 0;
226 }

```

#### 2.7.4 Rollback mo

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: ROLLBACK MO'S ALGORITHM
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Range queries", "Add is easy, Remove is impossible", "DSU on
   ranges".
6  * - Classic Scenarios: You are given a graph with N edges. You receive queries
   asking:
7  * "If we only use edges from index L to R, how many connected components are
   there?"
8  * - The Magic: Standard Mo's algorithm requires you to slide both the L and R
   pointers
9  * by adding AND removing elements. But you cannot "remove" an edge from a
   standard DSU!

```

```

10 * Rollback Mo sorts the queries so that the R pointer ONLY moves forward (add
   only).
11 * The L pointer moves backward temporarily, and we "undo" those L moves using
   a stack
12 * history (Rollback) before moving to the next query. NO `remove()` function
   needed!
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Define your state: You need a data structure that supports `add()`, `
   save_state()`,
16 * and `rollback(state)`. (I have included a Rollback DSU in this template).
17 * - Read Queries: Push all queries into a vector and pass them to the Mo's
   runner.
18 *
19 * - Complexity:
20 *     Time:  $O(Q * \sqrt{N} * \text{Cost}(\text{Add}))$ . If DSU, it's  $O(Q * \sqrt{N} * \log N)$ .
21 *     Space:  $O(N + Q)$  memory.
22 *
23 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
24 * - The Block Size (B):
25 * The math for Mo's dictates the block size should be roughly  $N / \sqrt{Q}$ .
26 * If the problem has  $N = 10^5$ , setting `B = 350` or `B = 400` is the standard
   sweet spot.
27 * - Changing the Data Structure:
28 * If the problem asks for "Max frequency in range", you don't need DSU. You
   just need
29 * an array tracking frequencies, and a variable `max_freq`. To rollback, you
   just
30 * remember the old `max_freq` and revert the array.
31 */
32
33 #include <bits/stdc++.h>
34 using namespace std;
35
36 // 1. The Rollback Data Structure
37 struct RollbackDSU
38 {
39     vector<int> parent, sz;
40     int components;
41
42     // We store the history of changes: {node_u, node_v, previous_components}
43     struct Change
44     {
45         int u, v, comps;
46     };
47     stack<Change> history;
48
49     RollbackDSU(int n)
50     {
51         parent.resize(n + 1);
52         sz.assign(n + 1, 1);
53         iota(parent.begin(), parent.end(), 0);

```

```

54     components = n;
55 }
56
57 // ! CRITICAL: No Path Compression allowed! Only Union by Size.
58 int find(int i)
59 {
60     while (i != parent[i])
61         i = parent[i];
62     return i;
63 }
64
65 void unite(int u, int v)
66 {
67     int root_u = find(u);
68     int root_v = find(v);
69
70     if (root_u == root_v)
71     {
72         // Even if they are already connected, we push a dummy change to keep
73         // the stack height aligned with the number of `add()` calls.
74         history.push({-1, -1, components});
75         return;
76     }
77
78     if (sz[root_u] < sz[root_v])
79         swap(root_u, root_v);
80
81     // Record the exact state BEFORE we change it
82     history.push({root_u, root_v, components});
83
84     // Apply changes
85     parent[root_v] = root_u;
86     sz[root_u] += sz[root_v];
87     components--;
88 }
89
90 // Returns the current "timestamp" or "height" of the history stack
91 int save()
92 {
93     return history.size();
94 }
95
96 // Undos all operations until the stack shrinks back to `target_time`
97 void rollback(int target_time)
98 {
99     while (history.size() > target_time)
100     {
101         auto [u, v, old_comps] = history.top();
102         history.pop();
103
104         if (u != -1)
105         {

```

```

106             sz[u] -= sz[v];
107             parent[v] = v;
108         }
109         components = old_comps;
110     }
111 }
112
113 void reset(int n)
114 {
115     while (!history.empty())
116         history.pop();
117     for (int i = 1; i <= n; i++)
118     {
119         parent[i] = i;
120         sz[i] = 1;
121     }
122     components = n;
123 }
124 };
125
126 // 2. Query Structure and Mo's Sorting logic
127 int BLOCK_SIZE;
128
129 struct Query
130 {
131     int l, r, id;
132
133     // Sort queries by their Left Block, then by their Right Endpoint
134     bool operator<(const Query &other) const
135     {
136         int block_own = l / BLOCK_SIZE;
137         int block_other = other.l / BLOCK_SIZE;
138         if (block_own != block_other)
139         {
140             return block_own < block_other;
141         }
142         // If in the same block, sort R strictly ascending
143         return r < other.r;
144     }
145 };
146
147 struct Edge
148 {
149     int u, v;
150 };
151
152 void solve()
153 {
154     int n, m, q;
155     // N vertices, M edges, Q queries
156     if (!(cin >> n >> m >> q))
157         return;

```

```

158
159 // Standard block size for  $10^5$  is roughly 350
160 BLOCK_SIZE = max(1, (int)sqrt(m));
161
162 vector<Edge> edges(m + 1); // 1-indexed edges
163 for (int i = 1; i <= m; i++)
164 {
165     cin >> edges[i].u >> edges[i].v;
166 }
167
168 vector<Query> queries(q);
169 for (int i = 0; i < q; i++)
170 {
171     cin >> queries[i].l >> queries[i].r;
172     queries[i].id = i;
173 }
174
175 // 1. Sort the queries
176 sort(queries.begin(), queries.end());
177
178 vector<int> ans(q);
179 RollbackDSU dsu(n);
180
181 int last_block = -1;
182 int current_r = 0;
183
184 for (const Query &query : queries)
185 {
186     int current_block = query.l / BLOCK_SIZE;
187
188     // If we entered a brand new block, completely wipe the DSU clean!
189     if (current_block != last_block)
190     {
191         dsu.reset(n);
192         // Set our R pointer to the very edge of the current block
193         current_r = min(m, (current_block + 1) * BLOCK_SIZE);
194         last_block = current_block;
195     }
196
197     // SCENARIO 1: The query is incredibly tiny and fits entirely inside one
198     // We answer it naively in  $O(\sqrt{N})$  and instantly roll it back.
199     if (query.r < (current_block + 1) * BLOCK_SIZE)
200     {
201         int time_stamp = dsu.save();
202         for (int i = query.l; i <= query.r; i++)
203         {
204             dsu.unite(edges[i].u, edges[i].v);
205         }
206         ans[query.id] = dsu.components;
207         dsu.rollback(time_stamp);
208         continue;

```

```

209     }
210
211     // SCENARIO 2: The query stretches outside the block.
212     // Step A: Expand our R pointer forward until it hits the query's R limit
213     // Notice we NEVER rollback these operations while staying in the same
214     // block!
215     while (current_r <= query.r)
216     {
217         dsu.unite(edges[current_r].u, edges[current_r].v);
218         current_r++;
219     }
220
221     // Step B: We temporarily drag the L pointer backward from the block edge
222     // to cover the left side of the query.
223     int time_stamp = dsu.save();
224     for (int i = query.l; i < (current_block + 1) * BLOCK_SIZE; i++)
225     {
226         dsu.unite(edges[i].u, edges[i].v);
227     }
228
229     // Step C: Record the answer!
230     ans[query.id] = dsu.components;
231
232     // Step D: Rollback the temporary L pointer movements.
233     // The R pointer progress remains saved for the next queries!
234     dsu.rollback(time_stamp);
235 }
236
237 for (int i = 0; i < q; i++)
238 {
239     cout << ans[i] << "\n";
240 }
241
242 int main()
243 {
244     // Fast I/O is mandatory for heavy Mo's algorithm problems
245     ios_base::sync_with_stdio(false);
246     cin.tie(NULL);
247     solve();
248     return 0;
249 }

```

### 2.7.5 sqrt decomposition

```

1
2 struct SQRT
3 {
4     vector<int> arr, blk;
5     int SQ;

```



```

6  Sqrt(vector<int> &a, int sq = 450)
7  {
8      SQ = sq;
9      arr = a;
10     int n = arr.size();
11     blk.resize(n / sq + 5);
12     for (int i = 0; i < n; i++)
13     {
14         blk[i / sq] += arr[i];
15     }
16 }
17 void update(int idx, int val)
18 {
19     int blkNum = idx / SQ;
20     blk[blkNum] -= arr[idx];
21     arr[idx] = val;
22     blk[blkNum] += arr[idx];
23 }
24
25 int query(int r)
26 {
27     int res = 0;
28     int blkNum = r / SQ;
29     for (int i = 0; i < blkNum; i++)
30     {
31         res += blk[i];
32     }
33     for (int i = blkNum * SQ; i <= r; i++)
34     {
35         res += arr[i];
36     }
37     return res;
38 }
39
40 int query(int l, int r)
41 {
42     return query(r) - query(l - 1);
43 }
44 /*
45 int query(int l, int r) {
46     int res = 0;
47     while (l <= r) {
48         if (l % SQ == 0 && l + SQ <= r) {
49             int blkNum = l / SQ;
50             // res = min(res, blk[blkNum]);
51             l += SQ;
52         }
53         else {
54             // res = min(res, arr[l]);
55             l++;
56         }
57     }

```

```

58     return res;
59 }
60 */
61 };

```

### 2.7.6 sqrt decomposition rangeUpdate rangeQuery

```

1 struct SqrtRangeUpdateRangeQuery
2 {
3     int n, S;
4     vector<long long> a, b, c; // b = lazy, c = sum of block
5
6     SqrtRangeUpdateRangeQuery(int n, const vector<long long> &arr)
7     {
8         this->n = n;
9         S = ceil(sqrt(n));
10        a = arr;
11        b.assign(S + 1, 0);
12        c.assign(S + 1, 0);
13        for (int i = 0; i < n; ++i)
14        {
15            c[i / S] += a[i];
16        }
17    }
18
19    // Helper to get block actual sum safely
20    long long get_block_sum(int k)
21    {
22        // Block sum = original sum + (lazy value * number of elements in block)
23        long long elements_count = min(n - k * S, S);
24        return c[k] + b[k] * elements_count;
25    }
26
27    // O(sqrt(N))
28    void add(int l, int r, long long delta)
29    {
30        int c_l = l / S, c_r = r / S;
31
32        if (c_l == c_r)
33        {
34            for (int i = l; i <= r; ++i)
35            {
36                a[i] += delta;
37                c[c_l] += delta;
38            }
39        }
40        else
41        {
42            for (int i = l, end = (c_l + 1) * S - 1; i <= end; ++i)
43            {
44                a[i] += delta;

```

```

45         c[c_l] += delta;
46     }
47     for (int k = c_l + 1; k <= c_r - 1; ++k)
48     {
49         b[k] += delta; // Lazy update
50     }
51     for (int i = c_r * S; i <= r; ++i)
52     {
53         a[i] += delta;
54         c[c_r] += delta;
55     }
56 }
57 }
58
59 // O(sqrt(N))
60 long long query(int l, int r)
61 {
62     long long sum = 0;
63     int c_l = l / S, c_r = r / S;
64
65     if (c_l == c_r)
66     {
67         for (int i = l; i <= r; ++i)
68             sum += a[i] + b[c_l];
69     }
70     else
71     {
72         for (int i = l, end = (c_l + 1) * S - 1; i <= end; ++i)
73             sum += a[i] + b[c_l];
74
75         for (int k = c_l + 1; k <= c_r - 1; ++k)
76             sum += get_block_sum(k); // Full block sum
77
78         for (int i = c_r * S; i <= r; ++i)
79             sum += a[i] + b[c_r];
80     }
81     return sum;
82 }
83 };

```

## 2.8 Segment Tree

### 2.8.1 2D segment tree

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: 2D SEGMENT TREE
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Subgrid queries", "Rectangle Sum/Max/Min", "Update a single cell
   in a grid".

```

```

6  * - Classic Scenarios: You are given an  $N \times M$  matrix. You need to repeatedly
   answer
7  * queries about a specific sub-rectangle (from top-left  $[R1, C1]$  to bottom-
   right  $[R2, C2]$ ).
8  * Between queries, you can update the value of any specific cell  $[R, C]$ .
9  * - The Magic: A standard Segment Tree handles a 1D array. A 2D Segment Tree is
   literally
10 * a "Segment Tree of Segment Trees". The outer tree splits the rows, and every
   single
11 * node in the outer tree contains an inner tree that splits the columns for
   that row-range.
12 *
13 * 2. HOW TO USE IT (THE BLACK BOX)
14 * - Initialization: Pass the row and column sizes to the constructor.
15 * Then, (optionally) build it using an existing 0-indexed 2D vector.
16 *     vector<vector<long long>> grid = {{1, 2}, {3, 4}};
17 *     SegTree2D st(2, 2);
18 *     st.build(grid);
19 *
20 * - Queries (All 0-indexed, Inclusive  $[r1, r2]$  and  $[c1, c2]$ ):
21 *     // Query the sum of the subgrid from top-left (0, 0) to bottom-right (1,
   1)
22 *     long long total = st.query(0, 0, 1, 1);
23 *
24 * - Updates (All 0-indexed):
25 *     // Change the value at row R, column C to X
26 *     st.update(R, C, X);
27 *
28 * - Complexity:
29 *     Time:  $O(N * M)$  to build.  $O(\log N * \log M)$  per update and query.
30 *     Space:  $O(N * M)$  memory. (Uses exactly  $16 * N * M$  elements).
31 *
32 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
33 * - Changing from Range Sum to Range Min/Max?
34 *     1. Change the `NEUTRAL` variable to `1e18` (for Min) or `-1e18` (for Max).
35 *     2. Change the `combine()` function to return `min(a, b)` or `max(a, b)`.
36 *
37 * - Memory Limit Exceeded (MLE) Warning:
38 *     If  $N = 1000$  and  $M = 1000$ , this tree uses 16,000,000 long longs (~128 MB).
39 *     This easily passes standard 256MB limits.
40 *     BUT, if  $N, M > 2000$ , you will MLE. If that happens and the problem only asks
   for SUM,
41 *     you must use a 2D Fenwick Tree (BIT) instead, which uses 1/16th of the
   memory!
42 */
43
44 #include <bits/stdc++.h>
45 using namespace std;
46
47 struct SegTree2D
48 {
49 private:

```

```

50  int n, m;
51  vector<vector<long long>> seg;
52
53  // ? Change these two dials to switch between Sum, Min, Max, GCD, etc.
54  const long long NEUTRAL = 0;
55
56  long long combine(long long a, long long b)
57  {
58      return a + b;
59  }
60
61  void build_y(int vx, int lx, int rx, int vy, int ly, int ry, const vector<
vector<long long>> &a)
62  {
63      if (ly == ry)
64      {
65          // We are at a leaf in the column tree.
66          if (lx == rx)
67          {
68              // We are ALSO at a leaf in the row tree. Base case!
69              seg[vx][vy] = a[lx][ly];
70          }
71          else
72          {
73              // We are a combination of two row-segments at the SAME exact
              column.
74              seg[vx][vy] = combine(seg[vx * 2][vy], seg[vx * 2 + 1][vy]);
75          }
76      }
77      else
78      {
79          int my = ly + (ry - ly) / 2;
80          build_y(vx, lx, rx, vy * 2, ly, my, a);
81          build_y(vx, lx, rx, vy * 2 + 1, my + 1, ry, a);
82          // Combine the left and right column-halves for the current row-
          segment
83          seg[vx][vy] = combine(seg[vx][vy * 2], seg[vx][vy * 2 + 1]);
84      }
85  }
86
87  void build_x(int vx, int lx, int rx, const vector<vector<long long>> &a)
88  {
89      if (lx != rx)
90      {
91          int mx = lx + (rx - lx) / 2;
92          build_x(vx * 2, lx, mx, a);
93          build_x(vx * 2 + 1, mx + 1, rx, a);
94      }
95      // Build the entire inner column-tree for the current row-segment
96      build_y(vx, lx, rx, 1, 0, m - 1, a);
97  }
98

```

```

99  void update_y(int vx, int lx, int rx, int vy, int ly, int ry, int x, int y,
long long val)
100 {
101     if (ly == ry)
102     {
103         if (lx == rx)
104         {
105             // Exact cell located!
106             seg[vx][vy] = val;
107         }
108         else
109         {
110             // Update the combined row-segment value for this specific column
111             seg[vx][vy] = combine(seg[vx * 2][vy], seg[vx * 2 + 1][vy]);
112         }
113     }
114     else
115     {
116         int my = ly + (ry - ly) / 2;
117         if (y <= my)
118         {
119             update_y(vx, lx, rx, vy * 2, ly, my, x, y, val);
120         }
121         else
122         {
123             update_y(vx, lx, rx, vy * 2 + 1, my + 1, ry, x, y, val);
124         }
125         seg[vx][vy] = combine(seg[vx][vy * 2], seg[vx][vy * 2 + 1]);
126     }
127 }
128
129 void update_x(int vx, int lx, int rx, int x, int y, long long val)
130 {
131     if (lx != rx)
132     {
133         int mx = lx + (rx - lx) / 2;
134         if (x <= mx)
135         {
136             update_x(vx * 2, lx, mx, x, y, val);
137         }
138         else
139         {
140             update_x(vx * 2 + 1, mx + 1, rx, x, y, val);
141         }
142     }
143     // Force the inner Y-tree to update itself using the new X data
144     update_y(vx, lx, rx, 1, 0, m - 1, x, y, val);
145 }
146
147 long long query_y(int vx, int vy, int ly, int ry, int qly, int qry)
148 {
149     if (qly > ry || qry < ly)

```

```

150         return NEUTRAL;
151     if (qly <= ly && ry <= qry)
152         return seg[vx][vy];
153
154     int my = ly + (ry - ly) / 2;
155     return combine(
156         query_y(vx, vy * 2, ly, my, qly, qry),
157         query_y(vx, vy * 2 + 1, my + 1, ry, qly, qry));
158 }
159
160 long long query_x(int vx, int lx, int rx, int qlx, int qrx, int qly, int qry)
161 {
162     if (qlx > rx || qrx < lx)
163         return NEUTRAL;
164
165     // If the row segment is fully inside our query, hand it off to the
166     // column query!
167     if (qlx <= lx && rx <= qrx)
168     {
169         return query_y(vx, 1, 0, m - 1, qly, qry);
170     }
171
172     int mx = lx + (rx - lx) / 2;
173     return combine(
174         query_x(vx * 2, lx, mx, qlx, qrx, qly, qry),
175         query_x(vx * 2 + 1, mx + 1, rx, qlx, qrx, qly, qry));
176 }
177 public:
178     SegTree2D(int rows, int cols)
179     {
180         n = rows;
181         m = cols;
182         // Allocate 4N rows, each containing a 4M column segment tree
183         seg.assign(4 * n, vector<long long>(4 * m, NEUTRAL));
184     }
185
186     void build(const vector<vector<long long>> &grid)
187     {
188         build_x(1, 0, n - 1, grid);
189     }
190
191     // Update cell at (row, col) to a new exact value
192     void update(int row, int col, long long val)
193     {
194         update_x(1, 0, n - 1, row, col, val);
195     }
196
197     // Query inclusive rectangle from top-left (r1, c1) to bottom-right (r2, c2)
198     long long query(int r1, int c1, int r2, int c2)
199     {
200         return query_x(1, 0, n - 1, r1, r2, c1, c2);

```

```

201     }
202 };

```

## 2.8.2 Dynamic segment tree

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DYNAMIC SEGMENT TREE
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Massive constraints", "109 or 1018 indices", "Sparse updates",
6  *   "Avoid Coordinate Compression".
7  * - Classic Scenarios: You need a Segment Tree over a massive range of
8  *   coordinates (e.g., from -109 to 109).
9  * - A standard segment tree would need an array of size 4 * 109, which
10  *   guarantees a Memory Limit Exceeded (MLE) crash.
11  * - The Magic: Instead of creating all nodes at the start, we only create nodes
12  *   *as we visit them*.
13  * - Initially, only the Root exists. When an update goes down the tree, it
14  *   allocates the exact  $O(\log(\text{Range}))$ 
15  *   children it needs to reach the bottom. Empty space simply doesn't exist in
16  *   memory!
17  *
18  * 2. HOW TO USE IT (THE BLACK BOX)
19  * - Initialization: Pass the absolute minimum and maximum possible coordinates
20  *   to the constructor.
21  *   // Create a tree covering the range [-109, 109]
22  *   DynamicSegTree st(-1e9, 1e9);
23  *
24  * - Updates (0-indexed point update):
25  *   // Add X to the value at coordinate `idx`
26  *   st.update(idx, X);
27  *
28  * - Queries (Inclusive [L, R]):
29  *   // Query the sum of elements from coordinate L to R
30  *   long long total = st.query(L, R);
31  *
32  * - Complexity:
33  *   Time:  $O(\log(\text{MAX\_VAL} - \text{MIN\_VAL}))$  per update and query.
34  *   Space:  $O(Q \log(\text{MAX\_VAL} - \text{MIN\_VAL}))$ . (Creates ~30
35  *   nodes per update for 109 ranges).
36  *
37  * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
38  * - Changing to Range Min/Max?
39  *   1. Change `Node::sum` to `val` and initialize to `1e18` (Min) or `-1e18` (
40  *     Max).
41  *   2. Change the `NEUTRAL` variable to `1e18` or `-1e18`.
42  *   3. In `query()`, return `min/max` instead of `+`.
43  * - "My code crashes randomly during updates!":
44  *   Never use references (`Node& curr = seg[node]`) before calling `.push_back()
45  *  `. Vectors reallocate memory when they

```

```

36 *   grow, which instantly invalidates all references to its elements, causing a
37 *   terrifying segmentation fault.
38 */
39
40 #include <bits/stdc++.h>
41 using namespace std;
42
43 struct DynamicSegTree
44 {
45 private:
46     struct Node
47     {
48         long long sum;
49         int lc; // Index of Left Child in the vector
50         int rc; // Index of Right Child in the vector
51
52         Node() : sum(0), lc(-1), rc(-1) {}
53     };
54
55     vector<Node> seg;
56     long long MIN_VAL, MAX_VAL;
57     const long long NEUTRAL = 0; // 0 for Sum. 1e18 for Min. -1e18 for Max.
58
59     // ? Creates children on-demand if they don't exist yet
60     void extend(int node)
61     {
62         if (seg[node].lc == -1)
63         {
64             seg[node].lc = seg.size();
65             seg.emplace_back(); // Safely allocates a new blank node
66         }
67         if (seg[node].rc == -1)
68         {
69             seg[node].rc = seg.size();
70             seg.emplace_back(); // Safely allocates a new blank node
71         }
72     }
73
74     void update(int node, long long lx, long long rx, long long idx, long long
75         val)
76     {
77         if (lx == rx)
78         {
79             seg[node].sum += val; // Point addition
80             return;
81         }
82         // ! We are going deeper, so we MUST ensure the children exist
83         extend(node);
84
85         // ? This safe midpoint calculation gracefully handles negative ranges
86         long long mid = lx + (rx - lx) / 2;
87
88         if (idx <= mid)
89         {
90             update(seg[node].lc, lx, mid, idx, val);
91         }
92         else
93         {
94             update(seg[node].rc, mid + 1, rx, idx, val);
95         }
96
97         // Pull up the answer from the children
98         seg[node].sum = seg[seg[node].lc].sum + seg[seg[node].rc].sum;
99     }
100
101     long long query(int node, long long lx, long long rx, long long ql, long long
102         qr)
103     {
104         // ! If this branch doesn't exist, it means nothing was ever added here!
105         // Return Neutral.
106         if (node == -1 || ql > rx || qr < lx)
107         {
108             return NEUTRAL;
109         }
110
111         // Total overlap
112         if (ql <= lx && rx <= qr)
113         {
114             return seg[node].sum;
115         }
116
117         long long mid = lx + (rx - lx) / 2;
118
119         // Partial overlap
120         long long left_ans = query(seg[node].lc, lx, mid, ql, qr);
121         long long right_ans = query(seg[node].rc, mid + 1, rx, ql, qr);
122
123         return left_ans + right_ans;
124     }
125
126 public:
127     DynamicSegTree(long long min_v, long long max_v)
128     {
129         MIN_VAL = min_v;
130         MAX_VAL = max_v;
131
132         // Pre-allocate some memory to minimize slow vector reallocations
133         seg.reserve(1000000);
134
135         // Create the root node at index 0
136         seg.emplace_back();

```

```

135     }
136
137     // Call this to add 'val' to coordinate 'idx'
138     void update(long long idx, long long val)
139     {
140         update(0, MIN_VAL, MAX_VAL, idx, val);
141     }
142
143     // Call this to get the sum of the range [l, r]
144     long long query(long long l, long long r)
145     {
146         return query(0, MIN_VAL, MAX_VAL, l, r);
147     }
148 };

```

### 2.8.3 Iterative segment tree

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: ITERATIVE SEGMENT TREE
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Point Updates", "Range Queries", "Strict Time Limits", "No Lazy
6  *   Propagation".
7  * - Classic Scenarios: You need a Segment Tree, but the time limit is extremely
8  *   tight (e.g., 1.0s for 5*105 queries),
9  *   or the memory limit is small. Recursive segment trees have overhead from
10  *   function calls and use 4*N memory.
11  *   This iterative version uses EXACTLY 2*N memory and runs in a tight `while`
12  *   loop, making it blindingly fast.
13  * - The Magic: The leaves of the tree are stored in indices `N` to `2N-1`. The
14  *   parents are stored in `1` to `N-1`.
15  *   To find the parent of node `x`, you just do `x / 2` (or `x >> 1`).
16  *   To find the sibling of node `x`, you just do `x ^ 1`.
17  *
18  * 2. HOW TO USE IT (THE BLACK BOX)
19  * - Initialization: Pass a 0-indexed vector to build the tree.
20  *   vector<long long> a = {1, 5, 2, 8, 3};
21  *   IterativeSegTree st(a);
22  *
23  * - Updates (0-indexed point SET):
24  *   // Replace the value at index `idx` with `X`
25  *   st.update(idx, X);
26  *
27  * - Queries (0-indexed, Inclusive [L, R]):
28  *   // Query the combined answer from index L to R
29  *   long long ans = st.query(L, R);
30  *
31  * - Complexity:
32  *   Time: O(N) to build. O(log N) per update and query (with a tiny constant
33  *   factor).
34  *   Space: O(N) memory. Uses exactly 2*N elements.

```

```

29 *
30 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
31 * - Changing from Range Sum to Min/Max/GCD?
32 * 1. Change the `NEUTRAL` variable to `1e18` (Min), `-1e18` (Max), or `0` (GCD)
33 *   ).
34 * 2. Update the `combine()` function to return `min(a, b)`, `max(a, b)`, or `
35 *   gcd(a, b)`.
36 * - Changing `update` from "Set" to "Add"?
37 * In `update()`, change `seg[idx += n] = val;` to `seg[idx += n] += val;`.
38 */
39 #include <bits/stdc++.h>
40 using namespace std;
41 struct IterativeSegTree
42 {
43 private:
44     int n;
45     vector<long long> seg;
46
47     // ? Dial 1: The Neutral Element (Identity)
48     // 0 for Sum, 1e18 for Min, -1e18 for Max, 0 for GCD
49     const long long NEUTRAL = 0;
50
51     // ? Dial 2: The Combination Logic
52     long long combine(long long a, long long b)
53     {
54         return a + b;
55     }
56 public:
57     IterativeSegTree(const vector<long long> &v)
58     {
59         n = v.size();
60         // Allocate exactly 2 * N memory. The 0-th index is unused dummy space.
61         seg.assign(2 * n, NEUTRAL);
62
63         // 1. Insert the leaf nodes directly into the second half of the array
64         for (int i = 0; i < n; i++)
65         {
66             seg[n + i] = v[i];
67         }
68
69         // 2. Build the parents from bottom to top
70         for (int i = n - 1; i > 0; i--)
71         {
72             // A node `i` has children at `2*i` and `2*i + 1`
73             seg[i] = combine(seg[i << 1], seg[i << 1 | 1]);
74         }
75     }
76
77     // Replace the value at 0-indexed position `idx` with `val`
78

```

```

79 void update(int idx, long long val)
80 {
81     // Jump straight to the leaf node
82     idx += n;
83     seg[idx] = val; // ! Change to += if problem requires Point Add instead
      of Point Set
84
85     // Climb to the root, updating parents
86     // `idx > 1` stops us before we update the unused 0-th index
87     for (idx >= 1; idx > 0; idx >= 1)
88     {
89         // `idx << 1` is left child, `idx << 1 | 1` is right child
90         seg[idx] = combine(seg[idx << 1], seg[idx << 1 | 1]);
91     }
92 }
93
94 // Query the inclusive range [l, r]
95 long long query(int l, int r)
96 {
97     long long res_left = NEUTRAL;
98     long long res_right = NEUTRAL;
99
100    // Jump straight to the leaves
101    // Note: We use inclusive [l, r] so our loop condition is l <= r
102    for (l += n, r += n; l <= r; l >= 1, r >= 1)
103    {
104
105        // ? If `l` is odd, it is a right child.
106        // This means its parent covers data OUTSIDE our [l, r] range!
107        // We must process this node independently, then move `l` to the
          right.
108        if (l & 1)
109        {
110            res_left = combine(res_left, seg[l++]);
111        }
112
113        // ? If `r` is even, it is a left child.
114        // This means its parent covers data OUTSIDE our [l, r] range!
115        // We must process this node independently, then move `r` to the left
116
117        if (!(r & 1))
118        {
119            res_right = combine(seg[r--], res_right);
120        }
121
122        // Combine the results from the left sweep and the right sweep.
123        // Doing it this way preserves the strict Left-to-Right order for non-
          commutative operations!
124        return combine(res_left, res_right);
125    }
126 };

```

```

127
128 void solve()
129 {
130     int n, q;
131     if (!(cin >> n >> q))
132         return;
133
134     vector<long long> a(n);
135     for (int i = 0; i < n; i++)
136         cin >> a[i];
137
138     IterativeSegTree st(a);
139
140     while (q--)
141     {
142         int type;
143         cin >> type;
144         if (type == 1)
145         {
146             int idx;
147             long long val;
148             cin >> idx >> val;
149             st.update(idx, val); // 0-indexed update
150         }
151         else
152         {
153             int l, r;
154             cin >> l >> r;
155             cout << st.query(l, r) << "\n"; // 0-indexed query
156         }
157     }
158 }
159
160 int main()
161 {
162     // Standard fast I/O
163     ios_base::sync_with_stdio(false);
164     cin.tie(NULL);
165     solve();
166     return 0;
167 }

```

## 2.8.4 Merge sort tree

```

1 struct Node
2 {
3     vector<int> v;
4 };
5
6 struct segTree
7 {

```

```

8   vector<Node> seg;
9   int sz;
10
11  segTree(int n)
12  {
13      sz = 1;
14      while (sz < n)
15          sz *= 2;
16      seg.assign(2 * sz, Node());
17  }
18
19  Node merge(Node &l, Node &r)
20  {
21      Node res;
22      res.v.reserve(l.v.size() + r.v.size());
23      std::merge(all(l.v), all(r.v), back_inserter(res.v));
24      return res;
25  }
26
27  void build(vector<int> &arr, int ni, int lx, int rx)
28  {
29      if (rx - lx == 1)
30      {
31          if (lx < (int)arr.size())
32          {
33              seg[ni].v = {arr[lx]};
34          }
35          return;
36      }
37      int mid = (lx + rx) / 2;
38      build(arr, ni * 2 + 1, lx, mid);
39      build(arr, ni * 2 + 2, mid, rx);
40      seg[ni] = merge(seg[ni * 2 + 1], seg[ni * 2 + 2]);
41  }
42
43  void build(vector<int> &arr)
44  {
45      build(arr, 0, 0, sz);
46  }
47
48  // 1. Elements Strictly Less Than K (< K)
49  int query_less(int l, int r, int k, int ni, int lx, int rx)
50  {
51      if (lx >= 1 && rx <= r)
52      {
53          return lower_bound(all(seg[ni].v), k) - seg[ni].v.begin();
54      }
55      if (lx >= r || rx <= 1)
56          return 0;
57      int mid = (lx + rx) / 2;
58      return query_less(l, r, k, ni * 2 + 1, lx, mid) + query_less(l, r, k, ni * 2 + 2, mid, rx);
59  }
60
61  // 2. Elements Less Than or Equal K (<= K)
62  int query_less_equal(int l, int r, int k, int ni, int lx, int rx)
63  {
64      if (lx >= 1 && rx <= r)
65      {
66          return upper_bound(all(seg[ni].v), k) - seg[ni].v.begin();
67      }
68      if (lx >= r || rx <= 1)
69          return 0;
70      int mid = (lx + rx) / 2;
71      return query_less_equal(l, r, k, ni * 2 + 1, lx, mid) + query_less_equal(l, r, k, ni * 2 + 2, mid, rx);
72  }
73
74  // 3. Elements Strictly Greater Than K (> K)
75  int query_greater(int l, int r, int k, int ni, int lx, int rx)
76  {
77      if (lx >= 1 && rx <= r)
78      {
79          return seg[ni].v.end() - upper_bound(all(seg[ni].v), k);
80      }
81      if (lx >= r || rx <= 1)
82          return 0;
83      int mid = (lx + rx) / 2;
84      return query_greater(l, r, k, ni * 2 + 1, lx, mid) + query_greater(l, r, k, ni * 2 + 2, mid, rx);
85  }
86
87  // 4. Elements Greater Than or Equal K (>= K)
88  int query_greater_equal(int l, int r, int k, int ni, int lx, int rx)
89  {
90      if (lx >= 1 && rx <= r)
91      {
92          return seg[ni].v.end() - lower_bound(all(seg[ni].v), k);
93      }
94      if (lx >= r || rx <= 1)
95          return 0;
96      int mid = (lx + rx) / 2;
97      return query_greater_equal(l, r, k, ni * 2 + 1, lx, mid) + query_greater_equal(l, r, k, ni * 2 + 2, mid, rx);
98  }
99  // 5. Elements in Range [X, Y]
100 int query_in_range(int l, int r, int x, int y)
101 {
102     // Total elements <= Y MINUS Total elements < X
103     return get_less_equal(l, r, y) - get_less(l, r, x);
104 }
105
106 // Helpers
107 int get_greater(int l, int r, int k) { return query_greater(l, r, k, 0, 0, sz);

```



```

    ); }
108 int get_greater_equal(int l, int r, int k) { return query_greater_equal(l, r,
    k, 0, 0, sz); }
109 int get_less(int l, int r, int k) { return query_less(l, r, k, 0, 0, sz); }
110 int get_less_equal(int l, int r, int k)
111 {
112     return query_less_equal(l, r, k, 0, 0, sz);
113 }
114 };

```

### 2.8.5 PST

```

1 // Distinct elements in range
2 // kth element
3 // counting or frequency of numbers in range
4
5 struct PST
6 {
7     struct Node
8     {
9         int l = 0;
10        int r = 0;
11        int val = 0;
12    };
13
14    vector<Node> tree;
15    vector<int> roots;
16    int n;
17    int neutral;
18
19    PST(int n, int neutral = 0) : n(n), neutral(neutral)
20    {
21        tree.reserve(1500000);
22        tree.push_back({0, 0, neutral});
23        roots.push_back(0);
24    }
25
26    int merge(int a, int b)
27    {
28        return a + b;
29    }
30
31    int build(const vector<int> &a, int lx, int rx)
32    {
33        int u = tree.size();
34        tree.push_back({0, 0, neutral});
35
36        if (rx - lx == 1)
37        {
38            if (lx < (int)a.size())
39                tree[u].val = a[lx];

```

```

40        return u;
41    }
42
43    int mid = lx + (rx - lx) / 2;
44    tree[u].l = build(a, lx, mid);
45    tree[u].r = build(a, mid, rx);
46    tree[u].val = merge(tree[tree[u].l].val, tree[tree[u].r].val);
47    return u;
48 }
49
50 void build_initial(const vector<int> &a)
51 {
52     roots.push_back(build(a, 0, n));
53 }
54
55 int update(int node, int idx, int val, int lx, int rx)
56 {
57     int u = tree.size();
58     tree.push_back(tree[node]);
59
60     if (rx - lx == 1)
61     {
62         tree[u].val += val;
63         return u;
64     }
65
66     int mid = lx + (rx - lx) / 2;
67     if (idx < mid)
68     {
69         int nxt = update(tree[node].l, idx, val, lx, mid);
70         tree[u].l = nxt;
71     }
72     else
73     {
74         int nxt = update(tree[node].r, idx, val, mid, rx);
75         tree[u].r = nxt;
76     }
77
78     tree[u].val = merge(tree[tree[u].l].val, tree[tree[u].r].val);
79     return u;
80 }
81
82 int update(int version, int idx, int val)
83 {
84     int new_root = update(roots[version], idx, val, 0, n);
85     roots.push_back(new_root);
86     return (int)roots.size() - 1;
87 }
88
89 int query(int node, int l, int r, int lx, int rx)
90 {
91     if (node == 0)

```

```

92         return neutral;
93     if (lx >= r || rx <= l)
94         return neutral;
95     if (lx >= l && rx <= r)
96         return tree[node].val;
97
98     int mid = lx + (rx - lx) / 2;
99     return merge(query(tree[node].l, l, r, lx, mid),
100                 query(tree[node].r, l, r, mid, rx));
101 }
102
103 int query(int version, int l, int r)
104 {
105     return query(roots[version], l, r, 0, n);
106 }
107
108 int findKth(int l, int r, int k, int lx, int rx)
109 {
110     if (rx - lx == 1)
111         return lx;
112     int cnt = tree[tree[r].l].val - tree[tree[l].l].val;
113     int mid = lx + (rx - lx) / 2;
114     if (cnt >= k)
115         return findKth(tree[l].l, tree[r].l, k, lx, mid);
116     return findKth(tree[l].r, tree[r].r, k - cnt, mid, rx);
117 }
118
119 int findKth(int l, int r, int k)
120 {
121     return findKth(roots[l], roots[r], k, 0, n);
122 }
123 };

```

## 2.8.6 PST Lazy

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: PERSISTENT LAZY SEGMENT TREE
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Time travel", "Historical versions", "Range updates on previous
   states".
6  * - Classic Scenarios: You are given an array. You need to perform Range
   Additions or Range Sets.
7  * Each update creates a new "version" of the array. You must be able to query
   the Range Sum
8  * of ANY past version of the array efficiently.
9  * - The Magic: Instead of modifying nodes in place, we create a clone of every
   node we touch.
10 * This means `update(v, L, R)` leaves version `v` completely untouched and
   returns the root ID

```

```

11 * of the brand new version. Because we only change  $O(\log N)$  nodes per update,
   memory and time
12 * remain highly efficient!
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization: Pass a 0-indexed vector<long long> to build Version 0.
16 *     vector<long long> a = {1, 5, 2, 8, 3};
17 *     PersistentLazySegTree pst(a);
18 *
19 * - Tracking Versions: Every update returns a NEW root index. You should store
   these!
20 *     vector<int> versions;
21 *     versions.push_back(pst.roots[0]); // Initial version
22 *
23 * - Updates (All 0-indexed, Inclusive [L, R]):
24 *     // Add X to range [L, R] on a specific version, and save the new version
   ID
25 *     int new_root = pst.updateAdd(versions.back(), L, R, X);
26 *     versions.push_back(new_root);
27 *
28 * - Queries (All 0-indexed, Inclusive [L, R]):
29 *     // Query the sum of range [L, R] exactly as it was in version 'V'
30 *     long long sum = pst.querySum(versions[V], L, R);
31 *
32 * - Complexity:
33 *     Time:  $O(N)$  to build.  $O(\log N)$  per update and query.
34 *     Space:  $O(N + Q \log N)$  memory.
35 *
36 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
37 * - Memory Limit Exceeded (MLE)? Persistent Lazy creates ~4 new nodes per update
   .
38 * If  $N = 10^5$  and  $Q = 10^5$ , you will generate roughly  $10^5 + 400,000$  nodes.
39 * The constructor already calls `seg.reserve(N * 40)` to prevent expensive
   vector reallocations.
40 * Increase this multiplier if Q is exceptionally large.
41 * - Changing from Range Add to Range Set?
42 * Range Add is special because we can just accumulate `lazy` tags during the
   query without
43 * ever modifying the tree. If you want Range Set, you must pass down a `
   lazySet` flag.
44 * In `query`, if you hit a node with an active `lazySet`, simply return `
   lazySet * (r - l + 1)`
45 * immediately, ignoring its children.
46 */
47
48 #include <bits/stdc++.h>
49 using namespace std;
50
51 struct PersistentLazySegTree
52 {
53 private:
54     struct Node

```

```

55 {
56     int lc, rc;    // Indices of the left and right children
57     long long sum; // The range sum
58     long long lazy; // The lazy tag (Range Add)
59
60     Node() : lc(0), rc(0), sum(0), lazy(0) {}
61     Node(int lc, int rc, long long sum, long long lazy)
62         : lc(lc), rc(rc), sum(sum), lazy(lazy) {}
63 };
64
65 int n;
66 vector<Node> seg;
67
68 // ? Creates a brand new node that is an exact copy of the given node ID
69 int clone(int node)
70 {
71     seg.push_back(seg[node]);
72     return seg.size() - 1;
73 }
74
75 int build(int l, int r, const vector<long long> &a)
76 {
77     int curr = seg.size();
78     seg.emplace_back(); // Allocate new node
79
80     if (l == r)
81     {
82         if (l < (int)a.size())
83         {
84             seg[curr].sum = a[l];
85         }
86         return curr;
87     }
88
89     int mid = l + (r - l) / 2;
90     int left_child = build(l, mid, a);
91     int right_child = build(mid + 1, r, a);
92
93     seg[curr].lc = left_child;
94     seg[curr].rc = right_child;
95     seg[curr].sum = seg[left_child].sum + seg[right_child].sum;
96
97     return curr;
98 }
99
100 int updateAdd(int node, int l, int r, int ql, int qr, long long val)
101 {
102     // ? Core Persistence Logic: NEVER modify the original node. Always work
103     //    on a clone!
104     int curr = clone(node);
105     // 1. Total Overlap

```

```

106     if (ql <= l && r <= qr)
107     {
108         seg[curr].sum += val * (r - l + 1);
109         seg[curr].lazy += val;
110         return curr;
111     }
112
113     int mid = l + (r - l) / 2;
114
115     // ? Lazy Propagation in Persistence:
116     // Before we branch down, we must push the current node's lazy tag to its
117     //    children.
118     // Because pushing modifies the children, we MUST clone the children
119     //    first!
120     if (seg[curr].lazy != 0)
121     {
122         if (l != r)
123         { // Don't push down if it's a leaf
124             int left_child = clone(seg[curr].lc);
125             seg[left_child].sum += seg[curr].lazy * (mid - l + 1);
126             seg[left_child].lazy += seg[curr].lazy;
127             seg[curr].lc = left_child;
128
129             int right_child = clone(seg[curr].rc);
130             seg[right_child].sum += seg[curr].lazy * (r - mid);
131             seg[right_child].lazy += seg[curr].lazy;
132             seg[curr].rc = right_child;
133         }
134         seg[curr].lazy = 0; // Cleared from the current node
135     }
136
137     // 2. Partial Overlap (Recursion)
138     if (ql <= mid)
139     {
140         seg[curr].lc = updateAdd(seg[curr].lc, l, mid, ql, qr, val);
141     }
142     if (qr > mid)
143     {
144         seg[curr].rc = updateAdd(seg[curr].rc, mid + 1, r, ql, qr, val);
145     }
146
147     // 3. Update Current Node Sum
148     seg[curr].sum = seg[seg[curr].lc].sum + seg[seg[curr].rc].sum;
149     return curr;
150 }
151
152 // ? IMPORTANT: We CANNOT push down lazy tags during a query. Pushing clones
153 //    nodes,
154 //    which would cause a read-only query to consume massive amounts of memory!
155 //    Instead, we just pass the accumulated lazy value down the recursion.
156 long long querySum(int node, int l, int r, int ql, int qr, long long lazy_acc)
157 {

```

```

154 {
155     if (!node || ql > r || qr < l)
156         return 0;
157
158     // 1. Total Overlap
159     if (ql <= l && r <= qr)
160     {
161         // Apply all the pending lazy additions from ancestors to this block
162         return seg[node].sum + lazy_acc * (r - l + 1);
163     }
164
165     int mid = l + (r - l) / 2;
166
167     // 2. Add the CURRENT node's lazy tag to the accumulator for its children
168     long long current_lazy = lazy_acc + seg[node].lazy;
169
170     // 3. Partial Overlap
171     long long res = 0;
172     if (ql <= mid)
173     {
174         res += querySum(seg[node].lc, l, mid, ql, qr, current_lazy);
175     }
176     if (qr > mid)
177     {
178         res += querySum(seg[node].rc, mid + 1, r, ql, qr, current_lazy);
179     }
180
181     return res;
182 }
183
184 public:
185     // Keeps track of the root node index for every version
186     vector<int> roots;
187
188     PersistentLazySegTree(const vector<long long> &a)
189     {
190         n = a.size();
191
192         // Pre-allocate memory to prevent expensive vector resizing.
193         // Formula:  $N + (Queries * 4 * \log_2(N))$ . 40 is a safe constant for  $10^5$ .
194         seg.reserve(n * 40);
195
196         // Create a dummy node 0 (Null Node)
197         seg.emplace_back();
198
199         // Build Version 0
200         roots.push_back(build(0, n - 1, a));
201     }
202
203     // Pass the root ID of the version you want to update, get the new root ID
204     // back
205     int updateAdd(int root_version, int l, int r, long long val)

```

```

205 {
206     return updateAdd(root_version, 0, n - 1, l, r, val);
207 }
208
209 // Pass the root ID of the version you want to query
210 long long querySum(int root_version, int l, int r)
211 {
212     return querySum(root_version, 0, n - 1, l, r, 0LL);
213 }
214 };

```

### 2.8.7 Segment tree

```

1 struct Node
2 {
3     int val;
4
5     Node()
6     {
7         val = 0;
8     }
9
10    Node(int val) : val(val)
11    {
12    }
13
14    void change(int x)
15    {
16        val = x;
17    }
18 };
19
20 struct segmentTree
21 {
22     int sz;
23     vector<Node> seg;
24
25     segmentTree(int n)
26     {
27         sz = 1;
28         while (sz < n)
29             sz *= 2;
30         seg.assign(2 * sz, Node());
31     }
32
33     void init(vector<int> &arr, int node, int lx, int rx)
34     {
35         if (rx - lx == 1)
36         {
37             if (lx < int(arr.size()))
38                 seg[node] = Node(arr[lx]);

```

```

39         return;
40     }
41     int mid = (lx + rx) / 2;
42     init(arr, 2 * node + 1, lx, mid);
43     init(arr, 2 * node + 2, mid, rx);
44     seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
45 }
46
47 void init(vector<int> &arr)
48 {
49     init(arr, 0, 0, sz);
50 }
51
52 Node merge(Node &lf, Node &rt)
53 {
54     Node ans = Node();
55     ans.val = lf.val + rt.val;
56     return ans;
57 }
58
59 void set(int idx, int val, int node, int lx, int rx)
60 {
61     if (rx - lx == 1)
62     {
63         seg[node].change(val);
64         return;
65     }
66     int mid = (lx + rx) / 2;
67     if (idx < mid)
68         set(idx, val, 2 * node + 1, lx, mid);
69     else
70         set(idx, val, 2 * node + 2, mid, rx);
71     seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
72 }
73
74 void set(int idx, int val)
75 {
76     set(idx, val, 0, 0, sz);
77 }
78
79 Node get(int l, int r, int node, int lx, int rx)
80 {
81     if (lx >= l && rx <= r)
82         return seg[node];
83     if (rx <= l || lx >= r)
84         return Node();
85     int mid = (lx + rx) / 2;
86     Node lf = get(l, r, 2 * node + 1, lx, mid);
87     Node ri = get(l, r, 2 * node + 2, mid, rx);
88     return merge(lf, ri);
89 }
90

```

```

91 int get(int l, int r)
92 {
93     return get(l, r, 0, 0, sz).val;
94 }
95 };

```

## 2.8.8 Segment tree beats

```

1  /**
2  *  THE ULTIMATE CP TEMPLATE BLUEPRINT: SEGMENT TREE BEATS
3  *
4  *  1.  WHAT PROBLEM DOES THIS SOLVE?
5  *  - Keywords: "Range chmin" ( $A[i] = \min(A[i], X)$ ), "Range chmax", "Range Sum
    with Min/Max updates".
6  *  - Classic Scenarios: Standard Segment Trees completely fail if you try to
    perform a range minimum
7  *  update (e.g., "for all elements from L to R, replace the element with min(
    element, X)")
8  *  while simultaneously asking for the Range Sum.
9  *  - The Magic: Segment Tree Beats tracks the *First Maximum* and *Second Maximum
    * in every node.
10 *  If your update X is between the First and Second Max, it only affects the
    First Max!
11 *  It uses this trick to update the sum mathematically in amortized  $O(\log^2 N)$ 
    time.
12 *
13 *  2.  HOW TO USE IT (THE BLACK BOX)
14 *  - Initialization: Pass a 0-indexed vector<long long> to the constructor.
15 *      vector<long long> a = {1, 5, 2, 8, 3};
16 *      SegmentTreeBeats seg(a);
17 *
18 *  - Queries (All 0-indexed, Inclusive [L, R]):
19 *      seg.updateMin(L, R, X) : Performs  $A[i] = \min(A[i], X)$  for  $i$  in  $[L, R]$ .
20 *      seg.updateMax(L, R, X) : Performs  $A[i] = \max(A[i], X)$  for  $i$  in  $[L, R]$ .
21 *      seg.updateAdd(L, R, X) : Performs  $A[i] = A[i] + X$  for  $i$  in  $[L, R]$ .
22 *      seg.updateSet(L, R, X) : Performs  $A[i] = X$  for  $i$  in  $[L, R]$ .
23 *      seg.querySum(L, R)      : Returns the sum of  $A[i]$  in  $[L, R]$ .
24 *      seg.queryMin(L, R)      : Returns the Minimum in  $[L, R]$ .
25 *      seg.queryMax(L, R)      : Returns the Maximum in  $[L, R]$ .
26 *
27 *  - Complexity:
28 *      Time:  $O(N)$  to build. Amortized  $O(\log^2 N)$  per query.
29 *      Space:  $O(N)$  memory.
30 *
31 *  3.  HOW TO ADAPT IT (THE DIALS & SWITCHES)
32 *  - 1-based indexing? Instead of changing the recursive logic, simply add a
    dummy element `0`
33 *  to the front of your vector before initializing: `v.insert(v.begin(), 0);`.
    Then query `[1, N]`.
34 *  - Numbers larger than  $10^{14}$ ? The `INF` value is set to  $2 * 10^{15}$ . If your
    problem features numbers

```

```

35 * larger than that, change the `const long long INF` variable at the top of
    the struct.
36 * - Need more speed? If the problem ONLY asks for `updateMin` and `querySum`,
    you can safely delete
37 * all variables related to `mn`, `secMn`, and `mnCnt` to cut memory and
    execution time in half.
38 */
39
40 #include <bits/stdc++.h>
41 using namespace std;
42
43 // ! Assumes you are using long long to avoid any overflow bugs.
44 struct SegmentTreeBeats
45 {
46 private:
47 // ! FIX: Added strict parentheses to prevent operator precedence bugs in math
48 #define L (2 * node + 1)
49 #define R (2 * node + 2)
50 #define mid ((l + r) >> 1)
51
52 // ! FIX: Safely typed INF to avoid macro expansion issues
53 const long long INF = 2e15 + 5;
54
55 struct Node
56 {
57     long long sum, mx, secMx, mxCnt, mn, mnCnt, secMn, lazyAdd, lazySet;
58
59     Node(long long summ, long long mxx, long long secMxx, long long mxCntt,
60         long long mnn, long long secMnn, long long mnCntt, long long add = 0, long long set = -1)
61         : sum(summ), mx(mxx), secMx(secMxx), mxCnt(mxCntt),
62         mn(mnn), secMn(secMnn), mnCnt(mnCntt),
63         lazyAdd(add), lazySet(set)
64     {
65     }
66 };
67
68 int sz;
69 vector<Node> seg;
70
71 Node merge(Node &a, Node &b)
72 {
73     Node ret(0, -INF, -INF, 0, INF, INF, 0, 0, -1);
74     ret.sum = a.sum + b.sum;
75
76     // ? Merge Maximums
77     ret.mx = max(a.mx, b.mx);
78     ret.secMx = max(a.secMx, b.secMx);
79     if (ret.mx == a.mx)
80         ret.mxCnt += a.mxCnt;
81     else
82         ret.secMx = max(ret.secMx, a.mx);
83
84     if (ret.mx == b.mx)
85         ret.mxCnt += b.mxCnt;
86     else
87         ret.secMx = max(ret.secMx, b.mx);
88
89     // ? Merge Minimums
90     ret.mn = min(a.mn, b.mn);
91     ret.secMn = min(a.secMn, b.secMn);
92     if (ret.mn == a.mn)
93         ret.mnCnt += a.mnCnt;
94     else
95         ret.secMn = min(ret.secMn, a.mn);
96     if (ret.mn == b.mn)
97         ret.mnCnt += b.mnCnt;
98     else
99         ret.secMn = min(ret.secMn, b.mn);
100
101     return ret;
102 }
103
104 void pushSet(int l, int r, int node, long long val)
105 {
106     seg[node] = Node(val * (r - l + 1), val, -INF, r - l + 1, val, INF, r - l
107         + 1, 0, val);
108 }
109
110 void pushMin(int l, int r, int node, long long mn)
111 {
112     if (seg[node].mn >= mn)
113     {
114         pushSet(l, r, node, mn);
115         return;
116     }
117     if (seg[node].mx > mn)
118     {
119         if (seg[node].secMn == seg[node].mx)
120             seg[node].secMn = mn;
121         long long val = seg[node].mx - mn;
122         seg[node].sum -= val * seg[node].mxCnt;
123         seg[node].mx = mn;
124     }
125 }
126
127 void pushMax(int l, int r, int node, long long mx)
128 {
129     if (seg[node].mx <= mx)
130     {
131         pushSet(l, r, node, mx);
132         return;
133     }
134     if (seg[node].mn < mx)
135     {

```

```

134         if (seg[node].secMx == seg[node].mn)
135             seg[node].secMx = mx;
136         long long val = mx - seg[node].mn;
137         seg[node].sum += val * seg[node].mnCnt;
138         seg[node].mn = mx;
139     }
140 }
141
142 void pushAdd(int l, int r, int node, long long val)
143 {
144     if (seg[node].mx == seg[node].mn)
145     {
146         pushSet(l, r, node, seg[node].mn + val);
147         return;
148     }
149     seg[node].mx += val;
150     if (seg[node].secMx != -INF)
151         seg[node].secMx += val;
152     seg[node].mn += val;
153     if (seg[node].secMn != INF)
154         seg[node].secMn += val;
155     seg[node].sum += (r - l + 1) * val;
156     seg[node].lazyAdd += val;
157 }
158
159 void propegate(int l, int r, int node)
160 {
161     if (l == r)
162         return;
163     if (seg[node].lazySet != -1)
164     {
165         pushSet(l, mid, L, seg[node].lazySet);
166         pushSet(mid + 1, r, R, seg[node].lazySet);
167         seg[node].lazySet = -1;
168     }
169     else
170     {
171         pushAdd(l, mid, L, seg[node].lazyAdd);
172         pushAdd(mid + 1, r, R, seg[node].lazyAdd);
173         seg[node].lazyAdd = 0;
174         pushMin(l, mid, L, seg[node].mx);
175         pushMin(mid + 1, r, R, seg[node].mx);
176         pushMax(l, mid, L, seg[node].mn);
177         pushMax(mid + 1, r, R, seg[node].mn);
178     }
179 }
180
181 void build(int l, int r, int node, vector<long long> &v)
182 {
183     if (l == r)
184     {
185         if (l < (int)v.size())

```

```

186         {
187             seg[node] = Node(v[l], v[l], -INF, 1, v[l], INF, 1);
188         }
189         return;
190     }
191     build(l, mid, L, v);
192     build(mid + 1, r, R, v);
193     seg[node] = merge(seg[L], seg[R]);
194 }
195
196 void updateAdd(int l, int r, int node, int lx, int rx, long long val)
197 {
198     if (l > rx || r < lx)
199         return;
200     if (l >= lx && r <= rx)
201     {
202         pushAdd(l, r, node, val);
203         return;
204     }
205     propegate(l, r, node);
206     updateAdd(l, mid, L, lx, rx, val);
207     updateAdd(mid + 1, r, R, lx, rx, val);
208     seg[node] = merge(seg[L], seg[R]);
209 }
210
211 void updateSet(int l, int r, int node, int lx, int rx, long long val)
212 {
213     if (l > rx || r < lx)
214         return;
215     if (l >= lx && r <= rx)
216     {
217         pushSet(l, r, node, val);
218         return;
219     }
220     propegate(l, r, node);
221     updateSet(l, mid, L, lx, rx, val);
222     updateSet(mid + 1, r, R, lx, rx, val);
223     seg[node] = merge(seg[L], seg[R]);
224 }
225
226 void updateMin(int l, int r, int node, int lx, int rx, long long mn)
227 {
228     // ? Core BEATS logic: If the segment max is already <= mn, do nothing.
229     if (r < lx || l > rx || seg[node].mx <= mn)
230         return;
231     // ? If the update affects the maximums, but NOT the second maximums, we
232     //    can apply it safely!
233     if (l >= lx && r <= rx && mn > seg[node].secMx)
234     {
235         pushMin(l, r, node, mn);
236         return;
237     }

```

```

237     propegate(l, r, node);
238     updateMin(l, mid, L, lx, rx, mn);
239     updateMin(mid + 1, r, R, lx, rx, mn);
240     seg[node] = merge(seg[L], seg[R]);
241 }
242
243 void updateMax(int l, int r, int node, int lx, int rx, long long mx)
244 {
245     if (l > rx || r < lx || mx <= seg[node].mn)
246         return;
247     if (l >= lx && r <= rx && mx < seg[node].secMn)
248     {
249         pushMax(l, r, node, mx);
250         return;
251     }
252     propegate(l, r, node);
253     updateMax(l, mid, L, lx, rx, mx);
254     updateMax(mid + 1, r, R, lx, rx, mx);
255     seg[node] = merge(seg[L], seg[R]);
256 }
257
258 long long querySum(int l, int r, int node, int lx, int rx)
259 {
260     if (l > rx || r < lx)
261         return 0;
262     if (l >= lx && r <= rx)
263         return seg[node].sum;
264     propegate(l, r, node);
265     return querySum(l, mid, L, lx, rx) + querySum(mid + 1, r, R, lx, rx);
266 }
267
268 long long queryMin(int l, int r, int node, int lx, int rx)
269 {
270     if (l > rx || r < lx)
271         return INF;
272     if (l >= lx && r <= rx)
273         return seg[node].mn;
274     propegate(l, r, node);
275     return min(queryMin(l, mid, L, lx, rx), queryMin(mid + 1, r, R, lx, rx));
276 }
277
278 long long queryMax(int l, int r, int node, int lx, int rx)
279 {
280     if (l > rx || r < lx)
281         return -INF;
282     if (l >= lx && r <= rx)
283         return seg[node].mx;
284     propegate(l, r, node);
285     return max(queryMax(l, mid, L, lx, rx), queryMax(mid + 1, r, R, lx, rx));
286 }
287
288 #undef L

```

```

289 #undef R
290 #undef mid
291
292 public:
293     SegmentTreeBeats(vector<long long> &v)
294     {
295         int n = v.size() + 5;
296         sz = 1;
297         while (sz <= n)
298             sz <<= 1;
299         seg = vector<Node>(sz << 1, Node(0, -INF, -INF, 0, INF, INF, 0, 0, -1));
300         build(0, sz - 1, 0, v);
301     }
302
303     void updateAdd(int l, int r, long long val) { updateAdd(0, sz - 1, 0, l, r, val); }
304     void updateSet(int l, int r, long long val) { updateSet(0, sz - 1, 0, l, r, val); }
305     void updateMin(int l, int r, long long mn) { updateMin(0, sz - 1, 0, l, r, mn); }
306     void updateMax(int l, int r, long long mx) { updateMax(0, sz - 1, 0, l, r, mx); }
307     long long queryMax(int l, int r) { return queryMax(0, sz - 1, 0, l, r); }
308     long long queryMin(int l, int r) { return queryMin(0, sz - 1, 0, l, r); }
309     long long querySum(int l, int r) { return querySum(0, sz - 1, 0, l, r); }
310 };

```

## 2.8.9 Segment tree lazy

```

1 struct Node
2 {
3     int val, lazy;
4     bool is_lazy;
5
6     Node()
7     {
8         val = 0, lazy = 0, is_lazy = false;
9     }
10
11     Node(int val) : val(val)
12     {
13         lazy = 0, is_lazy = false;
14     }
15
16     void change(int x, int lx, int rx)
17     {
18         val += x * (rx - lx);
19         lazy += x;
20         is_lazy = true;
21     }
22

```



```

23 void reset()
24 {
25     lazy = 0, is_lazy = false;
26 }
27 };
28
29 struct segmentTree
30 {
31     int sz;
32     vector<Node> seg;
33
34     segmentTree(int n)
35     {
36         sz = 1;
37         while (sz < n)
38             sz *= 2;
39         seg.assign(2 * sz, Node());
40     }
41
42     void init(vector<int> &arr, int node, int lx, int rx)
43     {
44         if (rx - lx == 1)
45         {
46             if (lx < int(arr.size()))
47                 seg[node] = Node(arr[lx]);
48             return;
49         }
50         int mid = (lx + rx) / 2;
51         init(arr, 2 * node + 1, lx, mid);
52         init(arr, 2 * node + 2, mid, rx);
53         seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
54     }
55
56     void init(vector<int> &arr)
57     {
58         init(arr, 0, 0, sz);
59     }
60
61     Node merge(Node &lf, Node &rt)
62     {
63         Node ans = Node();
64         ans.val = lf.val + rt.val;
65         return ans;
66     }
67
68     void push(int node, int lx, int rx)
69     {
70         if (rx - lx == 1 || !seg[node].is_lazy)
71             return;
72         int mid = (lx + rx) / 2;
73         seg[2 * node + 1].change(seg[node].lazy, lx, mid);
74         seg[2 * node + 2].change(seg[node].lazy, mid, rx);

```

```

75         seg[node].reset();
76     }
77
78     void update(int l, int r, int x, int node, int lx, int rx)
79     {
80         if (lx >= r || rx <= l)
81             return;
82         push(node, lx, rx);
83         if (lx >= l && rx <= r)
84         {
85             seg[node].change(x, lx, rx);
86             return;
87         }
88         int mid = (lx + rx) / 2;
89         update(l, r, x, 2 * node + 1, lx, mid);
90         update(l, r, x, 2 * node + 2, mid, rx);
91
92         seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
93     }
94
95     void update(int l, int r, int x)
96     {
97         return update(l, r, x, 0, 0, sz);
98     }
99
100     Node get(int l, int r, int node, int lx, int rx)
101     {
102         if (rx <= l || lx >= r)
103             return Node();
104         push(node, lx, rx);
105         if (lx >= l && rx <= r)
106             return seg[node];
107         int mid = (lx + rx) / 2;
108         Node lf = get(l, r, 2 * node + 1, lx, mid);
109         Node ri = get(l, r, 2 * node + 2, mid, rx);
110         return merge(lf, ri);
111     }
112
113     int get(int l, int r)
114     {
115         return get(l, r, 0, 0, sz).val;
116     }
117 };

```

### 2.8.10 Segment tree merging

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: SEGMENT TREE MERGING
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?

```

```

1 * - Keywords: "Subtree queries", "Colors in subtree", "Inversions in subtree", 44
   Tree DP".
2 * - Classic Scenarios: You are traversing a tree (or graph) bottom-up. Every 45 struct MergeableSegTree
   node needs
3 * to track a frequency array or sums of its subtree. 46 {
4 * - The Magic: Merging two Dynamic Segment Trees takes time proportional to 47 private:
   their
5 * *intersecting* nodes. If you merge all children into their parents bottom-up 48 struct Node
   ,
6 * the total time and memory for the entire process is mathematically bounded 49 {
   to
7 *  $O(N \log(\text{Range}))$ . It feels like magic because  $10^5$  independent trees merge 50     long long sum = 0;
   into
8 * one in a fraction of a second. 51     int lc = -1; // Left child index
9 * 2. HOW TO USE IT (THE BLACK BOX) 52     int rc = -1; // Right child index
10 * - Initialization: 53 };
11 * MergeableSegTree st(-1e9, 1e9); 54
12 * vector<int> roots(N + 1, -1); // Tracks the root ID for each node's 55 vector<Node> seg;
   personal tree 56 long long MIN_VAL, MAX_VAL;
13 * - Updates: 57
14 * // Add 'val' at coordinate 'idx' for node U's personal tree 58 int allocate()
15 * roots[U] = st.update(roots[U], idx, val); 59 {
16 * 23 * - Merging: 60     seg.emplace_back();
17 * // Merge the tree of child V into the tree of parent U 61     return seg.size() - 1;
18 * roots[U] = st.merge(roots[U], roots[V]); 62 }
19 * 24 * - Queries: 63
20 * // Query the sum in range [L, R] inside node U's tree 64 int update(int node, long long lx, long long rx, long long idx, long long val
21 * long long ans = st.query(roots[U], L, R); 65 )
22 * 66 {
23 * 67     // If the node doesn't exist yet, create it!
24 * 68     if (node == -1)
25 * 69         node = allocate();
26 * 70     if (lx == rx)
27 * 71     {
28 * 72         seg[node].sum += val;
29 * 73         return node;
30 * 74     }
31 * 75
32 * 76     long long mid = lx + (rx - lx) / 2;
33 * 77     if (idx <= mid)
34 * 78     {
35 * 79         seg[node].lc = update(seg[node].lc, lx, mid, idx, val);
36 * 80     }
37 * 81     else
38 * 82     {
39 * 83         seg[node].rc = update(seg[node].rc, mid + 1, rx, idx, val);
40 * 84     }
41 * 85
42 * 86     // Pull up answers from children
43 * 87     seg[node].sum = 0;
44 * 88     if (seg[node].lc != -1)
45 * 89         seg[node].sum += seg[seg[node].lc].sum;
46 * 90     if (seg[node].rc != -1)
47 * 91         seg[node].sum += seg[seg[node].rc].sum;
48 * 92
49 * 93     return node;
50 * 94 }
51 #include <bits/stdc++.h>
52 using namespace std;

```

```

95 // ? THE MAGIC: The Destructive Merge Function
96 int merge(int u, int v, long long lx, long long rx)
97 {
98     // 1. If one of the branches is empty, just point to the other! (O(1)
99     // shortcut)
100     if (u == -1)
101         return v;
102     if (v == -1)
103         return u;
104
105     // 2. Base case: Both trees have data at this leaf. Combine them!
106     if (lx == rx)
107     {
108         seg[u].sum += seg[v].sum;
109         return u;
110     }
111
112     // 3. Recursive case: Traverse overlapping paths and merge children
113     long long mid = lx + (rx - lx) / 2;
114     seg[u].lc = merge(seg[u].lc, seg[v].lc, lx, mid);
115     seg[u].rc = merge(seg[u].rc, seg[v].rc, mid + 1, rx);
116
117     // Pull up the new combined sum
118     seg[u].sum = 0;
119     if (seg[u].lc != -1)
120         seg[u].sum += seg[seg[u].lc].sum;
121     if (seg[u].rc != -1)
122         seg[u].sum += seg[seg[u].rc].sum;
123
124     return u; // Return the absorbed parent node
125 }
126
127 long long query(int node, long long lx, long long rx, long long ql, long long
128 qr)
129 {
130     if (node == -1 || ql > rx || qr < lx)
131         return 0;
132
133     if (ql <= lx && rx <= qr)
134         return seg[node].sum;
135
136     long long mid = lx + (rx - lx) / 2;
137     return query(seg[node].lc, lx, mid, ql, qr) +
138         query(seg[node].rc, mid + 1, rx, ql, qr);
139 }
140 public:
141 MergeableSegTree(long long min_v, long long max_v)
142 {
143     MIN_VAL = min_v;
144     MAX_VAL = max_v;
145
146     // Pre-allocate to prevent slow resizing.
147     // Formula:  $N * \log(\text{Range})$ . Usually  $40 * N$  is safe for  $10^9$  ranges.
148     seg.reserve(4000000);
149 }
150
151 // Pass the root of the tree you want to update
152 int update(int root, long long idx, long long val)
153 {
154     return update(root, MIN_VAL, MAX_VAL, idx, val);
155 }
156
157 // Pass the roots of the two trees you want to merge
158 int merge(int root_u, int root_v)
159 {
160     return merge(root_u, root_v, MIN_VAL, MAX_VAL);
161 }
162
163 // Pass the root of the tree you want to query
164 long long query(int root, long long l, long long r)
165 {
166     return query(root, MIN_VAL, MAX_VAL, l, r);
167 };
168
169 void solve()
170 {
171     int n;
172     // ... Read tree structure ...
173     // vector<vector<int>> adj(n + 1);
174
175     // Create the global Mergeable Segment Tree
176     MergeableSegTree st(1, 1e9);
177
178     // Array to track the root of the segment tree belonging to each node
179     vector<int> roots(n + 1, -1);
180
181     // Example: Add value '1' to coordinate 'color[u]' for node U's tree
182     // roots[u] = st.update(roots[u], color[u], 1);
183
184     // Example Tree DP logic:
185     // for (int v : adj[u]) {
186     //     dfs(v, u);
187     //     roots[u] = st.merge(roots[u], roots[v]);
188     // }
189
190     // Now roots[u] contains the fully merged frequency array of its entire
191     // subtree!
192 }
193
194 int main()
195 {
196     ios_base::sync_with_stdio(false);

```

```

196     cin.tie(NULL);
197     // solve();
198     return 0;
199 }

```

## 2.9 Sparse Table

### 2.9.1 2D sparse table

```

1  /**
2  *  THE ULTIMATE CP TEMPLATE BLUEPRINT: 2D SPARSE TABLE
3  *
4  *  1.  WHAT PROBLEM DOES THIS SOLVE?
5  *  - Keywords: "Static 2D Grid", "Maximum in subgrid", "Minimum in subgrid", "
    Huge number of queries".
6  *  - Classic Scenarios: You are given an  $N \times M$  grid of numbers. The grid NEVER
    changes.
7  *  You need to answer  $Q$  queries (up to  $10^6$ ) asking for the max/min in a
    rectangle bounded
8  *  by top-left  $(r1, c1)$  and bottom-right  $(r2, c2)$ .
9  *  - The Magic: A 2D Segment tree answers queries in  $O(\log N * \log M)$ , which can
    Time Limit Exceed
10 *  (TLE) if  $Q$  is massive. A 2D Sparse Table builds the answers for all powers
    of 2 in advance,
11 *  allowing it to answer ANY rectangle query in absolute  $O(1)$  time by
    overlapping 4 smaller rectangles!
12 *
13 *  2.  HOW TO USE IT (THE BLACK BOX)
14 *  - Initialization: Pass a 0-indexed 2D vector to the constructor.
15 *      vector<vector<int>> grid = {{1, 5}, {2, 8}};
16 *      SparseTable2D st(grid);
17 *
18 *  - Queries (All 0-indexed, Inclusive  $[r1, r2]$  and  $[c1, c2]$ ):
19 *      // Query the max in the subgrid from top-left  $(0, 0)$  to bottom-right  $(1, 1)$ 
20 *      int max_val = st.query(0, 0, 1, 1);
21 *
22 *  - Complexity:
23 *      Time:  $O(N * M * \log N * \log M)$  to build. Pure  $O(1)$  per query.
24 *      Space:  $O(N * M * \log N * \log M)$  memory.
25 *
26 *  3.  HOW TO ADAPT IT (THE DIALS & SWITCHES)
27 *  - Changing from Max to Min or GCD?
28 *  1. Change the `combine()` function to return `min(a, b)` or `std::gcd(a, b)`.
29 *  - Memory Limit Exceeded (MLE) Warning:
30 *  If  $N = 1000$  and  $M = 1000$ ,  $\log(N) = 10$ . The table requires  $1000 * 1000 * 10$ 
    10 =  $10^8$  ints.
31 *  This is ~400 MB of memory. This passes on most platforms (512MB limits), but
    if the limit

```

```

32 *  is strictly 256MB, you will MLE. Always check the memory limit before
    choosing this!
33 *  - Idempotent Operations ONLY:
34 *  Sparse tables ONLY work for operations where  $X + X = X$  (like Min, Max, GCD,
    Bitwise OR/AND).
35 *  Do NOT use this for Sum queries! (Use a 2D Prefix Sum array for  $O(1)$  sums
    instead).
36 */
37
38 #include <bits/stdc++.h>
39 using namespace std;
40
41 // struct SparseTable2D {
42 private:
43     int n, m;
44     int log_n, log_m;
45     // table[ir][ic][r][c]
46     // Represents the rectangle of size  $2^{ir} \times 2^{ic}$  starting at top-left  $(r, c)$ 
47     vector<vector<vector<vector<int>>>> table;
48
49     // Precomputed logs for pure  $O(1)$  queries
50     vector<int> log_table;
51
52     // ? Dial 1: The Combination Logic (Change to min, gcd, /, &, etc.)
53     inline int combine(int a, int b)
54     {
55         return max(a, b);
56     }
57
58     // public:
59     SparseTable2D(const vector<vector<int>> &grid)
60     {
61         if (grid.empty() || grid[0].empty())
62             return;
63         n = grid.size();
64         m = grid[0].size();
65
66         // Precompute logarithms
67         int max_dim = max(n, m);
68         log_table.assign(max_dim + 1, 0);
69         for (int i = 2; i <= max_dim; i++)
70         {
71             log_table[i] = log_table[i / 2] + 1;
72         }
73
74         log_n = log_table[n] + 1;
75         log_m = log_table[m] + 1;
76
77         // Allocate memory carefully: ir -> ic -> row -> col
78         // This specific ordering prevents massive cache-miss penalties during the
        build phase
79         table.assign(log_n, vector<vector<vector<int>>>>(log_m,

```

```

80         vector<vector<int>>>(n, vector<int>(m)));
81
82 // 1. Base Case: 1x1 rectangles (2^0 x 2^0)
83 for (int i = 0; i < n; i++)
84 {
85     for (int j = 0; j < m; j++)
86     {
87         table[0][0][i][j] = grid[i][j];
88     }
89 }
90
91 // 2. Build purely horizontally (1D Sparse Table for each row)
92 for (int ir = 0; ir < log_n; ir++)
93 {
94     for (int ic = 0; ic < log_m; ic++)
95     {
96         // Skip the base case we already filled
97         if (ir == 0 && ic == 0)
98             continue;
99
100     for (int i = 0; i + (1 << ir) <= n; i++)
101     {
102         for (int j = 0; j + (1 << ic) <= m; j++)
103         {
104
105             if (ir == 0)
106             {
107                 // Merge two adjacent horizontal blocks
108                 table[ir][ic][i][j] = combine(
109                     table[ir][ic - 1][i][j],
110                     table[ir][ic - 1][i][j + (1 << (ic - 1))]);
111             }
112             else
113             {
114                 // Merge two adjacent vertical blocks
115                 table[ir][ic][i][j] = combine(
116                     table[ir - 1][ic][i][j],
117                     table[ir - 1][ic][i + (1 << (ir - 1))][j]);
118             }
119         }
120     }
121 }
122 }
123 }
124
125 // // Query inclusive rectangle from top-left (r1, c1) to bottom-right (r2, c2)
126 int query(int r1, int c1, int r2, int c2)
127 {
128     // Find the largest power of 2 that fits inside the row and column lengths
129     int k_r = log_table[r2 - r1 + 1];
130     int k_c = log_table[c2 - c1 + 1];
131
132 // ? THE MAGIC: We perfectly cover the requested rectangle using 4 smaller
133 // overlapping
134 // rectangles of size (2^k_r) x (2^k_c).
135 // Because the operation is Idempotent (max/min), overlapping data doesn't
136 // change the answer!
137 int top_left = table[k_r][k_c][r1][c1];
138 int top_right = table[k_r][k_c][r1][c2 - (1 << k_c) + 1];
139 int bottom_left = table[k_r][k_c][r2 - (1 << k_r) + 1][c1];
140 int bottom_right = table[k_r][k_c][r2 - (1 << k_r) + 1][c2 - (1 << k_c) + 1];
141 return combine(combine(top_left, top_right), combine(bottom_left,
142     bottom_right));
143 }
144 ;
145
146 // void solve() {
147 int n, m, q;
148 // Example Input: N rows, M cols, Q queries
149 if (!(cin >> n >> m >> q))
150     return;
151
152 vector<vector<int>>> grid(n, vector<int>(m));
153 for (int i = 0; i < n; i++)
154 {
155     for (int j = 0; j < m; j++)
156     {
157         cin >> grid[i][j];
158     }
159 }
160
161 SparseTable2D st(grid);
162
163 while (q--)
164 {
165     int r1, c1, r2, c2;
166     // Assume 0-indexed coordinates are provided
167     cin >> r1 >> c1 >> r2 >> c2;
168     cout << st.query(r1, c1, r2, c2) << "\n";
169 }
170 }
171
172 int main()
173 {
174     // Fast I/O is mandatory for heavy 2D queries
175     ios_base::sync_with_stdio(false);
176     cin.tie(NULL);
177     solve();
178     return 0;

```

## 2.9.2 sparse table

```

179 }

1 struct SparseTable {
2     int n, LG;
3     vector<vector<int> > T;
4     SparseTable(const vector<int> &arr) {
5         n = arr.size();
6         LG = __lg(n) + 2;
7         T.assign(LG, vector<int>(n));
8         for (int i = 0; i < n; i++) {
9             T[0][i] = arr[i];
10        }
11        for (int pw = 1; (1 << pw) <= n; pw++) {
12            for (int i = 0; i + (1 << pw) <= n; i++) {
13                T[pw][i] = merge(T[pw - 1][i], T[pw - 1][i + (1 << (pw - 1))]);
14            }
15        }
16    }
17
18    int merge(int a, int b) {
19        return min(a, b);
20    }
21
22    int query(int l, int r) {
23        int len = r - l + 1;
24        int pw = __lg(len);
25        return merge(T[pw][l], T[pw][r - (1 << pw) + 1]);
26    }
27
28    int query1(int l, int r) { // o(log)
29        int res = 1e9; // change
30        int sz = r-l+1;
31        for (int i = 0; l<=r;i++) {
32            if (sz>>i&1) {
33                res = merge(res,T[i][l]);
34                l+=(1<<i);
35            }
36        }
37        return res;
38    }
39
40 };

```

## 2.10 Treap

### 2.10.1 implicit treap

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: IMPLICIT TREAP (CARTESIAN TREE)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Dynamic Array", "Range Reverse", "Insert/Delete at index", "Cut
   and Paste".
6  * - Classic Scenarios: You have an array and need to perform operations that a
   Segment
7  * Tree CANNOT do, such as:
8  * - Insert/Delete elements at specific indices (shifting the whole array).
9  * - Reversing a whole subsegment [L, R] dynamically.
10 * - Moving a block [L, R] to the front or end.
11 * - The Magic: An Implicit Treap acts like a balanced BST, but instead of
   sorting by
12 * a "Key", it sorts by the "Implicit Index" (the size of the left subtree).
   This allows
13 * us to split and merge arbitrary subarrays in O(log N) time.
14 *
15 * 2. HOW TO USE IT (THE BLACK BOX)
16 * - Initialization: Root = nullptr. Use `insert` to populate.
17 * - Range Operations:
18 *     // Reverse range [L, R] (0-indexed)
19 *     reverse_range(root, L, R);
20 *     // Query sum of range [L, R]
21 *     long long ans = query_sum(root, L, R);
22 * - Complexity:
23 *     Time: O(log N) per operation.
24 *     Space: O(N) memory.
25 *
26 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
27 * - Need Range Add? Add a `lazy` field to the Node and push it in the `push()`
   function.
28 * - Need Max/Min instead of Sum? Update `update()` and `get_sum()` to maintain
   `max`/`min` fields instead of `sum`.
29 *
30 */
31
32 #include <bits/stdc++.h>
33 using namespace std;
34
35 struct Node
36 {
37     int val, priority, size;
38     long long sum;
39     bool rev; // Flag for range reversal (Lazy Propagation)
40     Node *left, *right;
41
42     Node(int v) : val(v), priority(rand()), size(1), sum(v), rev(false), left(
   nullptr), right(nullptr) {}
43 };
44
45 int get_size(Node *t) { return t ? t->size : 0; }
46 long long get_sum(Node *t) { return t ? t->sum : 0; }

```

```

47
48 // Update subtree size and sum based on children
49 void update(Node *t)
50 {
51     if (!t)
52         return;
53     t->size = 1 + get_size(t->left) + get_size(t->right);
54     t->sum = t->val + get_sum(t->left) + get_sum(t->right);
55 }
56
57 // Push lazy propagation (reverse flag) down to children
58 void push(Node *t)
59 {
60     if (t && t->rev)
61     {
62         t->rev = false;
63         swap(t->left, t->right);
64         if (t->left)
65             t->left->rev ^= true;
66         if (t->right)
67             t->right->rev ^= true;
68     }
69 }
70
71 // Split treap t into l (first k nodes) and r (remaining nodes)
72 void split(Node *t, int k, Node *&l, Node *&r)
73 {
74     if (!t)
75     {
76         l = r = nullptr;
77         return;
78     }
79     push(t);
80     int left_size = get_size(t->left);
81     if (left_size < k)
82     {
83         split(t->right, k - left_size - 1, t->right, r);
84         l = t;
85     }
86     else
87     {
88         split(t->left, k, l, t->left);
89         r = t;
90     }
91     update(t);
92 }
93
94 // Merge treap l and treap r
95 Node *merge(Node *l, Node *r)
96 {
97     push(l);
98     push(r);

```

```

99     if (!l || !r)
100         return l ? l : r;
101     if (l->priority > r->priority)
102     {
103         l->right = merge(l->right, r);
104         update(l);
105         return l;
106     }
107     else
108     {
109         r->left = merge(l, r->left);
110         update(r);
111         return r;
112     }
113 }
114
115 // Reverse subarray [l, r] (0-indexed)
116 void reverse_range(Node *&root, int l, int r)
117 {
118     Node *t1, *t2, *t3;
119     split(root, r + 1, t2, t3);
120     split(t2, l, t1, t2);
121     if (t2)
122         t2->rev ^= true;
123     root = merge(t1, merge(t2, t3));
124 }
125
126 // Query sum in subarray [l, r]
127 long long query_sum(Node *&root, int l, int r)
128 {
129     Node *t1, *t2, *t3;
130     split(root, r + 1, t2, t3);
131     split(t2, l, t1, t2);
132     long long res = get_sum(t2);
133     root = merge(t1, merge(t2, t3));
134     return res;
135 }
136
137 // Insert value at index pos
138 void insert(Node *&root, int pos, int val)
139 {
140     Node *t1, *t2;
141     split(root, pos, t1, t2);
142     root = merge(merge(t1, new Node(val)), t2);
143 }
144
145 int main()
146 {
147     srand(time(0));
148     Node *root = nullptr;
149     // Example: Insert values
150     insert(root, 0, 10);

```

```

151     insert(root, 1, 20);
152     insert(root, 2, 30);
153
154     // Example: Sum query [0, 1]
155     // cout << query_sum(root, 0, 1) << endl;
156
157     return 0;
158 }

```

## 2.11 Trie

### 2.11.1 BinaryTrie Using Vector

```

1 class BinaryTrie {
2     struct Node {
3         int ch[2];
4         int frq[2];
5         Node() {
6             ch[0] = ch[1] = -1;
7             frq[0] = frq[1] = 0;
8         }
9     };
10    vector<Node> tree;
11 public:
12    BinaryTrie() {
13        tree.emplace_back();
14        insert(0);
15    }
16
17    void insert(int n) {
18        int cur = 0;
19        for (int i = 30; i >= 0; i--) {
20            int idx = (n >> i) & 1;
21
22            if (tree[cur].ch[idx] == -1) {
23                tree[cur].ch[idx] = tree.size();
24                tree.emplace_back();
25            }
26
27            tree[cur].frq[idx]++;
28            cur = tree[cur].ch[idx];
29        }
30    }
31
32    void erase(int n) {
33        int cur = 0;
34        for (int i = 30; i >= 0; i--) {
35            int idx = (n >> i) & 1;
36            tree[cur].frq[idx]--;
37            cur = tree[cur].ch[idx];
38        }

```

```

39    }
40
41    bool search(int n) {
42        int cur = 0;
43        for (int i = 30; i >= 0; i--) {
44            int idx = (n >> i) & 1;
45            if (tree[cur].ch[idx] == -1 || tree[cur].frq[idx] == 0) {
46                return false;
47            }
48            cur = tree[cur].ch[idx];
49        }
50        return true;
51    }
52
53    int maxXor(int x) {
54        int cur = 0;
55        int res = 0;
56        for (int i = 30; i >= 0; i--) {
57            int idx = (x >> i) & 1;
58            int target = 1 - idx;
59
60            if (tree[cur].ch[target] != -1 && tree[cur].frq[target] > 0) {
61                res |= (1 << i);
62                cur = tree[cur].ch[target];
63            } else {
64                cur = tree[cur].ch[idx];
65            }
66        }
67        return res;
68    }
69
70    void clear() {
71        tree.clear();
72        tree.emplace_back();
73        insert(0);
74    }
75 };

```

### 2.11.2 Binary trie

```

1 struct BitTrie
2 {
3 private:
4     BitTrie *child[2];
5     int cnt{};
6
7 public:
8     BitTrie()
9     {
10         child[0] = child[1] = nullptr;
11     }

```



```

12
13 ~BitTrie()
14 {
15     delete child[0];
16     delete child[1];
17 }
18
19 void insert(int num)
20 {
21     BitTrie *curr = this;
22     for (int i = 31; i >= 0; i--)
23     {
24         int bit = num >> i & 1;
25         if (!curr->child[bit])
26             curr->child[bit] = new BitTrie();
27         curr = curr->child[bit];
28         curr->cnt++;
29     }
30 }
31
32 void erase(int num)
33 {
34     BitTrie *curr = this;
35     for (int i = 31; i >= 0; i--)
36     {
37         int bit = num >> i & 1;
38         curr = curr->child[bit];
39         curr->cnt--;
40     }
41 }
42
43 int getMaxXor(int num)
44 {
45     BitTrie *curr = this;
46     int maxXor = 0;
47     for (int i = 31; i >= 0; i--)
48     {
49         int need = 1 - (num >> i & 1);
50         if (curr->child[need] && curr->child[need]->cnt > 0)
51         {
52             maxXor |= 1ll << i;
53             curr = curr->child[need];
54         }
55         else
56             curr = curr->child[1 - need];
57     }
58     return maxXor;
59 }
60
61 // count Xors with X less than K
62 int countXorLessThan(int x, int k)
63 {

```

```

64     int ret{};
65     BitTrie *curr = this;
66     for (int i = 31; i >= 0; i--)
67     {
68         if (!curr)
69             break;
70         int bitX = x >> i & 1;
71         int bitK = k >> i & 1;
72         if (bitK == 0)
73             curr = curr->child[bitX];
74         else
75         {
76             if (curr->child[bitX])
77                 ret += curr->child[bitX]->cnt;
78             curr = curr->child[1 - bitX];
79         }
80     }
81     return ret;
82 }
83 };
84
85 // get the Max Xor of two numbers in the array
86 int getMaxXorOfTwoNumbersInArray(vector<int> &v)
87 {
88     if (v.empty())
89         return 0;
90     int n = int(v.size());
91     BitTrie trie;
92     trie.insert(v[0]);
93     int max{};
94     for (int i = 1; i < n; i++)
95     {
96         max = std::max(max, trie.getMaxXor(v[i]));
97         trie.insert(v[i]);
98     }
99     return max;
100 }
101
102 struct Query
103 {
104     int x, a, id;
105
106     bool operator<(const Query &other) const
107     {
108         return a < other.a;
109     }
110 };
111
112 // max xor queries for a number with array with threshold A [The element i pick
113 // from array must be <= A]
114 void getMaxXorOfNumWithArrayLessThanThresholdQueries(vector<int> &v)

```

```

115     int n = int(v.size());
116     sort(v.begin(), v.end());
117     int q;
118     cin >> q;
119     vector<Query> queries(q);
120     for (int i = 0; i < q; i++)
121     {
122         cin >> queries[i].x >> queries[i].a;
123         queries[i].id = i;
124     }
125     sort(queries.begin(), queries.end());
126     BitTrie trie;
127     int idx{};
128     vector<int> ans(q);
129     for (int i = 0; i < q; i++)
130     {
131         auto [x, a, id] = queries[i];
132         while (idx < n && v[idx] <= a)
133             trie.insert(v[idx++]);
134         if (idx == 0)
135             ans[id] = -1;
136         else
137             ans[id] = trie.getMaxXor(x);
138     }
139     for (int i = 0; i < q; i++)
140         cout << ans[i] << "\n";
141 }
142
143 // Max xor subarray
144 int getMaxXorSubarray(vector<int> &a)
145 {
146     int n = int(a.size());
147     BitTrie trie;
148     trie.insert(0);
149     int pre{}, ret{};
150     for (int i = 0; i < n; i++)
151     {
152         pre ^= a[i];
153         ret = max(ret, trie.getMaxXor(pre));
154         trie.insert(pre);
155     }
156     return ret;
157 }
158
159 int numberOfSubarraysWithXorLessThanK(vector<int> &a, int k)
160 {
161     int n = int(a.size());
162     BitTrie trie;
163     trie.insert(0);
164     int pre{}, ret{};
165     for (int i = 0; i < n; i++)
166     {

```

```

167         pre ^= a[i];
168         ret += trie.countXorLessThan(pre, k);
169         trie.insert(pre);
170     }
171     return ret;
172 }

```

### 2.11.3 Persistent Trie

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: PERSISTENT BINARY TRIE (XOR MAXIMIZER)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Maximum XOR in a subarray", "Range XOR", "Subsegment XOR with X".
6  * - Classic Scenarios: You are given an array of N integers. You need to answer
7  *   Q queries.
8  *   Each query gives you a range [L, R] and a value X. You need to find an
9  *   element Y
10  *   in the subarray from L to R such that (Y ^ X) is maximized.
11  * - The Magic: A standard Trie can find the max XOR in a whole array. By making
12  *   it Persistent,
13  *   we save a new "version" of the Trie after inserting every single element. To
14  *   query the
15  *   range [L, R], we take the Trie at version R and mathematically subtract the
16  *   paths from
17  *   version L-1. It does this in blindingly fast O(30) time!
18  *
19  * 2. HOW TO USE IT (THE BLACK BOX)
20  * - Initialization: Pass a 0-indexed vector to build the historical versions.
21  *   vector<long long> a = {3, 10, 5, 25, 2};
22  *   PersistentTrie ptrie(a);
23  * - Queries (All 0-indexed, Inclusive [L, R]):
24  *   // Find the maximum XOR of 7 with any element in the subarray from index
25  *   1 to 3
26  *   long long ans = ptrie.queryMaxXor(1, 3, 7);
27  *
28  * - Complexity:
29  *   Time: O(N * BITS) to build. O(BITS) per query.
30  *   Space: O(N * BITS) memory.
31  *
32  * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
33  * - Dealing with huge numbers (up to 1018)?
34  *   Change the `BITS` constant from 30 to 60. (30 is enough for numbers up to
35  *   109).
36  * - Minimum XOR instead of Maximum XOR?
37  *   I have included a `queryMinXor` function right below the Max one! It simply
38  *   tries to
39  *   take the path of the SAME bit instead of the FLIPPED bit.
40  * - Need to dynamically add elements to the back of the array?

```

```

34 * Just call `ptrie.insertBack(X);`. It will add the element and create a new 84 // The OTHER child pointer (b ^ 1) was naturally copied over during
    version the
35 * that you can instantly query. 85 // clone() of the parent, so it still safely points to the historical
36 */ branch!
37 86
38 #include <bits/stdc++.h> 87 curr = trie[curr].child[b];
39 using namespace std; 88 prev = trie[prev].child[b];
40 89
41 struct PersistentTrie 90 trie[curr].cnt++; // Increment the count of elements that pass
42 { through this path
43 private: 91 }
44 // ? Dial 1: Set to 30 for <= 10^9. Set to 60 for <= 10^18. 92
45 const int BITS = 30; 93 return new_root;
46 94 }
47 struct Node 95
48 { 96 public:
49     int child[2]; 97 // roots[i] represents the Trie containing all elements from index 0 to i-1.
50     int cnt; // How many elements exist in this subtree 98 // roots[0] is the empty Trie.
51 99 vector<int> roots;
52     Node() 100
53     { 101 PersistentTrie(const vector<long long> &a = {})
54         child[0] = child[1] = 0; 102 {
55         cnt = 0; 103 // Pre-allocate memory to prevent slow vector reallocations.
56     } 104 // N elements * roughly (BITS + 1) nodes per element
57 }; 105 trie.reserve(a.size() * (BITS + 2) + 100);
58 106
59 vector<Node> trie; 107 // Node 0 is the Dummy/Empty node
60 108 trie.emplace_back();
61 // ? Creates a brand new node that is an exact copy of the given node ID 109
62 int clone(int node) 110 // Version 0 is the empty tree
63 { 111 roots.push_back(0);
64     trie.push_back(trie[node]); 112
65     return trie.size() - 1; 113 for (long long x : a)
66 } 114 {
67 115     insertBack(x);
68 int insert(int prev_root, long long val) 116 }
69 { 117 }
70     int curr = clone(prev_root); 118
71     int new_root = curr; 119 // Add an element to the end of the array dynamically
72     int prev = prev_root; 120 void insertBack(long long val)
73 121 {
74     trie[curr].cnt++; // Increment count for the root of this version 122     int new_root = insert(roots.back(), val);
75 123     roots.push_back(new_root);
76     for (int i = BITS; i >= 0; i--) 124 }
77     { 125
78         int b = (val >> i) & 1; 126 // Find the MAXIMUM value of (val ^ Y) where Y is an element in a[l...r]
79 127 long long queryMaxXor(int l, int r, long long val)
80 // ? We are taking path 'b'. To safely modify it without destroying 128 {
81 // the historical version, we MUST clone the 'b' child of the 129 // Because roots[i] covers [0...i-1], the range [L, R] is extracted by
    previous version! 130 // taking Version (R + 1) and subtracting Version L.
82 trie[curr].child[b] = clone(trie[prev].child[b]); 131 int curr_R = roots[r + 1];
83 132 int curr_L = roots[l];

```

```

133     long long ans = 0;
134
135     for (int i = BITS; i >= 0; i--)
136     {
137         int b = (val >> i) & 1;
138         int flipped = b ^ 1; // To MAXIMIZE XOR, we want the opposite bit!
139
140         // How many numbers in our specific range actually take the 'flipped
141         path?
142         int count_flipped = trie[trie[curr_R].child[flipped]].cnt -
143             trie[trie[curr_L].child[flipped]].cnt;
144
145         if (count_flipped > 0)
146         {
147             // The path exists! We take it.
148             ans |= (1LL << i);
149             curr_R = trie[curr_R].child[flipped];
150             curr_L = trie[curr_L].child[flipped];
151         }
152         else
153         {
154             // The opposite bit doesn't exist in this range. We are forced
155             take the same bit.
156             curr_R = trie[curr_R].child[b];
157             curr_L = trie[curr_L].child[b];
158         }
159     }
160     return ans;
161 }
162
163 // Find the MINIMUM value of (val ^ Y) where Y is an element in a[l...r]
164 long long queryMinXor(int l, int r, long long val)
165 {
166     int curr_R = roots[r + 1];
167     int curr_L = roots[l];
168     long long ans = 0;
169
170     for (int i = BITS; i >= 0; i--)
171     {
172         int b = (val >> i) & 1;
173
174         // To MINIMIZE XOR, we want the EXACT SAME bit!
175         int count_same = trie[trie[curr_R].child[b]].cnt -
176             trie[trie[curr_L].child[b]].cnt;
177
178         if (count_same > 0)
179         {
180             // The path exists! We take it. (Ans bit remains 0 because b ^ b
181             = 0)
182             curr_R = trie[curr_R].child[b];
183             curr_L = trie[curr_L].child[b];
184         }
185         else
186         {
187             // Forced to take the flipped bit, which increases our XOR
188             penalty.
189             int flipped = b ^ 1;
190             ans |= (1LL << i);
191             curr_R = trie[curr_R].child[flipped];
192             curr_L = trie[curr_L].child[flipped];
193         }
194     }
195     return ans;
196 }
197
198 void solve()
199 {
200     int n, q;
201     if (!(cin >> n >> q))
202         return;
203
204     vector<long long> a(n);
205     for (int i = 0; i < n; i++)
206         cin >> a[i];
207
208     PersistentTrie ptrie(a);
209
210     while (q--)
211     {
212         int l, r;
213         long long x;
214         // 0-indexed L and R, and the XOR target X
215         cin >> l >> r >> x;
216
217         cout << ptrie.queryMaxXor(l, r, x) << "\n";
218     }
219 }
220
221 int main()
222 {
223     ios_base::sync_with_stdio(false);
224     cin.tie(NULL);
225     solve();
226     return 0;
227 }

```

#### 2.11.4 Trie (String)

```

1 class Trie {
2     struct Node {
3         map<string, int> ch;
4         Node() {

```

```

5     }
6 };
7 vector<Node>root;
8 public:
9     Trie() {
10         root.emplace_back();
11     }
12     void insert(vector<string> &words) {
13         int cur = 0;
14         for (auto s : words) {
15             if (!root[cur].ch[s]) {
16                 root[cur].ch[s] = root.size();
17                 root.emplace_back();
18             }
19             cur = root[cur].ch[s];
20         }
21     }
22 };

```

### 2.11.5 Trie Using Vector

```

1 class Trie {
2     struct Node {
3         int ch[26];
4         int prefix, end;
5         Node() {
6             prefix = end = 0;
7             memset(ch, -1, sizeof(ch));
8         }
9     };
10    vector<Node> tree;
11 public:
12     Trie() {
13         tree.emplace_back();
14     }
15
16     void insert(string &s) {
17         int cur = 0;
18         for (auto i : s) {
19             int idx = i - 'a';
20
21             if (tree[cur].ch[idx] == -1) {
22                 tree[cur].ch[idx] = tree.size();
23                 tree.emplace_back();
24             }
25             cur = tree[cur].ch[idx];
26             tree[cur].prefix++;
27         }
28         tree[cur].end++;
29     }
30 }

```

```

31 bool search(string &s) {
32     int cur = 0;
33     for (auto i : s) {
34         int idx = i - 'a';
35         if (tree[cur].ch[idx] == -1) {
36             return false;
37         }
38         cur = tree[cur].ch[idx];
39     }
40     return tree[cur].end > 0;
41 }
42
43 int countWords(string &s) {
44     int cur = 0;
45     for (auto i : s) {
46         int idx = i - 'a';
47         if (tree[cur].ch[idx] == -1) {
48             return 0;
49         }
50         cur = tree[cur].ch[idx];
51     }
52     return tree[cur].end;
53 }
54
55 int countPrefix(string &s) {
56     int cur = 0;
57     for (auto i : s) {
58         int idx = i - 'a';
59         if (tree[cur].ch[idx] == -1) {
60             return 0;
61         }
62         cur = tree[cur].ch[idx];
63     }
64     return tree[cur].prefix;
65 }
66
67 // DFS in Trie
68 void dfs(int u) {
69     for (int v = 0; v < 26; v++) {
70         if (root[u].ch[v] != -1) {
71             int x = root[u].ch[v];
72             if (root[u].prefix != root[x].prefix) {
73                 ans+=root[x].prefix;
74             }
75             dfs(x);
76         }
77     }
78 }
79
80 void clear() {
81     tree.clear();
82     tree.emplace_back();

```

```
83     }
84 };
```

### 2.11.6 Trie

```
1 vector<int> ans;
2
3 class Trie
4 {
5 private:
6     Trie *child[26];
7     int word_cnt{}, prefix_cnt{};
8
9 public:
10    Trie()
11    {
12        memset(child, 0, sizeof child);
13    }
14
15    void insert(const string &str, int idx = 0)
16    {
17        prefix_cnt++;
18        if (idx == int(str.size()))
19        {
20            word_cnt++;
21            return;
22        }
23        int curr = str[idx] - 'a';
24        if (!child[curr])
25            child[curr] = new Trie();
26        child[curr]->insert(str, idx + 1);
27    }
28
29    void erase(const string &str, int idx = 0)
30    {
31        if (idx == 0 && !wordSearch(str))
32            return;
33        prefix_cnt--;
34        if (idx == int(str.size()))
35        {
36            word_cnt--;
37            return;
38        }
39        int curr = str[idx] - 'a';
40        child[curr]->erase(str, idx + 1);
41    }
42
43    void print(int idx = 0)
44    {
45        for (int i = 0; i < 26; i++)
46        {
```

```
47        if (!child[i] || !child[i]->prefix_cnt)
48            continue;
49        for (int k = 0; k < idx; k++)
50            cout << " ";
51        cout << char(i + 'a') << "\n";
52        child[i]->print(idx + 1);
53    }
54 }
55
56 // count number of chars needed to be typed by the user for each word [using
57 //   auto-completion]
58 // trie.insert("") at the beginning because the user needs to write the first
59 //   letter
60 void count(int cnt = 0)
61 {
62     if (word_cnt)
63         ans.push_back(cnt);
64     int num{};
65     for (int i = 0; i < 26; i++)
66     {
67         if (child[i])
68             num++;
69     }
70     for (int i = 0; i < 26; i++)
71     {
72         if (!child[i] || !child[i]->prefix_cnt)
73             continue;
74         if (num == 1 && word_cnt == 0)
75             child[i]->count(cnt);
76         else
77             child[i]->count(cnt + 1);
78     }
79 }
80
81 bool wordSearch(const string &str, int idx = 0)
82 {
83     if (idx == int(str.size()))
84         return word_cnt;
85     int curr = str[idx] - 'a';
86     if (!child[curr] || child[curr]->prefix_cnt == 0)
87         return false;
88     return child[curr]->wordSearch(str, idx + 1);
89 }
90
91 bool prefixSearch(const string &str, int idx = 0)
92 {
93     if (idx == int(str.size()))
94         return prefix_cnt;
95     int curr = str[idx] - 'a';
96     if (!child[curr] || child[curr]->prefix_cnt == 0)
97         return false;
98     return child[curr]->prefixSearch(str, idx + 1);
99 }
```

```

97     }
98
99     int countWordsEqualTo(const string &str, int idx = 0)
100     {
101         if (idx == int(str.size()))
102             return word_cnt;
103         int curr = str[idx] - 'a';
104         if (!child[curr] || child[curr]->prefix_cnt == 0)
105             return 0;
106         return child[curr]->countWordsEqualTo(str, idx + 1);
107     }
108
109     int countWordsStartingWith(const string &str, int idx = 0)
110     {
111         if (idx == int(str.size()))
112             return prefix_cnt;
113         int curr = str[idx] - 'a';
114         if (!child[curr] || child[curr]->prefix_cnt == 0)
115             return 0;
116         return child[curr]->countWordsStartingWith(str, idx + 1);
117     }
118
119     void printWordsStartingWith(const string &str, int idx = 0, string res = "")
120     {
121         if (idx < int(str.size()))
122         {
123             int curr = str[idx] - 'a';
124             if (!child[curr] || child[curr]->prefix_cnt == 0)
125                 return;
126             child[curr]->printWordsStartingWith(str, idx + 1, res + str[idx]);
127         }
128         else
129         {
130             if (word_cnt && idx != int(str.size()))
131                 // ans.push_back(res); /// uncomment this <-----
132                 for (int i = 0; i < 26; i++)
133                 {
134                     if (!child[i] || child[i]->prefix_cnt == 0)
135                         continue;
136                     child[i]->printWordsStartingWith(str, idx + 1, res + char('a'
137                                     + i));
138                 }
139         }
140
141         // longest word with all prefixes problem
142         bool checkCompleteness(const string &str, int idx = 0)
143         {
144             if (idx == int(str.size()))
145                 return word_cnt > 0;
146             int curr = str[idx] - 'a';
147             if (!child[curr] || child[curr]->word_cnt == 0)

```

```

148             return false;
149             return child[curr]->checkCompleteness(str, idx + 1);
150         }
151
152         string longestCompleteString(vector<string> &v)
153         {
154             int n = int(v.size());
155             string longest{};
156             for (int i = 0; i < n; i++)
157             {
158                 if (checkCompleteness(v[i]))
159                 {
160                     if (v[i].length() > longest.length() || (v[i].length() == longest
161                             .length() && v[i] < longest))
162                         longest = v[i];
163                 }
164             }
165             return longest.empty() ? "No Complete String" : longest;
166         }
167
168         int countNumberOfNodes()
169         {
170             int ret{};
171             for (int i = 0; i < 26; i++)
172             {
173                 if (child[i])
174                     ret += 1 + child[i]->countNumberOfNodes();
175             }
176             return ret;
177         }
178
179         // count number of distinct substrings
180         int countNumberOfDistinctSubStrings(string &str)
181         {
182             int n = int(str.size());
183             Trie trie;
184             for (int i = 0; i < n; i++)
185                 trie.insert(str.substr(i));
186             return trie.countNumberOfNodes();
187         }

```

## 3 Graphs

### 3.1 2-Sat

#### 3.1.1 2sat

```

1 const int N = 1e5 + 9;
2

```

```

3 int SCC_id[N];
4 GRAPH graph, graph_reverse, cond_graph;
5 int n, m;
6 vector<int> order, vis, vis_reverse, order_reverse;
7
8 void dfs(int node)
9 {
10     vis[node] = true;
11     for (auto &x : graph[node])
12     {
13         if (!vis[x])
14             dfs(x);
15     }
16     order.emplace_back(node);
17 }
18
19 void dfs_reverse(int node)
20 {
21     vis_reverse[node] = true;
22     for (auto &x : graph_reverse[node])
23     {
24         if (!vis_reverse[x])
25             dfs_reverse(x);
26     }
27     order_reverse.emplace_back(node);
28 }
29
30 void SCC()
31 {
32     graph.assign(n + 1, {}), graph_reverse.assign(n + 1, {}), vis.assign(n + 1, {}), vis_reverse.assign(n + 1, {}), cond_graph.assign(n + 1, {});
33     for (int i = 0; i < m; i++)
34     {
35         int x, y;
36         cin >> x >> y;
37         graph[x].emplace_back(y), graph_reverse[y].emplace_back(x);
38     }
39     order.clear();
40     for (int i = 1; i <= n; i++)
41     {
42         if (!vis[i])
43             dfs(i);
44     }
45     reverse(order.begin(), order.end());
46     vector<vector<int>> scc;
47     scc.push_back({});
48     for (auto &x : order)
49     {
50         if (!vis_reverse[x])
51         {
52             order_reverse.clear();
53             dfs_reverse(x);
54             scc.emplace_back(order_reverse);
55             for (auto &val : order_reverse)
56                 SCC_id[val] = int(scc.size()) - 1;
57         }
58     }
59     for (int u = 1; u <= n; u++)
60     {
61         for (int v : graph[u])
62         {
63             if (SCC_id[u] != SCC_id[v])
64                 cond_graph[SCC_id[u]].emplace_back(SCC_id[v]);
65         }
66     }
67 }
68
69 // Add this helper to add edges easily
70 void add_clause(int i, bool val_i, int j, bool val_j)
71 {
72     // i, j are variable indices (1 to n)
73     // val_i, val_j are the boolean requirements (true for X, false for !X)
74
75     // Logic: (X or Y) is equivalent to (!X -> Y) AND (!Y -> X)
76
77     // Map to 2*i and 2*i+1
78     int u = 2 * i + (val_i ? 0 : 1);
79     int v = 2 * j + (val_j ? 0 : 1);
80
81     // !u is u^1, !v is v^1
82     graph[u ^ 1].emplace_back(v);
83     graph[v ^ 1].emplace_back(u);
84
85     // Keep reversed graph for Kosaraju
86     graph_reverse[v].emplace_back(u ^ 1);
87     graph_reverse[u].emplace_back(v ^ 1);
88 }
89
90 void solve()
91 {
92     // 1. Read input
93     cin >> n >> m;
94     // Remember to adjust N to 2*N+1 if N was 1e5
95
96     // 2. Clear graph from your SCC template
97     // ... (Use your SCC setup code here)
98
99     for (int k = 0; k < m; k++)
100     {
101         int i, j;
102         bool val_i, val_j;
103         // Example: Reading "1 True 2 False" means (X1 or !X2)
104         cin >> i >> val_i >> j >> val_j;
105         add_clause(i, val_i, j, val_j);

```



```

106     }
107
108     // 3. Run Kosaraju
109     SCC();
110
111     // 4. Check for Impossibility
112     vector<int> assignment(n + 1);
113     for (int i = 1; i <= n; i++)
114     {
115         if (SCC_id[2 * i] == SCC_id[2 * i + 1])
116         {
117             cout << "IMPOSSIBLE\n";
118             return;
119         }
120         // 5. The Assignment Logic (The "Higher ID is True" strategy)
121         assignment[i] = (SCC_id[2 * i] > SCC_id[2 * i + 1]);
122     }
123
124     // 6. Print result
125     cout << "POSSIBLE\n";
126     for (int i = 1; i <= n; i++)
127     {
128         cout << assignment[i] << " ";
129     }
130 }

```

## 3.2 DAG

### 3.2.1 Dag reachability(bitset trick)

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: BITSET REACHABILITY
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Transitive Closure", "DAG Reachability", "Bitset Optimization".
9  * - Classic Scenarios: You have a Directed Acyclic Graph (DAG) and need to
   answer Q
10 * queries of the form "Can I reach node Y from node X?".
11 * - The Magic: Since a DAG has a topological order, we process nodes in REVERSE
12 * topological order. For each node U, its reachability set is the Bitwise OR
13 * of the reachability sets of all its neighbors. Because bitsets are 64x
   faster than
14 * arrays, this brings an  $O(N^3)$  problem down to an  $O(N^3 / 64)$  complexity.
15 *
16 * 2. HOW TO USE IT (THE BLACK BOX)
17 * - Initialization:
18 *     compute_reachability(N); // Run this once after building adjacency list
19 *

```

```

20 * - Query:
21 *     if (reach[start].test(target)) { ... }
22 *
23 * - Complexity:
24 *     Time:  $O(N * (N + M) / 64)$ .
25 *     Space:  $O(N^2 / 8)$  bytes. (For  $N=20,000$ , this is ~50MB).
26 *
27 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
28 * - Memory Limit Warning:
29 *     The current `MAXN = 20005` uses ~50MB. If  $N > 20,000$ , you will exceed
   standard 256MB limits.
30 *     If  $N > 20,000$ , you MUST use the "Block Reachability" approach (process bits
   in
31 * chunks of 1024 or 2048) rather than holding the entire  $N \times N$  matrix in
   memory.
32 */
33
34 // ? Dial 1: Adjust MAXN based on memory limits ( $N^2 / 8$  bytes)
35 const int MAXN = 20005;
36
37 // reach[u][v] = 1 if there is a path from u to v
38 bitset<MAXN> reach[MAXN];
39 vector<int> adj[MAXN];
40 vector<int> topo_order;
41 int in_degree[MAXN];
42
43 // Standard Kahn's Algorithm for Topological Sort
44 void compute_topological_sort(int n)
45 {
46     queue<int> q;
47     for (int i = 1; i <= n; i++)
48     {
49         if (in_degree[i] == 0)
50             q.push(i);
51     }
52
53     while (!q.empty())
54     {
55         int u = q.front();
56         q.pop();
57         topo_order.push_back(u);
58
59         for (int v : adj[u])
60         {
61             in_degree[v]--;
62             if (in_degree[v] == 0)
63                 q.push(v);
64         }
65     }
66 }
67
68 void compute_reachability(int n)

```

```

69 {
70     compute_topological_sort(n);
71
72     // ? THE MAGIC: Process in REVERSE topological order
73     // This ensures that when we process node u, all its neighbors v
74     // are already fully computed.
75     for (int i = n - 1; i >= 0; i--)
76     {
77         int u = topo_order[i];
78
79         reach[u].set(u); // A node can always reach itself
80
81         for (int v : adj[u])
82         {
83             // Bitwise OR is the secret sauce:
84             // This ORs all bits of reach[v] into reach[u] at once
85             reach[u] |= reach[v];
86         }
87     }
88 }
89
90 int main()
91 {
92     ios_base::sync_with_stdio(false);
93     cin.tie(NULL);
94
95     int n, m;
96     cin >> n >> m;
97
98     for (int i = 0; i < m; i++)
99     {
100         int u, v;
101         cin >> u >> v;
102         adj[u].push_back(v);
103         in_degree[v]++;
104     }
105
106     compute_reachability(n);
107
108     // Example Query: Can I reach node 'target' from node 'start'?
109     int start, target;
110     cin >> start >> target;
111
112     if (reach[start].test(target))
113     {
114         cout << "Reachable!\n";
115     }
116     else
117     {
118         cout << "Not reachable.\n";
119     }
120 }

```

```

121     return 0;
122 }

```

### 3.3 Eulerian path

#### 3.3.1 Eulerian path

```

1 const int N = 1e5, M = 1e5;
2
3 struct Edge
4 {
5     int to, id;
6 };
7
8 vector<Edge> graph[N];
9 bool vis[M];
10 vector<int> ans;
11
12 void dfs(int u)
13 {
14     while ((int)graph[u].size() > 0)
15     {
16         auto [v, id] = graph[u].back();
17         graph[u].pop_back();
18         if (!vis[id])
19         {
20             vis[id] = true;
21             dfs(v);
22         }
23     }
24     ans.emplace_back(u); // Eulerian path reversed
25 }

```

### 3.4 Flows

#### 3.4.1 Gomory-Hu Tree

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: GOMORY-HU TREE
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "All-pairs Min-Cut", "Minimum cut tree", "Undirected Graph Cut".
9  * - Classic Scenarios: You have an undirected graph and need to know the minimum
10 * cut between *every* possible pair of vertices.
11 * - The Magic: Normally, finding the min-cut between all pairs requires  $O(N^2)$ 

```

```

12 * Max Flow calls. The Gomory-Hu Tree (specifically Gusfield's algorithm
13 * implementation)
14 * builds a tree representing all min-cuts using only exactly N-1 Max Flow
15 * calls!
16 * The min-cut between any two nodes (u, v) is simply the minimum weight edge
17 * on
18 * the path between them in this resulting tree.
19 *
20 * 2. HOW TO USE IT (THE BLACK BOX)
21 * - Initialization:
22 *   GomoryHu gh(N); // N is the number of nodes (0-indexed)
23 * - Add Edge:
24 *   gh.add_edge(u, v, cap); // Undirected edge
25 * - Build:
26 *   gh.build(); // Runs the N-1 Max Flow calls to construct the tree
27 * - Query:
28 *   long long ans = gh.min_cut(u, v);
29 *
30 * - Complexity:
31 *   Time: O(N * MaxFlow(V, E)) to build. O(N) per query (or O(1) if you add
32 *   LCA).
33 *   Space: O(N + E)
34 *
35 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
36 * - Dinic Dependency: This template assumes you have your `Dinic` struct
37 *   available
38 *   in the same file or included. Ensure `Dinic` has `add_edge(u, v, cap)` and
39 *   `max_flow(s, t)`.
40 * - 1-based indexing: If your nodes are 1 to N, instantiate `GomoryHu gh(N + 1)`
41 *   and
42 *   just ignore index 0.
43 */
44
45 struct GomoryHu
46 {
47     int n;
48     struct Edge
49     {
50         int u, v;
51         long long cap;
52     };
53     vector<Edge> edges;
54
55     // Parent array for the tree (Gomory-Hu Tree structure)
56     vector<int> p;
57     // Weight array (min-cut value between i and p[i])
58     vector<long long> weight;
59     // Depth array for simple O(N) queries without complex LCA
60     vector<int> depth;
61
62     GomoryHu(int n) : n(n), p(n), weight(n), depth(n, 0) {}
63
64     void add_edge(int u, int v, long long cap)
65     {
66         edges.push_back({u, v, cap});
67     }
68
69     void build()
70     {
71         // 1. Initialize parents: everyone initially points to node 0
72         fill(p.begin(), p.end(), 0);
73         fill(weight.begin(), weight.end(), 0);
74
75         // 2. Iterate to build the tree (Gusfield's Algorithm)
76         for (int i = 1; i < n; i++)
77         {
78             // We rebuild the graph for Dinic each time.
79             // (Assuming `Dinic` is your standard Dinic template)
80             Dinic dinic(n);
81             for (auto &e : edges)
82             {
83                 dinic.add_edge(e.u, e.v, e.cap);
84                 dinic.add_edge(e.v, e.u, e.cap); // Gomory-Hu requires Undirected
85                 edges!
86             }
87
88             // Find min s-t cut between current node `i` and its parent `p[i]`
89             long long flow = dinic.max_flow(i, p[i]);
90             weight[i] = flow;
91
92             // Find which nodes are on the source side of the cut
93             // (reachable from s in the residual graph)
94             vector<bool> reachable(n, false);
95             dfs_residual(i, reachable, dinic);
96
97             // Update parents for next iterations
98             // If any node j > i points to p[i] but is actually on i's side of
99             // the cut,
100             // we redirect its parent to i.
101             for (int j = i + 1; j < n; j++)
102             {
103                 if (reachable[j] && p[j] == p[i])
104                 {
105                     p[j] = i;
106                 }
107             }
108         }
109
110         // 3. Precalculate depths to allow simple climbing during queries
111         for (int i = 1; i < n; i++)
112         {
113             depth[i] = depth[p[i]] + 1;
114         }
115     }
116 }

```

```

107 void dfs_residual(int u, vector<bool> &reachable, Dinic &dinic)
108 {
109     reachable[u] = true;
110     for (auto &e : dinic.adj[u])
111     {
112         // If capacity remains (cap - flow > 0), it is not saturated,
113         // so we can walk across it in the residual graph.
114         if (e.cap - e.flow > 0 && !reachable[e.to])
115         {
116             dfs_residual(e.to, reachable, dinic);
117         }
118     }
119 }
120
121 // Query min cut between any two nodes
122 long long min_cut(int u, int v)
123 {
124     if (u == v)
125         return 0; // Or INF depending on problem definition
126
127     long long res = 2e18; // Use large INF
128
129     // Walk up the tree from u and v simultaneously to find their LCA,
130     // maintaining the minimum edge weight seen on the path.
131     while (u != v)
132     {
133         if (depth[u] > depth[v])
134         {
135             res = min(res, weight[u]);
136             u = p[u];
137         }
138         else
139         {
140             res = min(res, weight[v]);
141             v = p[v];
142         }
143     }
144     return res;
145 }
146 }
147 };

```

### 3.4.2 MCMF

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: MIN COST MAX FLOW (MCMF)
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?

```

```

8  * - Keywords: "Cheapest Flow", "Matching with Costs", "Assignment Problem".
9  * - Classic Scenarios: You have workers and tasks with different costs. You need
10 * to assign every worker a task such that the total cost is minimized.
11 * - The Magic: It works by finding the "shortest" path (where "shortest" is
12 *   edge cost) in the residual graph. It keeps augmenting flow along these
13 *   shortest
14 *   paths until no path from source to sink exists.
15 * 2. HOW TO USE IT (THE BLACK BOX)
16 * - Initialization:
17 *     MCMF solver(N);
18 * - Add Edge:
19 *     solver.add_edge(u, v, capacity, cost);
20 * - Query:
21 *     pair<long long, long long> result = solver.min_cost_max_flow(source,
22 *       sink);
23 *     // result.first = max flow, result.second = min cost
24 * - Complexity:
25 *     O(F * E * log V) using Dijkstra + Potentials.
26 *     O(F * E * V) using SPFA (included below for simplicity with negative
27 *     edges).
28 */
29 const long long INF = 1e18;
30
31 struct Edge
32 {
33     int to;
34     long long cap, flow, cost;
35     int rev; // Reverse edge index
36 };
37
38 struct MCMF
39 {
40     int n;
41     vector<vector<Edge>> adj;
42     vector<long long> dist, parent_edge, parent_node;
43
44     MCMF(int n) : n(n), adj(n), dist(n), parent_edge(n), parent_node(n) {}
45
46     void add_edge(int from, int to, long long cap, long long cost)
47     {
48         adj[from].push_back({to, cap, 0, cost, (int)adj[to].size()});
49         adj[to].push_back({from, 0, 0, -cost, (int)adj[from].size() - 1});
50     }
51
52     // SPFA is used here to handle negative costs.
53     // If all costs are positive, Dijkstra with potentials is faster.
54     bool spfa(int s, int t, long long &flow, long long &cost)
55     {

```

```

56     dist.assign(n, INF);
57     parent_node.assign(n, -1);
58     parent_edge.assign(n, -1);
59     vector<bool> in_queue(n, false);
60     queue<int> q;
61
62     dist[s] = 0;
63     q.push(s);
64     in_queue[s] = true;
65
66     while (!q.empty())
67     {
68         int u = q.front();
69         q.pop();
70         in_queue[u] = false;
71
72         for (int i = 0; i < (int)adj[u].size(); i++)
73         {
74             Edge &e = adj[u][i];
75             if (e.cap - e.flow > 0 && dist[e.to] > dist[u] + e.cost)
76             {
77                 dist[e.to] = dist[u] + e.cost;
78                 parent_node[e.to] = u;
79                 parent_edge[e.to] = i;
80                 if (!in_queue[e.to])
81                 {
82                     q.push(e.to);
83                     in_queue[e.to] = true;
84                 }
85             }
86         }
87     }
88
89     if (dist[t] == INF)
90         return false;
91
92     // Push flow
93     long long push = INF;
94     int curr = t;
95     while (curr != s)
96     {
97         int prev = parent_node[curr];
98         int idx = parent_edge[curr];
99         push = min(push, adj[prev][idx].cap - adj[prev][idx].flow);
100        curr = prev;
101    }
102
103    flow += push;
104    cost += push * dist[t];
105    curr = t;
106    while (curr != s)
107    {

```

```

108        int prev = parent_node[curr];
109        int idx = parent_edge[curr];
110        adj[prev][idx].flow += push;
111        int rev_idx = adj[prev][idx].rev;
112        adj[curr][rev_idx].flow -= push;
113        curr = prev;
114    }
115
116    return true;
117 }
118
119 pair<long long, long long> min_cost_max_flow(int s, int t)
120 {
121     long long flow = 0, cost = 0;
122     while (spfa(s, t, flow, cost))
123     ;
124     return {flow, cost};
125 }
126 };

```

### 3.4.3 Max flow

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DINIC'S ALGORITHM
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Max Flow", "Min-Cut", "Bipartite Matching", "Maximum Closure".
9  * - Classic Scenarios: You have a network of pipes with capacities and need to
   find
10  * the maximum amount of "stuff" (flow) that can move from a Source to a Sink.
11  * - The Magic:
12  *   a) BFS: Builds a Level Graph (only allows flow from distance d to d+1).
13  *   b) DFS: Pushes flow along the Level Graph.
14  *   c) ptr: The "pointer optimization" ensures we don't re-scan edges that are
15  * already saturated in the current DFS phase (the secret to  $O(V^2 E)$ ).
16  *
17  * 2. HOW TO USE IT (THE BLACK BOX)
18  * - Initialization:
19  *   Dinic flow(N);
20  * - Add Edge:
21  *   flow.add_edge(u, v, capacity);
22  * - Query:
23  *   long long max_f = flow.max_flow(source, sink);
24  *
25  * - Complexity:
26  *   General:  $O(V^2 * E)$ .
27  *   Unit Networks (e.g., Bipartite Matching):  $O(E * \text{sqrt}(V))$ .
28  */

```

```

29
30 struct Dinic
31 {
32     struct Edge
33     {
34         int to;
35         long long cap, flow;
36         int rev; // Index of the reverse edge in adj[to]
37     };
38
39     int n;
40     vector<vector<Edge>> adj;
41     vector<int> level, ptr;
42
43     Dinic(int n) : n(n), adj(n), level(n), ptr(n) {}
44
45     void add_edge(int from, int to, long long cap)
46     {
47         // Forward edge: capacity 'cap', flow 0
48         adj[from].push_back({to, cap, 0, (int)adj[to].size()});
49         // Backward edge: capacity 0, flow 0
50         adj[to].push_back({from, 0, 0, (int)adj[from].size() - 1});
51     }
52
53     // BFS to build the Level Graph
54     bool bfs(int s, int t)
55     {
56         fill(level.begin(), level.end(), -1);
57         level[s] = 0;
58         queue<int> q;
59         q.push(s);
60         while (!q.empty())
61         {
62             int v = q.front();
63             q.pop();
64             for (auto &edge : adj[v])
65             {
66                 if (edge.cap - edge.flow > 0 && level[edge.to] == -1)
67                 {
68                     level[edge.to] = level[v] + 1;
69                     q.push(edge.to);
70                 }
71             }
72         }
73         return level[t] != -1;
74     }
75
76     // DFS to push flow through the Level Graph
77     long long dfs(int v, int t, long long pushed)
78     {
79         if (pushed == 0)
80             return 0;

```

```

81         if (v == t)
82             return pushed;
83
84         // ? Dial 1: The Pointer Optimization
85         // ptr[v] ensures we don't look at edges that we already know are dead
86         ends
87         for (int &cid = ptr[v]; cid < (int)adj[v].size(); cid++)
88         {
89             auto &edge = adj[v][cid];
90             int tr = edge.to;
91             if (level[v] + 1 != level[tr] || edge.cap - edge.flow == 0)
92                 continue;
93
94             long long tr_pushed = dfs(tr, t, min(pushed, edge.cap - edge.flow));
95             if (tr_pushed == 0)
96                 continue;
97
98             edge.flow += tr_pushed;
99             adj[tr][edge.rev].flow -= tr_pushed;
100             return tr_pushed;
101         }
102         return 0;
103     }
104
105     long long max_flow(int s, int t)
106     {
107         long long flow = 0;
108         while (bfs(s, t))
109         {
110             fill(ptr.begin(), ptr.end(), 0);
111             while (long long pushed = dfs(s, t, 1e18))
112             {
113                 flow += pushed;
114             }
115             return flow;
116         }
117     };

```

## 3.5 Graph Traverse

### 3.5.1 Bipartite Graph

```

1 // Check Odd Cycle
2 // using in problems teams or groups or colors
3 vector<int> teams;
4 Graph adj;
5 int n, m;
6 bool valid_team = true;
7 void bfs(int start) {
8     queue<int> q;

```

```

9    q.push(start);
10   if (teams[start] == 0)
11       teams[start] = 1;
12   while (!q.empty())
13   {
14       int cur = q.front(); q.pop();
15       for (auto ch : adj[cur])
16       {
17           if (!teams[ch])
18           {
19               q.push(ch);
20               teams[ch] = 3 - teams[cur];
21           }
22           else if (teams[ch] == teams[cur])
23           {
24               valid_team = false;
25               return;
26           }
27       }
28   }
29
30 }
```

### 3.5.2 Euler tour

```

1  const int N = 1e5 + 5;
2  GRAPH graph(N);
3  int tin[N], tout[N], value[N], flat[N];
4  int timer{1};
5
6  void dfs(int node, int p)
7  {
8      flat[timer] = value[node];
9      tin[node] = timer++;
10     for (auto &x : graph[node])
11     {
12         if (x != p)
13             dfs(x, node);
14     }
15     tout[node] = timer;
16     // [tin, tout)
17 }
```

### 3.5.3 Top sort

```

1  vector<int> res;
2
3  void topo_DFS(int node)
4  {
```

```

5      vis[node] = true;
6      for (auto &neighbor : graph[node])
7      {
8          if (!vis[neighbor])
9              topo_DFS(neighbor);
10     }
11     res.push_back(node);
12 }
13
14 // Then reverse
15
16 vector<int> topSort()
17 {
18     int n = int(graph.size());
19     vector<int> inDegree(n);
20     for (int i = 0; i < n; i++)
21     {
22         for (auto &neighbor : graph[i])
23             inDegree[neighbor]++;
24     }
25     queue<int> ready;
26     vector<int> ordering;
27     for (int i = 0; i < n; i++)
28     {
29         if (!inDegree[i])
30             ready.push(i);
31     }
32     while (!ready.empty())
33     {
34         int node = ready.front();
35         ready.pop();
36         ordering.push_back(node);
37         for (auto &neighbor : graph[node])
38         {
39             if (--inDegree[neighbor])
40                 ready.push(neighbor);
41         }
42     }
43     if ((int)ordering.size() != int(graph.size()))
44         ordering.clear();
45     return ordering;
46 }
```

## 3.6 Graph connectivity

### 3.6.1 Articulations

```

1  #include <bits/stdc++.h>
2  using namespace std;
3
4  const int MAXN = 1e5 + 5;
```

```

5 vector<int> adj[MAXN];
6 int disc[MAXN], low[MAXN], timer;
7 vector<pair<int, int>> bridges;
8 set<int> articulation_points;
9
10 void dfs(int u, int p = -1)
11 {
12     disc[u] = low[u] = ++timer;
13     int children = 0; // Needed for the root case
14
15     for (int v : adj[u])
16     {
17         if (v == p)
18             continue; // Don't go back to parent
19
20         if (disc[v])
21         {
22             // Back-edge found!
23             low[u] = min(low[u], disc[v]);
24         }
25         else
26         {
27             // Tree-edge
28             dfs(v, u);
29             low[u] = min(low[u], low[v]);
30
31             // BRIDGE CONDITION
32             if (low[v] > disc[u])
33                 bridges.push_back({u, v});
34
35             // ARTICULATION POINT CONDITION
36             if (low[v] >= disc[u] && p != -1)
37                 articulation_points.insert(u);
38
39             children++;
40         }
41     }
42
43     // ROOT SPECIAL CASE: If root has > 1 child, it's an AP
44     if (p == -1 && children > 1)
45         articulation_points.insert(u);
46 }

```

### 3.6.2 SCC

```

1 const int N = 1e5 + 9;
2
3 int SCC_id[N];
4 GRAPH graph, graph_reverse, cond_graph;
5 int n, m;
6 vector<int> order, vis, vis_reverse, order_reverse;

```

```

7
8 void dfs(int node)
9 {
10     vis[node] = true;
11     for (auto &x : graph[node])
12     {
13         if (!vis[x])
14             dfs(x);
15     }
16     order.emplace_back(node);
17 }
18
19 void dfs_reverse(int node)
20 {
21     vis_reverse[node] = true;
22     for (auto &x : graph_reverse[node])
23     {
24         if (!vis_reverse[x])
25             dfs_reverse(x);
26     }
27     order_reverse.emplace_back(node);
28 }
29
30 void SCC()
31 {
32     graph.assign(n + 1, {}), graph_reverse.assign(n + 1, {}), vis.assign(n + 1, {}),
33     vis_reverse.assign(n + 1, {}),
34     cond_graph.assign(n + 1, {});
35     for (int i = 0; i < m; i++)
36     {
37         int x, y;
38         cin >> x >> y;
39         graph[x].emplace_back(y), graph_reverse[y].emplace_back(x);
40     }
41     order.clear();
42     for (int i = 1; i <= n; i++)
43     {
44         if (!vis[i])
45             dfs(i);
46     }
47     reverse(order.begin(), order.end());
48     vector<vector<int>>> scc;
49     scc.push_back({});
50     for (auto &x : order)
51     {
52         if (!vis_reverse[x])
53         {
54             order_reverse.clear();
55             dfs_reverse(x);
56             scc.emplace_back(order_reverse);
57             for (auto &val : order_reverse)
58                 SCC_id[val] = int(scc.size()) - 1;

```



```

58     }
59 }
60 for (int u = 1; u <= n; u++)
61 {
62     for (int v : graph[u])
63     {
64         if (SCC_id[u] != SCC_id[v])
65             cond_graph[SCC_id[u]].emplace_back(SCC_id[v]);
66     }
67 }
68 }

```

## 3.7 Graph counting

### 3.7.1 BEST

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: BEST THEOREM
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Count Eulerian Circuits", "Directed Graph Cycles".
9  * - Classic Scenarios: You have a directed graph where every node's in-degree
10 *   equals its out-degree, and you need to know how many distinct Eulerian circuits it
11 *   has.
12 * - The Magic: The number of Eulerian circuits =  $t_w(G) * \prod ((deg(v) - 1)!)$ 
13 *   where  $t_w(G)$  is the number of arborescences rooted at any vertex  $w$ .
14 */
15 const int MOD = 1e9 + 7;
16
17 // Gaussian Elimination for Determinant (MOD)
18 long long power(long long base, long long exp)
19 {
20     long long res = 1;
21     base %= MOD;
22     while (exp > 0)
23     {
24         if (exp % 2 == 1)
25             res = (res * base) % MOD;
26         base = (base * base) % MOD;
27         exp /= 2;
28     }
29     return res;
30 }
31
32 long long modInverse(long long n) { return power(n, MOD - 2); }
33

```

```

34 long long determinant(vector<vector<long long>> &mat, int n)
35 {
36     long long det = 1;
37     for (int i = 0; i < n; i++)
38     {
39         int pivot = i;
40         for (int j = i + 1; j < n; j++)
41             if (abs(mat[j][i]) > abs(mat[pivot][i]))
42                 pivot = j;
43         swap(mat[i], mat[pivot]);
44         if (i != pivot)
45             det = (MOD - det) % MOD;
46         if (mat[i][i] == 0)
47             return 0;
48         det = (det * mat[i][i]) % MOD;
49         long long inv = modInverse(mat[i][i]);
50         for (int j = i + 1; j < n; j++)
51         {
52             long long factor = (mat[j][i] * inv) % MOD;
53             for (int k = i + 1; k < n; k++)
54                 mat[j][k] = (mat[j][k] - factor * mat[i][k] % MOD + MOD) % MOD;
55         }
56     }
57     return det;
58 }
59
60 long long best_theorem(int n, vector<vector<int>> &adj)
61 {
62     vector<int> out_degree(n, 0);
63     vector<vector<long long>> L(n, vector<long long>(n, 0));
64
65     // 1. Build Directed Laplacian
66     for (int u = 0; u < n; ++u)
67     {
68         for (int v : adj[u])
69         {
70             out_degree[u]++;
71             L[u][v]--;
72         }
73     }
74     for (int i = 0; i < n; ++i)
75         L[i][i] += out_degree[i];
76
77     // 2. Compute arborescences ( $t_w$ )
78     // Delete row 0 and col 0 (root at 0)
79     vector<vector<long long>> minor(n - 1, vector<long long>(n - 1));
80     for (int i = 1; i < n; i++)
81         for (int j = 1; j < n; j++)
82             minor[i - 1][j - 1] = (L[i][j] % MOD + MOD) % MOD;
83
84     long long t_w = determinant(minor, n - 1);
85

```

```

86 // 3. Multiply by prod((deg(v) - 1)!)
87 long long fact_prod = 1;
88 vector<long long> fact(n + 1, 1);
89 for (int i = 2; i <= n; i++)
90     fact[i] = (fact[i - 1] * i) % MOD;
91
92 for (int i = 0; i < n; i++)
93 {
94     if (out_degree[i] > 0)
95         fact_prod = (fact_prod * fact[out_degree[i] - 1]) % MOD;
96 }
97
98 return (t_w * fact_prod) % MOD;
99 }
100
101 int main()
102 {
103     ios_base::sync_with_stdio(false);
104     cin.tie(NULL);
105     // int n; cin >> n; ... adj list setup ...
106     return 0;
107 }

```

### 3.7.2 Prufer code

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: PRUFER CODE
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Labeled Trees", "Cayley's Formula", "Tree Counting".
9  * - Classic Scenarios: Bijectively mapping a labeled tree to a sequence of
   length N-2.
10 * - The Magic: It allows you to count trees with specific constraints (like
   degrees)
11 * by counting sequences instead, which is mathematically trivial.
12 */
13
14 struct Prufer
15 {
16     // Converts a labeled tree (adj list) to a Prufer sequence of length N-2
17     // Complexity: O(N log N)
18     static vector<int> tree_to_prufer(int n, vector<vector<int>> &adj)
19     {
20         if (n == 2)
21             return {}; // A 2-node tree has no Prufer sequence
22
23         vector<int> degree(n + 1);
24         priority_queue<int, vector<int>, greater<int>> leaves;

```

```

25
26         for (int i = 1; i <= n; i++)
27         {
28             degree[i] = adj[i].size();
29             if (degree[i] == 1)
30                 leaves.push(i);
31         }
32
33         vector<int> prufer;
34         vector<bool> removed(n + 1, false);
35
36         for (int i = 0; i < n - 2; i++)
37         {
38             int leaf = leaves.top();
39             leaves.pop();
40             removed[leaf] = true;
41
42             // Find the neighbor of this leaf
43             int neighbor = -1;
44             for (int v : adj[leaf])
45             {
46                 if (!removed[v])
47                 {
48                     neighbor = v;
49                     break;
50                 }
51             }
52
53             prufer.push_back(neighbor);
54             degree[neighbor]--;
55             if (degree[neighbor] == 1)
56                 leaves.push(neighbor);
57         }
58
59         return prufer;
60     }
61
62     // Converts a Prufer sequence of length N-2 back to an edge list of the tree
63     // Complexity: O(N log N)
64     static vector<pair<int, int>> prufer_to_tree(int n, const vector<int> &prufer)
65     {
66         vector<int> degree(n + 1, 1);
67         for (int x : prufer)
68             degree[x]++;
69
70         priority_queue<int, vector<int>, greater<int>> leaves;
71         for (int i = 1; i <= n; i++)
72         {
73             if (degree[i] == 1)
74                 leaves.push(i);
75         }

```

```

76
77     vector<pair<int, int>> edges;
78     for (int x : prufer)
79     {
80         int leaf = leaves.top();
81         leaves.pop();
82         edges.push_back({leaf, x});
83
84         degree[x]--;
85         if (degree[x] == 1)
86             leaves.push(x);
87     }
88
89     // Connect the last two remaining nodes
90     int u = leaves.top();
91     leaves.pop();
92     int v = leaves.top();
93     leaves.pop();
94     edges.push_back({u, v});
95
96     return edges;
97 }
98 };
99
100 void solve()
101 {
102     int n;
103     cin >> n;
104     // Example: Tree to Prufer
105     // vector<vector<int>> adj(n+1);
106     // ... fill adj ...
107     // vector<int> p = Prufer::tree_to_prufer(n, adj);
108 }
109
110 int main()
111 {
112     ios_base::sync_with_stdio(false);
113     cin.tie(NULL);
114     return 0;
115 }

```

## 3.8 LCA

### 3.8.1 LCA(Hassan)

```

1 struct LCA {
2     vector<vector<pair<int,int>> > adj;
3     vector<vector<int>> up, cost[2];
4     vector<int> depth;
5     int LOG = 30;
6     int ign = 1e9;

```

```

7     void build(int n, vector<vector<pair<int,int>> > &G, int root = 1) {
8         adj = G;
9         up = vector<vector<int>> (LOG + 5, vector<int>(n + 5));
10        cost[0] = vector<vector<int>> (LOG + 5, vector<int>(n + 5));
11        cost[1] = vector<vector<int>> (LOG + 5, vector<int>(n + 5));
12        depth = vector<int>(n + 5);
13        dfs(root, -1);
14    }
15    void dfs(int u, int p) {
16        for (auto [v,w]: adj[u]) {
17            if (v == p) continue;
18            depth[v] = depth[u] + 1;
19
20            up[0][v] = u;
21            cost[0][0][v] = w;
22            cost[1][0][v] = w;
23            for (int i = 1; i < LOG; i++) {
24                up[i][v] = up[i - 1][up[i - 1][v]];
25                cost[0][i][v] = min(cost[0][i - 1][v], cost[0][i - 1][up[i - 1][v]]);
26                cost[1][i][v] = max(cost[1][i - 1][v], cost[1][i - 1][up[i - 1][v]]);
27            }
28            dfs(v, u);
29        }
30    }
31 }
32
33 int kth_ancestor(int u, int k) {
34     for (int i = 0; i < LOG; i++) {
35         if (k >> i & 1) {
36             u = up[i][u];
37         }
38     }
39     return u;
40 }
41
42 int getLca(int u, int v) {
43     if (depth[u] < depth[v]) swap(u, v);
44     int k = depth[u] - depth[v];
45     u = kth_ancestor(u, k);
46     if (u == v) return u;
47
48     for (int i = LOG - 1; i >= 0; --i) {
49         if (up[i][u] != up[i][v]) {
50             u = up[i][u];
51             v = up[i][v];
52         }
53     }
54     return up[0][v];
55 }
56 pair<int,int> query(int u,int v) {

```

```

57     int mn = ign, mx = -ign;
58     int lca = getLca(u, v);
59     int k = depth[u] - depth[lca];
60     for (int i=0; i<LOG; i++) {
61         if (k>>i&1) {
62             mn = min(mn, cost[0][i][u]);
63             mx = max(mx, cost[1][i][u]);
64             u = up[i][u];
65         }
66     }
67     k = depth[v] - depth[lca];
68     for (int i=0; i<LOG; i++) {
69         if (k>>i&1) {
70             mn = min(mn, cost[0][i][v]);
71             mx = max(mx, cost[1][i][v]);
72             v = up[i][v];
73         }
74     }
75     return {mn, mx};
76 }
77
78
79 };

```

### 3.8.2 LCA

```

1 struct LCA
2 {
3     int n, LG, SKIP;
4     vector<vector<int>>> T, Cost;
5     vector<int> par, dep;
6
7     LCA(int nodes_count, const GRAPH &graph, int root = 1, int skip_val = 00)
8     {
9         n = nodes_count;
10        LG = __lg(n) + 2;
11        SKIP = skip_val;
12        T.assign(LG, vector<int>(n + 1, -1));
13        Cost.assign(LG, vector<int>(n + 1, SKIP));
14        par.assign(n + 1, -1);
15        dep.assign(n + 1, 0);
16
17        Cost[0][root] = SKIP;
18        dfs(root, -1, 0, graph);
19
20        for (int i = 1; i <= n; i++)
21            T[0][i] = par[i];
22
23        for (int pw = 1; (1 << pw) <= n; pw++)
24        {
25            for (int i = 1; i <= n; i++)

```

```

26        {
27            if (T[pw - 1][i] != -1)
28            {
29                T[pw][i] = T[pw - 1][T[pw - 1][i]];
30                Cost[pw][i] = merge(Cost[pw - 1][i], Cost[pw - 1][T[pw - 1][i]]);
31            }
32        }
33    }
34 }
35
36 int merge(int a, int b)
37 {
38     return min(a, b);
39 }
40
41 void dfs(int node, int p, int d, const GRAPH &graph)
42 {
43     dep[node] = d;
44     par[node] = p;
45     for (auto &[i, w] : graph[node])
46     {
47         if (i != p)
48         {
49             Cost[0][i] = w;
50             dfs(i, node, d + 1, graph);
51         }
52     }
53 }
54
55 int get_kth(int x, int k)
56 {
57     while (k)
58     {
59         if (x == -1)
60             break;
61         int pw = __builtin_ctz(k);
62         x = T[pw][x];
63         k &= (k - 1);
64     }
65     return x;
66 }
67
68 int get_lca(int u, int v)
69 {
70     if (dep[u] < dep[v])
71         swap(u, v);
72     u = get_kth(u, dep[u] - dep[v]);
73     if (u == v)
74         return u;
75     for (int j = LG - 1; j >= 0; j--)
76     {

```

```

77         if (T[j][u] != T[j][v])
78         {
79             u = T[j][u];
80             v = T[j][v];
81         }
82     }
83     return par[u];
84 }
85
86 int get_cost(int u, int v)
87 {
88     if (dep[u] < dep[v])
89         swap(u, v);
90     int k = dep[u] - dep[v], x = u;
91     int cost = SKIP;
92     while (k)
93     {
94         if (x == -1)
95             break;
96         int pw = __builtin_ctz(k);
97         cost = merge(cost, Cost[pw][x]);
98         x = T[pw][x];
99         k &= (k - 1);
100     }
101     return cost;
102 }
103
104 int query_path(int u, int v)
105 {
106     int lca = get_lca(u, v);
107     return merge(get_cost(u, lca), get_cost(v, lca));
108 }
109
110 int get_dist(int u, int v)
111 {
112     int lca = get_lca(u, v);
113     return dep[u] + dep[v] - 2 * dep[lca];
114 }
115 };

```

## 3.9 MST

### 3.9.1 Directed MST

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DIRECTED MST (CHU-LIU/EDMONDS)
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?

```

```

8  * - Keywords: "Minimum Spanning Arborescence", "Directed MST".
9  * - Classic Scenarios: Finding the cheapest way to connect all nodes in a
    DIRECTED graph
10 * rooted at a specific node, such that every node (except the root) has
    exactly 1 incoming edge.
11 * - The Magic: It greedily picks the minimum incoming edge for every node. If
    this creates
12 * cycles, it "contracts" those cycles into a single super-node, updates the
    weights of edges
13 * entering the cycle, and recurses.
14 *
15 * 2. HOW TO USE IT
16 * - edge: {u, v, cost}
17 * - roots: The node that acts as the root of the arborescence.
18 * - Complexity:  $O(V * E)$ 
19 */
20
21 struct Edge
22 {
23     int u, v;
24     long long cost;
25 };
26
27 long long edmonds(int n, int root, vector<Edge> &edges)
28 {
29     long long total_cost = 0;
30     vector<int> min_edge(n + 1);
31     vector<int> parent(n + 1);
32     vector<int> id(n + 1);
33     vector<int> vis(n + 1);
34
35     while (true)
36     {
37         fill(min_edge.begin(), min_edge.end(), 2e18); // Use large INF
38         for (const auto &e : edges)
39         {
40             if (e.u != e.v && e.cost < min_edge[e.v])
41             {
42                 min_edge[e.v] = e.cost;
43                 parent[e.v] = e.u;
44             }
45         }
46
47         for (int i = 1; i <= n; i++)
48         {
49             if (i != root && min_edge[i] == 2e18)
50                 return -1; // Impossible
51         }
52
53         int cnt_nodes = 0;
54         fill(id.begin(), id.end(), -1);
55         fill(vis.begin(), vis.end(), -1);

```

```

56     min_edge[root] = 0;
57
58     for (int i = 1; i <= n; i++)
59     {
60         total_cost += min_edge[i];
61         int v = i;
62         while (vis[v] != i && id[v] == -1 && v != root)
63         {
64             vis[v] = i;
65             v = parent[v];
66         }
67         if (v != root && id[v] == -1)
68         {
69             for (int u = parent[v]; u != v; u = parent[u])
70                 id[u] = cnt_nodes;
71             id[v] = cnt_nodes++;
72         }
73     }
74
75     if (cnt_nodes == 0)
76         break;
77     for (int i = 1; i <= n; i++)
78     {
79         if (id[i] == -1)
80             id[i] = cnt_nodes++;
81     }
82
83     for (auto &e : edges)
84     {
85         long long v_cost = min_edge[e.v];
86         e.u = id[e.u];
87         e.v = id[e.v];
88         if (e.u != e.v)
89             e.cost -= v_cost;
90     }
91     n = cnt_nodes;
92     root = id[root];
93 }
94 return total_cost;
95 }

```

### 3.9.2 Kirchoff

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: KIRCHHOFF'S THEOREM
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Count spanning trees", "Matrix Tree Theorem".

```

```

9  * - Classic Scenarios: You have a graph and need to know exactly how many
10 *   different
11 *   spanning trees can be formed.
12 * - The Magic: It converts a graph problem into a linear algebra problem.
13 *   It constructs the Laplacian Matrix and computes a determinant using
14 *   Gaussian Elimination.
15 * 2. HOW TO USE IT
16 * - Build the Laplacian matrix (L = Degree - Adjacency).
17 * - Remove the last row and last column.
18 * - Compute the determinant modulo MOD.
19 * - Complexity:  $O(N^3)$  (due to Gaussian Elimination).
20 */
21
22 const int MOD = 1e9 + 7;
23
24 long long power(long long base, long long exp)
25 {
26     long long res = 1;
27     base %= MOD;
28     while (exp > 0)
29     {
30         if (exp % 2 == 1)
31             res = (res * base) % MOD;
32         base = (base * base) % MOD;
33         exp /= 2;
34     }
35     return res;
36 }
37
38 long long modInverse(long long n)
39 {
40     return power(n, MOD - 2);
41 }
42
43 // Gaussian Elimination to find Determinant
44 long long determinant(vector<vector<long long>> &mat, int n)
45 {
46     long long det = 1;
47     for (int i = 0; i < n; i++)
48     {
49         int pivot = i;
50         for (int j = i + 1; j < n; j++)
51         {
52             if (abs(mat[j][i]) > abs(mat[pivot][i]))
53                 pivot = j;
54         }
55         swap(mat[i], mat[pivot]);
56         if (i != pivot)
57             det = (MOD - det) % MOD;
58         if (mat[i][i] == 0)
59             return 0;

```

```

60
61     det = (det * mat[i][i]) % MOD;
62     long long inv = modInverse(mat[i][i]);
63     for (int j = i + 1; j < n; j++)
64     {
65         long long factor = (mat[j][i] * inv) % MOD;
66         for (int k = i + 1; k < n; k++)
67         {
68             mat[j][k] = (mat[j][k] - factor * mat[i][k] % MOD + MOD) % MOD;
69         }
70     }
71 }
72 return det;
73 }
74
75 void solve()
76 {
77     int n, m;
78     cin >> n >> m;
79
80     // Laplacian Matrix
81     vector<vector<long long>> L(n, vector<long long>(n, 0));
82
83     for (int i = 0; i < m; i++)
84     {
85         int u, v;
86         cin >> u >> v;
87         --u;
88         --v; // 0-indexed
89
90         L[u][u]++;
91         L[v][v]++;
92         L[u][v]--;
93         L[v][u]--;
94     }
95
96     // Convert to positive modulo
97     for (int i = 0; i < n; i++)
98         for (int j = 0; j < n; j++)
99             L[i][j] = (L[i][j] % MOD + MOD) % MOD;
100
101     // Remove last row and column to get the Minor
102     vector<vector<long long>> minor(n - 1, vector<long long>(n - 1));
103     for (int i = 0; i < n - 1; i++)
104     {
105         for (int j = 0; j < n - 1; j++)
106         {
107             minor[i][j] = L[i][j];
108         }
109     }
110
111     cout << determinant(minor, n - 1) << "\n";

```

```

112 }
113
114 int main()
115 {
116     ios_base::sync_with_stdio(false);
117     cin.tie(NULL);
118     solve();
119     return 0;
120 }

```

### 3.9.3 Kruskal

```

1 class DSU
2 {
3 private:
4     vector<int> parent, rank;
5     int forests{};
6
7     void link(int x, int y)
8     {
9         if (rank[x] > rank[y])
10             swap(x, y);
11         parent[x] = y;
12         if (rank[x] == rank[y])
13             rank[y]++;
14     }
15
16 public:
17     DSU(int n)
18     {
19         parent = vector<int>(n), rank = vector<int>(n);
20         forests = n;
21         for (int i = 0; i < n; i++)
22             parent[i] = i, rank[i] = 1;
23     }
24
25     int find_set(int x)
26     {
27         if (x == parent[x])
28             return x;
29         return parent[x] = find_set(parent[x]);
30     }
31
32     int union_sets(int x, int y)
33     {
34         x = find_set(x), y = find_set(y);
35         if (x != y)
36         {
37             link(x, y);
38             forests--;
39         }

```

```

40     return x != y;
41 }
42 };
43
44 struct edge
45 {
46     int from, to, w;
47
48     edge(int from, int to, int w) : from(from), to(to), w(w)
49     {
50     }
51
52     bool operator<(const edge &e) const
53     {
54         return w < e.w;
55     }
56 };
57
58 int Kruskal(vector<edge> &edgeList, int n)
59 {
60     DSU uf(n + 1);
61     vector<edge> edges;
62     int mnCost{};
63     sort(edgeList.begin(), edgeList.end());
64     for (auto &[from, to, cost] : edgeList)
65     {
66         if (uf.union_sets(from, to))
67         {
68             edges.emplace_back(from, to, cost);
69             mnCost += cost;
70         }
71     }
72     return mnCost;
73 }

```

### 3.9.4 Prim

```

1 int n;
2 GRAPH graph;
3
4 void prim(int node)
5 {
6     int total{};
7     priority_queue<pair<int, int>, vector<pair<int, int>>, greater<pair<int, int>>> pq;
8     pq.emplace(0, node);
9     vector<bool> visited(n + 1, 0);
10    vector<int> dist(n + 1, 00);
11    dist[node] = 0;
12    while (!pq.empty())
13    {

```

```

14        auto [w, node] = pq.top();
15        pq.pop();
16        if (visited[node])
17            continue;
18        visited[node] = true;
19        total += w;
20        for (auto &[to, c] : graph[node])
21        {
22            if (!visited[to] && dist[to] > c)
23            {
24                dist[to] = c;
25                pq.push({dist[to], to});
26            }
27        }
28    }
29 }

```

## 3.10 Matching

### 3.10.1 Bipartite matching

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: HOPCROFT-KARP ALGORITHM
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Maximum Bipartite Matching", "Maximum Independent Set", "Minimum
   Vertex Cover".
9  * - Classic Scenarios: You have two distinct groups (e.g., Workers and Jobs, or
   Boys and Girls).
10 * Each worker can only do specific jobs. You want to pair up as many workers
   with jobs as possible
11 * such that no worker gets two jobs, and no job gets two workers.
12 * - The Magic: It finds the maximum matching in a bipartite graph. It works by
   using BFS to find
13 * multiple shortest augmenting paths simultaneously, and then uses DFS to push
   flow along them.
14 *
15 * 2. HOW TO USE IT (THE BLACK BOX)
16 * - Initialization: Pass the size of the Left partition (N) and Right partition
   (M).
17 * (1-indexed nodes!).
18 * HopcroftKarp hk(N, M);
19 *
20 * - Add Edge:
21 *     hk.add_edge(u, v); // Worker `u` (1 to N) can do Job `v` (1 to M).
22 *
23 * - Query:
24 *     int max_pairs = hk.max_matching();

```



```

25 *
26 * - Check Pairs:
27 * // After running max_matching(), pair_u[u] gives the job assigned to
   worker u.
28 * // If pair_u[u] == 0, worker u is unmatched.
29 *
30 * - Complexity:
31 * Time:  $O(E * \sqrt{V})$ . Blazing fast for up to  $10^5$  nodes and edges.
32 * Space:  $O(V + E)$  memory.
33 *
34 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
35 * - Knig's Theorem (Min Vertex Cover):
36 * To find the minimum number of nodes needed to "cover" all edges:
37 * 1. Run Max Matching. Min Vertex Cover = Max Matching.
38 * 2. To find the ACTUAL nodes: Run a DFS from all UNMATCHED nodes in the Left
   partition
39 * using alternating paths. The Vertex Cover = (Unvisited nodes in Left) +
   Visited nodes in Right).
40 *
41 * - Maximum Independent Set:
42 * The maximum number of nodes where no two nodes share an edge.
43 * Formula: Max Independent Set = (Total Nodes) - (Min Vertex Cover).
44 */
45
46 struct HopcroftKarp
47 {
48     int n; // Size of the Left partition (U)
49     int m; // Size of the Right partition (V)
50
51     vector<vector<int>> adj;
52
53     // pair_u[i] = the node in V matched with node i in U
54     // pair_v[j] = the node in U matched with node j in V
55     vector<int> pair_u, pair_v;
56
57     // distance array used by BFS to find shortest augmenting paths
58     vector<int> dist;
59
60     const int INF = 1e9;
61
62     HopcroftKarp(int n, int m) : n(n), m(m)
63     {
64         adj.resize(n + 1);
65         pair_u.assign(n + 1, 0);
66         pair_v.assign(m + 1, 0);
67         dist.assign(n + 1, 0);
68     }
69
70     void add_edge(int u, int v)
71     {
72         adj[u].push_back(v); // u is in Left (1...N), v is in Right (1...M)
73     }
74
75     // BFS builds the level graph, finding the shortest paths to unmatched nodes
   in V
76     bool bfs()
77     {
78         queue<int> q;
79
80         for (int u = 1; u <= n; u++)
81         {
82             // If u is unmatched, it's a starting point for an augmenting path
83             if (pair_u[u] == 0)
84             {
85                 dist[u] = 0;
86                 q.push(u);
87             }
88             else
89             {
90                 dist[u] = INF; // Otherwise, mark as unvisited
91             }
92         }
93
94         // Node 0 acts as a dummy node for all unmatched nodes in V
95         dist[0] = INF;
96
97         while (!q.empty())
98         {
99             int u = q.front();
100             q.pop();
101
102             // If this node has a distance less than the dummy node, we can
   explore it
103             if (dist[u] < dist[0])
104             {
105                 for (int v : adj[u])
106                 {
107                     // If the node matched to v is unvisited, we can walk there
108                     if (dist[pair_v[v]] == INF)
109                     {
110                         dist[pair_v[v]] = dist[u] + 1;
111                         q.push(pair_v[v]);
112                     }
113                 }
114             }
115         }
116
117         // If dist[0] is no longer INF, we found at least one augmenting path!
118         return dist[0] != INF;
119     }
120
121     // DFS greedily pushes "flow" along the paths discovered by BFS
122     bool dfs(int u)
123     {

```

```

124     if (u != 0)
125     {
126         for (int v : adj[u])
127         {
128             // Ensure we only walk along the shortest path DAG built by BFS
129             if (dist[pair_v[v]] == dist[u] + 1)
130             {
131                 if (dfs(pair_v[v]))
132                 {
133                     // Augment the path! Flip the matching status.
134                     pair_v[v] = u;
135                     pair_u[u] = v;
136                     return true;
137                 }
138             }
139         }
140         // If no augmenting path could be found from u, mark it as dead end
141         dist[u] = INF;
142         return false;
143     }
144     // Base case: Reached the dummy node 0 (an unmatched node in V)
145     return true;
146 }
147
148 // Repeatedly runs BFS and DFS to build the maximum matching
149 int max_matching()
150 {
151     int matching = 0;
152
153     // While there is an augmenting path...
154     while (bfs())
155     {
156         // Try to augment from every unmatched node in U
157         for (int u = 1; u <= n; u++)
158         {
159             if (pair_u[u] == 0 && dfs(u))
160             {
161                 matching++;
162             }
163         }
164     }
165     return matching;
166 }
167 };
168
169 void solve()
170 {
171     int n, m, edges;
172     // Example: N workers, M jobs, and E compatible worker-job pairs
173     if (!(cin >> n >> m >> edges))
174         return;
175

```

```

176     HopcroftKarp hk(n, m);
177
178     for (int i = 0; i < edges; i++)
179     {
180         int u, v;
181         cin >> u >> v;
182         hk.add_edge(u, v);
183     }
184
185     cout << "Maximum Pairs: " << hk.max_matching() << "\n";
186
187     // Print the actual assignments
188     for (int u = 1; u <= n; u++)
189     {
190         if (hk.pair_u[u] != 0)
191         {
192             cout << "Worker " << u << " is assigned Job " << hk.pair_u[u] << "\n"
193                 << "\n";
194         }
195     }
196
197 int main()
198 {
199     ios_base::sync_with_stdio(false);
200     cin.tie(NULL);
201     // solve();
202     return 0;
203 }

```

### 3.10.2 Hungarian algorithm

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: HUNGARIAN ALGORITHM
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Optimal Assignment", "Bipartite Weighted Matching", "Min-Cost
9  *   Perfect Matching".
10 * - Classic Scenarios: You have N workers and M jobs. Every worker-job pair has
11 *   a specific
12 *   cost or time to complete. You want to assign exactly one job to each worker
13 *   such that
14 *   the total cost is minimized.
15 * - The Magic: While MCMF works, Hungarian runs in a strict, highly-optimized  $O(N * M^2)$  time,
16 *   making it the undisputed king for dense assignment matrices.
17 *
18 * 2. HOW TO USE IT (THE BLACK BOX)

```

```

16 * - Initialization:
17 *     // Create a 1-indexed cost matrix of size (N+1) x (M+1)
18 *     // Important: N must be <= M! (Rows <= Cols)
19 *     Hungarian solver(n, m, cost_matrix);
20 *
21 * - Query:
22 *     long long min_cost = solver.solve();
23 *
24 * - Extract Assignments:
25 *     // solver.ans[i] will contain the job (1 to M) assigned to worker i.
26 *
27 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
28 * - Maximum Profit Assignment?
29 *   If you want to MAXIMIZE the score, simply negate all values in your matrix
   cost = -cost)
30 *   before passing it in. Then negate the final answer returned by `solve()`.
31 * - Forbidden Assignments?
32 *   If worker `i` cannot do job `j`, set `cost[i][j] = 1e15` (Infinity).
33 */
34
35 struct Hungarian
36 {
37     int n, m;
38     vector<vector<long long>> a;
39     vector<long long> u, v, minv;
40     vector<int> p, way, ans;
41     const long long INF = 1e15;
42
43     Hungarian(int n, int m, const vector<vector<long long>> &cost_matrix)
44     {
45         this->n = n;
46         this->m = m;
47         // 1-based indexing logic
48         a = cost_matrix;
49         u.assign(n + 1, 0);
50         v.assign(m + 1, 0);
51         p.assign(m + 1, 0);
52         way.assign(m + 1, 0);
53         ans.assign(n + 1, 0);
54     }
55
56     /* STREAMING_CHUNK: Core O(N * M^2) solver using augmenting paths */
57     long long solve()
58     {
59         // Iterate over every worker (row) to find them a job
60         for (int i = 1; i <= n; ++i)
61         {
62             p[0] = i;
63             int j0 = 0; // Dummy column 0
64
65             minv.assign(m + 1, INF);
66             vector<bool> used(m + 1, false);

```

```

67
68         do
69         {
70             used[j0] = true;
71             int i0 = p[j0], j1 = 0;
72             long long delta = INF;
73
74             // Update minv array and find the next column to explore
75             for (int j = 1; j <= m; ++j)
76             {
77                 if (!used[j])
78                 {
79                     long long cur = a[i0][j] - u[i0] - v[j];
80                     if (cur < minv[j])
81                     {
82                         minv[j] = cur;
83                         way[j] = j0;
84                     }
85                     if (minv[j] < delta)
86                     {
87                         delta = minv[j];
88                         j1 = j;
89                     }
90                 }
91             }
92
93             // Update potentials (dual variables)
94             for (int j = 0; j <= m; ++j)
95             {
96                 if (used[j])
97                 {
98                     u[p[j]] += delta;
99                     v[j] -= delta;
100                 }
101                 else
102                 {
103                     minv[j] -= delta;
104                 }
105             }
106             j0 = j1;
107         } while (p[j0] != 0); // Stop when we reach an unmatched column
108
109         do
110         {
111             int j1 = way[j0];
112             p[j0] = p[j1];
113             j0 = j1;
114         } while (j0 != 0);
115     }
116
117     // The optimal job assigned to worker `i`
118     for (int j = 1; j <= m; ++j)

```

```

119     {
120         if (p[j] != 0)
121         {
122             ans[p[j]] = j;
123         }
124     }
125
126     // The final minimum cost is stored negated in the dummy column potential
127     v[0]
128     return -v[0];
129 };
130
131 void example_usage()
132 {
133     int n, m;
134     // Example: N workers, M jobs (ensure N <= M)
135     if (!(cin >> n >> m))
136         return;
137
138     // Build the 1-indexed cost matrix
139     vector<vector<long long>> cost(n + 1, vector<long long>(m + 1, 0));
140     for (int i = 1; i <= n; i++)
141     {
142         for (int j = 1; j <= m; j++)
143         {
144             cin >> cost[i][j];
145         }
146     }
147
148     Hungarian solver(n, m, cost);
149     cout << "Minimum Total Cost: " << solver.solve() << "\n";
150
151     for (int i = 1; i <= n; i++)
152     {
153         cout << "Worker " << i << " assigned to Job " << solver.ans[i] << "\n";
154     }
155 }

```

### 3.10.3 Stable Marriage Problem

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: STABLE MARRIAGE (GALE-SHAPLEY)
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Stable Matching", "Preference Lists", "Gale-Shapley".
9  * - Classic Scenarios: Matching N men to N women, or N students to N colleges,
10  * based on ranked preference lists. A matching is "stable" if there is no

```

```

11  * pair (Man A, Woman B) who both prefer each other over their current partners
12  * - The Magic: The Gale-Shapley algorithm guarantees a stable matching in  $O(N^2)$ 
13  * time.
14  * 2. HOW TO USE IT (THE BLACK BOX)
15  * - Initialization:
16  *     StableMarriage sm(N);
17  *     sm.set_preferences(men_prefs, women_prefs);
18  * - Query:
19  *     sm.solve();
20  * - Extract Assignments:
21  *     // match_proposer[i] contains the receiver assigned to proposer i.
22  *     // match_receiver[j] contains the proposer assigned to receiver j.
23  *
24  * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
25  * - The Proposer Advantage:
26  *     Gale-Shapley strongly favors the group doing the proposing!
27  *     If you want the optimal outcome for Group X, pass Group X as the "Proposers"
28  *     (the first argument in set_preferences).
29  * - Unequal sizes: If Proposers > Receivers, add "dummy" receivers that rank
30  *     everyone equally at the bottom.
31  */
32
33 struct StableMarriage
34 {
35     int n;
36
37     // pref_proposer[i][k] = the k-th preferred receiver by proposer i
38     vector<vector<int>> pref_proposer;
39
40     // To make rejection  $O(1)$ , we invert the receiver's preferences.
41     // rank_receiver[j][i] = the rank position (1 to N) of proposer i in receiver
42     // j's list.
43     // Lower rank number means highly preferred!
44     vector<vector<int>> rank_receiver;
45
46     // The current matches. 0 means unassigned.
47     vector<int> match_proposer;
48     vector<int> match_receiver;
49
50     StableMarriage(int n) : n(n)
51     {
52         pref_proposer.assign(n + 1, vector<int>(n + 1, 0));
53         rank_receiver.assign(n + 1, vector<int>(n + 1, 0));
54         match_proposer.assign(n + 1, 0);
55         match_receiver.assign(n + 1, 0);
56     }
57
58     // Both 2D vectors should be 1-indexed: [1..n][1..n]
59     void set_preferences(const vector<vector<int>> &proposers, const vector<
60         vector<int>> &receivers)

```

```

59 {
60     pref_proposer = proposers;
61
62     // Invert the receiver's preference list for O(1) comparison later
63     for (int j = 1; j <= n; ++j)
64     {
65         for (int rank = 1; rank <= n; ++rank)
66         {
67             int proposer_id = receivers[j][rank];
68             rank_receiver[j][proposer_id] = rank;
69         }
70     }
71 }
72
73 void solve()
74 {
75     queue<int> free_proposers;
76     // next_proposal[i] tracks the index of the next receiver that proposer
77     // will ask
78     vector<int> next_proposal(n + 1, 1);
79
80     for (int i = 1; i <= n; ++i)
81     {
82         free_proposers.push(i);
83     }
84
85     // Loop until every proposer is matched
86     while (!free_proposers.empty())
87     {
88         int p = free_proposers.front();
89
90         // The receiver this proposer is going to ask next
91         int r = pref_proposer[p][next_proposal[p]];
92         next_proposal[p]++; // Increment pointer so he doesn't ask her again
93         // if rejected
94
95         if (match_receiver[r] == 0)
96         {
97             // Scenario 1: The receiver is entirely free. She happily accepts
98             !
99             match_receiver[r] = p;
100             match_proposer[p] = r;
101             free_proposers.pop(); // The proposer is no longer free
102         }
103         else
104         {
105             // Scenario 2: The receiver is already engaged to someone else
106             // current_p.
107             int current_p = match_receiver[r];
108
109             // Does she prefer the new proposer (p) over her current partner
110             // (current_p)?
111             if (rank_receiver[r][p] < rank_receiver[r][current_p])
112             {
113                 // She dumps her current partner!
114                 match_proposer[current_p] = 0;
115                 free_proposers.push(current_p); // The dumped partner is now
116                 // free and goes back in line
117                 // She accepts the new proposer
118                 match_receiver[r] = p;
119                 match_proposer[p] = r;
120                 free_proposers.pop();
121             }
122             // If she prefers her current partner, she rejects 'p', and 'p'
123             // stays in the queue.
124         }
125     }
126 }
127
128 void solve_example()
129 {
130     int n;
131     // Read N pairs
132     if (!(cin >> n))
133         return;
134
135     // 1-indexed preference lists
136     vector<vector<int>> men_prefs(n + 1, vector<int>(n + 1));
137     vector<vector<int>> women_prefs(n + 1, vector<int>(n + 1));
138
139     // Read Men's preferences (Proposers)
140     for (int i = 1; i <= n; ++i)
141     {
142         for (int j = 1; j <= n; ++j)
143         {
144             cin >> men_prefs[i][j];
145         }
146     }
147
148     // Read Women's preferences (Receivers)
149     for (int i = 1; i <= n; ++i)
150     {
151         for (int j = 1; j <= n; ++j)
152         {
153             cin >> women_prefs[i][j];
154         }
155     }
156
157     StableMarriage sm(n);
158     sm.set_preferences(men_prefs, women_prefs);
159     sm.solve();

```

```

156     cout << "Stable_Matches_ (Man_ -> Woman):\n";
157     for (int i = 1; i <= n; ++i)
158     {
159         cout << "Man_" << i << "_matches_with_Woman_" << sm.match_proposer[i] << 35 }
160         "\n";
161     }
162 }
163 int main()
164 {
165     ios_base::sync_with_stdio(false);
166     cin.tie(NULL);
167     // solve_example();
168     return 0;
169 }

```

## 3.11 Misc

### 3.11.1 Check Cycle

```

1 // Check Cycle in Undirected
2 Graph graph ;
3 vector<bool> vis;
4 bool isCycle = 0;
5 void dfs(int u,int prv )
6 {
7     vis[u] = 1;
8     for (auto& n : graph[u])
9     {
10         if (n == prv) continue;
11         if (!vis[n])
12             dfs(n, u);
13         else
14             isCycle = 1;
15     }
16 }
17
18 // Check Cycle in directed
19 Graph graph ;
20 vector<int> vis;
21 bool isCycle = 0;
22 void dfs(int u )
23 {
24     vis[u] = 2; // mark node that exist in recursion stack
25     for (auto& n : graph[u])
26     {
27         if (!vis[n]) // if node not vis & not exist inf recursion stack
28             dfs(n);
29         else if (vis[n] == 2) // node is visited and exist in recursion stack
30             isCycle = 1;
31     }

```

```

32
33     vis[u] = 1; // mark node that visited and not exist in recursion stack
34
35 }

```

### 3.11.2 Degree Sequences

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DEGREE SEQUENCES
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Given an array of degrees", "Can we form a simple graph", "
9  *   Construct a graph".
10 * - Classic Scenarios: You are given N integers representing the number of
11 *   friends each person has.
12 * You need to verify if this network is mathematically possible, and if so,
13 *   pair them up.
14 *
15 * 2. HOW TO USE IT (THE BLACK BOX)
16 * - Validation Only (Fast):
17 *   bool possible = DegreeSequence::erdos_gallai(degrees);
18 *   // Complexity: O(N log N). Use this when N = 10^5 and you only need YES/
19 *   NO.
20 *
21 * - Construction (Slower, but gives edges):
22 *   vector<pair<int, int>> edges = DegreeSequence::havel_hakimi(degrees);
23 *   // Complexity: O(N^2 log N). Use this when you must PRINT the graph.
24 *   // Returns an empty vector if construction is impossible.
25 *
26 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
27 * - 0-based vs 1-based indexing for Construction:
28 *   The `havel_hakimi` function returns edges using 1-based indexing by default.
29 *   If you need 0-based indexing, change `nodes.push_back({d[i], i + 1});` to `i
30 *   `
31 */
32
33 struct DegreeSequence
34 {
35
36     static bool erdos_gallai(vector<long long> d)
37     {
38         int n = d.size();
39         long long sum = 0;
40
41         // A valid graph MUST have an even sum of degrees (Handshaking Lemma)
42         for (long long x : d)
43             sum += x;
44         if (sum % 2 != 0)

```

```

40     return false;
41
42     // Sort degrees in descending order: d_1 >= d_2 >= ... >= d_n
43     sort(d.begin(), d.end(), greater<long long>());
44
45     // Build 1-indexed prefix sums for O(1) range queries
46     vector<long long> pref(n + 1, 0);
47     for (int i = 0; i < n; i++)
48     {
49         pref[i + 1] = pref[i] + d[i];
50     }
51
52     // Evaluate the Erds-Gallai inequality for every k from 1 to n:
53     // Sum(i=1 to k) d_i <= k(k-1) + Sum(i=k+1 to n) min(d_i, k)
54     for (int k = 1; k <= n; k++)
55     {
56         long long left_sum = pref[k];
57
58         // Binary search to find the split point in the suffix where d_i
59         // drops below k.
60         // This avoids an O(N) loop and keeps the overall complexity at O(N log N).
61         int p = max(k, (int)(lower_bound(d.begin(), d.end(), k, greater<long long>()) - d.begin()));
62
63         long long right_sum = 1LL * k * (k - 1);
64
65         // Elements from k to p-1 are >= k, so min(d_i, k) is always exactly k
66         right_sum += 1LL * k * (p - k);
67
68         // Elements from p to n-1 are < k, so min(d_i, k) is just d_i
69         right_sum += (pref[n] - pref[p]);
70
71         if (left_sum > right_sum)
72         {
73             return false; // Mathematical contradiction found!
74         }
75     }
76     return true;
77 }
78
79 static vector<pair<int, int>> havel_hakimi(const vector<int> &d)
80 {
81     int n = d.size();
82     long long sum = 0;
83     for (int x : d)
84         sum += x;
85     if (sum % 2 != 0)
86         return {}; // Fails Handshaking Lemma
87

```

```

88     // Store {degree, original_node_id}
89     vector<pair<int, int>> nodes;
90     for (int i = 0; i < n; i++)
91     {
92         if (d[i] > 0)
93         {
94             nodes.push_back({d[i], i + 1}); // 1-based indexing for nodes
95         }
96     }
97
98     vector<pair<int, int>> edges;
99
100    while (!nodes.empty())
101    {
102        // Sort descending by degree
103        sort(nodes.begin(), nodes.end(), greater<pair<int, int>>());
104
105        // The node with the highest degree demands to be connected
106        int current_degree = nodes[0].first;
107        int u = nodes[0].second;
108
109        // Remove it from the pool of available nodes
110        nodes.erase(nodes.begin());
111
112        // If it demands more connections than there are nodes left, it's
113        // impossible
114        if (current_degree > nodes.size())
115            return {};
116
117        // Connect 'u' greedily to the next 'current_degree' highest nodes
118        for (int i = 0; i < current_degree; i++)
119        {
120            edges.push_back({u, nodes[i].second});
121            nodes[i].first--; // They used up one of their connections
122
123            // If a degree drops below 0, it means we forced a connection
124            // that broke the graph
125            if (nodes[i].first < 0)
126                return {};
127        }
128
129        // Purge fully satisfied nodes (degree == 0) to speed up sorting
130        vector<pair<int, int>> remaining_nodes;
131        for (auto &node : nodes)
132        {
133            if (node.first > 0)
134            {
135                remaining_nodes.push_back(node);
136            }
137        }
138        nodes = remaining_nodes;
139    }

```

```

138
139     return edges;
140 }
141 };
142
143 void solve()
144 {
145     int n;
146     // Example: Read N, then N degrees
147     if (!(cin >> n))
148         return;
149
150     vector<int> d_int(n);
151     vector<long long> d_ll(n);
152     for (int i = 0; i < n; i++)
153     {
154         cin >> d_int[i];
155         d_ll[i] = d_int[i];
156     }
157
158     // 1.  $O(N \log N)$  Check (Best for large  $N$ )
159     if (DegreeSequence::erdos_gallai(d_ll))
160     {
161         cout << "YES\n";
162
163         // 2.  $O(N^2 \log N)$  Construction (Only run if you actually need the edges
164         // In CF/ICPC, if  $N > 10^4$ , they usually don't ask you to print the edges
165         vector<pair<int, int>> edges = DegreeSequence::havel_hakimi(d_int);
166         for (auto &edge : edges)
167         {
168             cout << edge.first << "□" << edge.second << "\n";
169         }
170     }
171     else
172     {
173         cout << "NO\n";
174     }
175 }
176
177 int main()
178 {
179     ios_base::sync_with_stdio(false);
180     cin.tie(NULL);
181     // solve();
182     return 0;
183 }

```

### 3.11.3 Tree hashing

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: TREE ISOMORPHISM (HASHING)
6  *
7  * Solves: Checks if two unrooted trees are structurally identical.
8  * Time Complexity:  $O(N)$ 
9  * Space Complexity:  $O(N)$ 
10 */
11
12 struct TreeIsomorphism
13 {
14     // Generate a random seed once per program execution to prevent anti-hash
15     // tests
16     static uint64_t get_seed()
17     {
18         mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
19         return rng();
20     }
21
22     // A highly robust non-linear permutation function (SplitMix64)
23     static uint64_t shift_hash(uint64_t x)
24     {
25         static const uint64_t SEED = get_seed();
26         x += SEED;
27         x ^= (x >> 33);
28         x *= 0xff51afd7ed558ccdULL;
29         x ^= (x >> 33);
30         x *= 0xc4ceb9fe1a85ec53ULL;
31         x ^= (x >> 33);
32         return x;
33     }
34
35     static vector<int> get_centers(int n, const vector<vector<int>> &adj)
36     {
37         vector<int> degree(n + 1);
38         queue<int> q;
39
40         for (int i = 1; i <= n; i++)
41         {
42             degree[i] = adj[i].size();
43             // Leaves have degree 1 (or 0 if n=1)
44             if (degree[i] <= 1)
45             {
46                 q.push(i);
47             }
48         }
49
50         int remaining_nodes = n;
51
52         // Peel away the leaves layer by layer like an onion

```



```

52     while (remaining_nodes > 2)
53     {
54         int layer_size = q.size();
55         remaining_nodes -= layer_size;
56
57         for (int i = 0; i < layer_size; i++)
58         {
59             int u = q.front();
60             q.pop();
61
62             for (int v : adj[u])
63             {
64                 if (--degree[v] == 1)
65                 {
66                     q.push(v);
67                 }
68             }
69         }
70     }
71
72     // The remaining 1 or 2 nodes are the absolute centers
73     vector<int> centers;
74     while (!q.empty())
75     {
76         centers.push_back(q.front());
77         q.pop();
78     }
79     return centers;
80 }
81
82 static uint64_t compute_hash(int u, int p, const vector<vector<int>> &adj)
83 {
84     uint64_t current_hash = 1; // Base value for a node
85
86     for (int v : adj[u])
87     {
88         if (v != p)
89         {
90             // To combine children, we hash the child's subtree, run it
91             // through
92             // our non-linear shift function, and sum them up.
93             // Using SUM guarantees that the order of children does not
94             // matter!
95             current_hash += shift_hash(compute_hash(v, u, adj));
96         }
97     }
98     return current_hash;
99 }
100 static bool are_isomorphic(int n, const vector<vector<int>> &adj1, const
    vector<vector<int>> &adj2)
    {

```

```

101     if (n == 1)
102         return true; // Trivial case
103
104     // 1. Find centers of both trees
105     vector<int> centers1 = get_centers(n, adj1);
106     vector<int> centers2 = get_centers(n, adj2);
107
108     // If the number of centers doesn't match, they physically cannot be
109     // isomorphic
110     if (centers1.size() != centers2.size())
111     {
112         return false;
113     }
114
115     // 2. Hash Tree 1 from its first center
116     uint64_t hash1 = compute_hash(centers1[0], 0, adj1);
117
118     // 3. Check Tree 2 against Tree 1's hash
119     for (int c2 : centers2)
120     {
121         uint64_t hash2 = compute_hash(c2, 0, adj2);
122         if (hash1 == hash2)
123         {
124             return true; // Match found!
125         }
126     }
127     return false;
128 }
129 };
130
131 void solve()
132 {
133     int n;
134     // Example: Read N, then N-1 edges for Tree 1, then N-1 edges for Tree 2
135     if (!(cin >> n))
136         return;
137
138     vector<vector<int>> tree1(n + 1);
139     for (int i = 0; i < n - 1; i++)
140     {
141         int u, v;
142         cin >> u >> v;
143         tree1[u].push_back(v);
144         tree1[v].push_back(u);
145     }
146
147     vector<vector<int>> tree2(n + 1);
148     for (int i = 0; i < n - 1; i++)
149     {
150         int u, v;
151         cin >> u >> v;

```

```

152     tree2[u].push_back(v);
153     tree2[v].push_back(u);
154 }
155
156 if (TreeIsomorphism::are_isomorphic(n, tree1, tree2))
157 {
158     cout << "Isomorphic_They_are_the_exact_same_tree\n";
159 }
160 else
161 {
162     cout << "Not_Isomorphic\n";
163 }
164 }
165
166 int main()
167 {
168     ios_base::sync_with_stdio(false);
169     cin.tie(NULL);
170     // solve();
171     return 0;
172 }

```

## 3.12 Shortest paths

### 3.12.1 BFS01

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 void zero_one_bfs(int start, int n, vector<vector<pair<int, int>>> &adj, vector<
    int> &dist)
5 {
6     dist.assign(n, 1e9);
7     deque<int> dq;
8     dist[start] = 0;
9     dq.push_front(start);
10
11     while (!dq.empty())
12     {
13         int u = dq.front();
14         dq.pop_front();
15
16         for (auto &edge : adj[u])
17         {
18             int v = edge.first;
19             int w = edge.second;
20             if (dist[v] > dist[u] + w)
21             {
22                 dist[v] = dist[u] + w;
23                 if (w == 0)
24                     dq.push_front(v);

```

```

25         else
26             dq.push_back(v);
27     }
28 }
29 }
30 }

```

### 3.12.2 Bellman ford

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 const long long INF = 1e18;
5
6 struct Edge
7 {
8     int u, v, w;
9 };
10
11 bool bellman_ford(int n, int start, vector<Edge> &edges, vector<long long> &dist)
12 {
13     dist.assign(n, INF);
14     dist[start] = 0;
15
16     for (int i = 0; i < n - 1; ++i)
17     {
18         for (auto &e : edges)
19         {
20             if (dist[e.u] != INF && dist[e.v] > dist[e.u] + e.w)
21                 dist[e.v] = dist[e.u] + e.w;
22         }
23     }
24
25     for (auto &e : edges)
26     {
27         if (dist[e.u] != INF && dist[e.v] > dist[e.u] + e.w)
28             return true;
29     }
30     return false;
31 }

```

### 3.12.3 Dijkstra

```

1 ll n;
2 GRAPH graph;
3
4 vector<ll> dijkstra(ll x)
5 {
6     vector<ll> dist(n + 1, 00);

```

```

7   dist[x] = 0;
8   priority_queue<pair<ll, ll>, vector<pair<ll, ll>>, greater<pair<ll, ll>>> pq;
9   pq.push({0, x});
10  while (!pq.empty())
11  {
12      auto [l, node] = pq.top();
13      pq.pop();
14      if (l > dist[node])
15          continue;
16      for (auto &[to, w] : graph[node])
17      {
18          if (dist[to] > dist[node] + w)
19          {
20              dist[to] = dist[node] + w;
21              pq.push({dist[to], to});
22          }
23      }
24  }
25  return dist;
26 }
27
28 // Path
29 ll n;
30 GRAPH graph;
31 vector<int> parent;
32 deque<int> path;
33 vector<ll> dijkstra(ll x)
34 {
35     parent.assign(n + 1, -1);
36     vector<ll> dist(n + 1, 00);
37     dist[x] = 0;
38     priority_queue<pair<ll, ll>, vector<pair<ll, ll>>, greater<pair<ll, ll>>> pq;
39     pq.push({0, x});
40     while (!pq.empty())
41     {
42         auto [l, node] = pq.top();
43         pq.pop();
44         if (l > dist[node])
45             continue;
46         for (auto &[to, w] : graph[node])
47         {
48             if (dist[to] > dist[node] + w)
49             {
50                 dist[to] = dist[node] + w;
51                 parent[to] = node;
52                 pq.push({dist[to], to});
53             }
54         }
55     }
56 }
57 return dist;
58
59 }
60
61 void build(int node)
62 {
63     path.push_front(node);
64     if (parent[node] == -1)
65         return;
66     build(parent[node]);
67 }
68
69 // Destination
70 ll n;
71 GRAPH graph;
72
73 vector<ll> dijkstra(ll x, ll d)
74 {
75     vector<ll> dist(n + 1, 00);
76     dist[x] = 0;
77     priority_queue<pair<ll, ll>, vector<pair<ll, ll>>, greater<pair<ll, ll>>> pq;
78     pq.push({0, x});
79     while (!pq.empty())
80     {
81         auto [l, node] = pq.top();
82         pq.pop();
83         if (node == d)
84             break;
85         if (l > dist[node])
86             continue;
87         for (auto &[to, w] : graph[node])
88         {
89             if (dist[to] > dist[node] + w)
90             {
91                 dist[to] = dist[node] + w;
92                 pq.push({dist[to], to});
93             }
94         }
95     }
96     return dist;
97 }
98
99 // Dijkstra indexed PQ
100 #include <bits/stdc++.h>
101
102 #include <ext/pb_ds/priority_queue.hpp>
103 using namespace std;
104
105 using namespace __gnu_pbds;
106 typedef long long ll;
107 const ll 00 = 1e18;
108 typedef vector<vector<pair<ll, ll>>> GRAPH;
109 typedef __gnu_pbds::priority_queue<pair<ll, ll>, greater<pair<ll, ll>>,

```

```

    pairing_heap_tag> indexedPQ;
111
112 ll n;
113 GRAPH graph;
114
115 vector<ll> dijkstra(ll x)
116 {
117     vector<ll> dist(n + 1, 00);
118     indexedPQ pq;
119     vector<indexedPQ::point_iterator> key(n + 1, nullptr);
120     dist[x] = 0;
121     key[x] = pq.push({0, x});
122     while (!pq.empty())
123     {
124         auto [l, node] = pq.top();
125         pq.pop();
126
127         for (auto &[to, w] : graph[node])
128         {
129             if (dist[to] > dist[node] + w)
130             {
131                 dist[to] = dist[node] + w;
132                 if (key[to] == nullptr)
133                     key[to] = pq.push({dist[to], to});
134                 else
135                     pq.modify(key[to], {dist[to], to});
136             }
137         }
138     }
139     return dist;
140 }
141
142 // Dijkstra remove
143 ll n;
144 GRAPH graph;
145
146 vector<ll> dijkstra(ll x)
147 {
148     vector<ll> dist(n + 1, 00);
149     dist[x] = 0;
150     set<pair<ll, ll>> s;
151     s.insert({0, x});
152     while (!s.empty())
153     {
154         auto [l, node] = *s.begin();
155         s.erase(s.begin());
156         if (l > dist[node])
157             continue;
158         for (auto &[to, w] : graph[node])
159         {
160             if (dist[to] > dist[node] + w)
161                 {

```

```

162             dist[to] = dist[node] + w;
163             s.insert({dist[to], to});
164         }
165     }
166 }
167 return dist;
168 }

```

### 3.12.4 Floyd

```

1 int n;
2 GRAPH graph;
3
4 vector<vector<int>> floyd()
5 {
6     vector<vector<int>> dist = graph;
7     for (int k = 0; k < n; k++)
8     {
9         for (int i = 0; i < n; i++)
10        {
11            for (int j = 0; j < n; j++)
12            {
13                if (dist[i][k] < 00 && dist[k][j] < 00)
14                    dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
15            }
16        }
17    }
18    return dist;
19 }
20
21 // real
22 int n;
23 GRAPH graph;
24
25 vector<vector<double>> floyd()
26 {
27     vector<vector<double>> dist = graph;
28     for (int k = 0; k < n; k++)
29     {
30         for (int i = 0; i < n; i++)
31         {
32             for (int j = 0; j < n; j++)
33             {
34                 if (dist[i][k] < 00 && dist[k][j] < 00)
35                 {
36                     if (dist[i][k] + dist[k][j] < dist[i][j] - DBL_EPSILON)
37                         dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
38                 }
39             }
40         }
41    }

```

```

42     return dist;
43 }
44
45 // path
46 int n;
47 GRAPH graph;
48 vector<int> path;
49 vector<vector<int>> phase;
50
51 vector<vector<int>> floyd()
52 {
53     phase.assign(n + 1, vector<int>(n + 1, -1));
54     vector<vector<int>> dist = graph;
55     for (int k = 0; k < n; k++)
56     {
57         for (int i = 0; i < n; i++)
58         {
59             for (int j = 0; j < n; j++)
60             {
61                 if (dist[i][k] < 00 && dist[k][j] < 00)
62                 {
63                     if (dist[i][j] > dist[i][k] + dist[k][j])
64                     {
65                         dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
66                         phase[i][j] = k;
67                     }
68                 }
69             }
70         }
71     }
72     return dist;
73 }
74
75 // push the start node manually
76 void build(int i, int j)
77 {
78     if (phase[i][j] == -1)
79     {
80         path.push_back(j);
81         return;
82     }
83     int k = phase[i][j];
84     build(i, k);
85     build(k, j);
86 }

```

### 3.12.5 Floyd Warshall(Hassan)

```

1 void hassan() {
2     int n,m;cin >>n>>m;
3     vector<vector<int>> dist(n+1, vector<int>(n+1, oo)); // oo >= n * maxCost

```

```

4
5     for (int i=0;i<m;i++) {
6         int u,v,w;cin>>u>>v>>w;
7         dist[u][v] = min(dist[u][v], w);
8     }
9     for (int i=1;i<=n;i++) {
10        dist[i][i] = 0;
11    }
12
13    for (int k = 1; k <= n; k++) {
14        for (int from = 1; from <= n; from++) {
15            for (int to = 1; to <= n; to++) {
16                if (dist[from][k] < oo && dist[k][to] < oo) {
17                    dist[from][to] = min(dist[from][to], dist[from][k] + dist[k][
18                        to]);
19                }
20            }
21        }
22        for (int i=1;i<=n;i++) {
23            for (int j=1;j<=n;j++) {
24                cout << "from" << i << "to" << j << "Dist" << dist[i][j] << "\n";
25            }
26        }
27    }
28
29 void reachability(int n, int m) {
30     vector<vector<bool>> canReach(n+1, vector<bool>(n+1, false));
31     // ... input u, v ... canReach[u][v] = true;
32     for (int i=1; i<=n; i++) canReach[i][i] = true;
33
34     for (int k=1; k<=n; k++) {
35         for (int i=1; i<=n; i++) {
36             for (int j=1; j<=n; j++) {
37                 canReach[i][j] = canReach[i][j] || (canReach[i][k] && canReach[k
38                     ][j]);
39             }
40         }
41     }
42
43     /// Find Minimax Or Maxmini Path //////////
44 void hassan() {
45     int n,m;cin >>n>>m;
46     vector<vector<int>> dist(n+1, vector<int>(n+1, oo)); // oo >= n * maxCost
47     for (int i=1;i<=n;i++) {
48         dist[i][i] = 0;
49     }
50     for (int i=0;i<m;i++) {
51         int u,v,w;cin>>u>>v>>w;
52         dist[u][v] = min(dist[u][v], w);
53     }

```

```

54
55     for (int k = 1; k <= n; k++) {
56         for (int from = 1; from <= n; from++) {
57             for (int to = 1; to <= n; to++) {
58                 if (dist[from][k] < oo && dist[k][to] < oo) {
59                     dist[from][to] = min(dist[from][to], max(dist[from][k], dist[k][to]));
60                     //dist[from][to] = max(dist[from][to], min(dist[from][k], dist[k][to]));
61                 }
62             }
63         }
64         for (int i=1;i<=n;i++) {
65             for (int j=1;j<=n;j++) {
66                 cout << "from_" << i << "_to_" << j << "_Dist_" <<dist[i][j] << "\n";
67             }
68         }
69     }
70 }
71
72 ////// Counting Shortest Paths //////
73 void hassan_counting() {
74     int n, m; cin >> n >> m;
75     long long oo = 1e15;
76
77     vector<vector<long long>> dist(n + 1, vector<long long>(n + 1, oo));
78     vector<vector<long long>> pathsCount(n + 1, vector<long long>(n + 1, 0));
79
80     for (int i = 1; i <= n; i++) {
81         dist[i][i] = 0;
82         pathsCount[i][i] = 1;
83     }
84
85     for (int i = 0; i < m; i++) {
86         int u, v, w; cin >> u >> v >> w;
87         if (w < dist[u][v]) {
88             dist[u][v] = w;
89             pathsCount[u][v] = 1;
90         } else if (w == dist[u][v]) {
91             pathsCount[u][v]++;
92         }
93     }
94
95     for (int k = 1; k <= n; k++) {
96         for (int i = 1; i <= n; i++) {
97             for (int j = 1; j <= n; j++) {
98                 if (dist[i][k] + dist[k][j] < dist[i][j]) {
99                     dist[i][j] = dist[i][k] + dist[k][j];
100                     pathsCount[i][j] = pathsCount[i][k] * pathsCount[k][j];
101                 }
102                 else if (dist[i][k] + dist[k][j] == dist[i][j] && i != k && k != j) {
103                     pathsCount[i][j] += (pathsCount[i][k] * pathsCount[k][j]);
104                 }
105             }
106         }
107     }
108
109     for (int i = 1; i <= n; i++) {
110         for (int j = 1; j <= n; j++) {
111             if (dist[i][j] == oo) continue;
112             cout << "From_" << i << "_to_" << j << " :_Min_Dist=_ " << dist[i][j]
113                 << "_|_Ways=_ " << pathsCount[i][j] << "\n";
114         }
115     }
116 }
117
118 //////// Diameter and Radius ////////
119 int diameter = 0;
120 int radius = oo;
121
122 for (int i=1; i<=n; i++) {
123     int max_dist_for_i = 0;
124     for (int j=1; j<=n; j++) {
125         if (dist[i][j] != oo)
126             max_dist_for_i = max(max_dist_for_i, dist[i][j]);
127     }
128     // Diameter: " "
129     diameter = max(diameter, max_dist_for_i);
130     // Radius: " "
131     radius = min(radius, max_dist_for_i);
132 }
133 }

```

### 3.12.6 MaxMin Cost Using Dijkstra

```

1 /*
2  * Problem Description:
3  * The goal is to find a path from the source node (Node 1) to all other nodes in
   the graph
4  * such that the minimum edge weight along this path is maximized (Maximize the
   minimum weight).
5  */
6 struct node {
7     int u,w;
8
9     bool operator<(const node &b) const {
10         return w > b.w;
11     }
12 };
13
14 void hassan() {

```

```

15     int n, m;
16     cin >> n >> m;
17     vector<vector<pair<int, int> > > adj(n + 1);
18     for (int i = 1; i <= m; i++) {
19         int u, v, w;
20         cin >> u >> v >> w;
21         adj[u].push_back({v, w});
22         adj[v].push_back({u, w});
23     }
24     vector<int> maxMin(n + 1, -1);
25     priority_queue<node> pq;
26     pq.push({1, oo});
27     maxMin[1] = oo;
28     while (!pq.empty()) {
29         auto [u,w] = pq.top();
30         pq.pop();
31
32         if (w < maxMin[u]) continue;
33
34         for (auto [v,c]: adj[u]) {
35             if (min(maxMin[u], c) > maxMin[v]) {
36                 maxMin[v] = min(maxMin[u], c);
37                 pq.push({v, maxMin[v]});
38             }
39         }
40     }
41     for (int i = 1; i <= n; i++) {
42         cout << maxMin[i] << " ";
43     }
44 }

```

### 3.12.7 MinMax Cost Using Dijkstra

```

1  /*
2   * Problem Description:
3   * The goal is to find a path from the source node (Node 1) to all other nodes in
4     the graph
5   * such that the Maximum edge weight along this path is Minimized (Minimize the
6     maximum weight).
7   */
8
9  struct node {
10     int u, w;
11
12     bool operator<(const node &b) const {
13         return w > b.w;
14     }
15 };
16
17 void hassan() {
18     int n, m;
19     cin >> n >> m;

```

```

17     vector<vector<pair<int, int> > > adj(n + 1);
18     for (int i = 1; i <= m; i++) {
19         int u, v, w;
20         cin >> u >> v >> w;
21         adj[u].push_back({v, w});
22         adj[v].push_back({u, w});
23     }
24     priority_queue<node> pq;
25     pq.push({1, 0});
26     vector<int> minMax(n + 1, oo);
27     minMax[1] = 0;
28     while (!pq.empty()) {
29         auto [u,w] = pq.top();
30         pq.pop();
31
32         if (w > minMax[u]) continue;
33
34         for (auto [v,c]: adj[u]) {
35             if (max(minMax[u], c) < minMax[v]) {
36                 minMax[v] = max(minMax[u], c);
37                 pq.push({v, minMax[v]});
38             }
39         }
40     }
41     for (int i = 1; i <= n; i++) {
42         cout << minMax[i] << " ";
43     }
44 }

```

## 3.13 Trees

### 3.13.1 Rerooting

```

1  int n;
2  vector<int> sz, dist, ans;
3  GRAPH graph;
4
5  void dfs(int u, int p)
6  {
7      sz[u] = 1;
8      for (auto &v : graph[u])
9      {
10         if (v == p)
11             continue;
12         dfs(v, u);
13         sz[u] += sz[v];
14         dist[u] += dist[v] + sz[v];
15     }
16 }
17
18 void dfs2(int u, int p)

```

```

19 {
20     for (auto &v : graph[u])
21     {
22         if (v == p)
23             continue;
24         ans[v] = ans[u] - sz[v] + (n - sz[v]);
25         dfs2(v, u);
26     }
27 }
28
29 void solve()
30 {
31     cin >> n;
32     sz.assign(n + 1, {}), dist.assign(n + 1, {}), graph.assign(n + 1, {}), ans.
        assign(n + 1, {});
33     for (int i = 1; i < n; i++)
34     {
35         int a, b;
36         cin >> a >> b;
37         graph[a].emplace_back(b), graph[b].emplace_back(a);
38     }
39     dfs(1, 0);
40     ans[1] = dist[1];
41     dfs2(1, 0);
42     for (int i = 1; i <= n; i++)
43         cout << ans[i] << " ";
44 }

```

### 3.13.2 Virtual tree

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 /**
5  * THE ULTIMATE CP TEMPLATE BLUEPRINT: VIRTUAL TREE (AUXILIARY TREE)
6  *
7  * 1. WHAT PROBLEM DOES THIS SOLVE?
8  * - Keywords: "Paths between subset nodes", "DP on subset of tree", "Sparse
        subset queries".
9  * - Classic Scenarios: A tree has  $N=10^5$  nodes. You have  $Q$  queries. Each query
        gives you a subset
10 * of  $K$  nodes. You need to perform a Tree DP or path calculation on these  $K$ 
        nodes, but running
11 * an  $O(N)$  DFS per query is impossible.
12 * - The Magic: The Virtual Tree builds a smaller tree containing only the  $K$ 
        subset nodes
13 * AND their necessary LCAs to maintain tree structure. This new tree has  $O(K)$ 
        nodes.
14 *
15 * 2. HOW TO USE IT (THE BLACK BOX)
16 * - Initialization:

```

```

17 *      1. Build the full tree using standard DFS.
18 *      2. For each query, collect the subset of  $K$  nodes.
19 *      3. Call `build_virtual_tree(subset)`.
20 *      4. Perform your DP on `virtual_adj` (the edges of the virtual tree).
21 * - Complexity:
22 *      Time:  $O(K \log K)$  per query to build.
23 *      Space:  $O(N)$  to store the full tree +  $O(K)$  for the virtual tree.
24 */
25
26 const int MAXN = 2e5 + 5;
27 const int LOG = 20;
28
29 // adj: Original tree, virtual_adj: The sparse tree we build for the subset
30 vector<int> adj[MAXN], virtual_adj[MAXN];
31 int tin[MAXN], tout[MAXN], depth[MAXN];
32 int up[MAXN][LOG];
33 int timer;
34
35 // Standard DFS to precalculate depths, entry times, and binary lifting table
36 void dfs(int u, int p)
37 {
38     tin[u] = ++timer;
39     depth[u] = depth[p] + 1;
40     up[u][0] = p;
41     for (int i = 1; i < LOG; i++)
42         up[u][i] = up[up[u][i - 1]][i - 1];
43
44     for (int v : adj[u])
45     {
46         if (v != p)
47             dfs(v, u);
48     }
49     tout[u] = timer;
50 }
51
52 // Find LCA using binary lifting
53 int get_lca(int u, int v)
54 {
55     if (depth[u] < depth[v])
56         swap(u, v);
57     for (int i = LOG - 1; i >= 0; i--)
58     {
59         if (depth[u] - (1 << i) >= depth[v])
60             u = up[u][i];
61     }
62     if (u == v)
63         return u;
64     for (int i = LOG - 1; i >= 0; i--)
65     {
66         if (up[u][i] != up[v][i])
67         {
68             u = up[u][i], v = up[v][i];

```



```

69     }
70 }
71     return up[u][0];
72 }
73
74 // Ancestor check using Entry/Exit times
75 bool is_ancestor(int u, int v)
76 {
77     return tin[u] <= tin[v] && tout[u] >= tout[v];
78 }
79
80 // Main function to build the virtual tree from a subset of nodes
81 void build_virtual_tree(vector<int> &nodes)
82 {
83     // 1. Sort by entry time so we can process the tree in DFS order
84     sort(nodes.begin(), nodes.end(), [](int a, int b)
85         { return tin[a] < tin[b]; });
86
87     // 2. Inject LCAs: To keep the tree connected, we must add the LCA of every
88     // adjacent pair in our sorted list.
89     int sz = nodes.size();
90     for (int i = 0; i < sz - 1; i++)
91     {
92         nodes.push_back(get_lca(nodes[i], nodes[i + 1]));
93     }
94
95     // 3. Sort again by entry time and remove duplicates (LCAs might have been
96     // added multiple times)
97     sort(nodes.begin(), nodes.end(), [](int a, int b)
98         { return tin[a] < tin[b]; });
99     nodes.erase(unique(nodes.begin(), nodes.end()), nodes.end());
100
101     // 4. Connect the nodes using a Stack
102     // We maintain a stack representing the current path from the root.
103     vector<int> st;
104     st.push_back(nodes[0]);
105
106     for (size_t i = 1; i < nodes.size(); i++)
107     {
108         // Pop nodes that are not ancestors of the current node
109         while (st.size() > 1 && !is_ancestor(st.back(), nodes[i]))
110         {
111             virtual_adj[st[st.size() - 2]].push_back(st.back());
112             st.pop_back();
113         }
114         st.push_back(nodes[i]);
115     }
116     // Connect remaining nodes in the stack
117     while (st.size() > 1)
118     {
119         virtual_adj[st[st.size() - 2]].push_back(st.back());
120         st.pop_back();
121     }
122
123 void solve()
124 {
125     int n;
126     cin >> n;
127
128     // Clear graph before building for multiple test cases
129     for (int i = 0; i <= n; ++i)
130     {
131         adj[i].clear();
132         virtual_adj[i].clear();
133     }
134
135     // Example: Reading N-1 edges
136     for (int i = 0; i < n - 1; ++i)
137     {
138         int u, v;
139         cin >> u >> v;
140         adj[u].push_back(v);
141         adj[v].push_back(u);
142     }
143
144     // Precompute full tree properties
145     timer = 0;
146     dfs(1, 1);
147
148     int k;
149     cin >> k;
150     vector<int> subset(k);
151     for (int &x : subset)
152         cin >> x;
153
154     // Build the virtual tree for the K nodes
155     build_virtual_tree(subset);
156
157     // Now perform DP on `virtual_adj`!
158     // subset now contains all nodes in the virtual tree.
159 }
160
161 int main()
162 {
163     ios_base::sync_with_stdio(false);
164     cin.tie(NULL);
165     solve();
166     return 0;
167 }

```

## 4 Math

### 4.1 Calculus

#### 4.1.1 Differentiation integration

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: ADAPTIVE SIMPSON's RULE
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Area under curve", "Evaluate integral", "Continuous probability"
6  * - Classic Scenarios: You are given a mathematical function  $f(x)$  and need the
7  *   integral from A to B with a precision of  $10^{-6}$ .
8  * - The Magic: Instead of chopping the interval into  $10^6$  fixed pieces (which
9  *   can
10 *   TLE or lose precision), Adaptive Simpson's evaluates the error dynamically.
11 *   It recursively splits the interval only in areas where the curve is highly
12 *   irregular, guaranteeing extreme precision in logarithmic time.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Query:
16 *   double area = integrate(A, B, 1e-7, my_function);
17 *
18 * - Complexity:
19 *   Time:  $O(\text{Depends on curve complexity, practically instant})$ .
20 *   Space:  $O(\text{Recursion depth})$ .
21 */
22 double simpson(double l, double r, double f_l, double f_r, double f_mid)
23 {
24     return (f_l + 4 * f_mid + f_r) * (r - l) / 6.0;
25 }
26
27 double adaptive_simpson(double l, double r, double f_l, double f_r, double f_mid,
28     double S, double eps, const function<double(double)> &f)
29 {
30     double mid = (l + r) / 2.0;
31     double l_mid = (l + mid) / 2.0;
32     double r_mid = (mid + r) / 2.0;
33
34     double f_l_mid = f(l_mid);
35     double f_r_mid = f(r_mid);
36
37     double left_S = simpson(l, mid, f_l, f_mid, f_l_mid);
38     double right_S = simpson(mid, r, f_mid, f_r, f_r_mid);
39
40     if (abs(left_S + right_S - S) <= 15.0 * eps)
41     {
42         return left_S + right_S + (left_S + right_S - S) / 15.0;
43     }
44
45     return adaptive_simpson(l, mid, f_l, f_mid, f_l_mid, left_S, eps / 2.0, f) +

```

```

45         adaptive_simpson(mid, r, f_mid, f_r, f_r_mid, right_S, eps / 2.0, f);
46 }
47
48 double integrate(double a, double b, double eps, const function<double(double)> &
49     f)
50 {
51     double f_a = f(a);
52     double f_b = f(b);
53     double f_mid = f((a + b) / 2.0);
54     double S = simpson(a, b, f_a, f_b, f_mid);
55     return adaptive_simpson(a, b, f_a, f_b, f_mid, S, eps, f);
56 }

```

## 4.2 Convolution

### 4.2.1 FWHT

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: FAST WALSH-HADAMARD TRANSFORM
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Bitwise convolution", "XOR sum", "AND sum", "OR sum".
6  * - Classic Scenarios: Find array C where  $C[k] = \text{sum}(A[i] * B[j])$  for all
7  *   pairs  $(i, j)$  such that  $(i \text{ XOR } j) == k$ .
8  * - The Magic: It alters the NTT butterfly structure to map binary bitwise
9  *   operations into addition/subtraction, resolving the convolution in  $O(N \log N)$ .
10 *
11 * 2. HOW TO USE IT (THE BLACK BOX)
12 * - Query:
13 *   vector<long long> C = fwht_xor(A, B);
14 *   // Swap 'fwht_xor' for 'fwht_and' or 'fwht_or' as needed.
15 *
16 * - Complexity:
17 *   Time:  $O(N \log N)$ . Note that N MUST be a power of 2.
18 *   Space:  $O(N)$ .
19 *
20 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
21 * - Padding: The input arrays MUST be padded with zeros until their size
22 *   is exactly a power of 2 before passing them into these functions.
23 */
24
25 using ll = long long;
26 const ll MOD = 998244353;
27
28 ll binpow_fwht(ll a, ll b)
29 {
30     ll res = 1;
31     a %= MOD;
32     while (b > 0)
33     {

```

```

34     if (b & 1)
35         res = res * a % MOD;
36     a = a * a % MOD;
37     b >>= 1;
38 }
39 return res;
40 }
41
42 ll modInv(ll n) { return binpow_fwht(n, MOD - 2); }
43
44 void fwht_xor_base(vector<ll> &a, bool inv)
45 {
46     int n = a.size();
47     for (int len = 1; 2 * len <= n; len <= 1)
48     {
49         for (int i = 0; i < n; i += 2 * len)
50         {
51             for (int j = 0; j < len; j++)
52             {
53                 ll u = a[i + j];
54                 ll v = a[i + len + j];
55                 a[i + j] = (u + v) % MOD;
56                 a[i + len + j] = (u - v + MOD) % MOD;
57             }
58         }
59     }
60     if (inv)
61     {
62         ll invN = modInv(n);
63         for (ll &x : a)
64             x = x * invN % MOD;
65     }
66 }
67
68 void fwht_and_base(vector<ll> &a, bool inv)
69 {
70     int n = a.size();
71     for (int len = 1; 2 * len <= n; len <= 1)
72     {
73         for (int i = 0; i < n; i += 2 * len)
74         {
75             for (int j = 0; j < len; j++)
76             {
77                 ll u = a[i + j];
78                 ll v = a[i + len + j];
79                 if (!inv)
80                 {
81                     a[i + j] = (u + v) % MOD;
82                 }
83                 else
84                 {
85                     a[i + j] = (u - v + MOD) % MOD;

```

```

86         }
87     }
88 }
89 }
90 }
91
92 void fwht_or_base(vector<ll> &a, bool inv)
93 {
94     int n = a.size();
95     for (int len = 1; 2 * len <= n; len <= 1)
96     {
97         for (int i = 0; i < n; i += 2 * len)
98         {
99             for (int j = 0; j < len; j++)
100             {
101                 ll u = a[i + j];
102                 ll v = a[i + len + j];
103                 if (!inv)
104                 {
105                     a[i + len + j] = (u + v) % MOD;
106                 }
107                 else
108                 {
109                     a[i + len + j] = (v - u + MOD) % MOD;
110                 }
111             }
112         }
113     }
114 }
115
116 vector<ll> fwht_xor(vector<ll> a, vector<ll> b)
117 {
118     fwht_xor_base(a, false);
119     fwht_xor_base(b, false);
120     for (size_t i = 0; i < a.size(); i++)
121         a[i] = a[i] * b[i] % MOD;
122     fwht_xor_base(a, true);
123     return a;
124 }
125
126 vector<ll> fwht_and(vector<ll> a, vector<ll> b)
127 {
128     fwht_and_base(a, false);
129     fwht_and_base(b, false);
130     for (size_t i = 0; i < a.size(); i++)
131         a[i] = a[i] * b[i] % MOD;
132     fwht_and_base(a, true);
133     return a;
134 }
135
136 vector<ll> fwht_or(vector<ll> a, vector<ll> b)
137 {

```

```

138     fwht_or_base(a, false);
139     fwht_or_base(b, false);
140     for (size_t i = 0; i < a.size(); i++)
141         a[i] = a[i] * b[i] % MOD;
142     fwht_or_base(a, true);
143     return a;
144 }

```

#### 4.2.2 GDC LCM Convolution

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: GCD & LCM CONVOLUTION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "gcd(i, j) = k", "lcm(i, j) = k", "Divisor transformation".
6  * - Classic Scenarios: Find array C where  $C[k] = \sum(A[i] * B[j])$  over all pairs
7  *   (i, j) such that gcd(i, j) == k (or lcm(i, j) == k).
8  * - The Magic: Bypasses polynomial rings entirely. It transforms the arrays into
9  *   prefix sums over their multiples (for GCD) or divisors (for LCM), multiplies
10  *   the point values, and reverses the transformation using Mobius-style
11  *   subtraction.
12  *
13  * 2. HOW TO USE IT (THE BLACK BOX)
14  * - Query:
15  *   vector<long long> C_gcd = gcd_convolution(A, B);
16  *   vector<long long> C_lcm = lcm_convolution(A, B);
17  *
18  * - Complexity:
19  *   Time:  $O(N \log \log N)$  utilizing a prime sieve optimization.
20  *   Space:  $O(N)$ .
21  *
22  * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
23  * - 1-Based Indexing: These arrays MUST be 1-indexed (i.e., A[1] is the first
24  *   element, A[0] is ignored). Ensure A and B are the same size before passing.
25  */
26 using ll = long long;
27 const ll MOD = 998244353;
28
29 vector<int> get_primes(int n)
30 {
31     vector<bool> is_prime(n + 1, true);
32     vector<int> primes;
33     for (int p = 2; p <= n; p++)
34     {
35         if (is_prime[p])
36         {
37             primes.push_back(p);
38             for (int i = p * 2; i <= n; i += p)
39                 is_prime[i] = false;
40         }

```

```

41     }
42     return primes;
43 }
44
45 vector<ll> gcd_convolution(vector<ll> a, vector<ll> b)
46 {
47     int n = a.size() - 1;
48     vector<int> primes = get_primes(n);
49
50     for (int p : primes)
51     {
52         for (int i = n / p; i > 0; i--)
53         {
54             a[i] = (a[i] + a[i * p]) % MOD;
55             b[i] = (b[i] + b[i * p]) % MOD;
56         }
57     }
58
59     vector<ll> c(n + 1);
60     for (int i = 1; i <= n; i++)
61     {
62         c[i] = a[i] * b[i] % MOD;
63     }
64
65     for (int p : primes)
66     {
67         for (int i = 1; i * p <= n; i++)
68         {
69             c[i] = (c[i] - c[i * p] + MOD) % MOD;
70         }
71     }
72     return c;
73 }
74
75 vector<ll> lcm_convolution(vector<ll> a, vector<ll> b)
76 {
77     int n = a.size() - 1;
78     vector<int> primes = get_primes(n);
79
80     for (int p : primes)
81     {
82         for (int i = 1; i * p <= n; i++)
83         {
84             a[i * p] = (a[i * p] + a[i]) % MOD;
85             b[i * p] = (b[i * p] + b[i]) % MOD;
86         }
87     }
88
89     vector<ll> c(n + 1);
90     for (int i = 1; i <= n; i++)
91     {
92         c[i] = a[i] * b[i] % MOD;

```

```

93     }
94
95     for (int p : primes)
96     {
97         for (int i = n / p; i > 0; i--)
98         {
99             c[i * p] = (c[i * p] - c[i] + MOD) % MOD;
100         }
101     }
102     return c;
103 }

```

### 4.2.3 NTT

```

1  /**
2   * THE ULTIMATE CP TEMPLATE BLUEPRINT: NUMBER THEORETIC TRANSFORM (NTT)
3   *
4   * 1. WHAT PROBLEM DOES THIS SOLVE?
5   * - Keywords: "Polynomial multiplication", "Convolution", "Sum of pairs".
6   * - Classic Scenarios: You are given two arrays A and B and need to find C
7   *   where  $C[k] = \sum(A[i] * B[k-i])$  modulo 998244353.
8   * - The Magic: A naive double loop takes  $O(N^2)$  and TLEs for  $N = 10^5$ .
9   * - NTT evaluates the polynomials at primitive roots of unity, multiplies
10  *   the point values in  $O(N)$ , and interpolates them back in  $O(N \log N)$ .
11  *
12  * 2. HOW TO USE IT (THE BLACK BOX)
13  * - Query:
14  *   vector<long long> C = multiply(A, B);
15  *
16  * - Complexity:
17  *   Time:  $O(N \log N)$ 
18  *   Space:  $O(N)$ 
19  *
20  * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
21  * - Modulo: This template is strictly hardcoded for MOD = 998244353 and
22  *   its primitive root 3. If the problem has a different modulo, you MUST
23  *   use an Arbitrary Modulo NTT (MTT) or find the primitive root of the new
24  *   modulo.
25  */
26 using ll = long long;
27 const ll MOD = 998244353;
28 const ll ROOT = 3;
29
30 ll binpow(ll a, ll b)
31 {
32     ll res = 1;
33     a %= MOD;
34     while (b > 0)
35     {
36         if (b & 1)

```

```

37         res = res * a % MOD;
38         a = a * a % MOD;
39         b >>= 1;
40     }
41     return res;
42 }
43
44 ll modInverse(ll n)
45 {
46     return binpow(n, MOD - 2);
47 }
48
49 void ntt(vector<ll> &a, bool invert)
50 {
51     int n = a.size();
52     for (int i = 1, j = 0; i < n; i++)
53     {
54         int bit = n >> 1;
55         for (; j & bit; bit >>= 1)
56             j ^= bit;
57         j ^= bit;
58         if (i < j)
59             swap(a[i], a[j]);
60     }
61     for (int len = 2; len <= n; len <= 1)
62     {
63         ll wlen = binpow(ROOT, (MOD - 1) / len);
64         if (invert)
65             wlen = modInverse(wlen);
66         for (int i = 0; i < n; i += len)
67         {
68             ll w = 1;
69             for (int j = 0; j < len / 2; j++)
70             {
71                 ll u = a[i + j];
72                 ll v = a[i + j + len / 2] * w % MOD;
73                 a[i + j] = (u + v < MOD ? u + v : u + v - MOD);
74                 a[i + j + len / 2] = (u - v >= 0 ? u - v : u - v + MOD);
75                 w = w * wlen % MOD;
76             }
77         }
78     }
79     if (invert)
80     {
81         ll n_inv = modInverse(n);
82         for (ll &x : a)
83             x = x * n_inv % MOD;
84     }
85 }
86
87 vector<ll> multiply(vector<ll> const &a, vector<ll> const &b)
88 {

```

```

89     vector<ll> fa(a.begin(), a.end()), fb(b.begin(), b.end());
90     int n = 1;
91     while (n < a.size() + b.size())
92         n <<= 1;
93     fa.resize(n);
94     fb.resize(n);
95     ntt(fa, false);
96     ntt(fb, false);
97     for (int i = 0; i < n; i++)
98         fa[i] = fa[i] * fb[i] % MOD;
99     ntt(fa, true);
100    while (fa.size() > 1 && fa.back() == 0)
101        fa.pop_back();
102    return fa;
103 }

```

## 4.3 Linear algebra

### 4.3.1 Determinant(composite mod)

### 4.3.2 Determinant(prime mod)

```

1  /**
2  *   THE ULTIMATE CP TEMPLATE BLUEPRINT: DETERMINANT (PRIME MODULO)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Matrix Tree Theorem", "Number of Spanning Trees".
6  * - Classic Scenarios: You have an  $N \times N$  matrix and need its determinant modulo
7  *   a prime number like  $1e9+7$  or  $998244353$ .
8  * - The Magic: Standard Gaussian elimination. It tracks row swaps because every
9  *   swap multiplies the final determinant by  $-1$ .
10 *
11 * 2. HOW TO USE IT (THE BLACK BOX)
12 * - Query:
13 *     long long ans = det_prime(matrix, MOD);
14 *
15 * - Complexity:
16 *     Time:  $O(N^3)$ 
17 *     Space:  $O(N^2)$ 
18 */
19
20 using ll = long long;
21
22 ll binpow_det(ll a, ll b, ll mod)
23 {
24     ll res = 1;
25     a %= mod;
26     while (b > 0)

```

```

27     {
28         if (b & 1)
29             res = res * a % mod;
30         a = a * a % mod;
31         b >>= 1;
32     }
33     return res;
34 }
35
36 ll modInverse_det(ll n, ll mod)
37 {
38     return binpow_det(n, mod - 2, mod);
39 }
40
41 ll det_prime(vector<vector<ll>> a, ll mod)
42 {
43     int n = a.size();
44     ll ans = 1;
45     for (int i = 0; i < n; ++i)
46     {
47         int pivot = i;
48         for (int j = i + 1; j < n; ++j)
49         {
50             if (a[j][i] > a[pivot][i])
51                 pivot = j;
52         }
53         if (a[pivot][i] == 0)
54             return 0;
55         if (i != pivot)
56         {
57             swap(a[i], a[pivot]);
58             ans = (mod - ans) % mod;
59         }
60         ans = ans * a[i][i] % mod;
61         ll inv = modInverse_det(a[i][i], mod);
62         for (int j = i + 1; j < n; ++j)
63         {
64             ll factor = a[j][i] * inv % mod;
65             for (int k = i; k < n; ++k)
66             {
67                 a[j][k] = (a[j][k] - factor * a[i][k]) % mod;
68                 if (a[j][k] < 0)
69                     a[j][k] += mod;
70             }
71         }
72     }
73     return ans;
74 }

```

### 4.3.3 Gaussian elimination

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: GAUSSIAN ELIMINATION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "System of equations", "Probabilities", "Markov Chains".
6  * - Classic Scenarios: Solving  $Ax = B$  for floating-point values.
7  * - The Magic: It reduces the matrix to row-echelon form to find the exact
8  *   values of the variables.
9  *
10 * 2. HOW TO USE IT (THE BLACK BOX)
11 * - Query:
12 *   vector<double> ans;
13 *   int solutions = gauss(matrix, ans);
14 * - Result Breakdown:
15 *   Returns 1 if there is exactly one solution (stored in 'ans').
16 *   Returns 0 if there are no solutions.
17 *   Returns 2 if there are infinitely many solutions.
18 *
19 * - Complexity:
20 *   Time:  $O(N^3)$ 
21 *   Space:  $O(N^2)$ 
22 */
23
24 const double EPS = 1e-9;
25
26 int gauss(vector<vector<double>> a, vector<double> &ans)
27 {
28     int n = a.size();
29     if (n == 0)
30         return 0;
31     int m = a[0].size() - 1;
32     vector<int> where(m, -1);
33
34     for (int col = 0, row = 0; col < m && row < n; ++col)
35     {
36         int sel = row;
37         for (int i = row; i < n; ++i)
38         {
39             if (abs(a[i][col]) > abs(a[sel][col]))
40                 sel = i;
41         }
42         if (abs(a[sel][col]) < EPS)
43             continue;
44
45         for (int i = col; i <= m; ++i)
46             swap(a[sel][i], a[row][i]);
47         where[col] = row;
48
49         for (int i = 0; i < n; ++i)
50         {
51             if (i != row)
52                 {

```

```

53                     double c = a[i][col] / a[row][col];
54                     for (int j = col; j <= m; ++j)
55                     {
56                         a[i][j] -= a[row][j] * c;
57                     }
58                 }
59             }
60         ++row;
61     }
62
63     ans.assign(m, 0);
64     for (int i = 0; i < m; ++i)
65     {
66         if (where[i] != -1)
67             ans[i] = a[where[i]][m] / a[where[i]][i];
68     }
69
70     for (int i = 0; i < n; ++i)
71     {
72         double sum = 0;
73         for (int j = 0; j < m; ++j)
74         {
75             sum += ans[j] * a[i][j];
76         }
77         if (abs(sum - a[i][m]) > EPS)
78             return 0;
79     }
80
81     for (int i = 0; i < m; ++i)
82     {
83         if (where[i] == -1)
84             return 2;
85     }
86     return 1;
87 }

```

#### 4.3.4 Gaussian elimination mod2

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: GAUSSIAN ELIMINATION MOD 2
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "XOR subset", "Boolean equations", "Lights out game".
6  * - Classic Scenarios: Finding if a specific XOR sum can be formed from an array
7  *   or solving  $Ax = B$  where all values and operations are modulo 2 (XOR).
8  * - The Magic: Replaces standard arrays with `std::bitset`, shrinking the time
9  *   constant by a factor of 64. Instead of division and subtraction, it simply
10 *   uses XOR.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)

```

```

13 * - Query:
14 *     bitset<2005> ans;
15 *     int solutions = gauss_mod2(matrix, N, M, ans);
16 *
17 * - Complexity:
18 *     Time:  $O(N^3 / 64)$ 
19 *     Space:  $O(N^2 / 64)$ 
20 */
21
22 const int MAX_V = 2005;
23
24 int gauss_mod2(vector<bitset<MAX_V>> a, int n, int m, bitset<MAX_V> &ans)
25 {
26     vector<int> where(m, -1);
27     for (int col = 0, row = 0; col < m && row < n; ++col)
28     {
29         for (int i = row; i < n; ++i)
30         {
31             if (a[i][col])
32             {
33                 swap(a[i], a[row]);
34                 break;
35             }
36         }
37         if (!a[row][col])
38             continue;
39         where[col] = row;
40         for (int i = 0; i < n; ++i)
41         {
42             if (i != row && a[i][col])
43             {
44                 a[i] ^= a[row];
45             }
46         }
47         ++row;
48     }
49
50     ans.reset();
51     for (int i = 0; i < m; ++i)
52     {
53         if (where[i] != -1)
54             ans[i] = a[where[i]][m];
55     }
56
57     for (int i = 0; i < n; ++i)
58     {
59         int sum = 0;
60         for (int j = 0; j < m; ++j)
61         {
62             if (ans[j] && a[i][j])
63                 sum ^= 1;
64         }

```

```

65         if (sum != a[i][m])
66             return 0;
67     }
68
69     for (int i = 0; i < m; ++i)
70     {
71         if (where[i] == -1)
72             return 2;
73     }
74     return 1;
75 }

```

#### 4.3.5 XOR basis

```

1 /**
2 * THE ULTIMATE CP TEMPLATE BLUEPRINT: XOR BASIS (LINEAR BASIS)
3 *
4 * 1. WHAT PROBLEM DOES THIS SOLVE?
5 * - Keywords: "Maximum XOR subset", "K-th XOR sum", "Can we form X".
6 * - Classic Scenarios: You are given an array of size  $10^5$  and need to find the
7 *   maximum possible XOR sum of any subset.
8 * - The Magic: It reduces the entire array into a set of at most B "basis"
9 *   vectors
10 *   (where B is the number of bits, usually 60 for long long). These basis
11 *   vectors
12 *   span the exact same XOR space as the original array.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization:
16 *     XORBasis basis;
17 *     for (ll x : arr) basis.insert(x);
18 * - Queries:
19 *     long long max_val = basis.max_xor();
20 *     bool exists = basis.contains(X);
21 *
22 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
23 * - Reduced Row Echelon Form (RREF) / K-th Smallest: To answer "K-th smallest
24 *   XOR sum"
25 *   (Topic 383/384), you MUST call `basis.build_rref()` first. Then use `basis.
26 *   kth(K)`.
27 *   (Note: K is 1-indexed. If K=1, it returns the smallest possible XOR sum > 0)
28 *
29 */
30
31 using ll = long long;
32
33 struct XORBasis
34 {
35     static const int BITS = 60; // Max bits for long long
36     vector<ll> basis;
37     vector<ll> rref; // For K-th queries

```



```

33  int sz;
34  bool is_rref_built;
35
36  XORBasis()
37  {
38      basis.assign(BITS, 0);
39      sz = 0;
40      is_rref_built = false;
41  }
42
43  bool insert(ll x)
44  {
45      for (int i = BITS - 1; i >= 0; i--)
46      {
47          if ((x >> i) & 1)
48          {
49              if (!basis[i])
50              {
51                  basis[i] = x;
52                  sz++;
53                  is_rref_built = false;
54                  return true;
55              }
56              x ^= basis[i];
57          }
58      }
59      return false;
60  }
61
62  bool contains(ll x)
63  {
64      for (int i = BITS - 1; i >= 0; i--)
65      {
66          if ((x >> i) & 1)
67          {
68              if (!basis[i])
69                  return false;
70              x ^= basis[i];
71          }
72      }
73      return true;
74  }
75
76  ll max_xor(ll current_xor = 0)
77  {
78      ll res = current_xor;
79      for (int i = BITS - 1; i >= 0; i--)
80      {
81          res = max(res, res ^ basis[i]);
82      }
83      return res;
84  }

```

```

85
86  // Call this strictly BEFORE querying kth()
87  void build_rref()
88  {
89      if (is_rref_built)
90          return;
91      rref.clear();
92      vector<ll> temp = basis;
93      for (int i = BITS - 1; i >= 0; i--)
94      {
95          if (!temp[i])
96              continue;
97          for (int j = i + 1; j < BITS; j++)
98          {
99              if ((temp[j] >> i) & 1)
100                  temp[j] ^= temp[i];
101          }
102      }
103      for (int i = 0; i < BITS; i++)
104      {
105          if (temp[i])
106              rref.push_back(temp[i]);
107      }
108      is_rref_built = true;
109  }
110
111  // Returns K-th smallest XOR sum (1-indexed, excluding 0)
112  // Returns -1 if K is out of bounds
113  ll kth(ll k)
114  {
115      build_rref();
116      if (k > (1LL << sz) - 1)
117          return -1;
118      ll res = 0;
119      for (int i = 0; i < rref.size(); i++)
120      {
121          if ((k >> i) & 1)
122              res ^= rref[i];
123      }
124      return res;
125  }
126 };

```

## 4.4 Linear recurrence

### 4.4.1 Berlekamp

```

1  /**
2  *  THE ULTIMATE CP TEMPLATE BLUEPRINT: BERLEKAMP-MASSEY
3  *
4  *  1.  WHAT PROBLEM DOES THIS SOLVE?

```

```

5  * - Keywords: "Find the recurrence", "Sequence guesser".
6  * - Classic Scenarios: You brute-forced the first 50 terms of a DP or counting
7  *   problem, but you cannot figure out the actual mathematical recurrence
   *   relation.
8  * - The Magic: It iteratively builds a polynomial that perfectly models the
9  *   input sequence. If the true recurrence has length  $N$ , giving BM the first
10 *    $2N$  terms guarantees it will perfectly reconstruct the formula.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     vector<ll> C = berlekamp_massey(S, MOD);
15 *     // S is the sequence you brute-forced.
16 *     // C contains the coefficients:  $S[i] = C[0]*S[i-1] + C[1]*S[i-2] + \dots$ 
17 *
18 * - Complexity:
19 *     Time:  $O(|S|^2)$ 
20 *     Space:  $O(|S|)$ 
21 */
22
23 using ll = long long;
24
25 ll binpow_bm(ll a, ll b, ll mod)
26 {
27     ll res = 1;
28     a %= mod;
29     while (b > 0)
30     {
31         if (b & 1)
32             res = res * a % mod;
33         a = a * a % mod;
34         b >>= 1;
35     }
36     return res;
37 }
38
39 ll modInverse_bm(ll n, ll mod)
40 {
41     return binpow_bm(n, mod - 2, mod);
42 }
43
44 vector<ll> berlekamp_massey(const vector<ll> &s, ll mod)
45 {
46     vector<ll> c = {1}, b = {1};
47     ll L = 0, m = 1, b_val = 1;
48     for (int i = 0; i < s.size(); i++)
49     {
50         ll d = 0;
51         for (int j = 0; j <= L; j++)
52         {
53             d = (d + c[j] * s[i - j]) % mod;
54         }
55         if (d == 0)

```

```

56     {
57         m++;
58     }
59     else
60     {
61         vector<ll> temp = c;
62         ll c_val = (d * modInverse_bm(b_val, mod)) % mod;
63         while (c.size() <= b.size() + m)
64             c.push_back(0);
65         for (int j = 0; j < b.size(); j++)
66         {
67             c[j + m] = (c[j + m] - c_val * b[j] % mod + mod) % mod;
68         }
69         if (2 * L <= i)
70         {
71             L = i + 1 - L;
72             b = temp;
73             b_val = d;
74             m = 1;
75         }
76         else
77         {
78             m++;
79         }
80     }
81 }
82 c.erase(c.begin());
83 for (ll &x : c)
84     x = (mod - x) % mod;
85 return c;
86 }

```

## 4.5 Matrix

### 4.5.1 Freivald's algorithm

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: FREIVALD'S ALGORITHM
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Verify Matrix Multiplication", " $A * B = C$ ".
6  * - Classic Scenarios: You are given three massive  $N \times N$  matrices ( $A$ ,  $B$ , and  $C$ ).
7  *   You need to determine if  $A * B$  equals  $C$  without TLEing.
8  * - The Magic: A full multiplication takes  $O(N^3)$ . Freivald's generates a random
9  *   1D vector ( $r$ ) of 0s and 1s, and checks if  $A * (B * r) == C * r$ .
10 *   Since matrix-vector multiplication is  $O(N^2)$ , this is drastically faster.
11 *   Repeating it  $K$  times drops the probability of a false positive to  $(1/2)^K$ .
12 *
13 * 2. HOW TO USE IT (THE BLACK BOX)
14 * - Query:

```

```

15 *      bool is_correct = freivalds(A, B, C, 15); // 15 iterations is perfectly
16 *      safe
17 * - Complexity:
18 *      Time:  $O(K * N^2)$ 
19 *      Space:  $O(N)$ 
20 */
21
22 #include <random>
23 #include <chrono>
24
25 using ll = long long;
26 const ll MOD = 1e9 + 7;
27
28 vector<ll> multiply_mat_vec(const vector<vector<ll>> &mat, const vector<ll> &vec)
29 {
30     int n = mat.size();
31     vector<ll> res(n, 0);
32     for (int i = 0; i < n; i++)
33     {
34         for (int j = 0; j < n; j++)
35         {
36             res[i] = (res[i] + mat[i][j] * vec[j]) % MOD;
37         }
38     }
39     return res;
40 }
41
42 bool freivalds(const vector<vector<ll>> &a, const vector<vector<ll>> &b, const
43     vector<vector<ll>> &c, int k_iterations = 15)
44 {
45     int n = a.size();
46     mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
47     uniform_int_distribution<int> dist(0, 1);
48
49     for (int k = 0; k < k_iterations; k++)
50     {
51         vector<ll> r(n);
52         for (int i = 0; i < n; i++)
53         {
54             r[i] = dist(rng);
55         }
56
57         vector<ll> br = multiply_mat_vec(b, r);
58         vector<ll> abr = multiply_mat_vec(a, br);
59         vector<ll> cr = multiply_mat_vec(c, r);
60
61         for (int i = 0; i < n; i++)
62         {
63             if (abr[i] != cr[i])
64                 return false;
65         }
66     }
67     return true;
68 }
69 }

```

#### 4.5.2 Matrix expo

```

1 typedef vector<vector<ll>> Matrix;
2
3 Matrix matrix_mul(Matrix &A, Matrix &B)
4 {
5     Matrix C;
6     ll n = ll(A.size());
7     ll m = ll(A[0].size());
8     ll l = ll(B[0].size());
9     C.resize(n, vector<ll>(l));
10    for (ll i = 0; i < n; i++)
11    {
12        for (ll k = 0; k < l; k++)
13        {
14            for (ll j = 0; j < m; j++)
15            {
16                C[i][k] += A[i][j] % MOD * B[j][k] % MOD;
17                C[i][k] %= MOD;
18            }
19        }
20    }
21    return C;
22 }
23
24 Matrix matrix_expo(Matrix &matrix, ll pow)
25 {
26     ll n = ll(matrix.size());
27     Matrix res;
28     res.resize(n, vector<ll>(n));
29     for (int i = 0; i < n; i++)
30         res[i][i] = 1;
31     while (pow)
32     {
33         if (pow & 1)
34             res = matrix_mul(matrix, res);
35         matrix = matrix_mul(matrix, matrix);
36         pow >>= 1;
37     }
38     return res;
39 }
40
41 Matrix I(int n)
42 {
43     Matrix T(n, vector<int>(n));

```

```

44     for (int i = 0; i < n; i++)
45         T[i][i] = 1;
46     return T;
47 }
48
49 // 2d Grid represent as graph
50 // count paths
51 short dx[8] = {1,1,-1,-1,2,2,-2,-2};
52 short dy[8] = {2,-2,2,-2,1,-1,1,-1};
53 void solve() {
54     int k;cin >>k;
55     Matrix T(65,Row(65));
56     for (int i = 0;i<8;i++) {
57         for (int j = 0;j<8;j++) {
58             int cur = i*8 + j;
59             for (int k = 0;k<8;k++) {
60                 short ni = i+dx[k];
61                 short nj = j+dy[k];
62                 if (ni < 0 || nj < 0 || ni > 7 || nj>7) continue;
63                 int nCur = ni*8+nj;
64                 T[nCur][cur] = 1;
65             }
66         }
67     }
68     for (int i = 0;i<65;i++) {
69         T[i][64] = 1;
70     }
71     T = fastPower(T,k+1);
72     int ans = T[0][64];
73     cout << ans;
74 }

```

#### 4.5.3 Matrix expo Queries

```

1
2 * THE ULTIMATE CP TEMPLATE BLUEPRINT: FAST PATH COUNTING (PRECOMPUTED MATRIX
  EXPONENTIATION)
3 * 1. WHAT PROBLEM DOES THIS SOLVE?
4 * - Keywords: "Number_of_paths_of_length_K", "Multiple_Queries(Q)", "State_
  Vector_multiplication".
5 * - Classic Scenarios: You need to find the number of valid paths of exactly
  length K between
6 * node U and node V in a directed graph, modulo some number. BUT, you have Q
  queries!
7 * Standard Matrix Exponentiation per query takes  $O(Q * N^3 * \log K)$ , which
  will Time Limit Exceed (TLE).
8 * - The Magic: "The_Vector-Matrix_Trick". We precompute the binary powers of the
  adjacency
9 * matrix  $A^{(2^0)}$ ,  $A^{(2^1)}$ , ...,  $A^{(2^{29})}$  ONCE in  $O(N^3 * \log K)$ .
10 * Then, for each query, we don't multiply  $N \times N$  matrices together. Instead, we
  start with

```

```

11 a_1x_N_state_vector (where only the starting node U is 1) and multiply this
  VECTOR
12 by the precomputed N x N matrices. This slashes the query time down to just
   $O(N^2 * \log K)!$ 
13
14 * HOW TO USE IT (THE BLACK BOX)
15 BaseState: Build your adjacency matrix `t`. If there is a directed edge u
  -> v, `t[u][v] = 1`.
16
17 * Precomputation:
18 pre[0] = t;
19 pre[i] = mul(pre[i-1], pre[i-1]); // Computes  $A^{(2^i)}$ 
20
21 * Queries:
22 Create a 1xN matrix `s` (state_vector). Set `s[0][u] = 1`.
23 For each set bit i in K, update `s = mul(s, pre[i])`.
24 The answer is simply `s[0][v]`.
25
26 * Complexity:
27 Time: Precomputation  $O(N^3 * \log K)$  + Queries  $O(Q * N^2 * \log K)$ .
28 Space:  $O(\log K * N^2)$  to store the 30 precomputed matrices.
29
30 const int mod = 1e9 + 7;
31 #define Row vector<int>
32 #define mat vector<Row>
33 mat mul(mat&a, mat&b) {
34     int n = a.size(), m = b[0].size(), k = a[0].size();
35     mat res(n, Row(m));
36     for (int i = 0; i < n; i++) {
37         for (int j = 0; j < m; j++) {
38             for (int o = 0; o < k; o++) {
39                 res[i][j] += a[i][o] * b[o][j] % mod;
40             }
41         }
42     }
43     return res;
44 }
45
46 mat power(mat a, int b) {
47     int n = a.size();
48     mat res(n, Row(n));
49     for (int i = 0; i < n; i++) res[i][i] = 1;
50     while (b) {
51         if (b & 1) res = mul(res, a);
52         a = mul(a, a);
53         b >>= 1;
54     }
55     return res;
56 }
57
58 void solve() {
59     int n, m, q; cin >> n >> m >> q;

```

```

60 mat_t(n, Row(n));
61 for(int i=0; i<m; i++){
62     int u, v; cin >> u >> v;
63     u--; v--;
64     mat[u][v] = 1;
65 }
66 vector<mat> pre(30, mat(n, Row(n)));
67 pre[0] = mat;
68 for(int i=1; i<30; i++){
69     pre[i] = mul(pre[i-1], pre[i-1]);
70 }
71 while(q--){
72     int u, v, k; cin >> u >> v >> k; --u, --v;
73     mat s(1, Row(n));
74     s[0][u] = 1;
75     for(int i=0; i<30; i++){
76         if(1<=i & k){
77             s = mul(s, pre[i]);
78         }
79     }
80     cout << s[0][v] << '\n';
81 }
82 }

```

#### 4.5.4 Min Path Using Matrix expo

```

1 * THE ULTIMATE CP TEMPLATE BLUEPRINT: MIN-PLUS MATRIX EXPONENTIATION
2 * 1. WHAT PROBLEM DOES THIS SOLVE?
3 * - Keywords: "Shortest_path_of_exactly_K_edges", "Paths_of_length_K", "K_is_up
4 * - Classic Scenarios: You are given a graph and need to find the minimum cost
5 *   to travel between
6 *   nodes using EXACTLY (or at most) K edges. Since K is huge, standard
7 *   algorithms like Dijkstra
8 *   or Bellman-Ford will Time Limit Exceed (TLE).
9 * - The Magic: We treat the adjacency matrix as paths of length 1. If we
10 *   multiply the matrix by itself,
11 *   we get paths of length 2. But instead of standard multiplication (+ and *),
12 *   we use the
13 *   Min-Plus Semiring (min instead of +, and + instead of *).
14 *   Using binary exponentiation, we can compute A^K to find paths of length K in
15 *   just O(N^3 log K)!
16 * 2. HOW TO USE IT (THE BLACK BOX)
17 * - Initialization: Create an N x N matrix filled with infinity (`oo`).
18 *   Matrix t(n, Row(n, oo));
19 * - Build Base State: Set the weight of directed edges.
20 *   t[u][v] = w; // Cost of 1 edge from u to v
21 * - Execution: Call `fastPower` to raise the matrix to the power of K.
22 *   t = fastPower(t, k);

```

```

20
21 #define Row vector<int>
22 #define Matrix vector<Row>
23 const int mod = 1e9 + 7;
24 const int oo = 2e18+5;
25
26 Matrix mul(Matrix &a, Matrix &b) {
27     int n = a.size();
28     int m = b[0].size();
29     int sz = a[0].size();
30     Matrix res(n, Row(m, oo));
31     for (int i = 0; i < n; i++) {
32         for (int j = 0; j < m; j++) {
33             for (int o = 0; o < sz; o++) {
34                 res[i][j] = min(res[i][j], a[i][o] + b[o][j]);
35             }
36         }
37     }
38     return res;
39 }
40
41 Matrix fastPower(Matrix &a, int b) {
42     int n = a.size();
43
44     Matrix res(n, Row(n, oo));
45     for (int i = 0; i < n; i++) {
46         res[i][i] = 0;
47     }
48
49     while (b) {
50         if (b & 1)
51             res = mul(res, a);
52         a = mul(a, a);
53         b /= 2;
54     }
55     return res;
56 }
57
58 void solve() {
59     int n, m, k;
60     cin >> n >> m >> k;
61     Matrix t(n, Row(n, oo));
62     for (int i = 0; i < m; i++) {
63         int u, v, w;
64         cin >> u >> v >> w;
65         --u;
66         --v;
67         t[u][v] = w;
68     }
69     t = fastPower(t, k);
70
71     int ans = oo;

```

```

72     for (int i = 0; i < n; i++) {
73         for (int j=0;j<n;j++) {
74             ans = min(ans, t[i][j]);
75         }
76     }
77     if (ans > 1e18) {
78         cout << "IMPOSSIBLE\n";
79     } else {
80         cout << ans << "\n";
81     }
82 }

```

#### 4.5.5 Permanent matrix

```

1  /**
2  *  THE ULTIMATE CP TEMPLATE BLUEPRINT: PERMANENT OF A MATRIX (RYSER'S FORMULA)
3  *
4  *  1.  WHAT PROBLEM DOES THIS SOLVE?
5  *  - Keywords: "Number of perfect matchings", "Bipartite graph matchings".
6  *  - Classic Scenarios: Given an  $N \times N$  adjacency matrix of a bipartite graph,
7  *  find exactly how many distinct perfect matchings exist.
8  *  - The Magic: Uses Inclusion-Exclusion (Ryser's Formula) to bypass the
9  *  factorial complexity of permutations, solving it in exactly  $O(N * 2^N)$ .
10 *
11 *  2.  HOW TO USE IT (THE BLACK BOX)
12 *  - Query:
13 *      long long ways = permanent(adj_matrix);
14 *
15 *  - Complexity:
16 *      Time:  $O(N * 2^N)$  Generally safe for  $N \leq 20$ .
17 *      Space:  $O(N)$ 
18 */
19
20 using ll = long long;
21 const ll MOD = 1e9 + 7;
22
23 ll permanent(const vector<vector<ll>> &a)
24 {
25     int n = a.size();
26     if (n == 0)
27         return 0;
28
29     ll total_perm = 0;
30     int max_mask = 1 << n;
31
32     for (int mask = 1; mask < max_mask; mask++)
33     {
34         ll current_prod = 1;
35         int set_bits = 0;
36
37         for (int i = 0; i < n; i++)

```

```

38     {
39         ll row_sum = 0;
40         int bit_count = 0;
41         for (int j = 0; j < n; j++)
42         {
43             if ((mask >> j) & 1)
44             {
45                 row_sum = (row_sum + a[i][j]) % MOD;
46                 if (i == 0)
47                     bit_count++;
48             }
49         }
50         if (i == 0)
51             set_bits = bit_count;
52         current_prod = (current_prod * row_sum) % MOD;
53     }
54
55     if ((n - set_bits) % 2 == 1)
56     {
57         total_perm = (total_perm - current_prod + MOD) % MOD;
58     }
59     else
60     {
61         total_perm = (total_perm + current_prod) % MOD;
62     }
63 }
64 return total_perm;
65 }

```

## 4.6 Optimization

### 4.6.1 Simplex

```

1  /**
2  *  THE ULTIMATE CP TEMPLATE BLUEPRINT: SIMPLEX ALGORITHM
3  *
4  *  1.  WHAT PROBLEM DOES THIS SOLVE?
5  *  - Keywords: "Maximize profit", "Minimize cost", "Subject to constraints".
6  *  - Classic Scenarios: A generic resource allocation problem where you need to
7  *  maximize an objective function  $C^T X$  subject to a system of inequalities  $AX \leq B$ .
8  *  - The Magic: Bypasses the need for complex min-cost max-flow graph modeling
9  *  or impossible greedy logic. You literally just pass the coefficients of your
10 *  equations into the matrix.
11 *
12 *  2.  HOW TO USE IT (THE BLACK BOX)
13 *  - Query:
14 *      vector<double> ans_vars;
15 *      double max_profit = simplex(A, b, c, ans_vars);
16 *  - Result Breakdown:
17 *      Returns numeric_limits<double>::infinity() if the problem is unbounded.

```

```

18 *      Returns -numeric_limits<double>::infinity() if there is no feasible
      solution.
19 *      Otherwise, returns the maximum possible value of the objective function
20 *      and populates `ans_vars` with the optimal values for variables  $X_1$  to
       $X_n$ .
21 *
22 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
23 * - Handling  $\geq$  constraints: If an equation is  $5x + 3y \geq 10$ , multiply the
      entire
24 * row by  $-1$  to make it  $-5x - 3y \leq -10$  before pushing it to matrix  $A$  and
      vector  $b$ .
25 * - Handling  $=$  constraints: Split it into two inequalities:
26 *  $(5x + 3y \leq 10)$  AND  $(-5x - 3y \leq -10)$ .
27 */
28
29 const double EPS = 1e-9;
30 const double INF = numeric_limits<double>::infinity();
31
32 double simplex(vector<vector<double>> A, vector<double> b, vector<double> c,
      vector<double> &x)
33 {
34     int m = b.size(), n = c.size();
35     vector<int> N(n + 1), B(m);
36     vector<vector<double>> D(m + 2, vector<double>(n + 2));
37
38     for (int i = 0; i < m; i++)
39     {
40         for (int j = 0; j < n; j++)
41             D[i][j] = A[i][j];
42         B[i] = n + i;
43         D[i][n] = -1;
44         D[i][n + 1] = b[i];
45     }
46     for (int j = 0; j < n; j++)
47     {
48         N[j] = j;
49         D[m][j] = -c[j];
50     }
51     N[n] = -1;
52     D[m + 1][n] = 1;
53
54     auto pivot = [&](int r, int s)
55     {
56         double inv = 1.0 / D[r][s];
57         for (int i = 0; i < m + 2; i++)
58         {
59             if (i != r && abs(D[i][s]) > EPS)
60             {
61                 double inv2 = D[i][s] * inv;
62                 for (int j = 0; j < n + 2; j++)
63                 {
64                     D[i][j] -= D[r][j] * inv2;

```

```

65         }
66         D[i][s] = D[r][s] * inv2;
67     }
68 }
69 for (int j = 0; j < n + 2; j++)
70 {
71     if (j != s)
72         D[r][j] *= inv;
73 }
74 D[r][s] = inv;
75 swap(B[r], N[s]);
76 };
77
78 auto phase = [&](int row)
79 {
80     while (true)
81     {
82         int s = -1;
83         for (int j = 0; j <= n; j++)
84         {
85             if (N[j] != -1 && (s == -1 || D[row][j] < D[row][s] - EPS))
86             {
87                 s = j;
88             }
89         }
90         if (D[row][s] > -EPS)
91             return true;
92
93         int r = -1;
94         for (int i = 0; i < m; i++)
95         {
96             if (D[i][s] > EPS && (r == -1 || D[i][n + 1] / D[i][s] < D[r][n +
                1] / D[r][s] - EPS))
97             {
98                 r = i;
99             }
100         }
101         if (r == -1)
102             return false;
103         pivot(r, s);
104     }
105 };
106
107 int r = 0;
108 for (int i = 1; i < m; i++)
109 {
110     if (D[i][n + 1] < D[r][n + 1])
111         r = i;
112 }
113
114 if (D[r][n + 1] < -EPS)
115 {

```

```

116     pivot(r, n);
117     if (!phase(m + 1) || D[m + 1][n + 1] < -EPS)
118         return -INF;
119     for (int i = 0; i < m; i++)
120     {
121         if (B[i] == -1)
122         {
123             int s = -1;
124             for (int j = 0; j <= n; j++)
125             {
126                 if (N[j] != -1 && (s == -1 || D[i][j] < D[i][s] - EPS))
127                     s = j;
128             }
129             pivot(i, s);
130         }
131     }
132 }
133
134 if (!phase(m))
135     return INF;
136
137 x.assign(n, 0);
138 for (int i = 0; i < m; i++)
139 {
140     if (B[i] < n)
141         x[B[i]] = D[i][n + 1];
142 }
143
144 return D[m][n + 1];
145 }

```

## 4.7 Polynomial

### 4.7.1 EXP POW Sqrt

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: POLYNOMIAL EXP, POW, & Sqrt
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Exponentiate Generating Function", " $e^{A(x)}$ ", " $A(x)^K$ ", " $\text{sqrt}(A(x))$ ".
6  * - Classic Scenarios: Solving combinatorial problems involving unlabelled sets
7  * or Euler transforms (Topic 335, 337, 338).
8  * - The Magic: Polynomial Exp uses Newton's method on logarithms:
9  *  $F_{k+1} = F_k * (1 - \ln(F_k) + A)$ . Polynomial Pow uses the identity
10 *  $A^k = \exp(k * \ln(A))$ .
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     vector<ll> expA = poly_exp(A, N);
15 *     vector<ll> powA = poly_pow(A, K, N);

```

```

16 *     vector<ll> sqrtA = poly_sqrt(A, N);
17 *
18 * - Complexity:
19 *     Time:  $O(N \log N)$ 
20 *     Space:  $O(N)$ 
21 *
22 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
23 * - Dependencies: Requires all functions from FPS Toolkit Part 1.
24 * - Constant Term Strictness:
25 *     - poly_exp: A[0] MUST be 0.
26 *     - poly_pow: A[0] MUST be 1 (if not, factor out the lowest degree term
27 *       first).
28 *     - poly_sqrt: A[0] MUST be 1.
29 */
30 vector<ll> poly_exp(vector<ll> a, int deg)
31 {
32     if (deg == 1)
33         return {1};
34     vector<ll> b = poly_exp(a, (deg + 1) / 2);
35     b.resize(deg);
36     vector<ll> lnb = poly_log(b, deg);
37     vector<ll> diff(deg);
38     for (int i = 0; i < deg; i++)
39     {
40         diff[i] = (a.size() > i ? a[i] : 0) - lnb[i];
41         if (diff[i] < 0)
42             diff[i] += MOD;
43     }
44     diff[0] = (diff[0] + 1) % MOD;
45     vector<ll> res = multiply(b, diff);
46     res.resize(deg);
47     return res;
48 }
49
50 vector<ll> poly_pow(vector<ll> a, ll k, int deg)
51 {
52     vector<ll> lna = poly_log(a, deg);
53     for (int i = 0; i < lna.size(); i++)
54     {
55         lna[i] = lna[i] * (k % MOD) % MOD;
56     }
57     return poly_exp(lna, deg);
58 }
59
60 vector<ll> poly_sqrt(vector<ll> a, int deg)
61 {
62     if (deg == 1)
63         return {1};
64     vector<ll> b = poly_sqrt(a, (deg + 1) / 2);
65     b.resize(deg);
66     vector<ll> inv_b = poly_inv(b, deg);

```



```

67     vector<ll> res = multiply(a, inv_b);
68     ll inv2 = modInverse(2);
69     res.resize(deg);
70     for (int i = 0; i < deg; i++)
71     {
72         res[i] = (res[i] + b[i]) % MOD * inv2 % MOD;
73     }
74     return res;
75 }

```

#### 4.7.2 Lagrange

```

1  /**
2   * THE ULTIMATE CP TEMPLATE BLUEPRINT: LAGRANGE INTERPOLATION O(K)
3   *
4   * 1. WHAT PROBLEM DOES THIS SOLVE?
5   * - Keywords: "N-th term of a sequence", "Sum of  $i^p$ ", "Polynomial fitting".
6   * - Classic Scenarios: You know a sequence is generated by a polynomial of
7   *   degree K.
8   *   You need the exact value at a massive index X (e.g.,  $X = 10^{18}$ ), but you can
9   *   only easily compute the first K+1 values (at  $x = 0, 1, 2, \dots, K$ ).
10  * - The Magic: Standard Lagrange takes  $O(K^2)$ . Because our evaluation points are
11  *   strictly contiguous integers (0 to K), the products in the numerator and
12  *   denominator become factorials and prefix/suffix products. This collapses the
13  *   complexity to pure  $O(K)$ , allowing you to evaluate degree  $10^6$  polynomials in
14  *   less than a second.
15  *
16  * 2. HOW TO USE IT (THE BLACK BOX)
17  * - Query:
18  *   long long ans = lagrange_eval(Y, X);
19  *   // Y is a vector of size K+1 containing P(0), P(1), ..., P(K).
20  *   // X is the massive point you want to evaluate at.
21  *
22  * - Complexity:
23  *   Time:  $O(K)$  where K is the degree of the polynomial (size of Y - 1).
24  *   Space:  $O(K)$  for prefix/suffix and factorial arrays.
25  *
26  * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
27  * - Modulo: Adjust the global MOD variable if the problem uses  $1e9+7$  instead
28  *   of 998244353.
29  */
30 using ll = long long;
31 const ll MOD = 998244353;
32
33 ll binpow(ll base, ll exp)
34 {
35     ll res = 1;
36     base %= MOD;
37     while (exp > 0)
38     {

```

```

39         if (exp % 2 == 1)
40             res = (res * base) % MOD;
41         base = (base * base) % MOD;
42         exp /= 2;
43     }
44     return res;
45 }
46
47 ll modInverse(ll n)
48 {
49     return binpow(n, MOD - 2);
50 }
51
52 ll lagrange_eval(const vector<ll> &y, ll x)
53 {
54     int k = y.size() - 1;
55     if (x <= k)
56         return y[x];
57
58     vector<ll> pref(k + 1, 1), suff(k + 1, 1);
59     vector<ll> fact(k + 1, 1), invFact(k + 1, 1);
60
61     for (int i = 1; i <= k; i++)
62         fact[i] = fact[i - 1] * i % MOD;
63     invFact[k] = modInverse(fact[k]);
64     for (int i = k - 1; i >= 0; i--)
65         invFact[i] = invFact[i + 1] * (i + 1) % MOD;
66
67     pref[0] = x % MOD;
68     for (int i = 1; i <= k; i++)
69         pref[i] = pref[i - 1] * (x - i) % MOD;
70
71     suff[k] = (x - k) % MOD;
72     for (int i = k - 1; i >= 0; i--)
73         suff[i] = suff[i + 1] * (x - i) % MOD;
74
75     ll ans = 0;
76     for (int i = 0; i <= k; i++)
77     {
78         ll num = (i == 0 ? 1 : pref[i - 1]) * (i == k ? 1 : suff[i + 1]) % MOD;
79         ll den = invFact[i] * invFact[k - i] % MOD;
80         ll term = y[i] * num % MOD * den % MOD;
81         if ((k - i) % 2 == 1)
82         {
83             ans = (ans - term + MOD) % MOD;
84         }
85         else
86         {
87             ans = (ans + term) % MOD;
88         }
89     }
90     return ans;

```

91 }

### 4.7.3 Multipoint evaluation

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: MULTIPOINT EVALUATION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Evaluate polynomial at multiple points", "P(x_i)".
6  * - Classic Scenarios: You have a massive polynomial P(x) of degree N, and you
   need
7  * to find its exact value at M different points (Topic 340).
8  * - The Magic: Horner's method takes O(N * M) which TLEs. This uses a Subproduct
   Tree
9  * (a segment tree of polynomials) and polynomial division with remainder to
   process
10 * all queries simultaneously in O(N log^2 N).
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     vector<ll> results = multipoint_eval(P, X);
15 *     // results[i] contains P(X[i]) mod 998244353.
16 *
17 * - Complexity:
18 *     Time: O(N log^2 N)
19 *     Space: O(N log N)
20 *
21 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
22 * - Polynomial Modulo: This struct includes `poly_mod(A, B)` which finds the
   remainder
23 * of A(x) / B(x). This is an incredibly powerful tool on its own.
24 */
25
26 vector<ll> poly_rev(vector<ll> a)
27 {
28     reverse(a.begin(), a.end());
29     return a;
30 }
31
32 vector<ll> poly_mod(vector<ll> a, vector<ll> b)
33 {
34     while (a.size() > 1 && a.back() == 0)
35         a.pop_back();
36     while (b.size() > 1 && b.back() == 0)
37         b.pop_back();
38     if (a.size() < b.size())
39         return a;
40
41     int n = a.size(), m = b.size();
42     vector<ll> rev_a = poly_rev(a);
43     vector<ll> rev_b = poly_rev(b);

```

```

44
45     rev_a.resize(n - m + 1);
46     vector<ll> inv_rev_b = poly_inv(rev_b, n - m + 1);
47     vector<ll> q = multiply(rev_a, inv_rev_b);
48     q.resize(n - m + 1);
49     q = poly_rev(q);
50
51     vector<ll> qb = multiply(q, b);
52     vector<ll> r(m - 1, 0);
53     for (int i = 0; i < m - 1; i++)
54     {
55         r[i] = (a[i] - (i < qb.size() ? qb[i] : 0)) % MOD;
56         if (r[i] < 0)
57             r[i] += MOD;
58     }
59     while (r.size() > 1 && r.back() == 0)
60         r.pop_back();
61     return r;
62 }
63
64 vector<vector<ll>> tree;
65
66 void build_subproduct_tree(int node, int l, int r, const vector<ll> &x)
67 {
68     if (l == r)
69     {
70         tree[node] = {(MOD - x[l]) % MOD, 1};
71         return;
72     }
73     int mid = l + (r - l) / 2;
74     build_subproduct_tree(2 * node, l, mid, x);
75     build_subproduct_tree(2 * node + 1, mid + 1, r, x);
76     tree[node] = multiply(tree[2 * node], tree[2 * node + 1]);
77 }
78
79 void eval_subproduct_tree(int node, int l, int r, vector<ll> p, vector<ll> &ans)
80 {
81     p = poly_mod(p, tree[node]);
82     if (l == r)
83     {
84         ans[l] = p.empty() ? 0 : p[0];
85         return;
86     }
87     int mid = l + (r - l) / 2;
88     eval_subproduct_tree(2 * node, l, mid, p, ans);
89     eval_subproduct_tree(2 * node + 1, mid + 1, r, p, ans);
90 }
91
92 vector<ll> multipoint_eval(const vector<ll> &p, const vector<ll> &x)
93 {
94     if (x.empty())
95         return {};

```

```

96     int m = x.size();
97     tree.resize(4 * m);
98     build_subproduct_tree(1, 0, m - 1, x);
99     vector<ll> ans(m);
100     eval_subproduct_tree(1, 0, m - 1, p, ans);
101     return ans;
102 }

```

#### 4.7.4 Poly inv log

```

1  /**
2   * THE ULTIMATE CP TEMPLATE BLUEPRINT: POLYNOMIAL INV & LOG
3   *
4   * 1. WHAT PROBLEM DOES THIS SOLVE?
5   * - Keywords: "Generating Functions", "Formal Power Series (FPS)", "Divide
6     Polynomials".
7   * - Classic Scenarios: You have a generating function expressed as a fraction  $A(x)/B(x)$ 
8     and need the first  $N$  coefficients.
9   * - The Magic: Polynomial Inverse (Topic 334) uses Newton's Method to find  $B^{-1}(x)$ 
10     such that  $B(x) * B^{-1}(x) = 1 \bmod x^n$  in  $O(N \log N)$ . Polynomial Log (Topic
11     336)
12     uses calculus:  $\ln(A(x)) = \text{integral}(A'(x) * A^{-1}(x))$ .
13   *
14   * 2. HOW TO USE IT (THE BLACK BOX)
15   * - Query:
16     vector<ll> invA = poly_inv(A, N);
17     vector<ll> logA = poly_log(A, N);
18   *
19   * - Complexity:
20     Time:  $O(N \log N)$ 
21     Space:  $O(N)$ 
22   *
23   * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
24   * - Dependencies: Requires `ntt`, `multiply`, and `modInverse` from your FFT/NTT
25     snippet.
26   * - Constant Term: For poly_inv,  $A[0]$  MUST NOT be 0. For poly_log,  $A[0]$  MUST be
27     1.
28   */
29 using ll = long long;
30 const ll MOD = 998244353;
31 vector<ll> poly_inv(vector<ll> a, int deg)
32 {
33     if (deg == 1)
34         return {modInverse(a[0])};
35     vector<ll> b = poly_inv(a, (deg + 1) / 2);
36     int n = 1;
37     while (n < deg * 2)

```

```

36         n <= 1;
37     vector<ll> fa(a.begin(), a.begin() + min((int)a.size(), deg));
38     fa.resize(n);
39     vector<ll> fb = b;
40     fb.resize(n);
41     ntt(fa, false);
42     ntt(fb, false);
43     for (int i = 0; i < n; i++)
44     {
45         fb[i] = fb[i] * (2 - fa[i] * fb[i] % MOD + MOD) % MOD;
46     }
47     ntt(fb, true);
48     fb.resize(deg);
49     return fb;
50 }
51
52 vector<ll> poly_deriv(const vector<ll> &a)
53 {
54     if (a.empty())
55         return {};
56     vector<ll> res(max(1, (int)a.size() - 1));
57     for (int i = 1; i < a.size(); i++)
58     {
59         res[i - 1] = a[i] * i % MOD;
60     }
61     return res;
62 }
63
64 vector<ll> poly_integr(const vector<ll> &a)
65 {
66     if (a.empty())
67         return {0};
68     vector<ll> res(a.size() + 1, 0);
69     for (int i = 0; i < a.size(); i++)
70     {
71         res[i + 1] = a[i] * modInverse(i + 1) % MOD;
72     }
73     return res;
74 }
75
76 vector<ll> poly_log(vector<ll> a, int deg)
77 {
78     vector<ll> deriv = poly_deriv(a);
79     vector<ll> inv = poly_inv(a, deg);
80     vector<ll> res = poly_integr(multiply(deriv, inv));
81     res.resize(deg);
82     return res;
83 }

```

## 4.8 Probability

### 4.8.1 Expected occurence time of substring

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: EXPECTED OCCURRENCE TIME OF SUBSTRING
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Expected number of coin flips", "Typing monkeys", "Expected time
6  * - Classic Scenarios: A random generator picks characters from an alphabet of
7  *   size A.
8  *   What is the expected number of generated characters until string S appears?
9  * - The Magic: Bypasses  $O(N^3)$  Gaussian elimination. By setting up a fair casino
10 *   Martingale where gamblers bet on consecutive characters, the expected time
11 *   simplifies
12 *   to a summation over the borders (prefix == suffix) of the string.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Query:
16 *   long long expected_time = expected_string_occurrence(S, ALPHABET_SIZE,
17 *   MOD);
18 *
19 * - Complexity:
20 *   Time:  $O(|S|)$ 
21 *   Space:  $O(|S|)$ 
22 *
23 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
24 * - Alphabet Size: If flipping a coin, alphabet_size = 2. If rolling a die, 6.
25 * - Array Form: This strictly uses the KMP prefix function. The condition `j ==
26 *   pi[j-1]`
27 *   efficiently finds all prefix-suffix borders of the entire string.
28 */
29
30 using ll = long long;
31
32 vector<int> compute_pi(const string &s)
33 {
34     int n = s.length();
35     vector<int> pi(n);
36     for (int i = 1; i < n; i++)
37     {
38         int j = pi[i - 1];
39         while (j > 0 && s[i] != s[j])
40             j = pi[j - 1];
41         if (s[i] == s[j])
42             j++;
43         pi[i] = j;
44     }
45     return pi;
46 }

```

```

44 ll expected_string_occurrence(const string &s, ll alphabet_size, ll mod)
45 {
46     vector<int> pi = compute_pi(s);
47     int n = s.length();
48
49     auto binpow = [&](ll base, ll exp)
50     {
51         ll res = 1;
52         base %= mod;
53         while (exp > 0)
54         {
55             if (exp % 2 == 1)
56                 res = (res * base) % mod;
57             base = (base * base) % mod;
58             exp /= 2;
59         }
60         return res;
61     };
62
63     ll expected_time = 0;
64     int current_border = n;
65
66     while (current_border > 0)
67     {
68         expected_time = (expected_time + binpow(alphabet_size, current_border)) %
69             mod;
70         current_border = pi[current_border - 1];
71     }
72
73     return expected_time;
74 }

```

## 5 Number Theory

### 5.1 Fibonacci

#### 5.1.1 Fast doubling fibonacci

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: FAST DOUBLING FIBONACCI
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "N-th Fibonacci", "Massive N".
6  * - Classic Scenarios: You need  $F_N$  modulo M where  $N = 10^{18}$ .
7  * - The Magic: Derives  $F_{2k}$  and  $F_{2k+1}$  directly from  $F_k$  and  $F_{k+1}$ .
8  *   It completely sidesteps matrix overhead, making it drastically faster
9  *   and much shorter to type than a robust matrix exponentiation struct.
10 *
11 * 2. HOW TO USE IT (THE BLACK BOX)
12 * - Query:

```

```

13 *      pair<ll, ll> ans = fast_doubling(N, M);
14 * - Result Breakdown:
15 *      ans.first = F_N
16 *      ans.second = F_{N+1}
17 *
18 * - Complexity:
19 *      Time: Strictly  $O(\log N)$ .
20 *      Space:  $O(\log N)$  for the recursion stack.
21 */
22
23 using ll = long long;
24
25 pair<ll, ll> fast_doubling(ll n, ll mod)
26 {
27     if (n == 0)
28         return {0, 1};
29     auto p = fast_doubling(n >> 1, mod);
30     ll c = p.first * (2 * p.second - p.first + mod) % mod;
31     ll d = (p.first * p.first + p.second * p.second) % mod;
32     if (n & 1)
33         return {d, (c + d) % mod};
34     return {c, d};
35 }

```

### 5.1.2 PHI field

```

1 /**
2 *  THE ULTIMATE CP TEMPLATE BLUEPRINT: PHI FIELD (Z[sqrt(5)])
3 *
4 * 1. WHAT PROBLEM DOES THIS SOLVE?
5 * - Keywords: "Fibonacci sum of powers", "Binet's formula modulo M".
6 * - Classic Scenarios: You need to evaluate algebraic expressions involving
7 *   Fibonacci numbers, such as  $\sum(F_i^K) \bmod M$ .
8 * - The Magic: It operates exactly like complex numbers, but instead of  $i^2 = -1$ ,
9 *   it maps  $a + b\sqrt{5}$  under modulo M. This allows you to directly use Binet's
10 *   formula natively even if 5 is a quadratic non-residue modulo M!
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *      Phi res = power_phi({A, B}, exp, MOD);
15 *
16 * - Complexity:
17 *      Time:  $O(\log(\text{exp}))$  per multiplication.
18 *      Space:  $O(1)$ 
19 */
20
21 using ll = long long;
22
23 struct Phi

```

```

24 {
25     ll a, b;
26 };
27
28 Phi mult_phi(Phi x, Phi y, ll mod)
29 {
30     return {
31         (x.a * y.a + 5 * x.b % mod * y.b) % mod,
32         (x.a * y.b + x.b * y.a) % mod};
33 }
34
35 Phi power_phi(Phi base, ll exp, ll mod)
36 {
37     Phi res = {1, 0};
38     base.a %= mod;
39     base.b %= mod;
40     while (exp > 0)
41     {
42         if (exp % 2 == 1)
43             res = mult_phi(res, base, mod);
44         base = mult_phi(base, base, mod);
45         exp /= 2;
46     }
47     return res;
48 }

```

## 5.2 Linear diophantine equation

### 5.2.1 Floor sum

```

1 /**
2 *  THE ULTIMATE CP TEMPLATE BLUEPRINT: FLOOR SUM
3 *
4 * 1. WHAT PROBLEM DOES THIS SOLVE?
5 * - Keywords: " $ax + by \leq c$ ", "Lattice points under line", "Sum of arithmetic
6 *   progression mod M".
7 * - Classic Scenarios: Compute  $\sum_{i=0}^{n-1} \text{floor}((a * i + b) / m)$ .
8 * - The Magic: It operates exactly like the Euclidean algorithm, trading
9 *   dimensions
10 *   and folding the geometry to evaluate the entire sum logarithmically.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *      long long ans = floor_sum(n, m, a, b);
15 *
16 * - Complexity:
17 *      Time:  $O(\log(\min(m, a)))$ 
18 *      Space:  $O(\log(\min(m, a)))$  due to recursion.
19 *
20 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)

```

```

19 * - Equivalent to  $ax + by \leq c$ : For a line equation in the first quadrant,
    rearrange
20 * it to  $y \leq (c - ax) / b$ , which directly translates to summing  $\text{floor}((c - a * x) / b)$ 
    over all valid  $x$ .
21 *
22 */
23
24 using ll = long long;
25
26 ll floor_sum(ll n, ll m, ll a, ll b)
27 {
28     ll ans = 0;
29     if (a >= m)
30     {
31         ans += (n - 1) * n / 2 * (a / m);
32         a %= m;
33     }
34     if (b >= m)
35     {
36         ans += n * (b / m);
37         b %= m;
38     }
39     ll y_max = (a * n + b) / m;
40     ll x_max = y_max * m - b;
41     if (y_max == 0)
42         return ans;
43     ans += (n - (x_max + a - 1) / a) * y_max;
44     ans += floor_sum(y_max, a, m, (a - x_max % a) % a);
45     return ans;
46 }

```

## 5.2.2 Generalized floor sum

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: GENERALIZED FLOOR SUM
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Sum of  $i * \text{floor}$ ", "Sum of squares of floors".
6  * - Classic Scenarios: A step above standard floor sum, used when evaluating
7  *   complex grid summations or polynomial constraints under a modulo line.
8  * - The Magic: Through a complex series of algebraic expansions and Euclidean
9  *   domain folding, it simultaneously resolves three interconnected sums in
10  *   pure logarithmic time.
11  *
12  * 2. HOW TO USE IT (THE BLACK BOX)
13  * - Query:
14  *     FSR ans = gen_floor_sum(n, a, b, c);
15  * - Result Breakdown:
16  *     ans.f =  $\sum_{i=0}^n \text{floor}((a*i+b)/c)$ 
17  *     ans.g =  $\sum_{i=0}^n i * \text{floor}((a*i+b)/c)$ 
18  *     ans.h =  $\sum_{i=0}^n \text{floor}((a*i+b)/c)^2$ 

```

```

19 *
20 * - Complexity:
21 *     Time:  $O(\log(\min(a, c)))$ 
22 *     Space:  $O(\log(\min(a, c)))$ 
23 *
24 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
25 * - Inclusive Bounds: This specifically calculates for  $i = 0$  to  $N$  (inclusive).
26 *   If your problem is 1-based, evaluate for  $N$  and subtract the  $i=0$  term, or
27 *   use  $N-1$  if the upper bound is exclusive.
28 */
29
30 using ll = long long;
31
32 struct FSR
33 {
34     ll f, g, h;
35 };
36
37 FSR gen_floor_sum(ll n, ll a, ll b, ll c)
38 {
39     ll ac = a / c, bc = b / c;
40     if (a == 0)
41     {
42         ll f = (n + 1) * bc;
43         ll g = bc * n * (n + 1) / 2;
44         ll h = (n + 1) * bc * bc;
45         return {f, g, h};
46     }
47     if (a >= c || b >= c)
48     {
49         FSR res = gen_floor_sum(n, a % c, b % c, c);
50         ll f = res.f + n * (n + 1) / 2 * ac + (n + 1) * bc;
51         ll g = res.g + ac * n * (n + 1) * (2 * n + 1) / 6 + bc * n * (n + 1) / 2;
52         ll h = res.h + ac * ac * n * (n + 1) * (2 * n + 1) / 6 + bc * bc * (n + 1) + 2 * ac * bc * n * (n + 1) / 2 + 2 * ac * res.g + 2 * bc * res.f;
53         return {f, g, h};
54     }
55     ll m = (a * n + b) / c;
56     FSR res = gen_floor_sum(m - 1, c, c - b - 1, a);
57     ll f = n * m - res.f;
58     ll g = (m * n * (n + 1) - res.f - res.h) / 2;
59     ll h = n * m * m - 2 * res.g - res.f;
60     return {f, g, h};
61 }

```

## 5.2.3 LDE

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: BOUNDED LDE2
3  *

```

```

4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Solutions in range", " $ax + by = c$ ", "Bounded variables".
6  * - Classic Scenarios: You need the exact count of integer pairs  $(x, y)$ 
   * satisfying
7  *  $ax + by = c$  such that  $\min_x \leq x \leq \max_x$  and  $\min_y \leq y \leq \max_y$ .
8  * - The Magic: It finds a base solution using Extended Euclidean Algorithm, then
9  * mathematically shifts the solution using  $(b/g)$  and  $(a/g)$  to find the
   * absolute
10 * minimum and maximum valid boundaries, intersecting them safely.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     long long total = count_solutions(a, b, c, min_x, max_x, min_y, max_y);
15 *
16 * - Complexity:
17 *     Time:  $O(\log(\min(a, b)))$ 
18 *     Space:  $O(1)$ 
19 *
20 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
21 * - Zero Handling: Safely handles edge cases where  $a = 0$  or  $b = 0$ , which often
22 * crash naive implementations.
23 */
24
25 using ll = long long;
26
27 ll extgcd(ll a, ll b, ll &x, ll &y)
28 {
29     if (b == 0)
30     {
31         x = 1;
32         y = 0;
33         return a;
34     }
35     ll x1, y1;
36     ll d = extgcd(b, a % b, x1, y1);
37     x = y1;
38     y = x1 - y1 * (a / b);
39     return d;
40 }
41
42 void shift_solution(ll &x, ll &y, ll a, ll b, ll cnt)
43 {
44     x += cnt * b;
45     y -= cnt * a;
46 }
47
48 ll count_solutions(ll a, ll b, ll c, ll min_x, ll max_x, ll min_y, ll max_y)
49 {
50     if (a == 0 && b == 0)
51     {
52         if (c == 0)
53             return (max_x - min_x + 1) * (max_y - min_y + 1);
54         return 0;
55     }
56     if (a == 0)
57     {
58         if (c % b == 0 && c / b >= min_y && c / b <= max_y)
59             return max_x - min_x + 1;
60         return 0;
61     }
62     if (b == 0)
63     {
64         if (c % a == 0 && c / a >= min_x && c / a <= max_x)
65             return max_y - min_y + 1;
66         return 0;
67     }
68
69     ll x, y;
70     ll g = extgcd(abs(a), abs(b), x, y);
71     if (c % g != 0)
72         return 0;
73
74     x *= c / g;
75     y *= c / g;
76     if (a < 0)
77         x = -x;
78     if (b < 0)
79         y = -y;
80
81     ll a_g = a / g, b_g = b / g;
82     ll sign_a = a > 0 ? +1 : -1;
83     ll sign_b = b > 0 ? +1 : -1;
84
85     shift_solution(x, y, a_g, b_g, (min_x - x) / b_g);
86     if (x < min_x)
87         shift_solution(x, y, a_g, b_g, sign_b);
88     if (x > max_x)
89         return 0;
90     ll lx1 = x;
91
92     shift_solution(x, y, a_g, b_g, (max_x - x) / b_g);
93     if (x > max_x)
94         shift_solution(x, y, a_g, b_g, -sign_b);
95     ll rx1 = x;
96
97     shift_solution(x, y, a_g, b_g, -(min_y - y) / a_g);
98     if (y < min_y)
99         shift_solution(x, y, a_g, b_g, -sign_a);
100    if (y > max_y)
101        return 0;
102    ll lx2 = x;
103
104    shift_solution(x, y, a_g, b_g, -(max_y - y) / a_g);
105    if (y > max_y)

```

```

106     shift_solution(x, y, a_g, b_g, sign_a);
107     ll rx2 = x;
108
109     if (lx2 > rx2)
110         swap(lx2, rx2);
111     ll lx = max(lx1, lx2);
112     ll rx = min(rx1, rx2);
113
114     if (lx > rx)
115         return 0;
116     return (rx - lx) / abs(b_g) + 1;
117 }

```

## 5.2.4 Pell equation

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: PELL'S EQUATION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: " $x^2 - d*y^2 = 1$ ", "Fundamental Solution".
6  * - Classic Scenarios: Find the smallest positive integer solution (x, y)
7  *   satisfying Pell's equation where d is a non-square positive integer.
8  * - The Magic: It generates the continued fraction expansion of  $\sqrt{d}$ .
9  * The convergents ( $h_i / k_i$ ) of this fraction are guaranteed to eventually
10 * satisfy the equation.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     pair<__int128, __int128> ans = solve_pell(d);
15 *     // Returns {x, y}. If d is a perfect square, returns {-1, -1}.
16 *
17 * - Complexity:
18 *     Time:  $O(\sqrt{d} * \log(d))$ 
19 *     Space:  $O(1)$ 
20 *
21 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
22 * - 128-bit Return Type: Fundamental solutions to Pell's equation grow
23 *   exponentially fast. Even for small d (e.g.,  $d = 61$ ), x and y can easily
24 *   overflow 64-bit integers. __int128 is highly recommended here.
25 * - Generating More Solutions: If (x1, y1) is the fundamental solution,
26 *   the next solution (x_k, y_k) is  $(x_1 + y_1*\sqrt{d})^k$ .
27 */
28
29 using u128 = __int128;
30 using ll = long long;
31
32 pair<u128, u128> solve_pell(ll d)
33 {
34     ll sq = sqrt(d);
35     if (sq * sq == d)
36         return {-1, -1};

```

```

37
38     ll m = 0, d_den = 1, a = sq;
39     u128 num1 = 1, num0 = a;
40     u128 den1 = 0, den0 = 1;
41
42     while (num0 * num0 - (u128)d * den0 * den0 != 1)
43     {
44         m = d_den * a - m;
45         d_den = (d - m * m) / d_den;
46         a = (sq + m) / d_den;
47
48         u128 num2 = num1;
49         num1 = num0;
50         num0 = (u128)a * num1 + num2;
51
52         u128 den2 = den1;
53         den1 = den0;
54         den0 = (u128)a * den1 + den2;
55     }
56     return {num0, den0};
57 }

```

## 5.3 Misc

### 5.3.1 Min X in Mod range

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: MINIMUM X IN MODULO RANGE
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Smallest x", " $L \leq ax \bmod m \leq R$ ", "Min of Mod of AP".
6  * - Classic Scenarios: Finding the first index where an arithmetic progression
7  *   wraps around a modulo into a specific target window.
8  * - The Magic: It uses a recursive Euclidean-style domain folding. Instead
9  *   of simulating the progression, it inverts the bounds and steps to find
10 *   the exact minimum x in  $O(\log M)$  time.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     long long x = min_x_in_mod_range(A, M, L, R);
15 *     // Returns -1 if no such x exists.
16 *
17 * - Complexity:
18 *     Time:  $O(\log M)$ 
19 *     Space:  $O(\log M)$  for recursion.
20 *
21 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
22 * - Min of Mod of AP (Topic 295): If you just need the absolute minimum of
23 *    $(A*x) \bmod M$  for x in  $[0, N]$ , you can binary search the answer V in  $[0, M-1]$ 
24 *   and use this function to check if a valid  $x \leq N$  exists for  $L=0, R=V$ .
25 */

```



```

26
27 using ll = long long;
28
29 ll min_x_in_mod_range(ll a, ll m, ll l, ll r)
30 {
31     if (l == 0)
32         return 0;
33     if (a == 0)
34         return -1;
35     ll step = (l + a - 1) / a;
36     if (step * a <= r)
37         return step;
38     ll b = m % a;
39     ll res = min_x_in_mod_range(b, a, a - r % a, a - l % a);
40     if (res == -1)
41         return -1;
42     return (l + res * b + a - 1) / a;
43 }

```

### 5.3.2 SIMPLEST FRACTION

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: STERN-BROCOT TREE (SIMPLEST FRACTION)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Stern-Brocot", "Simplest Fraction", "Rational Approximation".
6  * - Classic Scenarios: Find the fraction P/Q with the absolute smallest
7  * denominator Q that strictly lies between two other fractions A/B and C/D.
8  * - The Magic: Instead of simulating the binary search descent down the
9  * Stern-Brocot tree (which can TLE in  $O(\max(P, Q))$ ), it uses a continued
10 * fraction / Euclidean acceleration to jump down the branches.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     pair<long long, long long> ans = simplest_fraction(A, B, C, D);
15 *     // Returns {P, Q} such that  $A/B < P/Q < C/D$ .
16 *
17 * - Complexity:
18 *     Time:  $O(\log(\max(A, B, C, D)))$ 
19 *     Space:  $O(\log(\max))$  for recursion.
20 *
21 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
22 * - Fraction format: Ensure your input fractions A/B and C/D are simplified
23 * and that A/B is strictly less than C/D before passing them in.
24 */
25
26 using ll = long long;
27
28 pair<ll, ll> simplest_fraction(ll a, ll b, ll c, ll d)
29 {
30     if (a / b < c / d)

```

```

31     return {a / b + 1, 1};
32     if (a == 0)
33         return {1, d / c + 1};
34     auto res = simplest_fraction(d, c % d, b, a % b);
35     return {res.second + (a / b) * res.first, res.first};
36 }

```

## 5.4 Modular arithmetic

### 5.4.1 Baby step giant step

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: BABY-STEP GIANT-STEP (DISCRETE LOG)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Discrete Logarithm", "Find the exponent", " $A^x = B \pmod M$ ".
6  * - Classic Scenarios: You are given A, B, and M, and need to find the minimum
7  * non-negative integer x such that  $A^x \equiv B \pmod M$ .
8  * - The Magic: It trades space for time. By rewriting  $x = np - q$  (where  $n = \sqrt{M}$ ),
9  * the equation becomes  $A^{np} \equiv B \cdot A^{-q} \pmod M$ . We precompute the right side
10 * (Baby Steps) and store them in a hash map, then calculate the left side
11 * (Giant Steps) and check for collisions.
12 *
13 * 2. HOW TO USE IT (THE BLACK BOX)
14 * - Query:
15 *     long long ans = solve_bsgs(A, B, M);
16 *     // Returns -1 if no solution exists.
17 *
18 * - Complexity:
19 *     Time:  $O(\sqrt{M} \cdot \log(\sqrt{M}))$  due to the map.  $O(\sqrt{M})$  if using
20 * unordered_map.
21 *     Space:  $O(\sqrt{M})$  to store the baby steps.
22 *
23 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
24 * - Coprime Requirement: This standard version assumes  $\gcd(A, M) == 1$ . If A and
25 * M
26 * are not coprime, you need the Extended BSGS algorithm (which involves
27 * dividing
28 * out the GCDs until they are coprime).
29 */
30
31 using ll = long long;
32
33 ll solve_bsgs(ll a, ll b, ll m)
34 {
35     a %= m, b %= m;
36     if (b == 1 || m == 1)
37         return 0;
38     return 0;
39
40     ll n = sqrt(m) + 1;

```

```

37 map<ll, ll> vals;
38 ll cur = b;
39 for (ll q = 0; q <= n; ++q)
40 {
41     vals[cur] = q;
42     cur = (cur * a) % m;
43 }
44
45 ll a_n = 1;
46 for (ll i = 0; i < n; ++i)
47 {
48     a_n = (a_n * a) % m;
49 }
50
51 cur = 1;
52 for (ll p = 1; p <= n; ++p)
53 {
54     cur = (cur * a_n) % m;
55     if (vals.count(cur))
56     {
57         ll ans = n * p - vals[cur];
58         return ans;
59     }
60 }
61 return -1;
62 }

```

#### 5.4.2 Big Mod Power

```

1 /// if we need to calculate (n % m) if EX: n = 10^10000 and m = 10^9
2 ll Big_Mod_Power(string n, ll m)
3 {
4     ll x = 0;
5     for(ll i = 0; i < n.size(); i++)
6     {
7         x = ((x * 10) + (n[i] - '0'));
8         x %= m;
9     }
10    return x; /// return n % m
11 }

```

#### 5.4.3 CRT

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: CHINESE REMAINDER THEOREM (GENERAL)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "System of congruences", " $x = a_1 \bmod m_1$ ,  $x = a_2 \bmod m_2$ ".
6  * - The Magic: Unlike basic CRT which fails if the moduli are not coprime,

```

```

7  * this robust version iteratively combines the equations using the Extended
8  * Euclidean algorithm. It safely catches impossible systems.
9  *
10 * 2. HOW TO USE IT (THE BLACK BOX)
11 * - Query:
12 *     pair<ll, ll> ans = crt(A, M);
13 * - Result: Returns {remainder, modulo}. If no solution exists, returns {-1,
14 *     -1}.
15 */
16 using ll = long long;
17
18 ll extgcd_crt(ll a, ll b, ll &x, ll &y)
19 {
20     if (b == 0)
21     {
22         x = 1;
23         y = 0;
24         return a;
25     }
26     ll x1, y1;
27     ll d = extgcd_crt(b, a % b, x1, y1);
28     x = y1;
29     y = x1 - y1 * (a / b);
30     return d;
31 }
32
33 pair<ll, ll> crt(const vector<ll> &a, const vector<ll> &m)
34 {
35     if (a.empty())
36         return {0, 1};
37     ll rem = a[0], mod = m[0];
38     for (size_t i = 1; i < a.size(); i++)
39     {
40         ll x, y;
41         ll g = extgcd_crt(mod, m[i], x, y);
42         if ((a[i] - rem) % g != 0)
43             return {-1, -1};
44         ll target = (a[i] - rem) / g;
45         ll m_i_g = m[i] / g;
46         x = (x % m_i_g + m_i_g) % m_i_g;
47         x = (x * (target % m_i_g + m_i_g)) % m_i_g;
48         rem += mod * x;
49         mod *= m_i_g;
50         rem = (rem % mod + mod) % mod;
51     }
52     return {rem, mod};
53 }

```

#### 5.4.4 Discrete root

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DISCRETE ROOT (K-TH ROOT)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "K-th Root Modulo P", " $x^K = A \pmod P$ ".
6  * - Classic Scenarios: You need to reverse an exponentiation where the modulo is
   prime.
7  * - The Magic: It finds a primitive root g of P. Since x can be represented as
   g^y,
8  * the equation becomes  $(g^y)^K = A \pmod P \Rightarrow (g^K)^y = A \pmod P$ . This
   perfectly
9  * reduces the problem into a Discrete Logarithm (BSGS) problem!
10 *
11 * 2. HOW TO USE IT (THE BLACK BOX)
12 * - Query:
13 *     long long root = discrete_root(K, A, P);
14 *     // Returns -1 if no root exists.
15 *
16 * - Complexity:
17 *     Time:  $O(\sqrt{P} * \log(\sqrt{P}))$ .
18 *     Space:  $O(\sqrt{P})$ .
19 *
20 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
21 * - Dependencies: This snippet requires both `custom_pow` and `solve_bsgs` to be
22 * available in your code (from the snippets above).
23 */
24
25 using ll = long long;
26
27 ll primitive_root(ll p)
28 {
29     vector<ll> fact;
30     ll phi = p - 1, n = phi;
31     for (ll i = 2; i * i <= n; ++i)
32     {
33         if (n % i == 0)
34         {
35             fact.push_back(i);
36             while (n % i == 0)
37                 n /= i;
38         }
39     }
40     if (n > 1)
41         fact.push_back(n);
42
43     for (ll res = 2; res <= p; ++res)
44     {
45         bool ok = true;
46         for (size_t i = 0; i < fact.size() && ok; ++i)
47         {
48             ok &= custom_pow(res, phi / fact[i], p) != 1;
49         }

```

```

50         if (ok)
51             return res;
52     }
53     return -1;
54 }
55
56 ll discrete_root(ll k, ll a, ll p)
57 {
58     if (a == 0)
59         return 0;
60     if (k == 0)
61         return (a == 1) ? 1 : -1;
62
63     ll g = primitive_root(p);
64     if (g == -1)
65         return -1;
66
67     ll base = custom_pow(g, k, p);
68     ll y = solve_bsgs(base, a, p);
69     if (y == -1)
70         return -1;
71
72     return custom_pow(g, y, p);
73 }

```

#### 5.4.5 Distinct kth power

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DISTINCT K-TH POWERS &  $x^2 = 1 \pmod M$ 
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "How many perfect K-th powers modulo P", "Square roots of 1 modulo
   M".
6  * - The Magic: These are closed-form formula generators. Calculating the number
7  * of valid values for  $x^2 = 1 \pmod M$  depends heavily on the prime factorization
8  * of M (specifically treating the powers of 2 differently than odd primes).
9  *
10 * 2. HOW TO USE IT (THE BLACK BOX)
11 * - Query:
12 *     ll ans1 = distinct_kth_powers_prime_mod(K, P); // Requires P to be a
   prime
13 *     ll ans2 = solutions_x2_1_mod_m(M);
14 */
15
16 using ll = long long;
17
18 ll distinct_kth_powers_prime_mod(ll k, ll p)
19 {
20     return (p - 1) / __gcd(k, p - 1) + 1;
21 }
22

```

```

23 ll solutions_x2_1_mod_m(ll m)
24 {
25     if (m <= 1)
26         return 1;
27     ll count = 1;
28     ll temp = m;
29     int e2 = 0;
30     while (temp % 2 == 0)
31     {
32         e2++;
33         temp /= 2;
34     }
35     if (e2 == 2)
36         count *= 2;
37     else if (e2 >= 3)
38         count *= 4;
39
40     for (ll i = 3; i * i <= temp; i += 2)
41     {
42         if (temp % i == 0)
43         {
44             count *= 2;
45             while (temp % i == 0)
46                 temp /= i;
47         }
48     }
49     if (temp > 1)
50         count *= 2;
51     return count;
52 }

```

#### 5.4.6 Linear congruence

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: LINEAR CONGRUENCE EQUATION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Solve  $ax = b \pmod m$ ".
6  * - Classic Scenarios: You are given  $a$ ,  $b$ , and  $m$ , and need to find  $x$ .
7  * - The Magic: It uses the Extended Euclidean Algorithm to solve  $ax + my = b$ .
8  * It handles cases where  $\gcd(a, m) \neq 1$ . If a solution exists, it returns
9  * the base solution  $x_0$  and the number of distinct solutions modulo  $m$ .
10 *
11 * 2. HOW TO USE IT (THE BLACK BOX)
12 * - Query:
13 *     ll x, num_solutions;
14 *     bool possible = linear_congruence(a, b, m, x, num_solutions);
15 */
16
17 using ll = long long;
18

```

```

19 ll extgcd(ll a, ll b, ll &x, ll &y)
20 {
21     if (b == 0)
22     {
23         x = 1;
24         y = 0;
25         return a;
26     }
27     ll x1, y1;
28     ll d = extgcd(b, a % b, x1, y1);
29     x = y1;
30     y = x1 - y1 * (a / b);
31     return d;
32 }
33
34 bool linear_congruence(ll a, ll b, ll m, ll &x, ll &num_solutions)
35 {
36     ll x0, y0;
37     ll g = extgcd(abs(a), m, x0, y0);
38     if (b % g != 0)
39     {
40         x = 0;
41         num_solutions = 0;
42         return false;
43     }
44     x0 = (x0 % (m / g) + (m / g)) % (m / g);
45     x = (x0 * (b / g)) % (m / g);
46     num_solutions = g;
47     return true;
48 }

```

#### 5.4.7 MULTIPLICATIVE ORDER CARMICHAEL'S LAMBDA

#### 5.4.8 discrete sqrt

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: TONELLI-SHANKS ALGORITHM
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Quadratic Residue", "Modular Square Root", " $x^2 = N \pmod P$ ".
6  * - Classic Scenarios: You are given an integer  $N$  and an odd prime  $P$ , and you
7  * need to find  $x$  such that  $x^2 \equiv N \pmod P$ .
8  * - The Magic: It factors  $P-1$  as  $Q \cdot 2^S$ . By finding a quadratic non-residue  $Z$ ,
9  * it iteratively manipulates the exponents to isolate the square root. It is
10 * a beautiful but mathematically dense algorithm that you do not want to debug
    live.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)

```

```

13 * - Query:
14 *     long long root = tonelli_shanks(N, P);
15 *     // Returns -1 if no square root exists.
16 *     // If 'root' is a solution, 'P - root' is the other valid solution.
17 *
18 * - Complexity:
19 *     Time:  $O(\log^2 P)$  worst case,  $O(\log P)$  on average. Extremely fast.
20 *     Space:  $O(1)$ 
21 *
22 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
23 * - Prime Modulo Only: This specifically requires P to be an odd prime. If you
24 *   need to find roots modulo a composite number, you must use Tonelli-Shanks
25 *   on its prime factors and combine the results using the Chinese Remainder
26 *   Theorem (CRT).
27 */
28 using ll = long long;
29
30 ll custom_pow(ll base, ll exp, ll mod)
31 {
32     ll res = 1;
33     base %= mod;
34     while (exp > 0)
35     {
36         if (exp % 2 == 1)
37             res = (res * base) % mod;
38         base = (base * base) % mod;
39         exp /= 2;
40     }
41     return res;
42 }
43
44 ll tonelli_shanks(ll n, ll p)
45 {
46     if (n == 0)
47         return 0;
48     if (custom_pow(n, (p - 1) / 2, p) != 1)
49         return -1;
50
51     ll q = p - 1, s = 0;
52     while (q % 2 == 0)
53     {
54         s++;
55         q /= 2;
56     }
57
58     if (s == 1)
59         return custom_pow(n, (p + 1) / 4, p);
60
61     ll z = 2;
62     while (custom_pow(z, (p - 1) / 2, p) != p - 1)
63     {

```

```

64         z++;
65     }
66
67     ll c = custom_pow(z, q, p);
68     ll r = custom_pow(n, (q + 1) / 2, p);
69     ll t = custom_pow(n, q, p);
70     ll m = s;
71
72     while (t != 1)
73     {
74         ll t2 = (t * t) % p;
75         ll i = 1;
76         for (; i < m; i++)
77         {
78             if (t2 == 1)
79                 break;
80             t2 = (t2 * t2) % p;
81         }
82
83         ll b = c;
84         for (int j = 0; j < m - i - 1; j++)
85         {
86             b = (b * b) % p;
87         }
88
89         r = (r * b) % p;
90         c = (b * b) % p;
91         t = (t * c) % p;
92         m = i;
93     }
94     return r;
95 }

```

## 5.5 Multiplicative functions

### 5.5.1 Min 25 sieve

### 5.5.2 Power tower

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: POWER TOWER (GENERALIZED EULER)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Power Tower", " $A^B^C^D \bmod M$ ".
6  * - Classic Scenarios: You are given an array of numbers and asked to compute
7  *   their
8  *   power tower  $(a[0] \wedge (a[1] \wedge (a[2] \wedge \dots)))$  modulo M.

```

```

8  * - The Magic: It applies the Generalized Euler Theorem to safely reduce
   exponents
9  * using the Phi function, effectively capping the recursion depth at  $O(\log M)$ 
10 *
11 * 2. HOW TO USE IT (THE BLACK BOX)
12 * - Query:
13 *     long long ans = power_tower(arr, 0, M);
14 *
15 * - Complexity:
16 *     Time:  $O(\log M * \sqrt{M})$ 
17 *     Space:  $O(\log M)$  for the recursion stack.
18 */
19
20 using ll = long long;
21
22 ll phi(ll n)
23 {
24     ll res = n;
25     for (ll i = 2; i * i <= n; i++)
26     {
27         if (n % i == 0)
28         {
29             while (n % i == 0)
30                 n /= i;
31             res -= res / i;
32         }
33     }
34     if (n > 1)
35         res -= res / n;
36     return res;
37 }
38
39 ll custom_pow(ll base, ll exp, ll mod)
40 {
41     ll res = 1;
42     base = (base >= mod) ? (base % mod + mod) : base;
43     while (exp > 0)
44     {
45         if (exp % 2 == 1)
46         {
47             res *= base;
48             res = (res >= mod) ? (res % mod + mod) : res;
49         }
50         base *= base;
51         base = (base >= mod) ? (base % mod + mod) : base;
52         exp /= 2;
53     }
54     return res;
55 }
56
57 ll power_tower(const vector<ll> &arr, int idx, ll mod)
58 {

```

```

59     if (idx == arr.size() || mod == 1)
60         return 1;
61     if (arr[idx] == 1)
62         return 1;
63     ll next_mod = phi(mod);
64     ll exponent = power_tower(arr, idx + 1, next_mod);
65     return custom_pow(arr[idx], exponent, mod);
66 }

```

### 5.5.3 phi

```

1  // computes the number of coprimes number with n up to n
2
3  const int N = 5000006;
4  ll phi[N];
5
6  void sieve()
7  {
8      for (int i = 0; i < N; i++)
9      {
10         phi[i] = i;
11     }
12     for (int i = 2; i < N; i++)
13     {
14         if (phi[i] == i)
15         {
16             for (int j = i; j < N; j += i)
17             {
18                 phi[j] -= phi[j] / i;
19             }
20         }
21     }
22 }
23
24 ll phi(ll n)
25 {
26     ll ret = n;
27     for (ll i = 2; i * i <= n; i++)
28     {
29         if (n % i == 0)
30         {
31             while (n % i == 0)
32                 n /= i;
33             ret -= ret / i;
34         }
35     }
36     if (n > 1)
37         ret -= ret / n;
38     return ret;
39 }

```

## 5.6 Primes And Divisors

### 5.6.1 Basic

```

1 struct NumberTheory {
2     int n;
3     vector<int> p, spf, phi_arr, fac, inv, finv;
4     vector<bool> isPrime;
5     vector<vector<int>> divisor;
6
7     NumberTheory(int max_n = 1000000, bool pre_divs = false) {
8         n = max_n;
9         isPrime.assign(n + 1, true);
10        spf.assign(n + 1, 0);
11        phi_arr.assign(n + 1, 0);
12        isPrime[0] = isPrime[1] = false;
13
14        for (int i = 0; i <= n; i++) phi_arr[i] = i;
15
16        for (int i = 2; i <= n; i++) {
17            if (isPrime[i]) {
18                p.push_back(i);
19                spf[i] = i;
20                phi_arr[i] -= phi_arr[i] / i;
21            }
22            for (int j = 0; j < (int) p.size() && i * p[j] <= n; j++) {
23                isPrime[i * p[j]] = false;
24                spf[i * p[j]] = p[j];
25                if (i % p[j] == 0) {
26                    phi_arr[i * p[j]] = phi_arr[i] * p[j];
27                    break;
28                } else {
29                    phi_arr[i * p[j]] = phi_arr[i] * phi_arr[p[j]];
30                }
31            }
32        }
33
34        fac.assign(n + 1, 1);
35        inv.assign(n + 1, 1);
36        finv.assign(n + 1, 1);
37        for (int i = 1; i <= n; i++) fac[i] = mul(fac[i - 1], i);
38        for (int i = 2; i <= n; i++) inv[i] = (MOD - (MOD / i) * inv[MOD % i] %
39            MOD) % MOD;
40        for (int i = 2; i <= n; i++) finv[i] = mul(finv[i - 1], inv[i]);
41
42        if (pre_divs) {
43            divisor.resize(n + 1);
44            for (int i = 1; i <= n; i++) {
45                for (int j = i; j <= n; j += i) {
46                    divisor[j].push_back(i);
47                }
48            }
49        }
50    }
51
52    int add(int a, int b) { return ((a % MOD) + (b % MOD)) % MOD; }
53    int sub(int a, int b) { return (((a - b) % MOD) + MOD) % MOD; }
54    int mul(int a, int b) { return ((a % MOD) * (b % MOD)) % MOD; }
55
56    int power(int b, int pow) {
57        b %= MOD;
58        int res = 1;
59        while (pow) {
60            if (pow & 1) res = mul(res, b);
61            b = mul(b, b);
62            pow >>= 1;
63        }
64        return res;
65    }
66
67    int modInv(int x) { return power(x, MOD - 2); }
68    int divide(int a, int b) { return mul(a, modInv(b)); }
69
70    int nCr(int x, int y) {
71        if (x < 0 || y > x || y < 0) return 0;
72        return mul(fac[x], mul(finv[y], finv[x - y]));
73    }
74
75    int nPr(int x, int y) {
76        if (x < 0 || y > x || y < 0) return 0;
77        return mul(fac[x], finv[x - y]);
78    }
79
80    int catalan(int x) { return mul(nCr(2 * x, x), modInv(x + 1)); }
81    int StarsAndPars(int x, int k) { return nCr(x + k - 1, k - 1); }
82
83    int nCr_no_mod(int x, int r) {
84        if (r < 0 || x < 0 || r > x) return 0;
85        int ret = 1;
86        for (int i = 1; i <= min(r, x - r); i++) {
87            ret *= x - i + 1;
88            ret /= i;
89        }
90        return ret;
91    }
92
93    int nPr_no_mod(int x, int r) {
94        if (r < 0 || x < 0 || r > x) return 0;
95        int ret = 1;
96        for (int i = 0; i < r; i++) ret *= (x - i);
97        return ret;
98    }
99
100    int numberOfDivisors1ToN(int limit) {

```

```

48    }
49 }
50
51 int add(int a, int b) { return ((a % MOD) + (b % MOD)) % MOD; }
52 int sub(int a, int b) { return (((a - b) % MOD) + MOD) % MOD; }
53 int mul(int a, int b) { return ((a % MOD) * (b % MOD)) % MOD; }
54
55 int power(int b, int pow) {
56     b %= MOD;
57     int res = 1;
58     while (pow) {
59         if (pow & 1) res = mul(res, b);
60         b = mul(b, b);
61         pow >>= 1;
62     }
63     return res;
64 }
65
66 int modInv(int x) { return power(x, MOD - 2); }
67 int divide(int a, int b) { return mul(a, modInv(b)); }
68
69 int nCr(int x, int y) {
70     if (x < 0 || y > x || y < 0) return 0;
71     return mul(fac[x], mul(finv[y], finv[x - y]));
72 }
73
74 int nPr(int x, int y) {
75     if (x < 0 || y > x || y < 0) return 0;
76     return mul(fac[x], finv[x - y]);
77 }
78
79 int catalan(int x) { return mul(nCr(2 * x, x), modInv(x + 1)); }
80 int StarsAndPars(int x, int k) { return nCr(x + k - 1, k - 1); }
81
82 int nCr_no_mod(int x, int r) {
83     if (r < 0 || x < 0 || r > x) return 0;
84     int ret = 1;
85     for (int i = 1; i <= min(r, x - r); i++) {
86         ret *= x - i + 1;
87         ret /= i;
88     }
89     return ret;
90 }
91
92 int nPr_no_mod(int x, int r) {
93     if (r < 0 || x < 0 || r > x) return 0;
94     int ret = 1;
95     for (int i = 0; i < r; i++) ret *= (x - i);
96     return ret;
97 }
98
99 int numberOfDivisors1ToN(int limit) {

```

```

100     int sum = 0;
101     for (int i = 1; i <= limit; i++) sum += limit / i;
102     return sum;
103 }
104
105 bool is_prime(int x) {
106     if (x <= 1) return false;
107     if (x % 2 == 0) return x == 2;
108     for (int i = 3; i * i <= x; i += 2) {
109         if (x % i == 0) return false;
110     }
111     return true;
112 }
113
114 vector<int> factorization(int x) {
115     vector<int> primes_factors;
116     for (int i = 2; i * i <= x; i++) {
117         while (x % i == 0) {
118             primes_factors.push_back(i);
119             x /= i;
120         }
121     }
122     if (x > 1) primes_factors.push_back(x);
123     return primes_factors;
124 }
125
126 vector<int> spf_factorization(int x) {
127     vector<int> primes_factors;
128     while (x != 1) {
129         primes_factors.push_back(spf[x]);
130         x /= spf[x];
131     }
132     return primes_factors;
133 }
134
135 map<int, int> spf_factorization_freq(int x) {
136     map<int, int> frq;
137     while (x != 1) {
138         frq[spf[x]]++;
139         x /= spf[x];
140     }
141     return frq;
142 }
143
144 int phi(int x) {
145     int ret = x;
146     for (int i = 2; i * i <= x; i++) {
147         if (x % i == 0) {
148             while (x % i == 0) x /= i;
149             ret -= ret / i;
150         }
151     }

```

```

152     if (x > 1) ret -= ret / x;
153     return ret;
154 }
155 };

```

### 5.6.2 FastSumitionDivisor 1e16

```

1 * THE ULTIMATE CP TEMPLATE BLUEPRINT: FAST SUM OF PROPER DIVISORS
2 *
3 * 1. WHAT PROBLEM DOES THIS SOLVE?
4 * - Keywords: "Sum_of_divisors", "Proper_divisors", "Aliquot_sum", "N_up_to_
  10^16".
5 * - Classic Scenarios: You are given a massive number N (up to 10^16) and you
  need to find
6 * the sum of all its divisors EXCEPT the number itself.
7
8 vector<int> Primes;
9
10 void Sieve(int num)
11 {
12     vector<bool> is_prime(num + 1, true);
13     is_prime[0], is_prime[1] = false;
14     for (int i = 2; i * i <= num; i++)
15     {
16         if (is_prime[i])
17         {
18             for (int j = i * i; j <= num; j += i)
19                 is_prime[j] = false;
20         }
21     }
22     Primes.push_back(2);
23     for (int i = 3; i <= num; i += 2)
24     {
25         if (is_prime[i])
26             Primes.push_back(i);
27     }
28
29 }
30
31 void Hassan ()
32 {
33     //one integer between 1 and 1e16 inclusive
34     ll n; cin >> n;
35     ll ans = 1;
36     ll copy = n;
37     for (auto p : Primes)
38     {
39         if (1LL * p * p > n)
40             break;
41         if (n % p == 0)
42         {

```



```

43         ll sum = 1, pow = 1;
44         while (n % p == 0)
45         {
46             pow *= p;
47             sum += pow;
48             n /= p;
49         }
50         ans *= sum;
51     }
52 }
53 if (n > 1) ans *= (n + 1);
54 cout << ans - copy << '\n'; /// note
55 }
```

### 5.6.3 Linear sieve

```

1 const int N = 1e5 + 5;
2 vector<bool> isPrime(N, true);
3 vector<int> p;
4
5 void linear_sieve()
6 {
7     isPrime[0] = isPrime[1] = 0;
8     for (ll i = 2; i < N; i++)
9     {
10         if (isPrime[i] == 1)
11             p.push_back(i);
12         for (ll o = 0; o < p.size(); o++)
13         {
14             if (i * p[o] >= N)
15                 break;
16             isPrime[i * p[o]] = 0;
17             if (i % p[o] == 0)
18                 break;
19         }
20     }
21 }
```

### 5.6.4 Miller pollard

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: MILLER-RABIN & POLLARD'S RHO
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Primality Test", "Prime Factorization", "Count Divisors", "
    Massive N".
6  * - Classic Scenarios: You need to check if N is prime, find all prime factors
    of N,
7  * or count the total number of divisors where N is up to 1018. Standard O(
    sqrt(N))
```

```

8  * trial division will result in a Time Limit Exceeded (TLE) verdict.
9  * - The Magic: Miller-Rabin deterministically checks primality for 64-bit
    integers
10 * in O(log N) time using 7 specific mathematical bases. If it's composite,
    Pollard's Rho
11 * steps in to find a prime factor in expected O(N(1/4)) time. Together, they
    instantly
12 * break down 1018 numbers.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Primality Check:
16 *     bool is_prime = MillerRabin(n);
17 * - Prime Factorization:
18 *     map<uint64_t, int> prime_factors;
19 *     factorize(n, prime_factors); // Populates the map with {prime, frequency}
20 * - Count Total Divisors:
21 *     uint64_t total = count_divisors(n);
22 *
23 * - Complexity:
24 *     Time: O(log N) for Miller-Rabin. Expected O(N(1/4)) for Pollard's Rho.
25 *     Space: O(log N) for recursion depth and the frequency map.
26 *
27 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
28 * - Type Aliases: This template strictly uses `u64` (uint64_t) and `u128` (
    __uint128_t)
29 * to prevent silent overflows during modular multiplication of massive numbers
    .
30 * Do not change these to standard `long long`.
31 * - Randomness: `rand()` is used for the Rho polynomial `x^2 + c`. This is
    usually
32 * sufficient, but if you hit a malicious anti-hash test case, you can swap it
    for a faster, custom `mt19937_64` implementation.
33 *
34 */
35
36 using u64 = uint64_t;
37 using u128 = __uint128_t;
38
39 u64 binpower(u64 base, u64 e, u64 mod)
40 {
41     u64 result = 1;
42     base %= mod;
43     while (e)
44     {
45         if (e & 1)
46             result = (u128)result * base % mod;
47         base = (u128)base * base % mod;
48         e >>= 1;
49     }
50     return result;
51 }
52 }
```

```

53 bool check_composite(u64 n, u64 a, u64 d, int s)
54 {
55     u64 x = binpower(a, d, n);
56     if (x == 1 || x == n - 1)
57         return false;
58     for (int r = 1; r < s; r++)
59     {
60         x = (u128)x * x % n;
61         if (x == n - 1)
62             return false;
63     }
64     return true;
65 }
66
67 bool MillerRabin(u64 n)
68 {
69     if (n < 2)
70         return false;
71     int r = 0;
72     u64 d = n - 1;
73     while ((d & 1) == 0)
74     {
75         d >>= 1;
76         r++;
77     }
78     // These 7 bases are deterministically sufficient for all n < 2^64
79     for (u64 a : {2, 325, 9375, 28178, 450775, 9780504, 1795265022})
80     {
81         if (n == a)
82             return true;
83         if (check_composite(n, a, d, r))
84             return false;
85     }
86     return true;
87 }
88
89 u64 pollard_rho(u64 n)
90 {
91     if (n % 2 == 0)
92         return 2;
93     if (MillerRabin(n))
94         return n;
95
96     u64 x = 2, y = 2, d = 1, c = 1;
97     auto f = [&](u64 x, u64 n, u64 c)
98     {
99         return (u64)(((u128)x * x + c) % n);
100     };
101
102     while (d == 1)
103     {
104         x = f(x, n, c);

```

```

105         y = f(f(y, n, c), n, c);
106         d = __gcd(x > y ? x - y : y - x, n);
107         if (d == n)
108         {
109             x = rand() % (n - 2) + 2;
110             y = x;
111             c = rand() % (n - 1) + 1;
112             d = 1;
113         }
114     }
115     return d;
116 }
117
118 void factorize(u64 n, map<u64, int> &prime_factors)
119 {
120     if (n == 1)
121         return;
122     if (MillerRabin(n))
123     {
124         prime_factors[n]++;
125         return;
126     }
127     u64 divisor = pollard_rho(n);
128     factorize(divisor, prime_factors);
129     factorize(n / divisor, prime_factors);
130 }
131
132 u64 count_divisors(u64 n)
133 {
134     map<u64, int> prime_factors;
135     factorize(n, prime_factors);
136     u64 total_divisors = 1;
137     for (auto &pf : prime_factors)
138     {
139         total_divisors *= (pf.second + 1);
140     }
141     return total_divisors;
142 }

```

### 5.6.5 Number of divisors 1e18

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: FAST DIVISOR COUNTING (WHEEL
   FACTORIZATION)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Number of divisors of N", "N up to 10^18", "Count divisors
   without full factorization".
6  * - Classic Scenarios: You are given a massive number N (up to 10^14 or 10^18)
   and you need to

```

```

7  * find exactly how many divisors it has, but  $O(\sqrt{N})$  is too slow, and you
   * can't build a Sieve
8  * because  $N$  is too large.
9  * - The Magic: "Wheel Factorization". Instead of checking every number up to
   *  $\sqrt{N}$ , we manually
10 * factor out the first 8 primes (2, 3, 5, 7, 11, 13, 17, 19). The product of
   * these primes is ~9.6 million.
11 * For the remaining part of  $N$ , we ONLY check divisors that are coprime to
   * these 8 primes.
12 * This eliminates about ~83% of the useless checks in the  $O(\sqrt{N})$  loop!
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization: Create an instance globally or in `main`. It will precompute
   * the "wheel spokes" ONCE.
16 *     FastDivisorCounter fdc;
17 *
18 * - Queries: Pass your number  $N$  (up to  $10^{18}$ ) to get the number of divisors.
   *     ll total_divisors = fdc.count(N);
19 *
20 *
21 * - Complexity:
22 *     Time (Initialization): Precomputing the wheel takes  $O(P)$  where  $P$  is ~9.6
   *     *  $10^6$  (Done once in ~15ms).
23 *     Time (Per Query): roughly  $O(\sqrt{N} / 9.6 \times 10^6)$ . Lightning fast for huge
   *     queries!
24 *     Space: ~6 MB for the precomputed wheel.
25 *
26 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
27 * - If Memory Limit is EXTREMELY tight (<10 MB), you can remove the `spokes`
   * vector and calculate the
28 * coprime numbers inside the `count` function loop like your original code,
   * but only if  $Q=1$ .
29 * For  $Q > 1$ , the precomputed `spokes` vector is vastly superior.
30 */
31
32 #include <bits/stdc++.h>
33 using namespace std;
34 #define ll long long
35 struct FastDivisorCounter
36 {
37 private:
38     vector<int> primes;
39     ll wheel_size;
40     vector<ll> spokes; // Stores numbers from 1 to wheel_size that are coprime to
   * the base primes
41 public:
42     FastDivisorCounter()
43     {
44         primes = {2, 3, 5, 7, 11, 13, 17, 19};
45         wheel_size = 1;
46
47         for (int p : primes)
48         {

```

```

         wheel_size *= p;
     }
    // ? PRECOMPUTATION MAGIC: We find all valid "spokes" ONCE.
    // There are exactly 1,658,880 such numbers. By saving them, we skip
    millions
    // of useless iterations in future queries.
    for (ll i = 1; i < wheel_size; i++)
    {
        bool is_coprime = true;
        for (int p : primes)
        {
            if (i % p == 0)
            {
                is_coprime = false;
                break;
            }
        }
        if (is_coprime)
        {
            spokes.push_back(i);
        }
    }
}

ll count(ll n)
{
    ll ans = 1;
    // Step 1: Remove all factors of the base primes and count their
    contribution
    for (int p : primes)
    {
        int c = 0;
        while (n % p == 0)
        {
            n /= p;
            c++;
        }
        // Standard divisor formula: multiply by (power + 1)
        ans *= (c + 1);
    }
    // Step 2: The remaining part of 'n' is now completely coprime to our
    base primes.
    // We only check potential divisors using our precomputed wheel spokes.
    ll all = 0;
    for (ll spoke : spokes)
    {
        ll o = spoke;
        // ? THE SPEED JUMP: We jump by `wheel_size` (9,699,690) instead of
        +1 or +2!
        for (; o * o < n; o += wheel_size)
        {
            if (n % o == 0)
            {

```

```

97         all += 2; // Found pair of divisors: `o` and `n/o`
98     }
99 }
100
101 // Check perfect square
102 if (o * o == n)
103 {
104     all++;
105 }
106 }
107 // The total divisors of the original N is the product of the divisors
108 // of the base-prime part and the divisors of the remaining coprime part
109 return ans * all;
110 }
111 };
112 void solve()
113 {
114     int t;
115     if (!(cin >> t))
116         return;
117     // Initialize OUTSIDE the query loop so the wheel is built only once!
118     FastDivisorCounter fdc;
119     while (t--)
120     {
121         ll n;
122         cin >> n;
123         cout << fdc.count(n) << "\n";
124     }
125 }

```

### 5.6.6 Prime counting

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: PRIME COUNTING FUNCTION (LUCY-HEDGEHOG)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Count primes up to N", "Pi(x)", "Lucy-Hedgehog", "Min-25 Sieve".
6  * - Classic Scenarios: You are given N up to 10^11 and asked exactly how many
7  *   prime numbers are less than or equal to N.
8  * - The Magic: A standard sieve up to 10^11 will instantly crash with Memory
   Limit
9  * Exceeded and Time Limit Exceeded. The Lucy-Hedgehog dynamic programming
   approach
10 * computes this mathematically without iterating over or storing the actual
   primes!
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     uint64_t total_primes = prime_count(N);
15 *
16 * - Complexity:

```

```

17 * Time: O(N^(3/4)) Runs almost instantly for N = 10^10, and easily
18 * passes 1-second limits for N = 10^11.
19 * Space: O(sqrt(N)) Extremely memory-efficient.
20 *
21 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
22 * - Array Sizes: It only creates vectors of size sqrt(N). For N = 10^11, this is
23 *   just an array of size 3.16 * 10^5, taking mere megabytes of RAM.
24 */
25
26 using u64 = uint64_t;
27
28 u64 prime_count(u64 N)
29 {
30     if (N <= 1)
31         return 0;
32     u64 v = sqrt(N);
33     vector<u64> S0(v + 1), S1(v + 1);
34     for (u64 i = 1; i <= v; i++)
35     {
36         S0[i] = i - 1;
37         S1[i] = N / i - 1;
38     }
39     for (u64 p = 2; p <= v; p++)
40     {
41         if (S0[p] == S0[p - 1])
42             continue;
43         u64 p_cnt = S0[p - 1];
44         u64 q = p * p;
45         u64 end = min(v, N / q);
46         for (u64 i = 1; i <= end; i++)
47         {
48             u64 d = i * p;
49             if (d <= v)
50                 S1[i] -= S1[d] - p_cnt;
51             else
52                 S1[i] -= S0[N / d] - p_cnt;
53         }
54         for (u64 i = v; i >= q; i--)
55         {
56             S0[i] -= S0[i / p] - p_cnt;
57         }
58     }
59     return S1[1];
60 }

```

### 5.6.7 Sieve 1e9

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 // credit: min_25

```

```

5 // takes 0.5s for n = 1e9
6 vector<int> sieve(const int N, const int Q = 17, const int L = 1 << 15)
7 {
8     static const int rs[] = {1, 7, 11, 13, 17, 19, 23, 29};
9     struct P
10    {
11        P(int p) : p(p)
12        {
13        }
14
15        int p;
16        int pos[8];
17    };
18    auto approx_prime_count = [](const int N) -> int
19    {
20        return N > 60184
21            ? N / (log(N) - 1.1)
22            : max(1., N / (log(N) - 1.11)) + 1;
23    };
24
25    const int v = sqrt(N), vv = sqrt(v);
26    vector<bool> isp(v + 1, true);
27    for (int i = 2; i <= vv; ++i)
28        if (isp[i])
29        {
30            for (int j = i * i; j <= v; j += i)
31                isp[j] = false;
32        }
33
34    const int rsize = approx_prime_count(N + 30);
35    vector<int> primes = {2, 3, 5};
36    int psize = 3;
37    primes.resize(rsize);
38
39    vector<P> sprimes;
40    size_t pbeg = 0;
41    int prod = 1;
42    for (int p = 7; p <= v; ++p)
43    {
44        if (!isp[p])
45            continue;
46        if (p <= Q)
47            prod *= p, ++pbeg, primes[psize++] = p;
48        auto pp = P(p);
49        for (int t = 0; t < 8; ++t)
50        {
51            int j = (p <= Q) ? p : p * p;
52            while (j % 30 != rs[t])
53                j += p << 1;
54            pp.pos[t] = j / 30;
55        }
56        sprimes.push_back(pp);

```

```

57    }
58
59    vector<unsigned char> pre(prod, 0xFF);
60    for (size_t pi = 0; pi < pbeg; ++pi)
61    {
62        auto pp = sprimes[pi];
63        const int p = pp.p;
64        for (int t = 0; t < 8; ++t)
65        {
66            const unsigned char m = ~(1 << t);
67            for (int i = pp.pos[t]; i < prod; i += p)
68                pre[i] &= m;
69        }
70    }
71
72    const int block_size = (L + prod - 1) / prod * prod;
73    vector<unsigned char> block(block_size);
74    unsigned char *pblock = block.data();
75    const int M = (N + 29) / 30;
76
77    for (int beg = 0; beg < M; beg += block_size, pblock -= block_size)
78    {
79        int end = min(M, beg + block_size);
80        for (int i = beg; i < end; i += prod)
81        {
82            copy(pre.begin(), pre.end(), pblock + i);
83        }
84        if (beg == 0)
85            pblock[0] &= 0xFE;
86        for (size_t pi = pbeg; pi < sprimes.size(); ++pi)
87        {
88            auto &pp = sprimes[pi];
89            const int p = pp.p;
90            for (int t = 0; t < 8; ++t)
91            {
92                int i = pp.pos[t];
93                const unsigned char m = ~(1 << t);
94                for (; i < end; i += p)
95                    pblock[i] &= m;
96                pp.pos[t] = i;
97            }
98        }
99        for (int i = beg; i < end; ++i)
100        {
101            for (int m = pblock[i]; m > 0; m &= m - 1)
102            {
103                primes[psize++] = i * 30 + rs[__builtin_ctz(m)];
104            }
105        }
106    }
107    assert(psize <= rsize);
108    while (psize > 0 && primes[psize - 1] > N)

```

```

109     --psize;
110     primes.resize(psize);
111     return primes;
112 }
113
114 int32_t main()
115 {
116     ios_base::sync_with_stdio(0);
117     cin.tie(0);
118     int n, a, b;
119     cin >> n >> a >> b;
120     auto primes = sieve(n);
121     vector<int> ans;
122     for (int i = b; i < primes.size() && primes[i] <= n; i += a)
123         ans.push_back(primes[i]);
124     cout << primes.size() << ' ' << ans.size() << '\n';
125     for (auto x : ans)
126         cout << x << ' ';
127     cout << '\n';
128     return 0;
129 }

```

### 5.6.8 segmented sieve

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: SEGMENTED SIEVE
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Primes in Range [L, R]", "Massive Bounds".
6  * - Classic Scenarios: Find all primes or the number of primes between L and R.
7  * L and R can be as massive as  $10^{12}$ , but the difference  $(R - L)$  is small ( $\leq 10^6$ ).
8  * - The Magic: A regular sieve cannot allocate an array up to  $10^{12}$ . The
9  * segmented
10 * sieve only sieves up to  $\sqrt{R}$ , caches those base primes, and then "shifts"
11 * the sieving logic directly to the target window  $[L, R]$ , ignoring the void
12 * before L.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Query:
16 *     vector<uint64_t> primes = segmented_sieve(L, R);
17 *
18 * - Complexity:
19 *     Time:  $O((R - L + \sqrt{R}) * \log(\log(R)))$ 
20 *     Space:  $O(\sqrt{R} + (R - L))$ 
21 *
22 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
23 * - Edge Cases Guarded: Safely handles constraints where  $L = 0$  or  $L = 1$ 
24 * (which are not prime but often trip up basic segmented implementations).
25 */

```

```

25 using u64 = uint64_t;
26
27 vector<u64> segmented_sieve(u64 L, u64 R)
28 {
29     if (L > R)
30         return {};
31     u64 lim = sqrt(R);
32     vector<bool> mark(lim + 1, false);
33     vector<u64> primes;
34     for (u64 i = 2; i <= lim; ++i)
35     {
36         if (!mark[i])
37         {
38             primes.push_back(i);
39             for (u64 j = i * i; j <= lim; j += i)
40                 mark[j] = true;
41         }
42     }
43
44     vector<bool> isPrime(R - L + 1, true);
45     for (u64 p : primes)
46     {
47         u64 start = max(p * p, (L + p - 1) / p * p);
48         for (u64 j = start; j <= R; j += p)
49             isPrime[j - L] = false;
50     }
51
52     if (L == 0)
53     {
54         isPrime[0] = false;
55         if (R >= 1)
56             isPrime[1] = false;
57     }
58     else if (L == 1)
59     {
60         isPrime[0] = false;
61     }
62
63     vector<u64> range_primes;
64     for (u64 i = 0; i <= R - L; ++i)
65     {
66         if (isPrime[i])
67             range_primes.push_back(L + i);
68     }
69     return range_primes;
70 }

```

### 5.6.9 sum numberOfDivisors

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: CUBIC ROOT DIVISOR SUM

```

```

3 *
4 * 1. WHAT PROBLEM DOES THIS SOLVE?
5 * - Keywords: "Sum of Divisors", "Sigma0(i)", "Sum of floor(N/i)".
6 * - Classic Scenarios: You are asked to compute  $\sum_{i=1}^N d(i)$ , which is
7 * mathematically equivalent to  $\sum_{i=1}^N \text{floor}(N / i)$ , for a staggering  $N$ 
8 *  $= 10^{18}$ .
9 * - The Magic: The naive  $O(N)$  loop crashes. The standard Dirichlet hyperbola
10 * optimization takes  $O(\sqrt{N})$ , which runs in ~1 second for  $10^{12}$  but fails
11 * at  $10^{18}$ .
12 * This extremely advanced algorithm uses Stern-Brocot tree fraction
13 * approximation
14 * to step through the hyperbola in blocks, reducing the time to roughly  $O(N$ 
15 *  $\sqrt{1/3})$ !
16 *
17 * 2. HOW TO USE IT (THE BLACK BOX)
18 * - Query:
19 * __uint128_t ans = sum_sigma0(N);
20 * // Remember to write a custom print function for __uint128_t if
21 * outputting directly.
22 *
23 * - Complexity:
24 * Time:  $\sim O(N^{1/3})$ . Easily executes  $N = 10^{18}$  in < 100 milliseconds.
25 * Space:  $O(\log N)$  for the stack depth.
26 *
27 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
28 * - 128-bit Types: Be very careful if modifying the inner loop logic. The
29 * condition
30 * __uint128(x) * x * dy >= __uint128(n) * dx uses 128-bit integers
31 * because cubing boundaries near  $10^{18}$  guarantees overflow in standard
32 * __uint64_t.
33 */
34 using uint32 = unsigned int;
35 using uint64 = unsigned long long;
36 using uint128 = __uint128_t;
37
38 uint128 sum_sigma0(uint64 n)
39 {
40     auto out = [n](uint64 x, uint32 y)
41     { return x * y > n; };
42     auto cut = [n](uint64 x, uint32 dx, uint32 dy)
43     { return uint128(x) * x * dy >= uint128(n) * dx; };
44     const uint64 sn = sqrtl(n);
45     const uint64 cn = pow(n, 0.34);
46     uint64 x = n / sn;
47     uint32 y = n / x + 1;
48     uint128 ret = 0;
49     stack<pair<uint32, uint32>> st;
50     st.emplace(1, 0);
51     st.emplace(1, 1);
52     while (true)
53     {

```

```

48     uint32 lx, ly;
49     tie(lx, ly) = st.top();
50     st.pop();
51     while (out(x + lx, y - ly))
52     {
53         ret += x * ly + uint64(ly + 1) * (lx - 1) / 2;
54         x += lx, y -= ly;
55     }
56     if (y <= cn)
57         break;
58     uint32 rx = lx, ry = ly;
59     while (true)
60     {
61         tie(lx, ly) = st.top();
62         if (out(x + lx, y - ly))
63             break;
64         rx = lx, ry = ly;
65         st.pop();
66     }
67     while (true)
68     {
69         uint32 mx = lx + rx, my = ly + ry;
70         if (out(x + mx, y - my))
71         {
72             st.emplace(lx = mx, ly = my);
73         }
74         else
75         {
76             if (cut(x + mx, lx, ly))
77                 break;
78             rx = mx, ry = my;
79         }
80     }
81 }
82 for (--y; y > 0; --y)
83     ret += n / y;
84 return ret * 2 - sn * sn;
85 }

```

## 6 Combinatorics

### 6.1 Main

#### 6.1.1 Basic

```

1 // =====
2 // MODULE 1: MODULAR ARITHMETIC & CORE COMBINATORICS
3 // =====
4 struct Combinatorics
5 {

```

```

6   int N;
7   vector<ll> fact, invFact;
8
9   // Fast Modular Exponentiation: Computes (base^exp) % MOD in O(log exp)
10  ll power(ll base, ll exp)
11  {
12      ll res = 1;
13      base %= MOD;
14      while (exp > 0)
15      {
16          if (exp % 2 == 1)
17              res = (res * base) % MOD;
18          base = (base * base) % MOD;
19          exp /= 2;
20      }
21      return res;
22  }
23
24  // Modular Multiplicative Inverse: Computes (1 / n) % MOD in O(log MOD)
25  // Uses Fermat's Little Theorem (Requires MOD to be prime)
26  ll modInverse(ll n)
27  {
28      return power(n, MOD - 2);
29  }
30
31  // Initialize Factorials and Inverse Factorials in O(N)
32  // CALL THIS ONCE IN MAIN BEFORE SOLVING TEST CASES!
33  void init(int n)
34  {
35      N = n;
36      fact.resize(N + 1);
37      invFact.resize(N + 1);
38
39      fact[0] = 1;
40      invFact[0] = 1;
41      for (int i = 1; i <= N; i++)
42      {
43          fact[i] = (fact[i - 1] * i) % MOD;
44      }
45
46      // Compute the inverse of the largest factorial, then work backwards
47      invFact[N] = modInverse(fact[N]);
48      for (int i = N - 1; i >= 1; i--)
49      {
50          invFact[i] = (invFact[i + 1] * (i + 1)) % MOD;
51      }
52  }
53
54  // Choose: Number of ways to pick r items from n items. O(1)
55  ll nCr(int n, int r)
56  {
57      if (r < 0 || r > n)

```

```

58      return 0;
59      return fact[n] * invFact[r] % MOD * invFact[n - r] % MOD;
60  }
61
62  // Permutation: Number of ways to arrange r items out of n items. O(1)
63  ll nPr(int n, int r)
64  {
65      if (r < 0 || r > n)
66          return 0;
67      return fact[n] * invFact[n - r] % MOD;
68  }
69
70  // Stars and Bars: Ways to distribute n identical items into k distinct bins.
71  // O(1)
72  // Example: Solutions to  $x_1 + x_2 + \dots + x_k = n$  (where  $x_i \geq 0$ )
73  ll starsAndBars(int n, int k)
74  {
75      if (n == 0 && k == 0)
76          return 1;
77      return nCr(n + k - 1, k - 1);
78  }
79
80  // Stars and Bars (Positive): Distribute n identical items into k distinct
81  // bins,
82  // such that every bin gets AT LEAST ONE item. O(1)
83  // Example: Solutions to  $x_1 + x_2 + \dots + x_k = n$  (where  $x_i \geq 1$ )
84  ll starsAndBarsPositive(int n, int k)
85  {
86      if (n < k)
87          return 0;
88      return nCr(n - 1, k - 1);
89  }
90
91  // Catalan Numbers: O(1)
92  // Solves: Valid parenthesis expressions, binary trees, polygon
93  // triangulations.
94  // Formula:  $C(n) = (1 / (n + 1)) * (2n C n)$ 
95  ll catalan(int n)
96  {
97      if (n < 0)
98          return 0;
99      ll res = nCr(2 * n, n);
100     res = (res * modInverse(n + 1)) % MOD;
101     return res;
102 }
103
104 // =====
105 // MODULE 2: ADVANCED COUNTING (DERANGEMENTS, STIRLING, BELL)
106 // =====
107 struct AdvancedCombinatorics
108 {

```



```

107 // Derangements: Number of permutations of N elements where NO element is in
    // its original position.
108 // O(N) precomputation. Formula: D(n) = (n-1) * (D(n-1) + D(n-2))
109 vector<ll> derangement;
110
111 void initDerangements(int n)
112 {
113     derangement.assign(n + 1, 0);
114     if (n >= 0)
115         derangement[0] = 1;
116     if (n >= 1)
117         derangement[1] = 0;
118     for (int i = 2; i <= n; i++)
119     {
120         derangement[i] = ((i - 1) * (derangement[i - 1] + derangement[i - 2])
            % MOD) % MOD;
121     }
122 }
123
124 // Stirling Numbers of the First Kind: S1[n][k]
125 // Solves: Number of permutations of 'n' elements that contain exactly 'k'
    // disjoint cycles.
126 // O(N^2) precomputation.
127 vector<vector<ll>> S1;
128
129 void initStirling1(int n)
130 {
131     S1.assign(n + 1, vector<ll>(n + 1, 0));
132     S1[0][0] = 1;
133     for (int i = 1; i <= n; i++)
134     {
135         for (int j = 1; j <= i; j++)
136         {
137             S1[i][j] = (S1[i - 1][j - 1] + (i - 1) * S1[i - 1][j]) % MOD;
138         }
139     }
140 }
141
142 // Stirling Numbers of the Second Kind: S2[n][k]
143 // Solves: Number of ways to partition a set of 'n' DISTINCT elements into
    // exactly 'k' non-empty indistinguishable subsets.
144 // O(N^2) precomputation.
145 vector<vector<ll>> S2;
146
147 void initStirling2(int n)
148 {
149     S2.assign(n + 1, vector<ll>(n + 1, 0));
150     S2[0][0] = 1;
151     for (int i = 1; i <= n; i++)
152     {
153         for (int j = 1; j <= i; j++)
154             S2[i][j] = (S2[i - 1][j - 1] + j * S2[i - 1][j]) % MOD;
155     }
156 }
157
158 // Bell Numbers: Bell[n]
159 // Solves: Total number of ways to partition a set of 'n' elements into ANY
    // number of non-empty subsets.
160 // O(N^2) precomputation (Sum of Stirling2(n, k) for k = 0 to n).
161 vector<ll> bell;
162
163 void initBell(int n)
164 {
165     if (S2.empty())
166         initStirling2(n);
167     bell.assign(n + 1, 0);
168     for (int i = 0; i <= n; i++)
169     {
170         for (int j = 0; j <= i; j++)
171         {
172             bell[i] = (bell[i] + S2[i][j]) % MOD;
173         }
174     }
175 }
176
177 }
178 };
179
180 // =====
181 // MODULE 3: LUCAS THEOREM (LARGE N, SMALL PRIME MODULO)
182 // =====
183 // Solves: nCr % P when N and R are massive (up to 10^18) but P is small (up to
    // 10^5) and P is PRIME.
184 // Note: This requires a separate Combinatorics instance initialized up to P.
185 ll lucas_nCr(ll n, ll r, int p, Combinatorics &combo)
186 {
187     if (r < 0 || r > n)
188         return 0;
189     if (r == 0)
190         return 1;
191
192     // Split n and r into base P representation
193     ll ni = n % p;
194     ll ri = r % p;
195
196     if (ni < ri)
197         return 0; // Optimization: P divides nCr
198
199     // nCr = Product of (n_i C r_i) mod P
200     return (combo.nCr(ni, ri) * lucas_nCr(n / p, r / p, p, combo)) % p;
201 }

```

### 6.1.2 Pascal triangle

```

1 ll c[n][n]={};
2 c[0][0]=1;
3 for(int i=1;i<n;i++)
4 {
5     c[i][0]=c[i][i]=1;
6     for(int o=1;o<i;o++)
7     {
8         c[i][o]=c[i-1][o-1]+c[i-1][o];
9     }
10 }
```

## 6.2 PIE

### 6.2.1 multiples

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: INCLUSION-EXCLUSION (PIE) ON MULTIPLES
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Divisible by any", "Coprime to array", "Multiples below X".
6  * - Classic Scenarios: You are given a target number X, and an array of K
7  *   integers.
8  *   You need to find how many numbers from 1 to X are divisible by AT LEAST ONE
9  *   of the integers in the array.
10 * - The Magic: It systematically adds numbers divisible by 1 element, subtracts
11 *   those divisible by 2 elements, adds for 3, etc. A naive bitmask loop goes
12 *   out
13 *   of bounds (LCM overflow) quickly. This DFS safely prunes branches if the LCM
14 *   ever exceeds X, making it insanely fast and overflow-proof.
15 *
16 * 2. HOW TO USE IT (THE BLACK BOX)
17 * - Query:
18 *   long long ans = count_divisible(X, A);
19 *   // To find numbers COPRIME to the array: ans = X - count_divisible(X, A)
20 *
21 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
22 * - Array Cleaning: For maximum speed, it is highly recommended to sort the
23 *   array `A`,
24 *   remove duplicates, and remove any number that is a multiple of another
25 *   number
26 *   in the array before calling this function.
27 */
28 using ll = long long;
29 ll pie_dfs(int idx, ll current_lcm, int elements_chosen, ll max_val, const vector<ll> &a)
```

```

29     if (idx == a.size())
30     {
31         if (elements_chosen == 0)
32             return 0;
33         if (elements_chosen % 2 == 1)
34             return max_val / current_lcm;
35         return -(max_val / current_lcm);
36     }
37
38     ll ans = pie_dfs(idx + 1, current_lcm, elements_chosen, max_val, a);
39
40     ll g = __gcd(current_lcm, a[idx]);
41     if (current_lcm / g <= max_val / a[idx])
42     {
43         ans += pie_dfs(idx + 1, (current_lcm / g) * a[idx], elements_chosen + 1,
44             max_val, a);
45     }
46     return ans;
47 }
48
49 ll count_divisible(ll max_val, const vector<ll> &a)
50 {
51     return pie_dfs(0, 1, 0, max_val, a);
52 }
```

## 6.3 Partition

### 6.3.1 Integer partition

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: INTEGER PARTITION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Ways to sum to N", "Unordered integer partitions".
6  * - Classic Scenarios: Find the number of distinct ways to represent the integer
7  *   N
8  *   as a sum of positive integers (e.g., 4 = 4, 3+1, 2+2, 2+1+1, 1+1+1+1).
9  * - The Magic: A standard coin-change DP takes  $O(N^2)$ , which crashes for  $N = 10^5$ .
10 *   Euler's Pentagonal Number Theorem computes the partition function  $P(N)$  in  $O(\sqrt{N})$ 
11 *   by subtracting and adding previous  $P()$  values at specifically spaced
12 *   pentagonal
13 *   intervals:  $P(k) = P(k-1) + P(k-2) - P(k-5) - P(k-7) + P(k-12) + \dots$ 
14 *
15 * 2. HOW TO USE IT (THE BLACK BOX)
16 * - Query:
17 *   vector<long long> P = partition_function(N, MOD);
18 *   // P[i] contains the number of partitions for integer i.
19 */
```

```

18 * - Complexity:
19 *   Time:  $O(N * \sqrt{N})$ 
20 *   Space:  $O(N)$ 
21 *
22 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
23 * - Array Precomputation: It returns the entire array up to N. If the problem
   has
24 * multiple test cases querying different N's, run this function ONCE globally
   for
25 * the maximum possible N, then answer each query in  $O(1)$ .
26 */
27
28 using ll = long long;
29
30 vector<ll> partition_function(int n, ll mod)
31 {
32     vector<ll> p(n + 1, 0);
33     p[0] = 1;
34     for (int i = 1; i <= n; i++)
35     {
36         for (int k = 1;; k++)
37         {
38             int p1 = k * (3 * k - 1) / 2;
39             int p2 = k * (3 * k + 1) / 2;
40             if (p1 > i && p2 > i)
41                 break;
42
43             ll sign = (k % 2 == 1) ? 1 : -1;
44
45             if (p1 <= i)
46                 p[i] = (p[i] + sign * p[i - p1]) % mod;
47             if (p2 <= i)
48                 p[i] = (p[i] + sign * p[i - p2]) % mod;
49         }
50         p[i] = (p[i] + mod) % mod;
51     }
52     return p;
53 }

```

## 6.4 nCr

### 6.4.1 nCr Mod any mod

```

1 /**
2 * THE ULTIMATE CP TEMPLATE BLUEPRINT: nCr MODULO ANY MOD
3 *
4 * 1. WHAT PROBLEM DOES THIS SOLVE?
5 * - Keywords: "nCr", "Composite Modulo", "No Modular Inverse".
6 * - Classic Scenarios: Compute  $nCr(N, K)$  modulo M where M is NOT a prime number
7 *   (e.g.,  $M = 10^9$ , or some random composite).

```

```

8 * - The Magic: Since M is composite, you cannot divide by factorials (no inverse
   ).
9 * This algorithm factors M into prime powers  $p^k$ . It then counts the exact
10 * number of times p divides N!, K!, and (N-K)!, isolates the coprime parts,
11 * calculates the binomial coefficient modulo  $p^k$ , and finally uses the Chinese
12 * Remainder Theorem (CRT) to stitch them all back together into a single
   answer.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Query:
16 *   long long ans = nCr_arbitrary_mod(N, K, M);
17 *
18 * - Complexity:
19 *   Time:  $O(\sum(p^k) + \log N)$ . Extremely fast for  $M \leq 10^6$ .
20 *   Space:  $O(\sum(p^k))$  to store factorials modulo  $p^k$ .
21 *
22 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
23 * - Dependencies: This heavily relies on your existing `extgcd` and `crt`
   templates
24 * from your Modular Arithmetic section. Make sure they are included above this
   .
25 */
26
27 using ll = long long;
28
29 ll custom_pow(ll base, ll exp, ll mod)
30 {
31     ll res = 1;
32     base %= mod;
33     while (exp > 0)
34     {
35         if (exp % 2 == 1)
36             res = (res * base) % mod;
37         base = (base * base) % mod;
38         exp /= 2;
39     }
40     return res;
41 }
42
43 ll extgcd(ll a, ll b, ll &x, ll &y);
44 pair<ll, ll> crt(const vector<ll> &a, const vector<ll> &m);
45
46 ll modInverse(ll a, ll m)
47 {
48     ll x, y;
49     extgcd(a, m, x, y);
50     return (x % m + m) % m;
51 }
52
53 ll fact_mod_pk(ll n, ll p, ll pk, int &e)
54 {
55     e = 0;

```

```

56     if (n == 0)
57         return 1;
58     ll res = 1;
59     for (ll i = 2; i <= pk; i++)
60     {
61         if (i % p != 0)
62             res = (res * i) % pk;
63     }
64     res = custom_pow(res, n / pk, pk);
65     for (ll i = 2; i <= n % pk; i++)
66     {
67         if (i % p != 0)
68             res = (res * i) % pk;
69     }
70     int next_e;
71     res = (res * fact_mod_pk(n / p, p, pk, next_e)) % pk;
72     e += n / p + next_e;
73     return res;
74 }
75
76 ll nCr_pk(ll n, ll r, ll p, ll pk)
77 {
78     if (n < r || r < 0)
79         return 0;
80     int e1, e2, e3;
81     ll num = fact_mod_pk(n, p, pk, e1);
82     ll den1 = fact_mod_pk(r, p, pk, e2);
83     ll den2 = fact_mod_pk(n - r, p, pk, e3);
84     ll p_pow = custom_pow(p, e1 - e2 - e3, pk);
85     if (p_pow == 0)
86         return 0;
87     ll ans = (num * modInverse((den1 * den2) % pk, pk)) % pk;
88     return (ans * p_pow) % pk;
89 }
90
91 ll nCr_arbitrary_mod(ll n, ll r, ll m)
92 {
93     if (n < r || r < 0)
94         return 0;
95     if (m == 1)
96         return 0;
97     vector<ll> rem, mod;
98     ll temp = m;
99     for (ll i = 2; i * i <= temp; i++)
100     {
101         if (temp % i == 0)
102         {
103             ll pk = 1;
104             while (temp % i == 0)
105             {
106                 pk *= i;
107                 temp /= i;

```

```

108     }
109     rem.push_back(nCr_pk(n, r, i, pk));
110     mod.push_back(pk);
111 }
112 }
113 if (temp > 1)
114 {
115     rem.push_back(nCr_pk(n, r, temp, temp));
116     mod.push_back(temp);
117 }
118 return crt(rem, mod).first;
119 }

```

#### 6.4.2 nCr over congruence class

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: nCr OVER CONGRUENCE CLASS
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Sum of nCr where r mod M = K".
6  * - Classic Scenarios: You need to evaluate sum(nCr(N, i)) for all i where
7  *   i % M == K. N can be up to 1018, but M is small (M <= 100).
8  * - The Magic: It maps the combinatorial states into a matrix. If you have an
9  *   array tracking the sum mod M, multiplying it by a transition matrix
10  *   simulates
11  *   increasing N by 1. By exponentiating the matrix to the power of N in
12  *   O(M3 log N), you instantly compute the exact sum across the entire boundary
13  *   .
14  * 2. HOW TO USE IT (THE BLACK BOX)
15  * - Query:
16  *   long long ans = nCr_congruence(N, K, M, MOD);
17  * - Complexity:
18  *   Time: O(M3 * log N)
19  *   Space: O(M2) for the matrix.
20  */
21
22 using ll = long long;
23
24 using Matrix = vector<vector<ll>>;
25
26 Matrix multiply_matrix(const Matrix &A, const Matrix &B, ll mod)
27 {
28     int n = A.size();
29     Matrix C(n, vector<ll>(n, 0));
30     for (int i = 0; i < n; i++)
31     {
32         for (int k = 0; k < n; k++)
33         {
34             for (int j = 0; j < n; j++)

```

```

35     {
36         C[i][j] = (C[i][j] + A[i][k] * B[k][j]) % mod;
37     }
38 }
39 }
40 return C;
41 }
42
43 Matrix power_matrix(Matrix A, ll exp, ll mod)
44 {
45     int n = A.size();
46     Matrix res(n, vector<ll>(n, 0));
47     for (int i = 0; i < n; i++)
48         res[i][i] = 1;
49     while (exp > 0)
50     {
51         if (exp % 2 == 1)
52             res = multiply_matrix(res, A, mod);
53         A = multiply_matrix(A, A, mod);
54         exp /= 2;
55     }
56     return res;
57 }
58
59 ll nCr_congruence(ll n, int k, int m, ll mod)
60 {
61     if (m == 1)
62         return custom_pow(2, n, mod);
63     Matrix T(m, vector<ll>(m, 0));
64     for (int i = 0; i < m; i++)
65     {
66         T[i][i] = 1;
67         T[i][(i + m - 1) % m] = 1;
68     }
69     T = power_matrix(T, n, mod);
70     return T[k][0];
71 }

```

## 7 Geometry

### 7.1 Ball 3D Class

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: 3D CLOSEST PAIR OF POINTS (DIVIDE &
   CONQUER)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Minimum Euclidean distance in 3D", "Closest points in space", "N
   up to 105".

```

```

6  * - Classic Scenarios: You are given N points in a 3D coordinate system. Finding
   the closest
7  * pair naively takes O(N2), which will Time Limit Exceed (TLE) for N > 104.
8  * - The Magic: Divide and Conquer! We sort the points by the X-axis and
   recursively divide the
9  * space into halves. The tricky part in 3D is merging the halves (the "strip")
   . If we just
10 * check all points in the strip naively, it can still degrade to O(N2).
11 * To fix this, the algorithm groups the strip into 1D "columns" along the Z-
   axis using the
12 * current minimum distance `d` as the column width. This brilliant trick
   guarantees that each
13 * point only checks a strict O(1) number of neighboring points. The overall
   complexity
14 * beautifully drops to O(N log N).
15 *
16 * 2. HOW TO USE IT (THE BLACK BOX)
17 * - Initialization: Create a `BallManager` object. It automatically reads `N`
   from `std::cin`,
18 * followed by the `X Y Z` coordinates of the `N` points.
19 *     BallManager bm;
20 *
21 * - Execution: Call the main solver function to get the minimum distance.
22 *     long double min_distance = bm.ClosestPair();
23 *
24 * - Complexity:
25 *     Time: O(N log N) The X and Y sorting takes O(N log N), and the
   recursive strip
26 *     processing takes O(N) per level.
27 *     Space: O(N) to store the objects, vectors, and recursion stack.
28 *
29 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
30 * - Decoupling I/O: The constructor currently reads input directly. For better
   flexibility,
31 * you can modify the constructor to accept a `vector<vector<ll>>>` or `vector<
   Ball>` instead
32 * of using `cin` inside the class.
33 * - Precision Issues: The algorithm uses `long double` and `sqrtl`. While this
   is highly accurate,
34 * some problems require exact integer squared distances. You can return the
   squared distance
35 * at the very end by squaring `finalMinDist`, but do NOT remove `sqrtl` from `
   DistanceTo`,
36 * as the `SearchStrip` bucketing logic mathematically relies on the true
   linear distance.
37 */
38
39 #include <iostream>
40 #include <vector>
41 #include <cmath>
42 #include <algorithm>
43 #include <limits>

```

```

44
45 using namespace std;
46 #define ll long long
47 #define ld long double
48
49 class Ball
50 {
51 public:
52     ll posX, posY, posZ; //Position values in 3D space
53     ll columnNumber[3]; //ID number of the column which ball belongs to (left,
        middle, right)
54     ll columnIndex[3]; //Position inside the column (left, middle, right)
55
56     // Time Complexity: O(1)
57     //
58     ld DistanceTo(Ball* targetBall)
59     {
60         ll squareDistX = (posX - targetBall->posX) * (posX - targetBall->posX);
61         ll squareDistY = (posY - targetBall->posY) * (posY - targetBall->posY);
62         ll squareDistZ = (posZ - targetBall->posZ) * (posZ - targetBall->posZ);
63         return sqrtl(squareDistX + squareDistY + squareDistZ);
64     }
65 };
66
67 class BallManager
68 {
69     ll ballCount = 0; //Total number of balls
70     Ball* ballArray; //Dynamically allocated ball array which holds all balls
        inside
71
72     // Time Complexity: O(1)
73     static bool CompareX(Ball ball1, Ball ball2)
74     {
75         return (ball1.posX < ball2.posX);
76     }
77
78     // Time Complexity: O(1)
79     static bool CompareY(Ball* ball1, Ball* ball2)
80     {
81         return (ball1->posY < ball2->posY);
82     }
83
84     // Time Complexity: O(N^2)
85     // N (balls.size()).
86     ld NaiveMethod(std::vector<Ball*> balls)
87     {
88         int size = balls.size();
89         ld minDist = std::numeric_limits<double>::max();
90
91         //Take balls one by one
92         for (int i = 0; i < size - 1; ++i)
93         {
94             //Check other remaining balls
95             for (int j = i + 1; j < size; ++j)
96             {
97                 ld dist = balls[i]->DistanceTo(balls[j]);
98                 if (dist < minDist)
99                     minDist = dist;
100             }
101         }
102         return minDist;
103     }
104
105     // Time Complexity: O(S)
106     // S (stripSize).
107     // (Columns) Z
108     // ( ).
109     ld SearchStrip(std::vector<Ball*> strip, ld stripDist)
110     {
111         //Save the strip distance value as minimum distance known
112         ld minDist = stripDist;
113
114         //If ball count in strip is less than 2, return strip distance
115         int stripSize = strip.size();
116         if (stripSize < 2)
117             return minDist;
118
119         //If ball count in strip is 2 or 3, find distance with naive method
120         if (stripSize < 4)
121         {
122             ld newDist = NaiveMethod(strip);
123             if (newDist < minDist)
124                 minDist = newDist;
125             return minDist;
126         }
127
128         //If ball count is more than 3, start processing of creating columns
129         //Find the smallest z value among balls -> O(S)
130         int smallestZ = strip[0]->posZ;
131         for (int i = 1; i < stripSize; i++)
132             if (strip[i]->posZ < smallestZ)
133                 smallestZ = strip[i]->posZ;
134
135         //Find the largest z value among balls -> O(S)
136         int largestZ = strip[0]->posZ;
137         for (int i = 1; i < stripSize; i++)
138             if (strip[i]->posZ > largestZ)
139                 largestZ = strip[i]->posZ;
140
141         //Find range between z values and column count
142         int rangeZ = largestZ - smallestZ;
143         int columnCount = (int)(rangeZ / stripDist) + 1;
144
145         //Create an array of ball pointer vectors to put pointers of balls into

```

```

146     proper columns
147     std::vector<Ball*>* columns = new std::vector<Ball*>[columnCount];
148
149     //Put all balls into proper columns (Phase 1) -> O(S)
150     //Start picking balls one by one with the smallest Y first
151     for (int i = 0; i < stripSize; i++)
152     {
153         //Find middle(main) column of ball, put ball into it, save index
154         strip[i]->columnNumber[1] = (1l)((1d)(strip[i]->posZ - smallestZ) /
155             stripDist);
156         columns[strip[i]->columnNumber[1]].push_back(strip[i]);
157         strip[i]->columnIndex[1] = columns[strip[i]->columnNumber[1]].size()
158             - 1;
159
160         //Find and save left and right columns of ball
161         strip[i]->columnNumber[0] = strip[i]->columnNumber[1] - 1;
162         strip[i]->columnNumber[2] = strip[i]->columnNumber[1] + 1;
163
164         //There is no column at left of leftmost column, and right of
165         //rightmost column
166         //Save indices in left and right columns
167         if (strip[i]->columnNumber[1] != 0) //If not leftmost column
168             strip[i]->columnIndex[0] = columns[strip[i]->columnNumber[0]].
169                 size() - 1;
170
171         if (strip[i]->columnNumber[1] != columnCount - 1) //If not rightmos
172             strip[i]->columnIndex[2] = columns[strip[i]->columnNumber[2]].
173                 size() - 1;
174     }
175
176     //Find if such lesser distance exists between any pair of balls inside
177     //strip (Phase 2)
178     // O(S) overall. The inner loops run a constant number of times O(1) pe
179     ball.
180     //Start picking balls one by one with the smallest Y first
181     for (int i = 0; i < stripSize; i++)
182     {
183         //For left, middle and right columns of the selected ball
184         for (int j = 0; j < 3; j++)
185         {
186             //Skip invalid out of bounds columns
187             if ((strip[i]->columnNumber[j] == -1) || (strip[i]->columnNumber
188                 j] == columnCount))
189                 continue;
190
191             //Look through increasing Y value = O(1)
192             for (int k = strip[i]->columnIndex[j] + 1; k < (1l)columns[strip
193                 i]->columnNumber[j]].size(); k++)
194             {
195                 //If Y distance is already more than strip distance, no need
196                 to check remaning balls in column

```

```

186         if (columns[strip[i]->columnNumber[j]][k]->posY > strip[i]->
187             posY + stripDist)
188             break;
189
190         //Check distance and if it is smaller than current minimum
191         //distance, save it as new minimum distance
192         1d newDist = strip[i]->DistanceTo(columns[strip[i]->
193             columnIndex[j]][k]);
194         if (newDist < minDist)
195             minDist = newDist;
196     }
197 }
198
199 //Delete dynamically allocated columns array
200 delete[] columns;
201
202 //Return minimum distance found in strip or initially given strip
203 //distance
204 return minDist;
205 }
206
207 //Time Complexity: T(N) = 2*T(N/2) + O(N) => O(N log N)
208 // N (sortedY.size()). O(N).
209 ld RecursiveCall(int sortedXPos, std::vector<Ball*> sortedY)
210 {
211     //If size is small enough, find distance with naive method -> O(1)
212     int size = sortedY.size();
213     if (size < 4)
214         return NaiveMethod(sortedY);
215
216     //Get middle ball for X axis
217     int midIndex = sortedXPos + size / 2;
218     Ball* midBall = &ballArray[midIndex];
219
220     //Initilize left and right subarrays
221     std::vector<Ball*> leftVector;
222     std::vector<Ball*> rightVector;
223
224     //In case of there are other balls which share same X position as middle
225     //point
226     bool putToRightArr = true;
227
228     //Distribute balls into left and right vectors -> O(N)
229     for (int i = 0; i < size; i++)
230     {
231         if (sortedY[i]->posX < midBall->posX)
232             leftVector.push_back(sortedY[i]);
233         else if (sortedY[i]->posX > midBall->posX)
234             rightVector.push_back(sortedY[i]);
235         else //if (sortedY[i].posX == midBall.posX)
236         {

```

```

233         if (putToRightArr)
234             rightVector.push_back(sortedY[i]);
235         else
236             leftVector.push_back(sortedY[i]);
237         putToRightArr = !putToRightArr;
238     }
239 }
240
241 //Pass subarrays to same function recursively -> 2 * T(N/2)
242 ld minLeftDist = RecursiveCall(sortedXPos, leftVector);
243 ld minRightDist = RecursiveCall(sortedXPos + leftVector.size(),
    rightVector);
244
245 //Initiliaze strip vector
246 std::vector<Ball*> strip;
247 ld stripDist = std::min(minLeftDist, minRightDist);
248
249 //Put balls which are close enough to middle into strip -> O(N)
250 for (int i = 0; i < size; i++)
251 {
252     ll distToMid = sortedY[i]->posX - midBall->posX;
253     if (distToMid < 0)
254         distToMid *= -1;
255     if (distToMid < stripDist)
256         strip.push_back(sortedY[i]);
257 }
258
259 //Search in strip and return the minimum distance -> O(N) (Since strip
    size <= N)
260 return SearchStrip(strip, stripDist);
261 }
262
263 public:
264     // Time Complexity: O(N)
265     // N (ballCount)
266     BallManager(){
267         cin >> ballCount;
268         ballArray = new Ball[ballCount];
269         for (int i = 0; i < ballCount; ++i) {
270             cin >> ballArray[i].posX;
271             cin >> ballArray[i].posY;
272             cin >> ballArray[i].posZ;
273         }
274     }
275
276     // Time Complexity: O(1)
277     ~BallManager() { delete[] ballArray;};
278
279     // Time Complexity: O(N log N)
280     //
281     ld ClosestPair()
282     {

```

```

283         //Sort original ballArray by increasing X order = O(N log N)
284         std::sort(ballArray, ballArray + ballCount, &BallManager::CompareX);
285
286         //Initiliaze sortedY vector -> O(N)
287         std::vector<Ball*> sortedY;
288         for (int i = 0; i < ballCount; i++)
289             sortedY.push_back(&ballArray[i]);
290
291         //Sort sortedY by increasing Y order = O(N log N)
292         std::sort(sortedY.begin(), sortedY.end(), &BallManager::CompareY);
293
294         //Start recursive part of the program and get result -> O(N log N)
295         ld finalMinDist = RecursiveCall(0, sortedY);
296
297         //Return the final minimum distance
298         return finalMinDist;
299     }
300 };

```

## 7.2 allGemometry

```

1 #include <iostream>
2 #include <complex>
3 #include <vector>
4 #include <cmath>
5 #include <algorithm>
6 #include <set>
7 #include <cassert>
8 #include <deque>
9 #include <numeric>
10
11 using namespace std;
12
13 // =====
14 //                                     BASICS & TYPES
15 // =====
16
17 typedef long double ld;
18 typedef ld T;
19 typedef complex<T> pt;
20
21 const T EPS = 1e-9;
22 const long double PI = acos(-1.0);
23
24 #define x real()
25 #define y imag()
26
27 bool operator==(pt a, pt b) { return fabs(a.x - b.x) < EPS && fabs(a.y - b.y) <
    EPS; }
28 bool operator!=(pt a, pt b) { return !(a == b); }
29

```



```

30 int sgn(T val) {
31     if (val > EPS) return 1;
32     if (val < -EPS) return -1;
33     return 0;
34 }
35
36 T sq(pt p) { return p.x * p.x + p.y * p.y; }
37 ld abs(pt p) { return sqrt(sq(p)); }
38
39 pt perp(pt p) { return {-p.y, p.x}; }
40
41 T dot(pt v, pt w) { return v.x * w.x + v.y * w.y; }
42 T cross(pt v, pt w) { return v.x * w.y - v.y * w.x; }
43
44 bool isPerp(pt v, pt w) { return fabs(dot(v, w)) < EPS; }
45
46 pt unitVector(pt p) {
47     ld len = abs(p);
48     if (len < EPS) return {0, 0};
49     return p / len;
50 }
51
52 // =====
53 //                                TRANSFORMATIONS
54 // =====
55
56 pt translate(pt v, pt p) { return p + v; }
57
58 // Scale point p by a factor around center c
59 pt scale(pt c, T factor, pt p) {
60     return c + (p - c) * factor;
61 }
62
63 // Rotate point p by angle a around center c
64 pt rot(pt p, pt c, ld a) {
65     pt v = p - c;
66     pt rotate = {cos(a), sin(a)};
67     return c + rotate * v;
68 }
69
70 // If point p maps to fp and q maps to fq, find the image of r
71 pt linearTransfo(pt p, pt q, pt r, pt fp, pt fq) {
72     return fp + (r - p) * (fq - fp) / (q - p);
73 }
74
75 // =====
76 //                                ANGLES
77 // =====
78
79 // (AB X AC) --> relative to AB: if (C left) return pos, else if (C right) return
    neg
80 T orient(pt a, pt b, pt c) { return cross(b - a, c - a); }

81
82 bool inAngle(pt a, pt b, pt c, pt p) {
83     T abp = orient(a, b, p), acp = orient(a, c, p), abc = orient(a, b, c);
84     if (abc < 0) swap(abp, acp);
85     return (abp >= 0 && acp <= 0) ^ (abc < 0);
86 }
87
88 T angle(pt v, pt w) {
89     return atan2(fabs(cross(v, w)), dot(v, w));
90 }
91
92 ld orientedAngle(pt a, pt b, pt c) {
93     if (orient(a, b, c) >= 0)
94         return angle(b - a, c - a);
95     else
96         return 2 * PI - angle(b - a, c - a);
97 }
98
99 ld angleTravelled(pt a, pt p, pt q) {
100     ld ampli = angle(p - a, q - a);
101     if (orient(a, p, q) > 0) return ampli;
102     else return -ampli;
103 }
104
105 bool half(pt p) {
106     return p.y > EPS || (fabs(p.y) <= EPS && p.x < -EPS);
107 }
108
109 // =====
110 //                                LINES
111 // =====
112
113 struct line {
114     pt v; T c;
115
116     // From direction vector v and offset c
117     line(pt v, T c) : v(v), c(c) {}
118
119     // From equation ax+by=c
120     line(T a, T b, T _c) {
121         v = {b, -a};
122         c = _c;
123     }
124
125     // From points P and Q
126     line(pt p, pt q) {
127         v = q - p;
128         c = cross(v, p);
129     }
130
131     T side(pt p) { return cross(v, p) - c; }
132     double dist(pt p) { return abs(side(p)) / abs(v); }

```

```

133 double sqDist(pt p) { return side(p) * side(p) / (T)sq(v); }
134 line perpThrough(pt p) { return {p, p + perp(v)}; }
135 bool cmpProj(pt p, pt q) { return dot(v, p) < dot(v, q); }
136 line translate(pt t) { return {v, c + cross(v, t)}; }
137 line shiftLeft(double dist) { return {v, c + dist * abs(v)}; }
138 pt proj(pt p) { return p - perp(v) * side(p) / sq(v); }
139 pt refl(pt p) { return p - perp(v) * (T)2.0 * side(p) / sq(v); }
140 };
141
142 bool inter(line l1, line l2, pt &out) {
143     T d = cross(l1.v, l2.v);
144     if (fabs(d) <= EPS) return false;
145     out = (l2.v * l1.c - l1.v * l2.c) / d;
146     return true;
147 }
148
149 line bisector(line l1, line l2, bool interior) {
150     assert(cross(l1.v, l2.v) != 0); // Cannot be parallel
151     T sign = interior ? 1 : -1;
152     return {l2.v / (T)abs(l2.v) + l1.v / (T)abs(l1.v) * sign,
153            l2.c / abs(l2.v) + l1.c / abs(l1.v) * sign};
154 }
155
156 // =====
157 //                               SEGMENTS
158 // =====
159
160 bool inDisk(pt a, pt b, pt p) {
161     return dot(a - p, b - p) <= EPS;
162 }
163
164 bool onSegment(pt a, pt b, pt p) {
165     return fabs(orient(a, b, p)) <= EPS && inDisk(a, b, p);
166 }
167
168 bool properInter(pt a, pt b, pt c, pt d, pt &out) {
169     T oa = orient(c, d, a), ob = orient(c, d, b);
170     T oc = orient(a, b, c), od = orient(a, b, d);
171     if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0) {
172         out = (a * ob - b * oa) / (ob - oa);
173         return true;
174     }
175     return false;
176 }
177
178 set<pair<ld, ld>> inters(pt a, pt b, pt c, pt d) {
179     set<pair<ld, ld>> s;
180     pt out;
181     if (a == c || a == d) s.insert({a.x, a.y});
182     if (b == c || b == d) s.insert({b.x, b.y});
183     if (s.size()) return s;
184

```

```

185     if (properInter(a, b, c, d, out)) return {{out.x, out.y}};
186     if (onSegment(c, d, a)) s.insert({a.x, a.y});
187     if (onSegment(c, d, b)) s.insert({b.x, b.y});
188     if (onSegment(a, b, c)) s.insert({c.x, c.y});
189     if (onSegment(a, b, d)) s.insert({d.x, d.y});
190
191     return s;
192 }
193
194 ld segPoint(pt a, pt b, pt p) {
195     if (a != b) {
196         line l(a, b);
197         if (l.cmpProj(a, p) && l.cmpProj(p, b)) return l.dist(p);
198     }
199     return min(abs(p - a), abs(p - b));
200 }
201
202 ld rayPoint(pt a, pt b, pt p) {
203     pt ab = b - a, ap = p - a;
204     if (dot(ab, ap) < 0) return abs(ap);
205     line l(a, b);
206     return l.dist(p);
207 }
208
209 ld segSeg(pt a, pt b, pt c, pt d) {
210     pt dummy;
211     if (properInter(a, b, c, d, dummy)) return 0;
212     return min({segPoint(a, b, c), segPoint(a, b, d),
213               segPoint(c, d, a), segPoint(c, d, b)});
214 }
215
216 // =====
217 //                               CIRCLES
218 // =====
219
220 pair<pt, T> circumCircle(pt a, pt b, pt c) {
221     b = b - a, c = c - a;
222     assert(cross(b, c) != 0); // No circumcircle if aligned
223     return {a + perp(b * sq(c) - c * sq(b)) / cross(b, c) / (T)2,
224            abs(perp(b * sq(c) - c * sq(b)) / cross(b, c) / (T)2)};
225 }
226
227 int circleLine(pt o, double r, line l, pair<pt, pt> &out) {
228     double h2 = r * r - l.sqDist(o);
229     if (h2 >= 0) {
230         pt p = l.proj(o);
231         pt h = l.v * (T)(sqrt(h2) / abs(l.v));
232         out = {p - h, p + h};
233     }
234     return 1 + sgn(h2);
235 }
236

```

```

237 int circleCircle(pt o1, T r1, pt o2, T r2, pair<pt, pt> &out) {
238     pt d = o2 - o1; T d2 = sq(d);
239     if (d2 == 0) { assert(r1 != r2); return 0; }
240     T pd = (d2 + r1 * r1 - r2 * r2) / 2;
241     T h2 = r1 * r1 - pd * pd / d2;
242     if (h2 >= 0) {
243         pt p = o1 + d * pd / d2, h = perp(d) * sqrt(h2 / d2);
244         out = {p - h, p + h};
245     }
246     return 1 + sgn(h2);
247 }
248
249 int tangents(pt o1, T r1, pt o2, T r2, bool inner, vector<pair<pt, pt>> &out) {
250     if (inner) r2 = -r2;
251     pt d = o2 - o1;
252     T dr = r1 - r2, d2 = sq(d), h2 = d2 - dr * dr;
253     if (d2 == 0 || h2 < 0) { assert(h2 != 0); return 0; }
254     for (T sign : {-1, 1}) {
255         pt v = (d * dr + perp(d) * sqrt(h2) * sign) / d2;
256         out.push_back({o1 + v * r1, o2 + v * r2});
257     }
258     return 1 + (h2 > 0);
259 }
260
261 // =====
262 //          BASIC POLYGONS
263 // =====
264
265 ld areaTriangle(pt a, pt b, pt c) { return abs(cross(b - a, c - a)) / 2.0; }
266
267 ld areaPolygon(vector<pt> p) {
268     ld area = 0.0;
269     for (int i = 0, n = p.size(); i < n; i++)
270         area += cross(p[i], p[(i + 1) % n]);
271     return abs(area) / 2.0;
272 }
273
274 bool isConvex(vector<pt> p) {
275     bool hasPos = false, hasNeg = false;
276     for (int i = 0, n = p.size(); i < n; i++) {
277         int o = sgn(orient(p[i], p[(i + 1) % n], p[(i + 2) % n]));
278         if (o > 0) hasPos = true;
279         if (o < 0) hasNeg = true;
280     }
281     return !(hasPos && hasNeg);
282 }
283
284 bool above(pt a, pt p) { return p.y >= a.y; }
285
286 bool crossesRay(pt a, pt p, pt q) {
287     return (above(a, q) - above(a, p)) * orient(a, p, q) > 0;
288 }

```

```

289
290 bool inPolygon(vector<pt> p, pt a, bool strict = true) {
291     int numCrossings = 0;
292     for (int i = 0, n = p.size(); i < n; i++) {
293         if (onSegment(p[i], p[(i + 1) % n], a)) return !strict;
294         numCrossings += crossesRay(a, p[i], p[(i + 1) % n]);
295     }
296     return numCrossings & 1;
297 }
298
299 bool onPolygonSide(vector<pt> p, pt a) {
300     for (int i = 0, n = p.size(); i < n; i++) {
301         if (onSegment(p[i], p[(i + 1) % n], a)) return true;
302     }
303     return false;
304 }
305
306 pt polygonCentroid(const vector<pt>& p) {
307     int n = p.size();
308     ld area = 0.0, cx = 0.0, cy = 0.0;
309     for (int i = 0; i < n; i++) {
310         int next = (i + 1) % n;
311         ld cross_val = cross(p[i], p[next]);
312         area += cross_val;
313         cx += (p[i].x + p[next].x) * cross_val;
314         cy += (p[i].y + p[next].y) * cross_val;
315     }
316     area /= 2.0;
317     return {cx / (6.0 * area), cy / (6.0 * area)};
318 }
319
320 // =====
321 //          PICK'S THEOREM
322 // =====
323
324 int boundaryPoints(const vector<pt>& p) {
325     int b = 0, n = p.size();
326     for (int i = 0; i < n; i++) {
327         int next = (i + 1) % n;
328         int dx = abs(p[i].x - p[next].x);
329         int dy = abs(p[i].y - p[next].y);
330         b += std::gcd(dx, dy);
331     }
332     return b;
333 }
334
335 int doubleAreaPolygon(const vector<pt>& p) {
336     int doubleArea = 0, n = p.size();
337     for (int i = 0; i < n; i++) {
338         int next = (i + 1) % n;
339         doubleArea += cross(p[i], p[next]);
340     }

```

```

341     return abs(doubleArea);
342 }
343
344 int interiorPoints(const vector<pt>& p) {
345     return (doubleAreaPolygon(p) - boundaryPoints(p) + 2) / 2;
346 }
347
348 // =====
349 // ADVANCED POLYGONS
350 // =====
351
352 bool cw(pt a, pt b, pt c, bool include_collinear) {
353     int o = sgn(orient(a, b, c));
354     return o < 0 || (include_collinear && o == 0);
355 }
356
357 bool collinear(pt a, pt b, pt c) { return sgn(orient(a, b, c)) == 0; }
358
359 void convex_hull(vector<pt>& a, bool include_collinear = false) {
360     if (a.empty()) return;
361     pt p0 = *min_element(a.begin(), a.end(), [](pt a, pt b) {
362         return make_pair(a.y, a.x) < make_pair(b.y, b.x);
363     });
364     sort(a.begin(), a.end(), [&p0](const pt& a, const pt& b) {
365         int o = sgn(orient(p0, a, b));
366         if (o == 0) return sq(p0 - a) < sq(p0 - b);
367         return o < 0;
368     });
369     if (include_collinear) {
370         int i = (int)a.size() - 1;
371         while (i >= 0 && collinear(p0, a[i], a.back())) i--;
372         reverse(a.begin() + i + 1, a.end());
373     }
374     vector<pt> st;
375     for (int i = 0; i < (int)a.size(); i++) {
376         while (st.size() > 1 && !cw(st[st.size() - 2], st.back(), a[i],
377             include_collinear))
378             st.pop_back();
379         if (st.empty() || a[i] != st.back())
380             st.push_back(a[i]);
381     }
382     if (!include_collinear && st.size() == 2 && st[0] == st[1])
383         st.pop_back();
384     a = st;
385 }
386
387 void reorder_polygon(vector<pt> & P) {
388     size_t pos = 0;
389     for (size_t i = 1; i < P.size(); i++) {
390         if (P[i].y < P[pos].y || (P[i].y == P[pos].y && P[i].x < P[pos].x))
391             pos = i;
392     }
393     rotate(P.begin(), P.begin() + pos, P.end());
394 }
395 // P and Q must be counter clockwise
396 vector<pt> minkowski(vector<pt> P, vector<pt> Q) {
397     reorder_polygon(P);
398     reorder_polygon(Q);
399     P.push_back(P[0]); P.push_back(P[1]);
400     Q.push_back(Q[0]); Q.push_back(Q[1]);
401
402     vector<pt> result;
403     size_t i = 0, j = 0;
404     while (i < P.size() - 2 || j < Q.size() - 2) {
405         result.push_back(P[i] + Q[j]);
406         auto crs = cross(P[i + 1] - P[i], Q[j + 1] - Q[j]);
407         if (crs >= 0 && i < P.size() - 2) ++i;
408         if (crs <= 0 && j < Q.size() - 2) ++j;
409     }
410     return result;
411 }
412
413 double maximum_dist_from_polygon_to_polygon(vector<pt> &u, vector<pt> &v) {
414     int n = (int)u.size(), m = (int)v.size();
415     double ans = 0;
416     if (n < 3 || m < 3) {
417         for (int i = 0; i < n; i++)
418             for (int j = 0; j < m; j++)
419                 ans = max(ans, (double)sq(u[i] - v[j]));
420     }
421     return sqrt(ans);
422 }
423
424 if (u[0].x > v[0].x) { swap(n, m); swap(u, v); }
425
426 int i = 0, j = 0, step = n + m + 10;
427 while (j + 1 < m && v[j].x < v[j + 1].x) j++;
428 while (step--) {
429     if (cross(u[(i + 1) % n] - u[i], v[(j + 1) % m] - v[j]) >= 0) j = (j + 1) % m;
430     else i = (i + 1) % n;
431     ans = max(ans, (double)sq(u[i] - v[j]));
432 }
433 return sqrt(ans);
434 // =====
435 // HALF-PLANE INTERSECTION
436 // =====
437
438 struct Halfplane {
439     pt p, pq;
440     long double angle;
441
442     Halfplane() {}

```

[illegible]

```

546
547 vector<pair<int, int>> all_anti_podal(int n, vector<pt> &p) {
548     vector<pair<int, int>> result;
549     auto nx = [&](int i) { return (i + 1) % n; };
550     auto pv = [&](int i) { return (i - 1 + n) % n; };
551     vector<bool> vis(n, false);
552
553     for (int p1 = 0, p2 = 0; p1 < n; ++p1) {
554         pt base = p[nx(p1)] - p[p1];
555         while (p2 == p1 || p2 == nx(p1) || sgn(cross(base, p[nx(p2)] - p[p2])) == 0 ||
556             sgn(cross(base, p[p2] - p[pv(p2)]))) {
557             p2 = nx(p2);
558         }
559         if (vis[p1]) continue;
560         vis[p1] = true;
561         result.push_back({p1, p2});
562         result.push_back({nx(p1), p2});
563
564         if (sgn(cross(base, p[nx(p2)] - p[p2])) == 0) {
565             result.push_back({p1, nx(p2)});
566             result.push_back({nx(p1), nx(p2)});
567             vis[p2] = true;
568         }
569     }
570     return result;
571 }

```

### 7.3 geometryReferenceSheet

```

1 // =====
2 // BASICS & TYPES
3 // =====
4 int sgn(T val); // Returns -1 (negative), 1 (positive), or 0.
5 T sq(pt p); // Squared length of vector p.
6 ld abs(pt p); // Length of vector p.
7 pt perp(pt p); // Returns vector rotated 90 degrees CCW.
8 T dot(pt v, pt w); // Dot product.
9 T cross(pt v, pt w); // Cross product (v X w).
10 bool isPerp(pt v, pt w); // True if dot product is 0.
11 pt unitVector(pt p); // Vector of length 1 in the same direction.
12
13 // =====
14 // TRANSFORMATIONS
15 // =====
16 pt translate(pt v, pt p); // Moves point p by vector v.
17 pt scale(pt c, T factor, pt p); // Scales p away from center c by factor.
18 pt rot(pt p, pt c, ld a); // Rotates p around c by angle a (in radians).
19 pt linearTransfo(pt p, pt q, pt r, pt fp, pt fq); // Maps r given that p->fp and q->fq.
20
21 // =====
22 // ANGLES
23 // =====
24 T orient(pt a, pt b, pt c); // Relative to AB: > 0 (C is left), < 0 (C is right), 0 (collinear).
25 bool inAngle(pt a, pt b, pt c, pt p); // True if p is inside CCW angle BAC.
26 T angle(pt v, pt w); // Shortest positive non-oriented angle [0, PI].
27 ld orientedAngle(pt a, pt b, pt c); // CCW angle BAC in range [0, 2*PI].
28 ld angleTravelled(pt a, pt p, pt q); // Signed angle travelled from P to Q around A.
29
30 // =====
31 // LINES (struct line)
32 // =====
33 line(pt v, T c); // From direction vector and offset.
34 line(T a, T b, T _c); // From equation ax + by = c.
35 line(pt p, pt q); // From two points P and Q.
36 T side(pt p); // > 0 (left of line), < 0 (right of line), 0 (on line).
37 double dist(pt p); // Shortest distance from p to line.
38 line perpThrough(pt p); // Returns perpendicular line passing through p.
39 line translate(pt t); // Translates line by vector t.
40 line shiftLeft(double dist); // Shifts line to the left by given distance.
41 pt proj(pt p); // Projection of point p onto the line.
42 pt refl(pt p); // Reflection of point p across the line.
43
44 bool inter(line l1, line l2, pt &out); // Line-line intersection. Returns false if parallel.
45 line bisector(line l1, line l2, bool interior); // Angle bisector of l1 and l2.
46
47 // =====
48 // SEGMENTS & RAYS
49 // =====
50 bool onSegment(pt a, pt b, pt p); // True if p lies on segment AB.
51 bool properInter(pt a, pt b, pt c, pt d, pt &out); // True if segments AB and CD intersect at exactly one interior point.
52 set<pair<ld, ld>> inters(pt a, pt b, pt c, pt d); // Returns all intersection points of segments AB and CD.
53 ld segPoint(pt a, pt b, pt p); // Shortest distance from p to segment AB.
54 ld rayPoint(pt a, pt b, pt p); // Shortest distance from p to ray AB.
55 ld segSeg(pt a, pt b, pt c, pt d); // Shortest distance between segment AB and segment CD.
56
57 // =====
58 // CIRCLES
59 // =====
60 pair<pt, T> circumCircle(pt a, pt b, pt c); // Returns {center, radius} of triangle ABC.
61 int circleLine(pt o, double r, line l, pair<pt, pt> &out); // Intersections of line and circle. Returns count (0, 1, or 2).
62 int circleCircle(pt o1, T r1, pt o2, T r2, pair<pt, pt> &out); // Intersections of 2 circles. Returns count.
63 int tangents(pt o1, T r1, pt o2, T r2, bool inner, vector<pair<pt, pt>> &out); //

```

*Tangent lines between 2 circles.*

```

64
65 // =====
66 //                                     BASIC POLYGONS
67 // =====
68 ld areaPolygon(vector<pt> p);          // Area of any simple polygon.
69 bool isConvex(vector<pt> p);           // True if polygon is strictly convex.
70 bool inPolygon(vector<pt> p, pt a, bool strict = true); // O(N). True if a is
    inside. strict=true excludes boundaries.
71 pt polygonCentroid(const vector<pt>& p); // Center of gravity. Vertices must be
    ordered (CW/CCW).
72
73 // =====
74 //                                     PICK'S THEOREM (Integer Coordinates)
75 // =====
76 int boundaryPoints(const vector<pt>& p); // Number of integer points on
    boundary.
77 int doubleAreaPolygon(const vector<pt>& p); // 2 * Area (exact integer).
78 int interiorPoints(const vector<pt>& p); // Number of strictly interior integer
    points.
79
80 // =====
81 //                                     ADVANCED POLYGONS
82 // =====
83 void convex_hull(vector<pt>& a, bool include_collinear = false); // O(N log N).
    Modifies vector in-place.
84 vector<pt> minkowski(vector<pt> P, vector<pt> Q); // O(N+M). Minkowski sum. P and
    Q must be CCW convex.
85 double maximum_dist_from_polygon_to_polygon(vector<pt> &u, vector<pt> &v); // O(N
    +M). Max distance between two convex polygons.
86
87 // =====
88 //                                     HALF-PLANE INTERSECTION
89 // =====
90 Halfplane(const pt& a, const pt& b); // Halfplane to the LEFT of directed line AB
    .
91 vector<pt> hp_intersect(vector<Halfplane>& H); // O(N log N). Returns CCW convex
    polygon of the intersection. Empty vector if no intersection.
92
93 // =====
94 //                                     O(LOG N) CONVEX INSIDE
95 // =====
96 bool inConvex(vector<pt> &p, pt a); // O(log N). True if point inside convex
    polygon (including boundary).
97 bool StrictInConvex(vector<pt> &p, pt a); // O(log N). True if point strictly
    inside convex polygon.
98
99 // =====
100 //                                     ANTIPODAL POINTS
101 // =====
102 vector<pair<int, int>> all_anti_podal(int n, vector<pt> &p); // O(N). Returns
    pairs of indices of antipodal points in a convex polygon.

```

## 8 String

### 8.1 Aho Corasick

#### 8.1.1 Aho corasick

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: AHO-CORASICK
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Multiple patterns", "Dictionary matching", "String DP".
6  * - Classic Scenarios: Find all occurrences of an array of strings inside a
    massive text,
7  * or count how many strings of length N don't contain any of the dictionary
    words.
8  * - The Magic: Builds a Trie and connects nodes via "fail" links. If you
    mismatch,
9  * the fail link teleports you to the longest proper suffix that exists in the
    Trie.
10 *
11 * 2. HOW TO USE IT (THE BLACK BOX)
12 * - Initialization:
13 *     AhoCorasick ac;
14 *     for (string s : patterns) ac.insert(s);
15 *     ac.build();
16 * - Query:
17 *     int total_matches = ac.search(text);
18 *
19 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
20 * - DP Transitions: The `next_state(u, c)` transitions are precalculated during
    `build()`.
21 * You can use `ac.trie[u].next[c]` to transition seamlessly in Dynamic
    Programming.
22 * - Exit Link (dict_link): Fast-forwards to the next valid dictionary word
    ending to
23 * avoid O(N) worst-case walking up the failure tree.
24 */
25
26 const int K = 26;
27
28 struct Vertex
29 {
30     int next[K];
31     bool leaf = false;
32     int p = -1;
33     char pch;
34     int link = -1;
35     int exit_link = -1;
36     int matches = 0; // Count of dictionary words ending here
37
38     Vertex(int p = -1, char ch = '$') : p(p), pch(ch)
39     {

```

```

40     fill(begin(next), end(next), -1);
41 }
42 };
43
44 struct AhoCorasick
45 {
46     vector<Vertex> t;
47
48     AhoCorasick()
49     {
50         t.emplace_back();
51     }
52
53     void insert(const string &s)
54     {
55         int v = 0;
56         for (char ch : s)
57         {
58             int c = ch - 'a';
59             if (t[v].next[c] == -1)
60             {
61                 t[v].next[c] = t.size();
62                 t.emplace_back(v, ch);
63             }
64             v = t[v].next[c];
65         }
66         t[v].leaf = true;
67         t[v].matches++;
68     }
69
70     void build()
71     {
72         queue<int> q;
73         q.push(0);
74         t[0].link = 0;
75         t[0].exit_link = 0;
76
77         while (!q.empty())
78         {
79             int v = q.front();
80             q.pop();
81
82             for (int c = 0; c < K; c++)
83             {
84                 int u = t[v].next[c];
85                 if (u != -1)
86                 {
87                     t[u].link = (v == 0) ? 0 : t[t[v].link].next[c];
88                     t[u].exit_link = t[t[u].link].leaf ? t[u].link : t[t[u].link]
89                                     .exit_link;
90                     t[u].matches += t[t[u].link].matches;
91                     q.push(u);
92                 }
93             }
94         }
95     }
96
97     long long search(const string &text)
98     {
99         long long total = 0;
100         int v = 0;
101         for (char ch : text)
102         {
103             v = t[v].next[ch - 'a'];
104             total += t[v].matches;
105         }
106         return total;
107     }
108
109     long long search_non_overlapping(const string &text)
110     {
111         long long total = 0;
112         int v = 0;
113         for (char ch : text)
114         {
115             v = t[v].next[ch - 'a'];
116
117             if (t[v].matches > 0)
118             {
119                 total++;
120                 v = 0;
121             }
122         }
123         return total;
124     }
125 };

```

## 8.2 Hashing

### 8.2.1 2D string hashing

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: 2D STRING HASHING
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Subgrid matching", "2D Pattern matching".
6  * - Classic Scenarios: Find how many times an R x C pattern appears inside
7  *   an N x M grid.
8  * - The Magic: Extends the rolling hash to two dimensions using two different

```



```

9  * bases (one for rows, one for columns). Calculates 2D prefix sums of the hash
10 *
11 * 2. HOW TO USE IT (THE BLACK BOX)
12 * - Initialization:
13 *     Hash2D h(grid);
14 * - Query:
15 *     ll val = h.get(r1, c1, r2, c2); // 0-indexed, inclusive boundaries
16 */
17
18 using ll = long long;
19
20 struct Hash2D
21 {
22     const ll MOD = 1e9 + 7;
23     const ll BASE_R = 313, BASE_C = 317;
24     vector<vector<ll>> hash;
25     vector<ll> pR, pC;
26
27     Hash2D(const vector<string> &grid)
28     {
29         if (grid.empty())
30             return;
31         int n = grid.size();
32         int m = grid[0].size();
33
34         hash.assign(n + 1, vector<ll>(m + 1, 0));
35         pR.assign(n + 1, 1);
36         pC.assign(m + 1, 1);
37
38         for (int i = 1; i <= n; i++)
39             pR[i] = (pR[i - 1] * BASE_R) % MOD;
40         for (int i = 1; i <= m; i++)
41             pC[i] = (pC[i - 1] * BASE_C) % MOD;
42
43         for (int i = 1; i <= n; i++)
44         {
45             for (int j = 1; j <= m; j++)
46             {
47                 hash[i][j] = (hash[i - 1][j] * BASE_R + hash[i][j - 1] * BASE_C
48                     + grid[i - 1][j - 1] * BASE_R % MOD * BASE_C % MOD) % MOD;
49                 if (hash[i][j] < 0)
50                     hash[i][j] += MOD;
51             }
52         }
53
54         ll get(int r1, int c1, int r2, int c2)
55         {
56             r1++;
57             c1++;

```

```

58         r2++;
59         c2++;
60         ll h = hash[r2][c2];
61         h = (h - hash[r1 - 1][c2] * pR[r2 - r1 + 1]) % MOD;
62         h = (h - hash[r2][c1 - 1] * pC[c2 - c1 + 1]) % MOD;
63         h = (h + hash[r1 - 1][c1 - 1] * pR[r2 - r1 + 1] % MOD * pC[c2 - c1 + 1])
64             % MOD;
65         if (h < 0)
66             h += MOD;
67         return h;
68     };

```

### 8.2.2 Hash(Hassan)

```

1  const int mod1 = 1e9 + 7;
2  const int mod2 = 1e9 + 9;
3  const int base1 = 313;
4  const int base2 = 317;
5
6  const int MAXN = 3e4 + 100;
7  int pw1[MAXN], pw2[MAXN];
8  bool isCalPower = false;
9
10 void pre() {
11     if (isCalPower) return;
12
13     pw1[0] = pw2[0] = 1;
14     for (int i = 1; i < MAXN; i++) {
15         pw1[i] = (pw1[i - 1] * base1) % mod1;
16         pw2[i] = (pw2[i - 1] * base2) % mod2;
17     }
18     isCalPower = true;
19 }
20
21 struct Hash {
22     int n;
23     vector<pair<int, int>> pref;
24     vector<pair<int, int>> suf; // Used only if palindrome checking is needed
25
26     // Pass `compute_rev = true` if you need O(1) palindrome checks
27     Hash(const string& s, bool compute_rev = false) {
28         pre();
29         n = s.size();
30         pref.assign(n + 1, {0, 0});
31
32         for (int i = 0; i < n; i++) {
33             pref[i + 1].first = (pref[i].first * base1 + s[i]) % mod1;
34             pref[i + 1].second = (pref[i].second * base2 + s[i]) % mod2;
35         }
36

```

```

37     if (compute_rev) {
38         suf.assign(n + 1, {0, 0});
39         for (int i = n - 1; i >= 0; i--) {
40             suf[i].first = (suf[i + 1].first * base1 + s[i]) % mod1;
41             suf[i].second = (suf[i + 1].second * base2 + s[i]) % mod2;
42         }
43     }
44 }
45
46 // Get hash of substring s[l...r] (0-indexed) in O(1)
47 pair<int, int> get(int l, int r) {
48     if (l > r) return {0, 0};
49     int len = r - l + 1;
50
51     int h1 = (pref[r+1].first - (pref[l].first * pw1[len]) % mod1) % mod1;
52     if (h1 < 0) h1 += mod1;
53
54     int h2 = (pref[r+1].second - (pref[l].second * pw2[len]) % mod2) % mod2;
55     if (h2 < 0) h2 += mod2;
56
57     return {h1, h2};
58 }
59
60 // Get reverse hash of substring s[l...r] in O(1)
61 pair<int, int> get_rev(int l, int r) {
62     if (l > r || suf.empty()) return {0, 0};
63     int len = r - l + 1;
64     int h1 = (suf[l].first - suf[r + 1].first * pw1[len]) % mod1;
65     if (h1 < 0) h1 += mod1;
66
67     int h2 = (suf[l].second - suf[r + 1].second * pw2[len]) % mod2;
68     if (h2 < 0) h2 += mod2;
69
70     return {h1, h2};
71 }
72
73 // Check if s[l...r] is a palindrome in O(1)
74 bool is_pal(int l, int r) {
75     return get(l, r) == get_rev(l, r);
76 }
77 };
78
79
80 // --- USEFUL HASHING UTILITY FUNCTIONS ---
81
82 // Concatenate two hashes (h1 followed by h2 of length len2) in O(1)
83 pair<int, int> concat_hash(pair<int, int> h1, pair<int, int> h2, int len2) {
84     int ans1 = (h1.first * pw1[len2] + h2.first) % mod1;
85     int ans2 = (h1.second * pw2[len2] + h2.second) % mod2;
86     return {ans1, ans2};
87 }
88

```

```

89 // Find Longest Common Prefix (LCP) length of s1[l1...r1] and s2[l2...r2] in O(
    log N)
90 int get_lcp(Hash& sh1, int l1, int r1, Hash& sh2, int l2, int r2) {
91     int low = 1, high = min(r1 - l1 + 1, r2 - l2 + 1), ans = 0;
92     while (low <= high) {
93         int mid = low + (high - low) / 2;
94         if (sh1.get(l1, l1 + mid - 1) == sh2.get(l2, l2 + mid - 1)) {
95             ans = mid;
96             low = mid + 1;
97         } else {
98             high = mid - 1;
99         }
100     }
101     return ans;
102 }
103
104 // Compare s1[l1...r1] and s2[l2...r2] lexicographically in O(log N)
105 // Returns: -1 if s1 < s2, 0 if s1 == s2, 1 if s1 > s2
106 int compare_substrings(const string& s1, Hash& sh1, int l1, int r1,
107     const string& s2, Hash& sh2, int l2, int r2) {
108     int lcp_len = get_lcp(sh1, l1, r1, sh2, l2, r2);
109     int len1 = r1 - l1 + 1;
110     int len2 = r2 - l2 + 1;
111
112     // If one is completely a prefix of another
113     if (lcp_len == len1 && lcp_len == len2) return 0;
114     if (lcp_len == len1) return -1;
115     if (lcp_len == len2) return 1;
116
117     // Compare the first differing character
118     return s1[l1 + lcp_len] < s2[l2 + lcp_len] ? -1 : 1;
119 }

```

### 8.2.3 Hash

```

1 struct Hash
2 {
3     int base, mod1, mod2, inv1, inv2, n;
4     vector<int> pw1, pw2, invpw1, invpw2, pre1, pre2;
5
6     int power(int b, int p, int m)
7     {
8         int res{1};
9         b %= m;
10        while (p)
11        {
12            if (p & 1)
13                res = res * b % m;
14            b = b * b % m;
15            p >>= 1;
16        }

```

```

17     return res;
18 }
19
20 Hash(const string &s, int b)
21 {
22     base = b;
23     mod1 = 1e9 + 7;
24     mod2 = 1e9 + 9;
25     n = int(s.size());
26
27     inv1 = power(base, mod1 - 2, mod1);
28     inv2 = power(base, mod2 - 2, mod2);
29
30     pw1.assign(n + 1, 1), pw2.assign(n + 1, 1);
31     invpw1.assign(n + 1, 1), invpw2.assign(n + 1, 1);
32     pre1.assign(n + 1, {}), pre2.assign(n + 1, {});
33
34     for (int i = 0; i < n; i++)
35     {
36         pw1[i + 1] = pw1[i] * base % mod1;
37         pw2[i + 1] = pw2[i] * base % mod2;
38         invpw1[i + 1] = invpw1[i] * inv1 % mod1;
39         invpw2[i + 1] = invpw2[i] * inv2 % mod2;
40         pre1[i + 1] = (pre1[i] + pw1[i] * s[i]) % mod1;
41         pre2[i + 1] = (pre2[i] + pw2[i] * s[i]) % mod2;
42     }
43 }
44
45 pair<int, int> get(int l, int r)
46 {
47     return {(pre1[r + 1] - pre1[l] + mod1) * invpw1[l] % mod1, (pre2[r + 1] -
48         pre2[l] + mod2) * invpw2[l] % mod2};
49 };

```

## 8.2.4 MultisetHashing

```

1 struct MultisetHashing
2 {
3     int base, m, n;
4     vector<int> pre;
5
6     int power(int b, int p)
7     {
8         b %= m;
9         int res{1};
10        while (p)
11        {
12            if (p & 1)
13                res = res * b % m;
14            b = b * b % m;

```

```

15        p >>= 1;
16    }
17    return res;
18 }
19
20 MultisetHashing(vector<int> &arr, int b)
21 {
22     n = int(arr.size());
23     m = 1e9 + 7;
24     base = b;
25     pre.assign(n + 1, {});
26     for (int i = 0; i < n; i++)
27         pre[i + 1] = (pre[i] + power(base, arr[i])) % m;
28 }
29
30 int get(int l, int r)
31 {
32     // 0-based
33     return (pre[r + 1] - pre[l] + m) % m;
34 }
35 };

```

## 8.2.5 XOR hashing

```

1 struct XorHashing
2 {
3     int n;
4     map<int, int> h;
5     vector<int> pre;
6
7     XorHashing(vector<int> &arr)
8     {
9         n = int(arr.size());
10        pre.assign(n + 1, {});
11        for (int i = 0; i < n; i++)
12        {
13            if (!h.count(arr[i]))
14                h[arr[i]] = getRandom(1, 1e18);
15            pre[i + 1] = pre[i] ^ h[arr[i]];
16        }
17    }
18
19    int get(int l, int r)
20    {
21        // 0-based
22        return pre[r + 1] ^ pre[l];
23    }
24 };

```

## 8.3 LCS

### 8.3.1 Bitset LCS

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: BITSET LCS
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "LCS of massive strings", " $O(N^2)$  TLE".
6  * - Classic Scenarios: Find the Longest Common Subsequence of two strings where
7  *    $N$  and  $M$  are up to 50,000.
8  * - The Magic: C++ `std::bitset` does not support subtraction, which is required
9  *   for the classic bitwise LCS formula. This custom block engine maps the DP
10  *   states
11  *   directly onto native 64-bit unsigned integers (uint64_t). It computes 64 DP
12  *   cells simultaneously using carry-flag subtraction, slashing the  $O(N * M)$ 
13  *   time
14  *   by a factor of 64.
15  *
16  * 2. HOW TO USE IT (THE BLACK BOX)
17  * - Query:
18  *   int length = bitset_lcs(string_A, string_B);
19  *
20  * - Complexity:
21  *   Time:  $O(N * M / 64)$ 
22  *   Space:  $O(M / 64)$ 
23  */
24
25 #include <cstdint>
26
27 int bitset_lcs(const string &a, const string &b)
28 {
29     int n = a.length(), m = b.length();
30     if (n == 0 || m == 0)
31         return 0;
32
33     int words = (m >> 6) + 1;
34     vector<vector<uint64_t>> pos(256, vector<uint64_t>(words, 0));
35
36     for (int i = 0; i < m; i++)
37     {
38         pos[b[i]][i >> 6] |= (1ULL << (i & 63));
39     }
40
41     vector<uint64_t> D(words, 0);
42
43     for (int i = 0; i < n; i++)
44     {
45         uint64_t b_carry = 1;
46         uint64_t shift_carry = 0;
47
48         for (int j = 0; j < words; j++)

```

```

48     {
49         uint64_t P = D[j] | pos[a[i]][j];
50         uint64_t d_shl = (D[j] << 1) | shift_carry;
51         shift_carry = D[j] >> 63;
52
53         uint64_t sub = d_shl + b_carry;
54         uint64_t next_carry = (P < sub) || (sub < d_shl);
55         uint64_t diff = P - sub;
56
57         D[j] = P & (~P ^ diff);
58         b_carry = next_carry;
59     }
60
61     int lcs_length = 0;
62     for (int j = 0; j < words; j++)
63     {
64         uint64_t val = D[j];
65         while (val)
66         {
67             val &= (val - 1);
68             lcs_length++;
69         }
70     }
71
72     return lcs_length;
73 }

```

## 8.4 Matching

### 8.4.1 FFT wildcard matching

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: FFT WILDCARD MATCHING
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Wildcards in both text and pattern", "Massive strings".
6  * - Classic Scenarios: Both strings have '?' wildcards. Bitsets will TLE if
7  *    $N = 200,000$ .
8  * - The Magic: Uses the algebraic identity  $(T - P)^2 * T * P = 0$ . Maps wildcards
9  *   to 0 and letters to 1-26. Expands the polynomial into three separate NTT
10  *   convolutions. A result of 0 at index  $i$  means a perfect match.
11  *
12  * 2. HOW TO USE IT (THE BLACK BOX)
13  * - Query:
14  *   vector<int> matches = fft_wildcard_match(text, pattern);
15  *
16  * 3. HOW TO ADAPT IT
17  * - Dependencies: This heavily relies on your `multiply(A, B)` function from
18  *   your NTT template. Make sure it is included above this.
19  */

```

```

20
21 using ll = long long;
22
23 vector<ll> multiply(vector<ll> const &a, vector<ll> const &b);
24
25 vector<int> fft_wildcard_match(const string &text, const string &pattern)
26 {
27     int n = text.length(), m = pattern.length();
28     if (m > n)
29         return {};
30
31     vector<ll> t1(n), t2(n), t3(n);
32     vector<ll> p1(m), p2(m), p3(m);
33
34     for (int i = 0; i < n; i++)
35     {
36         ll val = (text[i] == '?' ? 0 : text[i] - 'a' + 1);
37         t1[i] = val;
38         t2[i] = val * val;
39         t3[i] = val * val * val;
40     }
41
42     for (int i = 0; i < m; i++)
43     {
44         ll val = (pattern[i] == '?' ? 0 : pattern[i] - 'a' + 1);
45         p1[m - 1 - i] = val;
46         p2[m - 1 - i] = val * val;
47         p3[m - 1 - i] = val * val * val;
48     }
49
50     vector<ll> res1 = multiply(t3, p1);
51     vector<ll> res2 = multiply(t2, p2);
52     vector<ll> res3 = multiply(t1, p3);
53
54     vector<int> matches;
55     for (int i = m - 1; i < n; i++)
56     {
57         ll diff = res1[i] - 2 * res2[i] + res3[i];
58         if (diff == 0)
59         {
60             matches.push_back(i - (m - 1));
61         }
62     }
63     return matches;
64 }

```

#### 8.4.2 KMP

```

1 vector<int> KMP(string s, string p = "")
2 {
3     if (!p.empty())

```

```

4         s = p + "$" + s;
5         int n = int(s.size());
6         vector<int> LPS(n);
7         int i{1}, j{};
8         while (i < n)
9         {
10             if (s[i] == s[j])
11             {
12                 j++;
13                 LPS[i] = j;
14                 i++;
15             }
16             else
17             {
18                 if (j == 0)
19                 {
20                     LPS[i] = 0;
21                     i++;
22                 }
23                 else
24                     j = LPS[j - 1];
25             }
26         }
27         return LPS;
28 }
29
30 int numberOfMatches(string &s, string &p)
31 {
32     vector<int> pi = KMP(s, p);
33     int ans{};
34     for (auto &val : pi)
35         ans += val == int(p.size());
36     return ans;
37 }
38
39 vector<int> countPrefixMatching(string &s)
40 {
41     int n = int(s.size());
42     vector<int> pi = KMP(s);
43     vector<int> frq(n);
44     for (auto &x : pi)
45         frq[x]++;
46     for (int i = n - 1; i; i--)
47         frq[pi[i - 1]] += frq[i];
48     return frq;
49 }
50
51 vector<vector<int>> KMPAutomaton(string &s)
52 {
53     vector<int> pi = KMP(s);
54     int n = int(s.size());
55     vector aut(n + 1, vector<int>(26));

```

```

56     aut[0][s[0] - 'a'] = 1;
57     for (int i = 1; i <= n; i++)
58     {
59         for (int j = 0; j < 26; j++)
60         {
61             if (i < n && s[i] - 'a' == j)
62                 aut[i][j] = i + 1;
63             else
64                 aut[i][j] = aut[pi[i - 1]][j];
65         }
66     }
67     return aut;
68 }

```

### 8.4.3 String matching using bitset

```

1  /**
2   * THE ULTIMATE CP TEMPLATE BLUEPRINT: BITSET STRING MATCHING
3   *
4   * 1. WHAT PROBLEM DOES THIS SOLVE?
5   * - Keywords: "Wildcards in text", "Small alphabet", "Multiple patterns".
6   * - Classic Scenarios: Text has wildcards. Find all occurrences of pattern.
7   * - The Magic: Represents occurrences of each character as a bitset. By shifting
8   *   and ANDing the bitsets corresponding to the pattern's characters, you
9   *   instantly
10  *   identify valid shift offsets in  $O(N * M / 64)$  time.
11  *
12  * 2. HOW TO USE IT (THE BLACK BOX)
13  * - Query:
14  *   bitset<MAX_N> matches = bitset_match(text, pattern);
15  *   // If matches[i] is 1, the pattern exists starting at text[i].
16  *
17  * 3. HOW TO ADAPT IT
18  * - MAX_N must be a constant (e.g., 100005) because bitsets require compile-time
19  *   sizes.
20  */
21 const int MAX_N = 100005;
22 bitset<MAX_N> bitset_match(const string &text, const string &pattern)
23 {
24     vector<bitset<MAX_N>> mask(26);
25     for (int i = 0; i < text.length(); i++)
26     {
27         if (text[i] != '?')
28         {
29             mask[text[i] - 'a'].set(i);
30         }
31         else
32         {
33             for (int c = 0; c < 26; c++)

```

```

34             mask[c].set(i);
35         }
36     }
37
38     bitset<MAX_N> ans;
39     ans.set();
40
41     for (int i = 0; i < pattern.length(); i++)
42     {
43         if (pattern[i] != '?')
44         {
45             ans &= (mask[pattern[i] - 'a'] >> i);
46         }
47     }
48     return ans;
49 }

```

### 8.4.4 Z

```

1 struct ZAlgorithm
2 {
3     vector<int> build(const string &s)
4     {
5         int n = s.size();
6         vector<int> z(n, 0);
7         int l = 0, r = 0;
8         for (int i = 1; i < n; i++)
9         {
10             if (i < r)
11                 z[i] = min(r - i, z[i - l]);
12             while (i + z[i] < n && s[z[i]] == s[i + z[i]])
13                 z[i]++;
14             if (i + z[i] > r)
15             {
16                 l = i;
17                 r = i + z[i];
18             }
19         }
20         return z;
21     }
22
23     int numberOfMatches(const string &s, const string &p)
24     {
25         if (p.empty() || s.empty())
26             return 0;
27         string combined = p + "$" + s;
28         vector<int> z = build(combined);
29         int ans = 0;
30         int p_len = p.size();
31         for (int i = p_len + 1; i < z.size(); i++)
32         {

```

```

33         if (z[i] == p_len)
34             ans++;
35     }
36     return ans;
37 }
38
39 vector<int> findOccurrences(const string &s, const string &p)
40 {
41     if (p.empty() || s.empty())
42         return {};
43     string combined = p + "$" + s;
44     vector<int> z = build(combined);
45     vector<int> occ;
46     int p_len = p.size();
47     for (int i = p_len + 1; i < z.size(); i++)
48     {
49         if (z[i] == p_len)
50             occ.push_back(i - p_len - 1);
51     }
52     return occ;
53 }
54 };

```

## 8.5 Misc

### 8.5.1 Debruijn sequence

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DE BRUIJN SEQUENCE
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Contains all permutations", "Shortest string with all substrings".
6  * - Classic Scenarios: Generate the shortest possible string that contains every
7  * possible sequence of length N using an alphabet of size K.
8  * - The Magic: Mathematical guarantees state this string will always have length
9  *  $K^N + N - 1$ . This uses a recursive DFS (essentially finding an Eulerian path)
10 * to generate the lexicographically smallest De Bruijn sequence in optimal
11 * time.
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     string sequence = de_bruijn(K, N);
15 * - Example:
16 *     de_bruijn(2, 3) generates "00010111".
17 *     (Note: To act as a true cyclic sequence, you conceptually wrap the ends.
18 *     If you need it strictly linear, append the first N-1 characters to the
19 *     end).
20 * - Complexity:

```

```

21 *     Time:  $O(K^N)$ 
22 *     Space:  $O(K * N)$ 
23 */
24
25 string de_bruijn(int k, int n)
26 {
27     if (k == 1)
28         return string(n, '0');
29     if (n == 1)
30     {
31         string res = "";
32         for (int i = 0; i < k; i++)
33             res += (char)('0' + i);
34         return res;
35     }
36
37     vector<int> a(k * n, 0);
38     string sequence;
39
40     auto db = [&](auto &self, int t, int p) -> void
41     {
42         if (t > n)
43         {
44             if (n % p == 0)
45             {
46                 for (int i = 1; i <= p; i++)
47                 {
48                     sequence += (char)('0' + a[i]);
49                 }
50             }
51         }
52         else
53         {
54             a[t] = a[t - p];
55             self(self, t + 1, p);
56             for (int j = a[t - p] + 1; j < k; j++)
57             {
58                 a[t] = j;
59                 self(self, t + 1, t);
60             }
61         }
62     };
63
64     db(db, 1, 1);
65     return sequence;
66 }

```

### 8.5.2 Expression parser

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: EXPRESSION PARSER

```

```

3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Evaluate expression", "Calculator", "Infix to Postfix".
6  * - Classic Scenarios: You are given a raw string like "3 + 5 * (2 - 8)" and
7  *   need the exact mathematical result.
8  * - The Magic: Uses Recursive Descent to strictly enforce PEMDAS (Parentheses,
9  *   Exponents, Multiplication/Division, Addition/Subtraction) in pure O(N) time
10 *
11 * 2. HOW TO USE IT (THE BLACK BOX)
12 * - Query:
13 *   string expr = "10 + 2 * (6 - 3)";
14 *   int pos = 0;
15 *   long long result = parse_expr(expr, pos);
16 *
17 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
18 * - Division by Zero: Currently, it performs integer division. If a problem
19 *   tests division by zero, add a check inside `parse_term`.
20 * - Modulo: If the expression must be evaluated modulo M, wrap all additions,
21 *   subtractions, and multiplications in the respective functions with `% MOD`.
22 */
23
24 using ll = long long;
25
26 ll parse_expr(const string &s, int &i);
27 ll parse_term(const string &s, int &i);
28 ll parse_factor(const string &s, int &i);
29 ll parse_number(const string &s, int &i);
30
31 void skip_spaces(const string &s, int &i)
32 {
33     while (i < s.length() && isspace(s[i]))
34         i++;
35 }
36
37 ll parse_expr(const string &s, int &i)
38 {
39     ll res = parse_term(s, i);
40     while (i < s.length())
41     {
42         skip_spaces(s, i);
43         if (i < s.length() && (s[i] == '+' || s[i] == '-'))
44         {
45             char op = s[i++];
46             ll next_term = parse_term(s, i);
47             if (op == '+')
48                 res += next_term;
49             else
50                 res -= next_term;
51         }
52         else
53         {
54             break;
55         }
56     }
57     return res;
58 }
59
60 ll parse_term(const string &s, int &i)
61 {
62     ll res = parse_factor(s, i);
63     while (i < s.length())
64     {
65         skip_spaces(s, i);
66         if (i < s.length() && (s[i] == '*' || s[i] == '/'))
67         {
68             char op = s[i++];
69             ll next_factor = parse_factor(s, i);
70             if (op == '*')
71                 res *= next_factor;
72             else
73                 res /= next_factor;
74         }
75         else
76         {
77             break;
78         }
79     }
80     return res;
81 }
82
83 ll parse_factor(const string &s, int &i)
84 {
85     skip_spaces(s, i);
86     if (i < s.length() && s[i] == '(')
87     {
88         i++;
89         ll res = parse_expr(s, i);
90         skip_spaces(s, i);
91         if (i < s.length() && s[i] == ')')
92             i++;
93         return res;
94     }
95     return parse_number(s, i);
96 }
97
98 ll parse_number(const string &s, int &i)
99 {
100     skip_spaces(s, i);
101     ll res = 0;
102     while (i < s.length() && isdigit(s[i]))
103     {
104         res = res * 10 + (s[i++] - '0');
105     }
106     return res;

```



### 8.5.3 Min cyclic shift

107 }

```

1  /**
2   *  THE ULTIMATE CP TEMPLATE BLUEPRINT: DUVAL'S & MIN CYCLIC SHIFT
3   *
4   *  1.  WHAT PROBLEM DOES THIS SOLVE?
5   *  - Keywords: "Lexicographically smallest rotation", "Lyndon words".
6   *  - Classic Scenarios: You can rotate a string S any number of times. Find the
7   *    rotation that comes first in the dictionary.
8   *  - The Magic: Duval's algorithm processes the string in strict  $O(N)$  time and
9   *     $O(1)$  auxiliary space, bypassing the need to build a heavy Suffix Automaton
10  *    on the concatenated string `S + S`.
11  *
12  *  2.  HOW TO USE IT (THE BLACK BOX)
13  *  - Query:
14  *    int best_index = min_cyclic_shift(S);
15  *    string smallest_rotation = S.substr(best_index) + S.substr(0, best_index);
16  *    vector<string> factorization = duval_factorization(S);
17  *
18  *  - Complexity:
19  *    Time:  $O(N)$ 
20  *    Space:  $O(1)$  for shift,  $O(N)$  for factorization return.
21  */
22
23 int min_cyclic_shift(string s)
24 {
25     s += s;
26     int n = s.length();
27     int i = 0, ans = 0;
28     while (i < n / 2)
29     {
30         ans = i;
31         int j = i + 1, k = i;
32         while (j < n && s[k] <= s[j])
33         {
34             if (s[k] < s[j])
35                 k = i;
36             else
37                 k++;
38             j++;
39         }
40         while (i <= k)
41             i += j - k;
42     }
43     return ans;
44 }
45
46 vector<string> duval_factorization(const string &s)

```

```

47 {
48     int n = s.length();
49     int i = 0;
50     vector<string> factorization;
51     while (i < n)
52     {
53         int j = i + 1, k = i;
54         while (j < n && s[k] <= s[j])
55         {
56             if (s[k] < s[j])
57                 k = i;
58             else
59                 k++;
60             j++;
61         }
62         while (i <= k)
63         {
64             factorization.push_back(s.substr(i, j - k));
65             i += j - k;
66         }
67     }
68     return factorization;
69 }

```

## 8.6 Palindromes

### 8.6.1 Manacher

```

1 struct Manacher
2 {
3     vector<int> p;
4     string s;
5
6     Manacher(const string &str)
7     {
8         s = str;
9         string t = "";
10        for (char c : s)
11        {
12            t += "#";
13            t += c;
14        }
15        t += "#";
16
17        int n = t.size();
18        string temp = "$" + t + "^";
19        vector<int> p_temp(n + 2, 0);
20        int l = 0, r = 1;
21
22        for (int i = 1; i <= n; i++)
23        {

```

```

24     if (i <= r)
25         p_temp[i] = min(r - i, p_temp[l + (r - i)]);
26     while (temp[i - p_temp[i]] == temp[i + p_temp[i]])
27         p_temp[i]++;
28     if (i + p_temp[i] > r)
29     {
30         l = i - p_temp[i];
31         r = i + p_temp[i];
32     }
33 }
34
35 p.assign(n, 0);
36 for (int i = 1; i <= n; i++)
37 {
38     p[i - 1] = p_temp[i] - 1;
39 }
40 }
41
42 bool is_palindrome(int l, int r)
43 {
44     if (l > r)
45         swap(l, r);
46     int len = r - l + 1;
47     int idx = l + r + 1;
48     return p[idx] >= len;
49 }
50
51 string longest_palindrome()
52 {
53     int max_len = 0, center_idx = 0;
54     for (int i = 0; i < p.size(); i++)
55     {
56         if (p[i] > max_len)
57         {
58             max_len = p[i];
59             center_idx = i;
60         }
61     }
62     int start = (center_idx - max_len) / 2;
63     return s.substr(start, max_len);
64 }
65 };

```

### 8.6.2 Palindromic tree

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: PALINDROMIC TREE (EERTREE)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Distinct palindromes", "Occurrences of palindromes".
6  * - Classic Scenarios: Find the number of distinct palindromic substrings, or

```

```

7  * find the palindrome P that maximizes length(P) * occurrences(P).
8  * - The Magic: It builds a graph containing exactly two roots: one for even-
9  * length
10 * palindromes and one for odd-length. Every node represents a distinct
11 * palindrome,
12 * and fail links point to the longest proper palindromic suffix.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Initialization:
16 *     Eertree pt;
17 *     for (char c : S) pt.add_char(c);
18 *     pt.count_occurrences(); // Must call this after building!
19 *
20 * - Complexity:
21 *     Time: Amortized O(N)
22 *     Space: O(N * ALPHABET_SIZE)
23 */
24
25 struct Eertree
26 {
27     static const int K = 26;
28     struct Node
29     {
30         int next[K];
31         int len;
32         int link;
33         long long cnt;
34
35         Node(int len = 0, int link = 0) : len(len), link(link), cnt(0)
36         {
37             fill(begin(next), end(next), 0);
38         }
39     };
40
41     vector<Node> tree;
42     string s;
43     int last;
44
45     Eertree()
46     {
47         tree.emplace_back(0, 1); // Node 0: Even length root
48         tree.emplace_back(-1, 1); // Node 1: Odd length root
49         last = 0;
50     }
51
52     void add_char(char ch)
53     {
54         s += ch;
55         int c = ch - 'a';
56         int curr = last;
57         int pos = s.length() - 1;

```

```

57     while (pos - 1 - tree[curr].len < 0 || s[pos - 1 - tree[curr].len] != ch)4
58     {
59         curr = tree[curr].link;
60     }
61
62     if (tree[curr].next[c])
63     {
64         last = tree[curr].next[c];
65         tree[last].cnt++;
66         return;
67     }
68
69     int now = tree.size();
70     tree.emplace_back(tree[curr].len + 2);
71     tree[curr].next[c] = now;
72
73     if (tree[now].len == 1)
74     {
75         tree[now].link = 0;
76     }
77     else
78     {
79         int fail = tree[curr].link;
80         while (pos - 1 - tree[fail].len < 0 || s[pos - 1 - tree[fail].len] !=
81             ch)
82         {
83             fail = tree[fail].link;
84         }
85         tree[now].link = tree[fail].next[c];
86     }
87
88     last = now;
89     tree[last].cnt++;
90 }
91
92 void count_occurrences()
93 {
94     for (int i = tree.size() - 1; i > 1; i--)
95     {
96         tree[tree[i].link].cnt += tree[i].cnt;
97     }
98 };

```

## 8.7 Suffix Structures

### 8.7.1 Suffix array

```

1 struct SuffixArray
2 {
3     string s;

```

```

4     int n;
5     vector<int> p;
6     vector<vector<int>>> C;
7     vector<int> lcp;
8     bool is_cyclic;
9
10    SuffixArray(string str, bool cyclic = false)
11    {
12        is_cyclic = cyclic;
13        s = str;
14        if (!is_cyclic)
15            s += '$';
16        n = s.size();
17        const int alphabet = 256;
18        p.assign(n, 0);
19        vector<int> c(n), cnt(max(n, alphabet), 0);
20
21        for (int i = 0; i < n; i++)
22            cnt[s[i]]++;
23        for (int i = 1; i < alphabet; i++)
24            cnt[i] += cnt[i - 1];
25        for (int i = 0; i < n; i++)
26            p[--cnt[s[i]]] = i;
27
28        c[p[0]] = 0;
29        int classes = 1;
30        for (int i = 1; i < n; i++)
31        {
32            if (s[p[i]] != s[p[i - 1]])
33                classes++;
34            c[p[i]] = classes - 1;
35        }
36        C.push_back(c);
37
38        vector<int> pn(n), cn(n);
39        for (int h = 0; (1 << h) < n; h++)
40        {
41            for (int i = 0; i < n; i++)
42            {
43                pn[i] = p[i] - (1 << h);
44                if (pn[i] < 0)
45                    pn[i] += n;
46            }
47            fill(cnt.begin(), cnt.begin() + classes, 0);
48            for (int i = 0; i < n; i++)
49                cnt[C.back()[pn[i]]]++;
50            for (int i = 1; i < classes; i++)
51                cnt[i] += cnt[i - 1];
52            for (int i = n - 1; i >= 0; i--)
53                p[--cnt[C.back()[pn[i]]]] = pn[i];
54
55            cn[p[0]] = 0;

```

```

56     classes = 1;
57     for (int i = 1; i < n; i++)
58     {
59         pair<int, int> curr = {C.back()[p[i]], C.back()[p[i] + (1 << h) % n]};
60         pair<int, int> prev = {C.back()[p[i - 1]], C.back()[p[i - 1] + (1 << h) % n]};
61         if (curr != prev)
62             classes++;
63         cn[p[i]] = classes - 1;
64     }
65     C.push_back(cn);
66 }
67
68 if (!is_cyclic)
69 {
70     p.erase(p.begin());
71     int orig_n = n - 1;
72     vector<int> rank(orig_n, 0);
73     for (int i = 0; i < orig_n; i++)
74         rank[p[i]] = i;
75     int k = 0;
76     lcp.assign(orig_n - 1, 0);
77     for (int i = 0; i < orig_n; i++)
78     {
79         if (rank[i] == orig_n - 1)
80         {
81             k = 0;
82             continue;
83         }
84         int j = p[rank[i] + 1];
85         while (i + k < orig_n && j + k < orig_n && s[i + k] == s[j + k])
86             k++;
87         lcp[rank[i]] = k;
88         if (k)
89             k--;
90     }
91 }
92 }
93
94 string smallestCyclicShift()
95 {
96     return s.substr(p[0]) + s.substr(0, p[0]);
97 }
98
99 vector<int> subStringInString(string t)
100 {
101     int m = t.size();
102     int orig_n = n - 1;
103     int l = 0, r = orig_n - 1;
104     int first = -1;
105     while (l <= r)
106     {
107         int mid = l + (r - l) / 2;
108         auto can = [&]()
109         {
110             for (int i = 0; i < m; i++)
111             {
112                 if (p[mid] + i >= orig_n)
113                     return 1LL;
114                 if (s[p[mid] + i] != t[i])
115                 {
116                     if (s[p[mid] + i] < t[i])
117                         return 1LL;
118                     if (s[p[mid] + i] > t[i])
119                         return 2LL;
120                 }
121             }
122             return 0LL;
123         };
124         int res = can();
125         if (res == 0)
126         {
127             first = mid;
128             r = mid - 1;
129         }
130         else if (res == 1)
131         {
132             l = mid + 1;
133         }
134         else
135         {
136             r = mid - 1;
137         }
138     }
139     if (first == -1)
140         return {};
141     l = 0, r = orig_n - 1;
142     int last = -1;
143     while (l <= r)
144     {
145         int mid = l + (r - l) / 2;
146         auto can = [&]()
147         {
148             for (int i = 0; i < m; i++)
149             {
150                 if (p[mid] + i >= orig_n)
151                     return 1LL;
152                 if (s[p[mid] + i] != t[i])
153                 {
154                     if (s[p[mid] + i] < t[i])
155                         return 1LL;
156                     if (s[p[mid] + i] > t[i])
157                         return 2LL;

```

```

158         }
159     }
160     return 0LL;
161 };
162 int res = can();
163 if (res == 0)
164 {
165     last = mid;
166     l = mid + 1;
167 }
168 else if (res == 1)
169 {
170     l = mid + 1;
171 }
172 else
173 {
174     r = mid - 1;
175 }
176 }
177 vector<int> ans;
178 for (int i = first; i <= last; i++)
179 {
180     ans.push_back(p[i]);
181 }
182 sort(ans.begin(), ans.end());
183 return ans;
184 }
185
186 int compareSubstrings(int i, int len1, int j, int len2)
187 {
188     int l = min(len1, len2);
189     if (l > 0)
190     {
191         int k = __lg(l);
192         pair<int, int> a = {C[k][i], C[k][i + 1 - (1 << k)]};
193         pair<int, int> b = {C[k][j], C[k][j + 1 - (1 << k)]};
194         if (a != b)
195             return a < b ? -1 : 1;
196     }
197     if (len1 == len2)
198         return 0;
199     return len1 < len2 ? -1 : 1;
200 }
201
202 int numberOfDifferentSubStrings()
203 {
204     int orig_n = n - 1;
205     return (orig_n * orig_n + orig_n) / 2 - accumulate(lcp.begin(), lcp.end(), 0LL);
206 }
207
208 string longestCommonSubstring(string t)

```

```

209 {
210     string orig_s = s;
211     if (!is_cyclic)
212         orig_s.pop_back();
213     int len1 = orig_s.size();
214     SuffixArray sa(orig_s + char(1) + t);
215     int max_len = 0;
216     int start_idx = -1;
217     for (int i = 0; i < (int)sa.lcp.size(); i++)
218     {
219         if ((sa.p[i] < len1 && sa.p[i + 1] > len1) || (sa.p[i] > len1 && sa.p
220             [i + 1] < len1))
221         {
222             if (sa.lcp[i] > max_len)
223             {
224                 max_len = sa.lcp[i];
225                 start_idx = sa.p[i];
226             }
227         }
228         if (max_len == 0)
229             return "";
230         return sa.s.substr(start_idx, max_len);
231     }
232
233     string longestRepeatedSubstring()
234     {
235         int max_len = 0;
236         int start_idx = -1;
237         for (int i = 0; i < (int)lcp.size(); i++)
238         {
239             if (lcp[i] > max_len)
240             {
241                 max_len = lcp[i];
242                 start_idx = p[i];
243             }
244         }
245         if (max_len == 0)
246             return "";
247         return s.substr(start_idx, max_len);
248     }
249 };

```

### 8.7.2 Suffix automaton

```

1 struct SuffixAutomaton
2 {
3     struct State
4     {
5         int len, link;
6         int first_pos;

```

```

7     bool is_clone;
8     int cnt;
9     long long paths;
10    map<char, int> next;
11    vector<int> inv_link;
12 };
13
14 vector<State> st;
15 int sz, last;
16 string s;
17
18 SuffixAutomaton()
19 {
20     st.assign(2, State());
21     st[0].len = 0;
22     st[0].link = -1;
23     st[0].cnt = 0;
24     sz = 1;
25     last = 0;
26 }
27
28 void extend(char c)
29 {
30     int cur = sz++;
31     if (cur >= st.size())
32         st.resize(st.size() * 2);
33     st[cur].len = st[last].len + 1;
34     st[cur].first_pos = st[cur].len - 1;
35     st[cur].is_clone = false;
36     st[cur].cnt = 1;
37
38     s += c;
39
40     int p = last;
41     while (p != -1 && !st[p].next.count(c))
42     {
43         st[p].next[c] = cur;
44         p = st[p].link;
45     }
46     if (p == -1)
47     {
48         st[cur].link = 0;
49     }
50     else
51     {
52         int q = st[p].next[c];
53         if (st[p].len + 1 == st[q].len)
54         {
55             st[cur].link = q;
56         }
57         else
58         {

```

```

59             int clone = sz++;
60             if (clone >= st.size())
61                 st.resize(st.size() * 2);
62             st[clone].len = st[p].len + 1;
63             st[clone].next = st[q].next;
64             st[clone].link = st[q].link;
65             st[clone].first_pos = st[q].first_pos;
66             st[clone].is_clone = true;
67             st[clone].cnt = 0;
68
69             while (p != -1 && st[p].next[c] == q)
70             {
71                 st[p].next[c] = clone;
72                 p = st[p].link;
73             }
74             st[q].link = st[cur].link = clone;
75         }
76     }
77     last = cur;
78 }
79
80 void build(const string &text)
81 {
82     for (char c : text)
83         extend(c);
84     precompute();
85 }
86
87 void precompute()
88 {
89     for (int v = 1; v < sz; v++)
90     {
91         st[st[v].link].inv_link.push_back(v);
92     }
93     vector<pair<int, int>> order(sz);
94     for (int i = 0; i < sz; i++)
95         order[i] = {st[i].len, i};
96     sort(order.rbegin(), order.rend());
97     for (int i = 0; i < sz - 1; i++)
98     {
99         int v = order[i].second;
100         if (st[v].link != -1)
101         {
102             st[st[v].link].cnt += st[v].cnt;
103         }
104     }
105 }
106
107 bool checkOccurrence(string P)
108 {
109     int v = 0;
110     for (char c : P)

```

```

111     {
112         if (!st[v].next.count(c))
113             return false;
114         v = st[v].next[c];
115     }
116     return true;
117 }
118
119 long long numberOfDifferentSubstrings()
120 {
121     long long tot = 0;
122     for (int i = 1; i < sz; i++)
123     {
124         tot += st[i].len - st[st[i].link].len;
125     }
126     return tot;
127 }
128
129 long long totalLengthOfDifferentSubstrings()
130 {
131     long long tot = 0;
132     for (int i = 1; i < sz; i++)
133     {
134         long long shortest = st[st[i].link].len + 1;
135         long long longest = st[i].len;
136         long long num_strings = longest - shortest + 1;
137         long long cur = num_strings * (longest + shortest) / 2;
138         tot += cur;
139     }
140     return tot;
141 }
142
143 void precomputePaths()
144 {
145     vector<pair<int, int>> order(sz);
146     for (int i = 0; i < sz; i++)
147         order[i] = {st[i].len, i};
148     sort(order.rbegin(), order.rend());
149     for (int i = 0; i < sz; i++)
150     {
151         int v = order[i].second;
152         st[v].paths = st[v].cnt;
153         for (auto &edge : st[v].next)
154         {
155             st[v].paths += st[edge.second].paths;
156         }
157     }
158 }
159
160 string kthSubstring(long long k)
161 {
162     precomputePaths();

```

```

163     int v = 0;
164     string ans = "";
165     while (k > 0)
166     {
167         for (auto &edge : st[v].next)
168         {
169             int w = edge.second;
170             if (st[w].paths >= k)
171             {
172                 ans += edge.first;
173                 k -= st[w].cnt;
174                 v = w;
175                 break;
176             }
177             else
178             {
179                 k -= st[w].paths;
180             }
181         }
182     }
183     return ans;
184 }
185
186 static string smallestCyclicShift(string S)
187 {
188     SuffixAutomaton sa;
189     sa.build(S + S);
190     int v = 0;
191     string ans = "";
192     for (int i = 0; i < S.size(); i++)
193     {
194         auto edge = *sa.st[v].next.begin();
195         ans += edge.first;
196         v = edge.second;
197     }
198     return ans;
199 }
200
201 int numberOfOccurrences(string P)
202 {
203     int v = 0;
204     for (char c : P)
205     {
206         if (!st[v].next.count(c))
207             return 0;
208         v = st[v].next[c];
209     }
210     return st[v].cnt;
211 }
212
213 int firstOccurrencePosition(string P)
214 {

```

```

215     int v = 0;
216     for (char c : P)
217     {
218         if (!st[v].next.count(c))
219             return -1;
220         v = st[v].next[c];
221     }
222     return st[v].first_pos - P.length() + 1;
223 }
224
225 void dfsOccurrences(int v, int P_length, vector<int> &res)
226 {
227     if (!st[v].is_clone)
228     {
229         res.push_back(st[v].first_pos - P_length + 1);
230     }
231     for (int u : st[v].inv_link)
232     {
233         dfsOccurrences(u, P_length, res);
234     }
235 }
236
237 vector<int> allOccurrencePositions(string P)
238 {
239     int v = 0;
240     for (char c : P)
241     {
242         if (!st[v].next.count(c))
243             return {};
244         v = st[v].next[c];
245     }
246     vector<int> res;
247     dfsOccurrences(v, P.length(), res);
248     sort(res.begin(), res.end());
249     return res;
250 }
251
252 string shortestNonAppearingString(char min_char = 'a', char max_char = 'z')
253 {
254     vector<int> d(sz, -1);
255     auto dfs = [&](auto &self, int v) -> int
256     {
257         if (d[v] != -1)
258             return d[v];
259         d[v] = 1e9;
260         for (char c = min_char; c <= max_char; c++)
261         {
262             if (!st[v].next.count(c))
263             {
264                 d[v] = 1;
265                 break;
266             }

```

```

267         else
268         {
269             d[v] = min(d[v], 1 + self(self, st[v].next[c]));
270         }
271     }
272     return d[v];
273 };
274 dfs(dfs, 0);
275 string ans = "";
276 int v = 0;
277 while (true)
278 {
279     for (char c = min_char; c <= max_char; c++)
280     {
281         if (!st[v].next.count(c))
282         {
283             ans += c;
284             return ans;
285         }
286     }
287     for (char c = min_char; c <= max_char; c++)
288     {
289         if (d[v] == 1 + d[st[v].next[c]])
290         {
291             ans += c;
292             v = st[v].next[c];
293             break;
294         }
295     }
296 }
297
298
299 string longestCommonSubstringTwo(string T)
300 {
301     int v = 0, l = 0, best = 0, bestpos = 0;
302     for (int i = 0; i < T.size(); i++)
303     {
304         while (v && !st[v].next.count(T[i]))
305         {
306             v = st[v].link;
307             l = st[v].len;
308         }
309         if (st[v].next.count(T[i]))
310         {
311             v = st[v].next[T[i]];
312             l++;
313         }
314         if (l > best)
315         {
316             best = l;
317             bestpos = i;
318         }

```



```

319     }
320     return T.substr(bestpos - best + 1, best);
321 }
322
323 static string longestCommonSubstringMultiple(vector<string> strings)
324 {
325     if (strings.empty())
326         return "";
327     if (strings.size() == 1)
328         return strings[0];
329
330     int min_idx = 0;
331     for (int i = 1; i < strings.size(); i++)
332     {
333         if (strings[i].size() < strings[min_idx].size())
334         {
335             min_idx = i;
336         }
337     }
338     swap(strings[0], strings[min_idx]);
339
340     SuffixAutomaton sa;
341     sa.build(strings[0]);
342
343     vector<int> order(sa.sz);
344     iota(order.begin(), order.end(), 0);
345     sort(order.begin(), order.end(), [&](int a, int b)
346         { return sa.st[a].len > sa.st[b].len; });
347
348     vector<int> min_max_l(sa.sz);
349     for (int i = 0; i < sa.sz; i++)
350     {
351         min_max_l[i] = sa.st[i].len;
352     }
353
354     for (int i = 1; i < strings.size(); i++)
355     {
356         vector<int> max_l(sa.sz, 0);
357         int v = 0, l = 0;
358         for (char c : strings[i])
359         {
360             while (v > 0 && !sa.st[v].next.count(c))
361             {
362                 v = sa.st[v].link;
363                 l = sa.st[v].len;
364             }
365             if (sa.st[v].next.count(c))
366             {
367                 v = sa.st[v].next[c];
368                 l++;
369             }
370             max_l[v] = max(max_l[v], l);

```

```

371     }
372
373     for (int u : order)
374     {
375         min_max_l[u] = min(min_max_l[u], max_l[u]);
376         int lnk = sa.st[u].link;
377         if (lnk != -1)
378         {
379             max_l[lnk] = max(max_l[lnk], min(max_l[u], sa.st[lnk].len));
380         }
381     }
382 }
383
384 int best_len = 0;
385 int best_v = 0;
386 for (int i = 1; i < sa.sz; i++)
387 {
388     if (min_max_l[i] > best_len)
389     {
390         best_len = min_max_l[i];
391         best_v = i;
392     }
393 }
394 if (best_len == 0)
395     return "";
396 return strings[0].substr(sa.st[best_v].first_pos - best_len + 1, best_len);
397 }
398 };

```

## 9 DP

### 9.1 DP Bitmask

#### 9.1.1 Bitmask

```

1 int n;
2 vector<vector<int>> v;
3 int dp[22][1 << 22];
4
5 int sol(int man, int mask)
6 {
7     if (man >= n)
8         return 1;
9     auto &ret = dp[man][mask];
10    if (~ret)
11        return ret;
12    ret = 0;
13    for (int i = 0; i < n; i++)
14    {

```

```

15     if (mask >> i & 1)
16         continue;
17     if (v[man][i])
18     {
19         ret += sol(man + 1, mask | 1 << i);
20         ret %= MOD;
21     }
22 }
23 return ret;
24 }

```

### 9.1.2 SOS

```

1 vector<int> SOS(int n, vector<int> &a)
2 {
3     vector<int> p = a;
4     for (int dim = 0; dim < n; dim++)
5     {
6         for (int i = 0; i < 1 << n; i++)
7         {
8             if (i >> dim & 1)
9                 p[i] += p[i - (1 << dim)];
10        }
11    }
12    return p;
13 }

```

### 9.1.3 Submask

```

1 int dp[1<<16];
2 int cost[1<<16];
3 int memo (int mask) {
4     if (mask == 0) return 0;
5     auto &ret = dp[mask];
6     if (~ret) return ret;
7     for (int i = mask; i > 0; i = (i-1) & mask) {
8         ret = max(ret, cost[i] + memo(mask ^ i));
9     }
10    return ret;
11 }
12 void hassan() {
13     int n; cin >> n;
14     int a[n][n];
15     for (int i = 0; i < n; i++) {
16         for (int j = 0; j < n; j++) {
17             cin >> a[i][j];
18         }
19     }
20     for (int mask = 0; mask < (1<<n); mask++) {

```

```

21         for (int i = 0; i < n; i++) {
22             if (mask & (1<<i)) {
23                 for (int j = 0; j < i; j++) {
24                     if (mask & (1<<j)) {
25                         cost[mask] += a[i][j];
26                     }
27                 }
28             }
29         }
30         // cout << mask << " >> " << cost[mask] << endl;
31     }
32     memset(dp, -1, sizeof(dp));
33     cout << memo((1<<n)-1);
34 }
35 }

```

### 9.1.4 subset convolution

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: SUBSET CONVOLUTION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Disjoint subset sum", "i OR j = k AND i AND j = 0".
6  * - Classic Scenarios: Combining two DP states where the masks must perfectly
7  *   interlock (share no common set bits) to form a target mask.
8  * - The Magic: By adding an extra dimension for the `__builtin_popcount`, we
9  *   guarantee that if two masks combine to form a mask of size K, their
10     individual
11     set bits must sum to exactly K, implicitly enforcing that they are disjoint.
12     It runs SOS DP on each popcount layer.
13
14  * 2. HOW TO USE IT (THE BLACK BOX)
15  * - Query:
16     vector<long long> C = subset_convolution(A, B, N, MOD);
17
18  * - Complexity:
19     Time:  $O(N^2 * 2^N)$ 
20     Space:  $O(N * 2^N)$ 
21 */
22 using ll = long long;
23
24 vector<ll> subset_convolution(const vector<ll> &a, const vector<ll> &b, int n, ll
    mod)
25 {
26     vector<vector<ll>> f_hat(n + 1, vector<ll>(1 << n, 0));
27     vector<vector<ll>> g_hat(n + 1, vector<ll>(1 << n, 0));
28     vector<vector<ll>> h_hat(n + 1, vector<ll>(1 << n, 0));
29
30     for (int mask = 0; mask < (1 << n); mask++)
31     {

```

```

32     int pc = __builtin_popcount(mask);
33     f_hat[pc][mask] = a[mask] % mod;
34     g_hat[pc][mask] = b[mask] % mod;
35 }
36
37 for (int i = 0; i <= n; i++)
38 {
39     for (int j = 0; j < n; j++)
40     {
41         for (int mask = 0; mask < (1 << n); mask++)
42         {
43             if (mask & (1 << j))
44             {
45                 f_hat[i][mask] = (f_hat[i][mask] + f_hat[i][mask ^ (1 << j)])
46                     % mod;
47                 g_hat[i][mask] = (g_hat[i][mask] + g_hat[i][mask ^ (1 << j)])
48                     % mod;
49             }
50         }
51     }
52     for (int mask = 0; mask < (1 << n); mask++)
53     {
54         for (int i = 0; i <= n; i++)
55         {
56             ll sum = 0;
57             for (int j = 0; j <= i; j++)
58             {
59                 sum = (sum + f_hat[j][mask] * g_hat[i - j][mask]) % mod;
60             }
61             h_hat[i][mask] = sum;
62         }
63     }
64 }
65 for (int i = 0; i <= n; i++)
66 {
67     for (int j = 0; j < n; j++)
68     {
69         for (int mask = 0; mask < (1 << n); mask++)
70         {
71             if (mask & (1 << j))
72             {
73                 h_hat[i][mask] = (h_hat[i][mask] - h_hat[i][mask ^ (1 << j)]
74                     + mod) % mod;
75             }
76         }
77     }
78 }
79 vector<ll> c(1 << n);
80 for (int mask = 0; mask < (1 << n); mask++)

```

```

81 {
82     c[mask] = h_hat[__builtin_popcount(mask)][mask];
83 }
84 return c;
85 }

```

## 9.2 Digit Dp

### 9.2.1 main

```

1 string s;
2 int d, k;
3 int dp[20][20][2][2];
4
5 int sol(int idx, int cnt, int tight, int num)
6 {
7     if (idx >= int(s.size()))
8     {
9         if (cnt == k)
10             return 1;
11         return 0;
12     }
13     auto &ret = dp[idx][cnt][tight][num];
14     if (~ret)
15         return ret;
16     ret = 0;
17     int limit = tight ? s[idx] - '0' : 9;
18     for (int i = 0; i <= limit; i++)
19     {
20         int nn = num || i;
21         int nt = tight && i == limit;
22         int nc = cnt + (d == i && nn);
23         ret += sol(idx + 1, nc, nt, nn);
24     }
25     return ret;
26 }
27
28 void solve()
29 {
30     cin >> s >> d >> k;
31     memset(dp, -1, sizeof dp);
32     cout << sol(0, 0, 1, 0);
33 }

```

## 9.3 Iterative

### 9.3.1 main

```

1 int answer(int n)
2 {
3     if (n == 0 || n == 1)
4         return n;
5     return dp[n];
6 }
7
8 void solve()
9 {
10     int n;
11     cin >> n;
12     for (int i = 2; i <= n; i++)
13         dp[i] = answer(i - 1) + answer(i - 2);
14     cout << answer(n);
15 }

```

## 9.4 Knapsack

### 9.4.1 main

```

1 int n, W;
2 int dp[1001][31];
3 vector<pair<int, int>> a;
4
5 int sol(int i, int w)
6 {
7     if (i >= n)
8         return 0;
9     auto &ret = dp[i][w];
10    if (~ret)
11        return ret;
12    ret = sol(i + 1, w);
13    if (w + a[i].second <= W)
14        ret = max(ret, a[i].first + sol(i + 1, w + a[i].second));
15    return ret;
16 }

```

## 9.5 LCS

### 9.5.1 LCS

```

1 vector<vector<int>> dp;
2 int n1, n2;
3 string s1, s2;
4
5 int LCS(int i1, int i2)
6 {
7     if (i1 >= n1 || i2 >= n2)
8         return 0;

```

```

9     auto &ret = dp[i1][i2];
10    if (ret != -1)
11        return ret;
12    int choice1 = LCS(i1 + 1, i2);
13    int choice2 = LCS(i1, i2 + 1);
14    int choice3{};
15    if (s1[i1] == s2[i2])
16        choice3 = 1 + LCS(i1 + 1, i2 + 1);
17    return ret = max({choice1, choice2, choice3});
18 }

```

## 9.6 LIS

### 9.6.1 Basic 2D

```

1 int dp1[2010][2010];
2 int dp2[2010][2010];
3
4 int LDS(int i, int prev)
5 {
6     if (i > n)
7         return 0;
8     auto &ret = dp1[i][prev];
9     if (~ret)
10        return ret;
11    ret = LDS(i + 1, prev);
12    if (prev == 0 || a[i] < a[prev])
13        ret = max(ret, 1 + LDS(i + 1, i));
14    return ret;
15 }
16
17 int LIS(int i, int prev)
18 {
19     if (i > n)
20         return 0;
21     auto &ret = dp2[i][prev];
22     if (~ret)
23         return ret;
24     ret = LIS(i + 1, prev);
25     if (prev == 0 || a[i] > a[prev])
26         ret = max(ret, 1 + LIS(i + 1, i));
27     return ret;
28 }

```

### 9.6.2 LIS1D

```

1 int LIS(int i)
2 {
3     if (i > n)

```

```

4     return 0;
5     auto &ret = dp[i];
6     if (dp[i] == -1)
7         return ret;
8     ret = 0;
9     for (int j = i + 1; j <= n; j++)
10    {
11        if (v[j] > v[i])
12            ret = max(ret, LIS(j));
13    }
14    ret++;
15    return ret;
16 }
17
18 // int res{};
19 // for (int i = 1; i <= n; i++)
20 // {
21 //     int num = LIS(i);
22 //     res = max(num, res);
23 // }

```

### 9.6.3 LIS BS

```

1 vector<int> LIS;
2 for (int i = 0; i < n; i++)
3 {
4     auto it = lower_bound(LIS.begin(), LIS.end(), ans[i]);
5     if (it == LIS.end())
6         LIS.push_back(ans[i]);
7     else
8         *it = ans[i];
9 }

```

### 9.6.4 LIS seg

```

1 struct Node
2 {
3     int val;
4
5     Node()
6     {
7         val = 0;
8     }
9
10    Node(int val)
11        : val(val)
12    {
13    }
14 }

```

```

15 void change(int x)
16 {
17     val = max(x, val);
18 }
19 };
20
21 struct segmentTree
22 {
23     int sz;
24     vector<Node> seg;
25
26     segmentTree(int n)
27     {
28         sz = 1;
29         while (sz < n)
30             sz *= 2;
31         seg.assign(2 * sz, Node());
32     }
33
34     void init(vector<int> &arr, int node, int lx, int rx)
35     {
36         if (rx - lx == 1)
37         {
38             if (lx < int(arr.size()))
39                 seg[node] = Node(arr[lx]);
40             return;
41         }
42         int mid = (lx + rx) / 2;
43         init(arr, 2 * node + 1, lx, mid);
44         init(arr, 2 * node + 2, mid, rx);
45         seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
46     }
47
48     void init(vector<int> &arr)
49     {
50         init(arr, 0, 0, sz);
51     }
52
53     Node merge(Node &lf, Node &rt)
54     {
55         Node ans = Node();
56         ans.val = max(lf.val, rt.val);
57         return ans;
58     }
59
60     void set(int idx, int val, int node, int lx, int rx)
61     {
62         if (rx - lx == 1)
63         {
64             seg[node].change(val);
65             return;
66         }

```

```

67     int mid = (lx + rx) / 2;
68     if (idx < mid)
69         set(idx, val, 2 * node + 1, lx, mid);
70     else
71         set(idx, val, 2 * node + 2, mid, rx);
72     seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
73 }
74
75 void set(int idx, int val)
76 {
77     set(idx, val, 0, 0, sz);
78 }
79
80 Node get(int l, int r, int node, int lx, int rx)
81 {
82     if (lx >= l && rx <= r)
83         return seg[node];
84     if (rx <= l || lx >= r)
85         return Node();
86     int mid = (lx + rx) / 2;
87     Node lf = get(l, r, 2 * node + 1, lx, mid);
88     Node ri = get(l, r, 2 * node + 2, mid, rx);
89     return merge(lf, ri);
90 }
91
92 int get(int l, int r)
93 {
94     return get(l, r, 0, 0, sz).val;
95 }
96 };
97
98 int LIS(int n, vector<int> &p)
99 {
100     segmentTree seg(n + 5);
101     vector<int> dp(n + 1);
102     int ans{};
103     for (int i = 0; i < n; i++)
104     {
105         dp[i] = seg.get(0, p[i]) + 1;
106         seg.set(p[i], dp[i]);
107         ans = max(ans, dp[i]);
108     }
109     return ans;
110 }

```

## 9.7 Lichao tree

### 9.7.1 lichao tree

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: LI CHAO TREE (EXTENDED)

```

```

3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Minimum of lines", "Linear cost over a range", " $y = mx + c$ ".
6  * - Classic Scenarios: You need to minimize a linear DP transition cost, but
7  *   certain transitions/lines are only valid within a specific interval  $[L, R]$ .
8  * - The Magic: A sparse segment tree where each node holds the line that
9  *   provides the optimal value at the midpoint of its interval. It dynamically
10 *   allocates memory (sparse) so it can comfortably handle  $x$ -coordinates up to
11    $10^{18}$ .
12
13 * 2. HOW TO USE IT (THE BLACK BOX)
14 * - Initialization:
15 *   LiChaoTree lct(-1e18, 1e18); // Pass the minimum and maximum possible X
16   queries
17 * - Operations:
18 *   lct.add_line(M, C); // Adds an infinite line  $y = Mx + C$ 
19 *   lct.add_segment(M, C, L, R); // Adds line  $y = Mx + C$  strictly
20   for  $x$  in  $[L, R]$ 
21 *   long long min_y = lct.query(X); // Gets the minimum  $y$  at point  $X$ 
22
23 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
24 * - Maximization: This template is strictly built to find the MINIMUM value.
25 *   If you need the MAXIMUM, insert your lines with inverted signs:
26 *   `lct.add_line(-M, -C)` and then negate the result:  $-lct.query(X)$ .
27 */
28
29 using ll = long long;
30 const ll INF = 4e18; // Safe upper bound to prevent overflow during additions
31
32 struct Line
33 {
34     ll m, c;
35     ll eval(ll x) const { return m * x + c; }
36 };
37
38 struct Node
39 {
40     Line line;
41     int lc = -1, rc = -1;
42     Node() : line({0, INF}) {}
43     Node(Line l) : line(l) {}
44 };
45
46 struct LiChaoTree
47 {
48     vector<Node> tree;
49     ll MIN_X, MAX_X;
50
51     LiChaoTree(ll min_x, ll max_x) : MIN_X(min_x), MAX_X(max_x)
52     {
53         tree.emplace_back();
54     }
55 }

```

```

52
53 void add_line(Line nw, int u, ll l, ll r)
54 {
55     ll mid = l + (r - 1) / 2;
56     bool left = nw.eval(l) < tree[u].line.eval(l);
57     bool middle = nw.eval(mid) < tree[u].line.eval(mid);
58
59     if (middle)
60         swap(tree[u].line, nw);
61     if (l == r)
62         return;
63
64     if (left != middle)
65     {
66         if (tree[u].lc == -1)
67         {
68             tree[u].lc = tree.size();
69             tree.emplace_back();
70         }
71         add_line(nw, tree[u].lc, l, mid);
72     }
73     else
74     {
75         if (tree[u].rc == -1)
76         {
77             tree[u].rc = tree.size();
78             tree.emplace_back();
79         }
80         add_line(nw, tree[u].rc, mid + 1, r);
81     }
82 }
83
84 void add_line(ll m, ll c)
85 {
86     add_line({m, c}, 0, MIN_X, MAX_X);
87 }
88
89 void add_segment(Line nw, ll ql, ll qr, int u, ll l, ll r)
90 {
91     if (ql <= l && r <= qr)
92     {
93         add_line(nw, u, l, r);
94         return;
95     }
96     ll mid = l + (r - 1) / 2;
97     if (ql <= mid)
98     {
99         if (tree[u].lc == -1)
100         {
101             tree[u].lc = tree.size();
102             tree.emplace_back();
103         }

```

```

104         add_segment(nw, ql, qr, tree[u].lc, l, mid);
105     }
106     if (qr > mid)
107     {
108         if (tree[u].rc == -1)
109         {
110             tree[u].rc = tree.size();
111             tree.emplace_back();
112         }
113         add_segment(nw, ql, qr, tree[u].rc, mid + 1, r);
114     }
115 }
116
117 void add_segment(ll m, ll c, ll ql, ll qr)
118 {
119     add_segment({m, c}, ql, qr, 0, MIN_X, MAX_X);
120 }
121
122 ll query(ll x, int u, ll l, ll r)
123 {
124     if (u == -1)
125         return INF;
126     ll res = tree[u].line.eval(x);
127     if (l == r)
128         return res;
129
130     ll mid = l + (r - 1) / 2;
131     if (x <= mid)
132         res = min(res, query(x, tree[u].lc, l, mid));
133     else
134         res = min(res, query(x, tree[u].rc, mid + 1, r));
135
136     return res;
137 }
138
139 ll query(ll x)
140 {
141     return query(x, 0, MIN_X, MAX_X);
142 }
143 };

```

## 9.8 Misc

### 9.8.1 Broken profile

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: BROKEN PROFILE DP (PLUG DP)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Tile the grid", "Dominoes", "Hamiltonian Path on Grid".
6  * - Classic Scenarios: Find the number of ways to completely cover an N x M grid

```

```

7  *   using 1x2 dominoes where N is huge (10^5) and M is very small (<= 15).      54
8  * - The Magic: It sweeps through the grid cell by cell (not row by row).      55
9  * It maintains a bitmask of length M representing the "profile" (the boundary 56
10 * of cells protruding into the current row). As we move to the next cell, we
11 * shift the mask and try placing dominoes horizontally or vertically.        57
12 *                                                                              58
13 * 2. HOW TO USE IT (THE BLACK BOX)                                           59
14 * - Query:                                                                    60
15 *     long long ways = plug_dp_dominoes(N, M, MOD);                          61
16 *                                                                              62
17 * - Complexity:                                                                63
18 *     Time: O(N * M * 2^M)                                                    64
19 *     Space: O(2^M) using rolling arrays.                                     65
20 *                                                                              66
21 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)                                  67
22 * - Obstacles: If the grid has blocked cells, simply check if the current cell
23 *   (i, j) is blocked. If it is, ensure the current bit in the mask is 0, shift
24 *   it to 1 (meaning it's "filled"), and move to the next cell without placing
25 *   dominoes.                                                                69
26 *                                                                              70
27 using ll = long long;                                                         71
28
29 ll plug_dp_dominoes(int n, int m, ll mod)                                     72
30 {
31     if ((n * m) % 2 != 0)                                                      73
32         return 0; // Odd number of cells cannot be tiled by 1x2
33
34     // We only need the current and next state, so we use a rolling array of size
35     // 2
36     vector<vector<ll>> dp(2, vector<ll>(1 << m, 0));                          75
37
38     int cur = 0, next = 1;
39     dp[cur][0] = 1; // Base case: completely empty profile
40
41     for (int i = 0; i < n; i++)
42     {
43         for (int j = 0; j < m; j++)
44         {
45             fill(dp[next].begin(), dp[next].end(), 0);
46
47             for (int mask = 0; mask < (1 << m); mask++)
48             {
49                 if (dp[cur][mask] == 0)
50                     continue;
51
52                 // Case 1: The current cell is already filled by a vertical
53                 // domino
54                 // from the previous row. We must leave it alone. The bit shifts
55                 // out.
56                 if (mask & (1 << (m - 1)))
57
58                 {
59                     int next_mask = (mask ^ (1 << (m - 1))) << 1;
60                     dp[next][next_mask] = (dp[next][next_mask] + dp[cur][mask]) %
61                     mod;
62                 }
63                 // Case 2: The cell is empty. We must place a domino.
64                 else
65                 {
66                     // Option A: Place a vertical domino (protrudes into the next
67                     // row).
68                     // We set the 0-th bit of the shifted mask to 1.
69                     int next_mask_vert = (mask << 1) | 1;
70                     dp[next][next_mask_vert] = (dp[next][next_mask_vert] + dp[cur]
71                     [mask]) % mod;
72
73                     // Option B: Place a horizontal domino.
74                     // Only valid if we aren't at the right edge AND the next
75                     // cell is empty.
76                     if (j + 1 < m && !(mask & 1))
77                     {
78                         // The horizontal domino fills the current and next cell
79                         // in THIS row.
80                         // We shift the mask by 1, but we don't protrude into the
81                         // next row,
82                         // so the 0-th bit remains 0. However, the cell to our
83                         // right is now filled,
84                         // which we must account for in our transition logic by
85                         // skipping it or
86                         // setting its bit. In this cell-by-cell state machine,
87                         // placing a horizontal
88                         // domino effectively sets the bit for the *next* cell in
89                         // the current row.
90                         int next_mask_horiz = (mask << 1) | 2;
91                         dp[next][next_mask_horiz] = (dp[next][next_mask_horiz] +
92                         dp[cur][mask]) % mod;
93                     }
94                 }
95                 swap(cur, next);
96             }
97         }
98     }
99
100    // The answer is the number of ways to have a completely flat profile (mask
101    // == 0) at the end.
102    return dp[cur][0];
103 }

```

## 9.9 More Tricks

### 9.9.1 Slope trick



```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: SLOPE TRICK
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Minimum operations to make array non-decreasing".
6  * - Classic Scenarios: You can increment or decrement elements of an array.
7  * Find the minimum total cost to make the array sorted (non-decreasing).
8  * - The Magic: A 2D DP takes  $O(N * MAX\_VAL)$  which TLEs. By observing that the
9  * DP cost function is convex and piecewise linear, we can just maintain the
10 * points where the slope changes using a max-heap.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     long long min_cost = make_non_decreasing(A);
15 *
16 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
17 * - Strictly Increasing: If the problem requires strictly increasing ( $A[i] < A[i+1]$ ),
18 * subtract  $i$  from every element initially:  $A[i] -= i$ , then run this exact
19 * function. It maps the strictly increasing problem directly to a non-
20 * decreasing one.
21 * - Complexity:
22 *     Time:  $O(N \log N)$ 
23 *     Space:  $O(N)$ 
24 */
25
26 using ll = long long;
27
28 ll make_non_decreasing(const vector<ll> &a)
29 {
30     priority_queue<ll> pq;
31     ll total_cost = 0;
32
33     for (int i = 0; i < a.size(); i++)
34     {
35         pq.push(a[i]);
36
37         // If the max element so far is strictly greater than the current element
38         // it means we have a slope violation. We "flatten" it to match  $A[i]$ .
39         if (pq.top() > a[i])
40         {
41             total_cost += pq.top() - a[i];
42             pq.pop();
43             pq.push(a[i]);
44         }
45     }
46
47     return total_cost;
48 }

```

## 9.10 Optimization

### 9.10.1 1D1D

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: 1D1D OPTIMIZATION (LINE CONTAINER)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Convex Hull Trick", " $y = mx + c$ ", "Li Chao Tree alternative".
6  * - Classic Scenarios:  $dp[i] = \min_{j < i} (dp[j] + M[j] * X[i] + C[j])$ .
7  * - The Magic: Maintains the upper/lower envelope of a set of lines. It supports
8  * adding lines dynamically and querying the extreme value at any  $X$  in  $O(\log N)$ .
9  * This specific implementation (KACTL's LineContainer) handles lines added
10 * with
11 * NON-MONOTONIC slopes, making it universally safe.
12 *
13 * 2. HOW TO USE IT (THE BLACK BOX)
14 * - Initialization:
15 *     LineContainer cht;
16 * - Operations:
17 *     cht.add(m, c); // Adds line  $y = mx + c$ 
18 *     ll max_y = cht.query(x);
19 *
20 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
21 * - Minimization: This natively computes the MAXIMUM value at  $x$ . If you need the
22 * MINIMUM value, insert lines with inverted signs:  $\text{cht.add}(-m, -c)$  and
23 * negate
24 * the queried result:  $-\text{cht.query}(x)$ .
25 */
26
27 using ll = long long;
28
29 struct Line
30 {
31     mutable ll k, m, p;
32     bool operator<(const Line &o) const { return k < o.k; }
33     bool operator<(ll x) const { return p < x; }
34 };
35
36 struct LineContainer : multiset<Line, less<>>
37 {
38     static const ll inf = LLONG_MAX;
39     ll div(ll a, ll b)
40     {
41         return a / b - ((a ^ b) < 0 && a % b);
42     }
43     bool isect(iterator x, iterator y)
44     {
45         if (y == end())
46             return x->p = inf, 0;
47         if (x->k == y->k)

```

```

46         x->p = x->m > y->m ? inf : -inf;
47     else
48         x->p = div(y->m - x->m, x->k - y->k);
49     return x->p >= y->p;
50 }
51 void add(ll k, ll m)
52 {
53     auto z = insert({k, m, 0}), y = z++, x = y;
54     while (isect(y, z))
55         z = erase(z);
56     if (x != begin() && isect(--x, y))
57         isect(x, y = erase(y));
58     while ((y = x) != begin() && (--x)->p >= y->p)
59         isect(x, erase(y));
60 }
61 ll query(ll x)
62 {
63     assert(!empty());
64     auto l = *lower_bound(x);
65     return l.k * x + l.m;
66 }
67 };

```

### 9.10.2 Aliens

```

1 #include <bits/stdc++.h>
2
3 using namespace std;
4
5 #define El_Hefnawys \
6     ios_base::sync_with_stdio(false); \
7     cin.tie(nullptr); \
8     cout.tie(nullptr)
9
10 typedef long long ll;
11 #define int ll
12
13 const int OO = 1e15;
14
15 // choose k subarrays from the array and trying to maximize the sum
16 // dp[i][0/1]
17
18 // The DP function returns a pair:
19 // 1. The maximum sum (including the penalties applied)
20 // 2. The NEGATIVE number of subarrays used (to break ties by preferring fewer
    subarrays)
21 pair<int, int> check(int penalty, int n, const vector<int> &a)
22 {
23     // dp[i][0] = max sum up to index i, currently NOT in a subarray
24     // dp[i][1] = max sum up to index i, currently INSIDE a subarray
25     vector<vector<pair<int, int>>> dp(n, vector<pair<int, int>>(2));

```

```

26
27     dp[0][0] = {0, 0};
28     dp[0][1] = {a[0] - penalty, -1}; // Starting an array costs 'penalty'
29
30     for (int i = 1; i < n; i++)
31     {
32         // Option 0: We are not in a subarray.
33         // We either were already not in one, or we just closed the previous one.
34         dp[i][0] = max(dp[i - 1][0], dp[i - 1][1]);
35
36         // Option 1: We are inside a subarray.
37         // Transition A: Start a brand NEW subarray today (costs penalty, adds -1
            to count)
38         pair<int, int> start_new = make_pair(
39             dp[i - 1][0].first - penalty + a[i],
40             dp[i - 1][0].second - 1);
41
42         // Transition B: Extend the EXISTING subarray (no penalty, count doesn't
            change)
43         pair<int, int> extend_existing = make_pair(
44             dp[i - 1][1].first + a[i],
45             dp[i - 1][1].second);
46
47         dp[i][1] = max(start_new, extend_existing);
48     }
49
50     return max(dp[n - 1][0], dp[n - 1][1]);
51 }
52
53 void solve()
54 {
55     int n, k;
56     if (!(cin >> n >> k))
57         return;
58
59     vector<int> a(n);
60     for (int i = 0; i < n; i++)
61     {
62         cin >> a[i];
63     }
64
65     // The penalty bounds. Can be negative depending on the problem!
66     int l = -1e14, r = 1e14;
67     int best_ans = 0;
68
69     while (l <= r)
70     {
71         int mid_penalty = l + (r - l) / 2;
72
73         pair<int, int> res = check(mid_penalty, n, a);
74         int arrays_used = -res.second; // Flip back to positive
75

```

```

76 // If we used at least K arrays, this penalty is valid to consider.
77 // We might have tied and used exactly K, or more.
78 if (arrays_used >= k)
79 {
80     // Restore the true sum by adding back the penalty for EXACTLY K
81     // arrays.
82     // Even if the DP used more than K due to ties (0-sum arrays),
83     // the mathematical projection at the hull tangent guarantees this
84     // formula is exact!
85     best_ans = res.first + (k * mid_penalty);
86
87     // Try to force it to use FEWER arrays by INCREASING the penalty
88     l = mid_penalty + 1;
89 }
90 else
91 {
92     // We used too few arrays. The penalty is too harsh. Decrease it.
93     r = mid_penalty - 1;
94 }
95 cout << best_ans << "\n";
96 }
97
98 signed main()
99 {
100     El_Hefnawys;
101     int t{1};
102     // cin >> t;
103     while (t--)
104     {
105         solve();
106     }
107     return 0;
108 }

```

### 9.10.3 DC

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DIVIDE AND CONQUER OPTIMIZATION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Partition array into K contiguous segments", "Minimize maximum
6  * cost".
7  * - Classic Scenarios:  $dp[i][j] = \min_{k < j} (dp[i-1][k] + cost(k, j))$ .
8  * - The Magic: If the optimal split point  $opt(i, j)$  satisfies the condition
9  *  $opt(i, j) \leq opt(i, j+1)$ , we don't need to check all  $k$ . We recursively
10 * compute the middle state and restrict the search bounds for the left and
11 * right halves, dropping  $O(K * N^2)$  to  $O(K * N \log N)$ .
12 * 2. HOW TO USE IT (THE BLACK BOX)

```

```

13 * - Setup: You need two 1D arrays: `dp_prev` (results from i-1 segments)
14 * and `dp_cur` (results for i segments).
15 * - Query:
16 *     for (int i = 2; i <= K; i++) {
17 *         compute_dc(1, N, 1, N, dp_prev, dp_cur);
18 *         dp_prev = dp_cur;
19 *     }
20 */
21
22 using ll = long long;
23 const ll INF = 1e18;
24
25 // Replace this with your actual O(1) cost function (usually via prefix sums)
26 ll cost(int l, int r)
27 {
28     if (l > r)
29         return 0;
30     return 0; // return actual cost
31 }
32
33 void compute_dc(int l, int r, int opt_l, int opt_r, const vector<ll> &dp_prev,
34                 vector<ll> &dp_cur)
35 {
36     if (l > r)
37         return;
38     int mid = l + (r - l) / 2;
39     pair<ll, int> best = {INF, -1};
40     for (int k = opt_l; k <= min(mid, opt_r); k++)
41     {
42         ll current_cost = dp_prev[k - 1] + cost(k, mid);
43         if (current_cost < best.first)
44         {
45             best = {current_cost, k};
46         }
47     }
48
49     dp_cur[mid] = best.first;
50     int opt = best.second;
51
52     compute_dc(l, mid - 1, opt_l, opt, dp_prev, dp_cur);
53     compute_dc(mid + 1, r, opt, opt_r, dp_prev, dp_cur);
54 }

```

### 9.10.4 Knuth

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DIVIDE AND CONQUER OPTIMIZATION
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?

```

```

5  * - Keywords: "Partition array into K contiguous segments", "Minimize maximum
   * cost".
6  * - Classic Scenarios:  $dp[i][j] = \min_{k < j} (dp[i-1][k] + cost(k, j))$ .
7  * - The Magic: If the optimal split point  $opt(i, j)$  satisfies the condition
8  *  $opt(i, j) \leq opt(i, j+1)$ , we don't need to check all  $k$ . We recursively
9  * compute the middle state and restrict the search bounds for the left and
10 * right halves, dropping  $O(K * N^2)$  to  $O(K * N \log N)$ .
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Setup: You need two 1D arrays:  $dp\_prev$  (results from  $i-1$  segments)
14 * and  $dp\_cur$  (results for  $i$  segments).
15 * - Query:
16 *     for (int i = 2; i <= K; i++) {
17 *         compute_dc(1, N, 1, N, dp_prev, dp_cur);
18 *         dp_prev = dp_cur;
19 *     }
20 */
21
22 using ll = long long;
23 const ll INF = 1e18;
24
25 // Replace this with your actual  $O(1)$  cost function (usually via prefix sums)
26 ll cost(int l, int r)
27 {
28     if (l > r)
29         return 0;
30     return 0; // return actual cost
31 }
32
33 void compute_dc(int l, int r, int opt_l, int opt_r, const vector<ll> &dp_prev,
34 vector<ll> &dp_cur)
35 {
36     if (l > r)
37         return;
38     int mid = l + (r - l) / 2;
39     pair<ll, int> best = {INF, -1};
40     for (int k = opt_l; k <= min(mid, opt_r); k++)
41     {
42         ll current_cost = dp_prev[k - 1] + cost(k, mid);
43         if (current_cost < best.first)
44         {
45             best = {current_cost, k};
46         }
47     }
48     dp_cur[mid] = best.first;
49     int opt = best.second;
50
51     compute_dc(l, mid - 1, opt_l, opt, dp_prev, dp_cur);
52     compute_dc(mid + 1, r, opt, opt_r, dp_prev, dp_cur);
53 }
54 }

```

### 9.10.5 Using data structures

```

1  struct Node
2  {
3      int val;
4
5      Node()
6      {
7          val = 0;
8      }
9
10     Node(int val)
11     : val(val)
12     {
13     }
14
15     void change(int x)
16     {
17         val = max(x, val);
18     }
19 };
20
21 struct segmentTree
22 {
23     int sz;
24     vector<Node> seg;
25
26     segmentTree(int n)
27     {
28         sz = 1;
29         while (sz < n)
30             sz *= 2;
31         seg.assign(2 * sz, Node());
32     }
33
34     void init(vector<int> &arr, int node, int lx, int rx)
35     {
36         if (rx - lx == 1)
37         {
38             if (lx < int(arr.size()))
39                 seg[node] = Node(arr[lx]);
40             return;
41         }
42         int mid = (lx + rx) / 2;
43         init(arr, 2 * node + 1, lx, mid);
44         init(arr, 2 * node + 2, mid, rx);
45         seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
46     }
47
48     void init(vector<int> &arr)
49     {
50         init(arr, 0, 0, sz);
51     }
52 }

```

```

51     }
52
53     Node merge(Node &lf, Node &rt)
54     {
55         Node ans = Node();
56         ans.val = max(lf.val, rt.val);
57         return ans;
58     }
59
60     void set(int idx, int val, int node, int lx, int rx)
61     {
62         if (rx - lx == 1)
63         {
64             seg[node].change(val);
65             return;
66         }
67         int mid = (lx + rx) / 2;
68         if (idx < mid)
69             set(idx, val, 2 * node + 1, lx, mid);
70         else
71             set(idx, val, 2 * node + 2, mid, rx);
72         seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
73     }
74
75     void set(int idx, int val)
76     {
77         set(idx, val, 0, 0, sz);
78     }
79
80     Node get(int l, int r, int node, int lx, int rx)
81     {
82         if (lx >= l && rx <= r)
83             return seg[node];
84         if (rx <= l || lx >= r)
85             return Node();
86         int mid = (lx + rx) / 2;
87         Node lf = get(l, r, 2 * node + 1, lx, mid);
88         Node ri = get(l, r, 2 * node + 2, mid, rx);
89         return merge(lf, ri);
90     }
91
92     int get(int l, int r)
93     {
94         return get(l, r, 0, 0, sz).val;
95     }
96 };
97
98 int LIS(int n, vector<int> &p)
99 {
100     segmentTree seg(n + 5);
101     vector<int> dp(n + 1);
102     int ans{};

```

```

103     for (int i = 0; i < n; i++)
104     {
105         dp[i] = seg.get(0, p[i]) + 1;
106         seg.set(p[i], dp[i]);
107         ans = max(ans, dp[i]);
108     }
109     return ans;
110 }

```

## 10 Misc

### 10.1 Big Integer

#### 10.1.1 Big integers

```

1  /**
2  *   THE ULTIMATE CP TEMPLATE BLUEPRINT: COMPRESSED BIGINT
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "No modulo given", "Massive addition/multiplication", "1000 digits"
6  * - Classic Scenarios: DP state sums or combinatorics where the answer easily
7  *   exceeds  $2^{64}$ , but no modulo is provided.
8  * - The Magic: C++ has no native BigInt. Using strings character-by-character
9  *   is  $O(N^2)$  with a massive constant factor. This struct bundles 9 decimal
10  *   digits
11  *   into a single integer, processing data 9x faster and using significantly
12  *   less memory.
13  *
14  * 2. HOW TO USE IT (THE BLACK BOX)
15  * - Initialization:
16  *     BigInt a = "12345678901234567890";
17  *     BigInt b = 987654321;
18  * - Operations:
19  *     BigInt c = a + b;
20  *     c.print();
21  */
22
23 const int BASE = 1e9;
24
25 struct BigInt
26 {
27     vector<int> a;
28
29     BigInt() {}
30     BigInt(long long v)
31     {
32         while (v > 0)
33             a.push_back(v % BASE);

```

```

33         v /= BASE;
34     }
35 }
36 BigInt(string s)
37 {
38     if (s.empty())
39         return;
40     for (int i = s.length(); i > 0; i -= 9)
41     {
42         if (i < 9)
43             a.push_back(stoll(s.substr(0, i)));
44         else
45             a.push_back(stoll(s.substr(i - 9, 9)));
46     }
47     trim();
48 }
49
50 void trim()
51 {
52     while (a.size() > 1 && a.back() == 0)
53         a.pop_back();
54 }
55
56 BigInt operator+(const BigInt &b) const
57 {
58     BigInt res;
59     int carry = 0;
60     for (int i = 0; i < max(a.size(), b.a.size()) || carry; i++)
61     {
62         long long cur = carry;
63         if (i < a.size())
64             cur += a[i];
65         if (i < b.a.size())
66             cur += b.a[i];
67         res.a.push_back(cur % BASE);
68         carry = cur / BASE;
69     }
70     return res;
71 }
72
73 BigInt operator*(const BigInt &b) const
74 {
75     BigInt res;
76     if (a.empty() || b.a.empty())
77         return res;
78     res.a.assign(a.size() + b.a.size(), 0);
79     for (int i = 0; i < a.size(); i++)
80     {
81         long long carry = 0;
82         for (int j = 0; j < b.a.size() || carry > 0; j++)
83         {
84             long long cur = res.a[i + j] + a[i] * 1LL * (j < b.a.size() ? b.a

```

```

85             [j] : 0) + carry;
86             res.a[i + j] = cur % BASE;
87             carry = cur / BASE;
88         }
89     }
90     res.trim();
91     return res;
92 }
93
94 bool operator<(const BigInt &b) const
95 {
96     if (a.size() != b.a.size())
97         return a.size() < b.a.size();
98     for (int i = a.size() - 1; i >= 0; i--)
99     {
100         if (a[i] != b.a[i])
101             return a[i] < b.a[i];
102     }
103     return false;
104 }
105
106 void print() const
107 {
108     if (a.empty())
109     {
110         cout << 0 << "\n";
111         return;
112     }
113     cout << a.back();
114     for (int i = (int)a.size() - 2; i >= 0; i--)
115     {
116         cout << setfill('0') << setw(9) << a[i];
117     }
118     cout << "\n";
119 }

```

## 10.2 Chess

### 10.2.1 Knight's tour

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: WARNSDORFF'S HEURISTIC
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Visit every square", "Knight's Tour", "Hamiltonian path on grid".
6  * - Classic Scenarios: Output a valid sequence of knight moves to visit all N x
7  *   N
8  *   cells exactly once without repeating.
9  * - The Magic: Backtracking has a massive branching factor and TLEs immediately.
10 * Warnsdorff's Rule states: always prioritize the next valid move that has the

```

```

10 *   LEAST number of subsequent valid moves. This local greedy choice finds the
11 *   global Hamiltonian path almost instantly.
12 *
13 * 2. HOW TO USE IT (THE BLACK BOX)
14 * - Query:
15 *     vector<pair<int, int>> tour;
16 *     bool success = knight_tour(START_X, START_Y, BOARD_SIZE, tour);
17 */
18
19 const int dx[] = {1, 1, 2, 2, -1, -1, -2, -2};
20 const int dy[] = {2, -2, 1, -1, 2, -2, 1, -1};
21
22 int get_degree(int x, int y, int n, const vector<vector<int>> &board)
23 {
24     int count = 0;
25     for (int i = 0; i < 8; i++)
26     {
27         int nx = x + dx[i], ny = y + dy[i];
28         if (nx >= 0 && ny >= 0 && nx < n && ny < n && board[nx][ny] == -1)
29         {
30             count++;
31         }
32     }
33     return count;
34 }
35
36 bool solve_kt(int x, int y, int move_i, int n, vector<vector<int>> &board, vector<
    <pair<int, int>> &tour)
37 {
38     board[x][y] = move_i;
39     tour.push_back({x, y});
40
41     if (move_i == n * n - 1)
42         return true;
43
44     vector<pair<int, int>> next_moves;
45     for (int i = 0; i < 8; i++)
46     {
47         int nx = x + dx[i], ny = y + dy[i];
48         if (nx >= 0 && ny >= 0 && nx < n && ny < n && board[nx][ny] == -1)
49         {
50             next_moves.push_back({get_degree(nx, ny, n, board), i});
51         }
52     }
53
54     // Warnsdorff's Heuristic: Sort by degree ascending
55     sort(next_moves.begin(), next_moves.end());
56
57     for (auto move : next_moves)
58     {
59         int i = move.second;
60         int nx = x + dx[i], ny = y + dy[i];

```

```

61         if (solve_kt(nx, ny, move_i + 1, n, board, tour))
62         {
63             return true;
64         }
65     }
66
67     // Backtrack
68     board[x][y] = -1;
69     tour.pop_back();
70     return false;
71 }
72
73 bool knight_tour(int start_x, int start_y, int n, vector<pair<int, int>> &tour)
74 {
75     vector<vector<int>> board(n, vector<int>(n, -1));
76     return solve_kt(start_x, start_y, 0, n, board, tour);
77 }

```

## 10.3 Matroids

### 10.3.1 Matroid intersection

```

1 /**
2 * THE ULTIMATE CP TEMPLATE BLUEPRINT: MATROID INTERSECTION
3 *
4 * 1. WHAT PROBLEM DOES THIS SOLVE?
5 * - Keywords: "Spanning tree with unique colors", "Maximize intersection of
    rules".
6 * - Classic Scenarios: Find the maximum number of elements from a set that
    satisfy
7 * TWO independent structural rules simultaneously (e.g., forms a forest AND
    has
8 * at most K edges from any specific set).
9 * - The Magic: It builds a bipartite graph between the currently picked elements
    (IN)
10 * and unpicked elements (OUT). It searches for an augmenting path that swaps
    elements
11 * while maintaining independence in BOTH matroids.
12 *
13 * 2. HOW TO USE IT (THE BLACK BOX)
14 * - Setup: You must define what "independence" means for your two rules by
    writing
15 * the `oracle1_can_add` and `oracle2_can_add` logic.
16 * - Query:
17 *     vector<int> independent_set = matroid_intersection(N);
18 *
19 * 3. ENGINEERING NOTE
20 * - The robust, modular design here perfectly mirrors the isolation of concerns
    required in high-tier backend systems engineering. The core augmenting
21 * engine

```

```

22 *   never cares about the underlying rules, only the true/false oracle response
23 */
24
25 vector<int> matroid_intersection(int n)
26 {
27     vector<bool> in_set(n, false);
28
29     // Abstract augmenting path loop
30     while (true)
31     {
32         // Build the bipartite exchange graph
33         vector<int> X1, X2; // X1: valid starts, X2: valid ends
34         vector<vector<int>> adj(n);
35
36         // --- ORACLE 1 (e.g., Graphic Matroid) ---
37         // For every out_node 'v', if {in_set U v} is independent, add 'v' to X1
38         // If dependent, find which in_node 'u' forms the circuit and add edge v
39         //   -> u.
40
41         // --- ORACLE 2 (e.g., Colorful Matroid) ---
42         // For every out_node 'v', if {in_set U v} is independent, add 'v' to X2
43         // If dependent, find which in_node 'u' forms the circuit and add edge v
44         //   -> u.
45
46         // Example placeholders:
47         /*
48         for (int i = 0; i < n; i++) {
49             if (!in_set[i]) {
50                 if (oracle1_can_add(i)) X1.push_back(i);
51                 if (oracle2_can_add(i)) X2.push_back(i);
52             }
53         }
54         */
55
56         // BFS to find the shortest augmenting path from X1 to X2
57         queue<int> q;
58         vector<int> dist(n, 1e9), parent(n, -1);
59
60         for (int x : X1)
61         {
62             dist[x] = 0;
63             q.push(x);
64         }
65
66         int end_node = -1;
67         while (!q.empty())
68         {
69             int u = q.front();
70             q.pop();
71
72             // Check if we hit the target set
73             if (find(X2.begin(), X2.end(), u) != X2.end())
74             {
75                 end_node = u;
76                 break;
77             }
78
79             for (int v : adj[u])
80             {
81                 if (dist[v] > dist[u] + 1)
82                 {
83                     dist[v] = dist[u] + 1;
84                     parent[v] = u;
85                     q.push(v);
86                 }
87             }
88         }
89
90         // If no augmenting path exists, we have reached the maximum independent
91         // set
92         if (end_node == -1)
93             break;
94
95         // Augment the independent set by flipping states along the path
96         int curr = end_node;
97         while (curr != -1)
98         {
99             in_set[curr] = !in_set[curr];
100             curr = parent[curr];
101         }
102
103         vector<int> result;
104         for (int i = 0; i < n; i++)
105         {
106             if (in_set[i])
107                 result.push_back(i);
108         }
109         return result;
110     }
111 }

```

## 10.4 Search

### 10.4.1 Ternary search

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 // The function we want to minimize (e.g., total cost, distance, time)
5 // For this example, let's pretend it's a parabola:  $f(x) = (x - 5)^2 + 10$ 
6 // The minimum should clearly be at  $x = 5$ .
7 double f_double(double x)

```



```

8 {
9     return (x - 5.0) * (x - 5.0) + 10.0;
10 }
11
12 long long f_int(long long x)
13 {
14     return (x - 5) * (x - 5) + 10;
15 }
16
17 // -----
18 // TEMPLATE 1: Continuous Ternary Search (Floating Point)
19 // -----
20 double ternary_search_double(double l, double r)
21 {
22     // THE CP TRICK: Never use `while (r - l > eps)`. Floating point precision
23     // errors can cause an infinite loop and TLE your submission.
24     // Instead, looping exactly 200 times guarantees immense precision
25     // (the range shrinks by  $2/3^{200}$ , which is basically 0).
26     for (int i = 0; i < 200; i++)
27     {
28         double m1 = l + (r - l) / 3.0;
29         double m2 = r - (r - l) / 3.0;
30
31         if (f_double(m1) < f_double(m2))
32         {
33             r = m2; // We are minimizing, so throw away the higher right side
34         }
35         else
36         {
37             l = m1; // Throw away the higher left side
38         }
39     }
40     return l; // l and r are essentially identical now
41 }
42
43 // -----
44 // TEMPLATE 2: Discrete Ternary Search (Integers)
45 // -----
46 long long ternary_search_int(long long l, long long r)
47 {
48     // We stop when the search space shrinks to 3 elements or less.
49     // If we try to split a range of size 2 into thirds with integers,
50     // m1 and m2 will evaluate to the same number and cause an infinite loop!
51     while (r - l > 2)
52     {
53         long long m1 = l + (r - l) / 3;
54         long long m2 = r - (r - l) / 3;
55
56         if (f_int(m1) < f_int(m2))
57         {
58             r = m2;
59         }

```

```

60     else
61     {
62         l = m1;
63     }
64 }
65
66 // Now we have a tiny range (l, l+1, r). Just brute force the answer!
67 long long best_x = l;
68 long long min_val = f_int(l);
69
70 for (long long i = l + 1; i <= r; i++)
71 {
72     if (f_int(i) < min_val)
73     {
74         min_val = f_int(i);
75         best_x = i;
76     }
77 }
78
79 return best_x;
80 }
81
82 void solve()
83 {
84     cout << "Continuous_Minimum_at_X=" << ternary_search_double(-1e9, 1e9) << "\n";
85     cout << "Discrete_Minimum_at_X=" << ternary_search_int(-1000000, 1000000)
86         << "\n";
87 }
88
89 int main()
90 {
91     ios_base::sync_with_stdio(false);
92     cin.tie(NULL);
93     solve();
94     return 0;
95 }

```

#### 10.4.2 fraction binary search

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: FRACTION BINARY SEARCH
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Best rational approximation", "Fraction with bounded denominator".
6  * - Classic Scenarios: You are given a target floating point value and must find
7  * a fraction P/Q exactly equal to it (or closest to it) such that Q <= MAX_D.
8  * - The Magic: The Stern-Brocot tree generates all positive rational numbers in
9  * simplest form. Given bounds a/b and c/d, the mediant (a+c)/(b+d) is strictly
10 * between them. We binary search down this tree.

```

```

11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     pair<ll, ll> best_fraction = fraction_search(target, MAX_D);
15 *
16 * - Complexity:
17 *     Time:  $O(\log(\text{MAX\_D}))$ 
18 */
19
20 using ll = long long;
21
22 // Returns {numerator, denominator}
23 pair<ll, ll> fraction_search(double target, ll max_d)
24 {
25     ll a = 0, b = 1; // Lower bound: 0/1
26     ll c = 1, d = 0; // Upper bound: 1/0 (Infinity)
27
28     ll best_num = 0, best_den = 1;
29     double min_diff = 1e18;
30
31     while (b + d <= max_d)
32     {
33         ll mid_num = a + c;
34         ll mid_den = b + d;
35
36         double mid_val = (double)mid_num / mid_den;
37         double diff = abs(mid_val - target);
38
39         if (diff < min_diff)
40         {
41             min_diff = diff;
42             best_num = mid_num;
43             best_den = mid_den;
44         }
45
46         if (mid_val == target)
47         {
48             return {mid_num, mid_den};
49         }
50         else if (mid_val < target)
51         {
52             a = mid_num;
53             b = mid_den;
54         }
55         else
56         {
57             c = mid_num;
58             d = mid_den;
59         }
60     }
61
62     return {best_num, best_den};

```

```

63 }

```

### 10.4.3 parallel binary search

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: PARALLEL BINARY SEARCH
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Earliest time all components connect", "Minimum weight to satisfy
6  *   Q queries".
7  * - Classic Scenarios: You are given an empty graph and an array of edges added
8  *   over time.
9  *   You have Q queries asking: "At what time do nodes U and V finally connect?"
10 * - The Magic: It runs a binary search for ALL queries simultaneously. It sweeps
11 *   through the timeline 1 to T exactly  $\log(T)$  times. At each step, it checks if
12 *   the current timeline state satisfies the query's midpoint.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Array Setup:
16 *     L[i] = 1, R[i] = T for all queries i.
17 *     ans[i] = -1.
18 * - Loop:
19 *     Run the `while` loop until no queries need updating.
20 *
21 * 3. HOW TO ADAPT IT (THE DIALS & SWITCHES)
22 * - State Machine: The `DSU` and `apply_event` logic inside the timeline sweep
23 *   MUST
24 *   be replaced with whatever data structure the specific problem demands
25 *   (e.g., Fenwick Tree for range sums, Segment Tree, etc.).
26 */
27
28 struct Query
29 {
30     int u, v; // Replace with problem-specific query parameters
31 };
32
33 void parallel_binary_search(int num_events, const vector<Query> &queries)
34 {
35     int q = queries.size();
36     vector<int> L(q, 1), R(q, num_events), ans(q, -1);
37     bool changed = true;
38
39     while (changed)
40     {
41         changed = false;
42         // check_at[t] stores a list of query indices that need to be evaluated
43         // at time t
44         vector<vector<int>> check_at(num_events + 1);
45
46         for (int i = 0; i < q; i++)
47         {

```

```

44     if (L[i] <= R[i])
45     {
46         int mid = L[i] + (R[i] - L[i]) / 2;
47         check_at[mid].push_back(i);
48         changed = true;
49     }
50 }
51
52 if (!changed)
53     break;
54
55 // --- RESET DATA STRUCTURE HERE ---
56 // e.g., DSU dsu(N);
57 // -----
58
59 for (int t = 1; t <= num_events; t++)
60 {
61     // --- APPLY EVENT 't' TO DATA STRUCTURE ---
62     // e.g., dsu.unite(edges[t].u, edges[t].v);
63     // -----
64
65     for (int query_idx : check_at[t])
66     {
67         // --- EVALUATE QUERY CONDITION ---
68         // e.g., bool ok = dsu.same(queries[query_idx].u, queries[
69             query_idx].v);
70         bool ok = true; // Replace with actual condition
71         // -----
72
73         if (ok)
74         {
75             ans[query_idx] = t;
76             R[query_idx] = t - 1; // Try to find an earlier time
77         }
78         else
79         {
80             L[query_idx] = t + 1; // Need more time/events
81         }
82     }
83 }
84
85 // 'ans' now contains the exact minimum time for every query in  $O((N+Q) \log t)$ 
86 }

```

## 10.5 Techniques

### 10.5.1 MANHATTAN CHEBYSHEV

1 /\*\*

```

2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: MANHATTAN <-> CHEBYSHEV
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Rotate coordinate system", "Simplify absolute value distance".
6  * - Classic Scenarios: A problem asks you to find the number of points within
7  *   Manhattan distance D from (X, Y). This is an "octagonal" area, which is
8  *   extremely hard to query.
9  * - The Magic:
10 *   Transformation: (x, y) -> (x+y, x-y)
11 *   Manhattan distance in the original plane is equivalent to Chebyshev distance
12 *   in the transformed plane.
13 *   Chebyshev distance:  $\max(|x1 - x2|, |y1 - y2|)$  creates a *square* area, which
14 *   is trivially solvable using standard 2D Segment Trees or Fenwick Trees.
15 *
16 * 2. HOW TO USE IT
17 * - Transform:
18 *     new_x = x + y;
19 *     new_y = x - y;
20 */

```

### 10.5.2 Meet in the middle

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 // This template solves the classic "Find if any subset sums to Target"
5 // Complexity:  $O(2^{(N/2)} * N)$ 
6
7 void generate_sums(const vector<long long> &arr, vector<long long> &sums)
8 {
9     int n = arr.size();
10    // Use bitmasking to generate all  $2^n$  subsets
11    for (int i = 0; i < (1 << n); i++)
12    {
13        long long current_sum = 0;
14        for (int j = 0; j < n; j++)
15        {
16            if ((i >> j) & 1)
17            {
18                current_sum += arr[j];
19            }
20        }
21        sums.push_back(current_sum);
22    }
23 }
24
25 void solve()
26 {
27     int n;
28     long long target;
29     cin >> n >> target;

```

```

30
31 vector<long long> arr(n);
32 for (int i = 0; i < n; i++)
33     cin >> arr[i];
34
35 // 1. Split into two halves
36 vector<long long> left_part, right_part;
37 for (int i = 0; i < n; i++)
38 {
39     if (i < n / 2)
40         left_part.push_back(arr[i]);
41     else
42         right_part.push_back(arr[i]);
43 }
44
45 // 2. Generate all subset sums for both halves
46 vector<long long> left_sums, right_sums;
47 generate_sums(left_part, left_sums);
48 generate_sums(right_part, right_sums);
49
50 // 3. Sort one half for binary searching
51 sort(left_sums.begin(), left_sums.end());
52
53 // 4. Search for the complement in the other half
54 bool found = false;
55 for (long long s : right_sums)
56 {
57     long long needed = target - s;
58     // Check if 'needed' exists in left_sums
59     if (binary_search(left_sums.begin(), left_sums.end(), needed))
60     {
61         found = true;
62         break;
63     }
64 }
65
66 if (found)
67     cout << "YES\n";
68 else
69     cout << "NO\n";
70 }
71
72 int main()
73 {
74     ios_base::sync_with_stdio(false);
75     cin.tie(NULL);
76     solve();
77     return 0;
78 }

```

### 10.5.3 Sparse table lsa trick

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: SPARSE TABLE (LSA TRICK)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Range Maximum Query", "Immutable range query".
6  * - Classic Scenarios: You have an array that never changes, and you need to
7  *   query the maximum/minimum in any range [L, R] in  $O(1)$ .
8  * - The Magic: Precomputes logs for all ranges of length  $2^k$ . Since `max` is
9  *   idempotent ( $\max(a, a) = a$ ), we can answer queries by taking the max of two
10  *   overlapping  $2^k$  blocks that cover the entire [L, R].
11  */
12
13 const int K = 20; // Enough for  $10^6$  elements ( $2^{20} > 10^6$ )
14 int st[K][100005];
15 int lg[100005];
16
17 void build_sparse_table(const vector<int> &a)
18 {
19     int n = a.size();
20     lg[1] = 0;
21     for (int i = 2; i <= n; i++)
22         lg[i] = lg[i / 2] + 1;
23     for (int i = 0; i < n; i++)
24         st[0][i] = a[i];
25
26     for (int j = 1; j < K; j++)
27     {
28         for (int i = 0; i + (1 << j) <= n; i++)
29         {
30             st[j][i] = max(st[j - 1][i], st[j - 1][i + (1 << (j - 1))]);
31         }
32     }
33 }
34
35 int query(int L, int R)
36 {
37     int j = lg[R - L + 1];
38     return max(st[j][L], st[j][R - (1 << j) + 1]);
39 }

```

## 10.6 min plus convolution.cpp

### 10.6.1 shawk

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: SMAWK ALGORITHM
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?

```

```

5  * - Keywords: "Totally monotone matrix", "Min-plus convolution", " $O(N$ 
    optimization".
6  * - Classic Scenarios: You need to find the minimum value in every row of an  $N \times M$ 
    matrix where the matrix satisfies the Monge property (or is totally monotone).
7  * - The Magic: Bypasses standard  $O(NM)$  brute force and even  $O(N \log M)$  Divide & Conquer. It recursively filters out columns that can never contain a row
    minimum,
10 * solving the entire problem in strictly  $O(N + M)$  time.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Query:
14 *     vector<int> min_indices = smawk(R, C, cost_function);
15 * - Result:
16 *     Returns an array `ans` of size R, where `ans[i]` is the column index
17 *     containing the minimum value for row i.
18 */
19
20 using ll = long long;
21
22 template <class F>
23 vector<int> smawk(int r, int c, F cost)
24 {
25     auto solve = [&](auto &self, const vector<int> &rows, const vector<int> &cols
        ) -> vector<int>
26     {
27         if (rows.empty())
28             return {};
29
30         // Reduce: filter columns
31         vector<int> st;
32         for (int c : cols)
33         {
34             while (!st.empty())
35             {
36                 if (cost(rows[st.size() - 1], st.back()) > cost(rows[st.size() -
                    1], c))
37                 {
38                     st.pop_back();
39                 }
40                 else if (st.size() < rows.size())
41                 {
42                     break;
43                 }
44                 else
45                 {
46                     goto next_col;
47                 }
48             }
49             st.push_back(c);
50             next_col;
51         }
52
53         vector<int> odd_rows;
54         for (int i = 1; i < rows.size(); i += 2)
55             odd_rows.push_back(rows[i]);
56
57         // Solve recursively for odd-indexed rows
58         vector<int> odd_ans = self(self, odd_rows, st);
59         vector<int> ans(rows.size());
60         for (int i = 0; i < odd_ans.size(); i++)
61             ans[2 * i + 1] = odd_ans[i];
62
63         // Interpolate for even-indexed rows
64         int c_idx = 0;
65         for (int i = 0; i < rows.size(); i += 2)
66         {
67             int end_c = (i + 1 == rows.size()) ? st.back() : ans[i + 1];
68             int best_c = st[c_idx];
69             ll best_val = cost(rows[i], best_c);
70
71             while (st[c_idx] != end_c)
72             {
73                 c_idx++;
74                 ll val = cost(rows[i], st[c_idx]);
75                 if (val < best_val)
76                 {
77                     best_val = val;
78                     best_c = st[c_idx];
79                 }
80             }
81             ans[i] = best_c;
82         }
83         return ans;
84     };
85
86     vector<int> rows(r), cols(c);
87     iota(rows.begin(), rows.end(), 0);
88     iota(cols.begin(), cols.end(), 0);
89     return solve(solve, rows, cols);
90 }

```

## 10.7 misc

### 10.7.1 Dynamic bitset

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: DYNAMIC BITSET
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Bitset DP with variable size", "Memory Limit Exceeded".
6  * - Classic Scenarios: You need the  $O(N / 64)$  speedup of std::bitset, but the

```

```

7  *   required size of the bitset is determined by the input (e.g., sum of array)
8  *   meaning you cannot use std::bitset<MAX_SIZE> without hitting Memory Limits.
9  * - The Magic: Wraps a std::vector<uint64_t> and manually implements the
10 *   left-shift (<<) and bitwise OR (|) operators with hardware-level carry logic.
11 *
12 * 2. HOW TO USE IT (THE BLACK BOX)
13 * - Initialization:
14 *   DynamicBitset dp(total_sum + 1);
15 *   dp.set(0);
16 * - Transitions:
17 *   dp |= (dp << item_weight);
18 */
19
20 #include <cstdint>
21
22 struct DynamicBitset
23 {
24     vector<uint64_t> blocks;
25     int bits;
26
27     DynamicBitset(int n = 0)
28     {
29         bits = n;
30         blocks.assign((n + 63) / 64, 0);
31     }
32
33     void set(int i)
34     {
35         if (i < bits)
36             blocks[i / 64] |= (1ULL << (i % 64));
37     }
38
39     bool test(int i) const
40     {
41         if (i >= bits)
42             return false;
43         return (blocks[i / 64] >> (i % 64)) & 1;
44     }
45
46     // Crucial for Knapsack / Reachability DP
47     DynamicBitset operator<<(int shift) const
48     {
49         DynamicBitset res(bits + shift);
50         int block_shift = shift / 64;
51         int bit_shift = shift % 64;
52         uint64_t carry = 0;
53
54         for (int i = 0; i < blocks.size(); i++)
55         {
56             res.blocks[i + block_shift] = (blocks[i] << bit_shift) | carry;
57             if (bit_shift > 0)

```

```

58         {
59             carry = blocks[i] >> (64 - bit_shift);
60         }
61         else
62         {
63             carry = 0;
64         }
65     }
66     if (carry && (blocks.size() + block_shift < res.blocks.size()))
67     {
68         res.blocks[blocks.size() + block_shift] = carry;
69     }
70     // Restore original logical size constraints if necessary
71     res.bits = bits + shift;
72     return res;
73 }
74
75 void operator|=(const DynamicBitset &other)
76 {
77     for (int i = 0; i < min(blocks.size(), other.blocks.size()); i++)
78     {
79         blocks[i] |= other.blocks[i];
80     }
81 }
82 };

```

### 10.7.2 LGV lemma

```

1 /**
2  * THE ULTIMATE CP TEMPLATE BLUEPRINT: LGV LEMMA (NON-INTERSECTING PATHS)
3  *
4  * 1. WHAT PROBLEM DOES THIS SOLVE?
5  * - Keywords: "Non-intersecting paths", "Multiple robots on a grid".
6  * - Classic Scenarios: You have K robots starting at points A_1...A_K and they
7  *   must reach B_1...B_K on a DAG (like a grid with obstacles). They cannot
8  *   share any vertices. How many valid routing configurations exist?
9  * - The Magic: You only need to calculate the standard number of paths from
10 *   every A_i to every B_j. You construct a K x K matrix of these path counts.
11 *   The determinant of this matrix magically cancels out all intersecting path
12 *   configurations via parity inversions.
13 *
14 * 2. HOW TO USE IT (THE BLACK BOX)
15 * - Dependencies: You MUST have your `det_prime` function (Template 371)
16 *   included above this from your Linear Algebra section.
17 * - Logic: Replace `calc_ways(start, end)` with a standard O(V+E) DP or nCr math.
18 */
19
20 using ll = long long;

```

```

21
22 // Forward declaration of your existing determinant template
23 ll det_prime(vector<vector<ll>> a, ll mod);
24
25 // Replace this with your specific problem's path counting logic
26 ll calc_ways(pair<int, int> start, pair<int, int> end, ll mod)
27 {
28     // Example: Grid paths without obstacles using Combinatorics:  $C(dx + dy, dx)$ 
29     // Example: Grid paths with obstacles using  $O(N^2)$  DP
30     return 0;
31 }
32
33 ll lgv_lemma(const vector<pair<int, int>> &starts, const vector<pair<int, int>> &
    ends, ll mod)
34 {
35     int k = starts.size();
36     if (k != ends.size())
37         return 0;
38
39     vector<vector<ll>> M(k, vector<ll>(k, 0));
40
41     // Build the path matrix
42     for (int i = 0; i < k; i++)
43     {
44         for (int j = 0; j < k; j++)
45         {
46             M[i][j] = calc_ways(starts[i], ends[j], mod);
47         }
48     }
49
50     // The answer is literally just the determinant of the path matrix
51     return det_prime(M, mod);
52 }

```

## 11 Helpers

### 11.1 Adhocs

```

1 void PratialSum()
2 {
3     int n; cin >> n;
4     vector<int> arr(n + 1);
5     vector<ll> pratial(n + 1, 0);
6
7     for (int i = 1; i <= n; i++) cin >> arr[i];
8
9     int l, r, value; cin >> l >> r >> value;    // sure  $l < r$ 
10
11     pratial[l] += value;
12     pratial[r + 1] -= value;

```

```

13
14     for (int i = 1; i <= n; i++)
15         pratial[i] += pratial[i - 1];
16
17     for (int i = 1; i <= n; i++)
18         pratial[i] += arr[i];
19
20     for (int i = 1; i <= n; i++)
21         cout << pratial[i] << " ";
22
23 }
24 // Function to get the sum of elements in the subarray defined by (x1, y1) to (x2
    , y2)
25 long long getSum(const vector<vector<ll>>& Prefix, int x1, int y1, int x2, int y2
    ) {
26     return Prefix[x2][y2] - Prefix[x1 - 1][y2] - Prefix[x2][y1 - 1] + Prefix[x1 -
    1][y1 - 1];
27 }
28 void PrefixSum_2D() {
29     int n, m, q;
30     cin >> n >> m >> q;
31     vector<vector<ll>> arr(n + 1, vector<ll>(m + 1, 0));
32     vector<vector<ll>> Prefix(n + 1, vector<ll>(m + 1, 0));
33     for (int i = 1; i <= n; i++) {
34         for (int j = 1; j <= m; j++) {
35             cin >> arr[i][j];
36         }
37     }
38     for (int i = 1; i <= n; i++) {
39         for (int j = 1; j <= m; j++) {
40             Prefix[i][j] = arr[i][j]
41                 + Prefix[i - 1][j]
42                 + Prefix[i][j - 1]
43                 - Prefix[i - 1][j - 1];
44         }
45     }
46     while (q--) {
47         int x1, y1, x2, y2;
48         cin >> x1 >> y1 >> x2 >> y2; // sure  $x1 < x2$  and  $y1 < y2$ 
49         long long result = getSum(Prefix, x1, y1, x2, y2);
50         cout << result << endl;
51     }
52 }
53
54 void PratialSum_2D()
55 {
56     int n, m, q;
57     cin >> n >> m >> q;
58     vector<vector<ll>> arr(n + 1, vector<ll>(m + 1, 0));
59     vector<vector<ll>> Pratial(n + 2, vector<ll>(m + 2, 0));
60     while (q--) {
61         int x1, y1, x2, y2, val;

```

```

62     cin >> x1 >> y1 >> x2 >> y2 >> val;
63     Pratial[x1][y1] += val;
64     Pratial[x1][y2 + 1] -= val;
65     Pratial[x2 + 1][y1] -= val;
66     Pratial[x2+1][y2+1] += val;
67 }
68 for (int i = 1; i <= n; i++) {
69     for (int j = 1; j <= m; j++) {
70         Pratial[i][j] = Pratial[i][j]
71             + Pratial[i - 1][j]
72             + Pratial[i][j - 1]
73             - Pratial[i - 1][j - 1];
74     }
75 }
76
77 for (int i = 1; i <= n; i++) {
78     for (int j = 1; j <= m; j++) {
79         cout << Pratial[i][j] << " ";
80     }
81 }
82 cout << '\n';
83 }
84
85 }

```

## 11.2 BitMask&Complete Search

```

1
2 vector<ll> getRepresentation(ll num, ll base) // convert any number to (any)base
3 {
4     vector<ll>ans;
5     while (num)
6     {
7         ans.push_back(num % base);
8         num /= base;
9     }
10    reverse(ans.begin(), ans.end());
11    return ans;
12 }
13
14 // 1. Check if the i-th bit is set (1) or not (0)
15 bool checkBit(long long mask, int i) {
16     return (mask & (1LL << i)) != 0;
17 }
18
19 // 2. Set the i-th bit to 1 (leave other bits unchanged)
20 long long setBit(long long mask, int i) {
21     return mask | (1LL << i);
22 }
23
24 // 3. Clear the i-th bit to 0 (leave other bits unchanged)

```

```

25 long long clearBit(long long mask, int i) {
26     return mask & ~(1LL << i);
27 }
28
29 // 4. Toggle the i-th bit (flip 1 to 0, or 0 to 1)
30 long long toggleBit(long long mask, int i) {
31     return mask ^ (1LL << i);
32 }
33
34 // 5. Get the value of the lowest set bit (e.g., 1010 -> 0010)
35 // Useful for Fenwick trees (Binary Indexed Trees)
36 long long lowestSetBit(long long mask) {
37     return mask & -mask;
38 }
39
40 // 6. Check if a number is a perfect power of 2
41 // A power of 2 has exactly one bit set (e.g., 8 is 1000 in binary)
42 bool isPowerOfTwo(long long n) {
43     return n > 0 && (n & (n - 1)) == 0;
44 }
45
46 // 7. Strip the lowest set bit (change the rightmost 1 to a 0)
47 long long stripLowestSetBit(long long mask) {
48     return mask & (mask - 1);
49 }
50 int LastBitValue(ll n)
51 {
52     return n & ~(n - 1);
53 }
54
55 // 1. Count the number of set bits (1s) in a 32-bit integer
56 int count = __builtin_popcount(mask);
57
58 // 2. Count the number of set bits (1s) in a 64-bit integer (long long)
59 int countLL = __builtin_popcountll(mask);
60
61 // 3. Count the number of leading zeros
62 int leadingZeros = __builtin_clz(mask);
63
64 // 4. Count the number of trailing zeros (useful to find the index of the first
65     set bit)
66 int trailingZeros = __builtin_ctz(mask);
67
68 // Ranges
69 void prefix_Xor()
70 {
71     int n; cin >> n; vector<ll>v(n + 1);
72     for (int i = 1; i <= n; i++) cin >> v[i];
73     vector<ll>pre_Xor(n + 1);
74     for (int i = 1; i <= n; i++)
75         pre_Xor[i] = pre_Xor[i - 1] ^ v[i];

```



```

76     int q; cin >> q; while (q--) {
77         int l, r; cin >> l >> r;
78         cout << (pre_Xor[r] ^ pre_Xor[l - 1]) << '\n';
79     }
80 }
81 void prefix_OR()
82 {
83     int n; cin >> n; vector<ll>v(n + 1);
84     vector<vector<ll>>>prefix_bit(64, vector<ll>(n + 1));
85     for (int i = 1; i <= n; i++) cin >> v[i];
86
87     for (int bit = 0; bit < 64; bit++)
88     {
89         for (int i = 1; i <= n; i++)
90         {
91             prefix_bit[bit][i] = prefix_bit[bit][i - 1] + ((v[i] >> bit) & 1);
92         }
93     }
94     int q; cin >> q; while (q--) {
95         int l, r; cin >> l >> r;
96         ll OR = 0;
97         for (int bit = 0; bit < 64; bit++)
98         {
99             int on_bits = prefix_bit[bit][r] - prefix_bit[bit][l - 1];
100             if (on_bits > 0)
101             {
102                 OR += (1LL << bit);
103             }
104         }
105         cout << OR << '\n';
106     }
107 }
108 void prefix_AND()
109 {
110     int n; cin >> n; vector<ll>v(n + 1);
111     vector<vector<ll>>>prefix_bit(64, vector<ll>(n + 1));
112     for (int i = 1; i <= n; i++) cin >> v[i];
113
114     for (int bit = 0; bit < 64; bit++)
115     {
116         for (int i = 1; i <= n; i++)
117         {
118             prefix_bit[bit][i] = prefix_bit[bit][i - 1] + ((v[i] >> bit) & 1);
119         }
120     }
121     int q; cin >> q; while (q--) {
122         int l, r; cin >> l >> r;
123         ll AND = 0;
124         for (int bit = 0; bit < 64; bit++)
125         {
126             int on_bits = prefix_bit[bit][r] - prefix_bit[bit][l - 1];
127             if (on_bits == r-l+1)

```

```

128         {
129             AND += (1LL << bit);
130         }
131     }
132     cout << AND << '\n';
133 }
134 }
135
136 void CompleteSearch()
137 {
138     int n; cin >> n;
139     for (int mask = 0; mask < (1 << n); mask++)
140     {
141         for (int i = 0; i < n; i++)
142         {
143             if ((mask >> i) & 1)
144                 //if (getBit(mask, i))
145                 cout << 1;
146             else
147                 cout << 0;
148         }
149         cout << '\n';
150     }
151 }
152
153 void iterateSubsets(int mask) {
154     // Start with the full mask, and keep stripping away bits
155     for (int sub = mask; sub > 0; sub = (sub - 1) & mask) {
156         // 'sub' is a valid subset of 'mask'
157         std::cout << sub << "\n";
158     }
159     // Note: The loop above misses the empty subset (0).
160     // If you need to process 0 as well, you can do it after the loop.
161 }
162
163 void BitSet() {
164     bitset<32> b = 15;
165
166     b[3] = 0;
167     cout << b.to_ulong() << '\n'; // 7
168     cout << b << '\n'; // 0000000000000000000000000000111
169     cout << b.size() << "\n"; //32 bit
170     cout << b.test(2) << '\n'; // 1 >> true
171     cout << b.test(3) << '\n'; // 0 >> false
172     cout << b.count() << '\n'; // Count number of 1 == 3
173     cout << b.flip() << '\n'; // 1111111111111111111111111111000
174     cout << b.set() << '\n'; //1111111111111111111111111111111
175     cout << b.reset() << '\n'; //0000000000000000000000000000000
176     b[4] = 1;
177     cout << b.to_string() << '\n'; //000000000000000000000000000010000
178
179

```

### 11.3 Bitmasks(AMR)

```

180 }

1 //
2 // Created by AMR on 4/16/2025.
3 //
4
5 #include <bits/stdc++.h>
6 using namespace std;
7 typedef long long ll;
8 #define El_Hefnawys ios_base::sync_with_stdio(false); cin.tie(nullptr); cout.tie(nullptr)
9
10 const int MOD = 1e9 + 7;
11
12
13
14 //  $A \setminus B = (A/B) - (A \cap B)$ 
15 //  $A \cap B = (A+B) - 2*(A \setminus B)$ 
16 //  $A \setminus B = 2*(A/B) - (A+B)$ 
17
18 bool is_power_2(int x) {
19     if (x == 0)
20         return 0;
21     return x&(x-1) == 0;
22 }
23
24 int MSB_value(int n) {
25     if (n == 0)
26         return 0;
27     int index = 31 - __builtin_clz(n);
28     return 1 << index;
29 }
30
31 int MSB_index(int n) {
32     if (n == 0)
33         return -1;
34     return 31 - __builtin_clz(n);
35 }
36
37 // you can use __builtin_ctz(n) to get the LSB also directly
38 int LSB_value(int n) {
39     if (n == 0)
40         return 0;
41     return n&(-n);
42 }
43
44 int LSB_index(int n) {
45     if (n == 0)
46         return -1;
47
48     int value = n&(-n);
49     return 31 - __builtin_clz(value);
50 }
51 bool check_bit(ll n, ll i) {
52     return n&(1ll<<i);
53 }
54
55 // Another way
56 bool check_bit_2(ll n, ll i) {
57     return 1ll&(n>>i);
58 }
59
60 ll toggle_bit(ll n, ll i) {
61     return n^(1ll<<i);
62 }
63
64 ll set_bit(ll n, ll i) {
65     return n | (1ll<<i);
66 }
67
68 ll clear_bit(ll n, ll i) {
69     return n & ~(1ll<<i);
70 }
71
72 bool check_pow_2(ll n) {
73     return __builtin_popcountll(n) == 1; // function to count the number of 1s in
74     // 0(1) function
75     // a number
76 }
77 ll num_modulo_pow_2(ll n, ll k) {
78     return n & ((1ll << k) - 1);
79 }
80
81 int main() {
82     El_Hefnawys;
83     cout << num_modulo_pow_2(25,4);
84     return 0;
85 }
86
87 //////////////////////////////////////
88
89 //
90 // Created by AMR on 4/17/2025.
91 //
92
93 #include <bits/stdc++.h>
94 using namespace std;
95 typedef long long ll;
96 #define El_Hefnawys \
97     ios_base::sync_with_stdio(false); \

```

```

98     cin.tie(nullptr);
99     cout.tie(nullptr)
100
101 const int MOD = 1e9 + 7;
102
103 int Xor_nums_from_1_to_n(int n)
104 {
105     if (!(n % 4))
106         return n;
107     if (n % 4 == 1)
108         return 1;
109     if (n % 4 == 2)
110         return n + 1;
111     return 0;
112 }

```

## 11.4 Compression

```

1 vector<int> compress(vector<int> &v)
2 {
3     vector<int> help = v;
4     sort(help.begin(), help.end());
5     help.erase(unique(help.begin(), help.end()), help.end());
6     // 1-based
7     // to get original value -> help[compressed value - 1];
8     for (auto &val : v)
9         val = distance(help.begin(), lower_bound(help.begin(), help.end(), val))
10            + 1;
11     return help;
12 }

```

## 11.5 Hashing segment tree

```

1 #define int long long
2 const int N = 1e5 + 5, mod1 = 1e9 + 7, mod2 = 2e9 + 11;
3 ll base1 = 31, base2 = 37, pw1[N + 1], pw2[N + 1], inv1[N + 1], inv2[N + 1];
4
5 ll powmod(ll a, ll b, ll m)
6 {
7     ll ans = 1;
8     while (b > 0)
9     {
10         if (b & 1)
11             ans = (ans * a) % m;
12         a = (a * a) % m;
13         b >>= 1;
14     }
15 }

```

```

17     return ans;
18 }
19
20 void init()
21 {
22     pw1[0] = pw2[0] = inv1[0] = inv2[0] = 1;
23     int temp1 = powmod(base1, mod1 - 2, mod1);
24     int temp2 = powmod(base2, mod2 - 2, mod2);
25     for (int i = 1; i < N; i++)
26     {
27         pw1[i] = (base1 * pw1[i - 1]) % mod1;
28         pw2[i] = (base2 * pw2[i - 1]) % mod2;
29         inv1[i] = (inv1[i - 1] * temp1) % mod1;
30         inv2[i] = (inv2[i - 1] * temp2) % mod2;
31     }
32 }
33
34 struct HashingSegmentTree
35 {
36 private:
37     vector<pair<int, int>> seg;
38     int sz;
39
40     pair<int, int> merge(pair<int, int> l, pair<int, int> r)
41     {
42         pair<int, int> ret = l;
43         ret.first = (ret.first + r.first) % mod1;
44         ret.second = (ret.second + r.second) % mod2;
45         return ret;
46     }
47
48     void update(int l, int r, int node, int idx, int ch)
49     {
50         if (l == r)
51         {
52             seg[node] = {(ch * pw1[idx]) % mod1, (ch * pw2[idx]) % mod2};
53             return;
54         }
55         int mid = l + r >> 1;
56         if (idx <= mid)
57             update(l, mid, 2 * node + 1, idx, ch);
58         else
59             update(mid + 1, r, 2 * node + 2, idx, ch);
60         seg[node] = merge(seg[2 * node + 1], seg[2 * node + 2]);
61     }
62
63     pair<int, int> query(int l, int r, int node, int lx, int rx)
64     {
65         if (l >= lx && r <= rx)
66         {
67             return seg[node];
68         }

```

```

69     if (l > rx || r < lx)
70         return {0, 0};
71     int mid = l + r >> 1;
72     pair<int, int> lft = query(l, mid, 2 * node + 1, lx, rx);
73     pair<int, int> rgt = query(mid + 1, r, 2 * node + 2, lx, rx);
74     return merge(lft, rgt);
75 }
76
77 public:
78     HashingSegmentTree(int n)
79     {
80         sz = 1;
81         while (sz <= n)
82             sz *= 2;
83         seg = vector<pair<int, int>>(sz * 2);
84     }
85
86     void update(int idx, char ch)
87     {
88         update(0, sz - 1, 0, idx, ch - 'a' + 1);
89     }
90
91     pair<int, int> query(int l, int r)
92     {
93         pair<int, int> ret = query(0, sz - 1, 0, l, r);
94         ret.first = (ret.first * inv1[l - 1]) % mod1;
95         ret.second = (ret.second * inv2[l - 1]) % mod2;
96         return ret;
97     }
98 };
99
100 bool isPalindrome(HashingSegmentTree &a, HashingSegmentTree &b, int &l, int &r,
101                  int &n)
102 {
103     return (a.query(l, r) == b.query(n - r + 1, n - l + 1));
104 }

```

## 11.6 Hashing treap

```

1 #include <bits/stdc++.h>
2 using namespace std;
3 #define ll long long
4 #define int long long
5 const int N = 1e6 + 5, mod = 1e9 + 7;
6 ll base = 31, pw[N + 1];
7
8 void init()
9 {
10     pw[0] = 1;
11     for (int i = 1; i < N; i++)
12     {

```

```

13         pw[i] = (base * pw[i - 1]) % mod;
14     }
15 }
16
17 std::mt19937_64 rnd(std::chrono::system_clock::now().time_since_epoch().count());
18
19 struct Treap
20 {
21     struct Node
22     {
23         int val, h = 0;
24         int pri = rnd(), sz = 1;
25         array<Node *, 2> c = {0, 0};
26
27         Node()
28         {
29         }
30
31         Node(int k)
32         {
33             val = h = k;
34         }
35     };
36
37     Node *root = 0;
38     int getSize(Node *t)
39     {
40         return t ? t->sz : 0;
41     }
42
43     int getHash(Node *t)
44     {
45         return t ? t->h : 0;
46     }
47
48     Treap(string &s)
49     {
50         for (int i = 0; i < s.size(); i++)
51         {
52             root = merge(root, new Node(s[i] - 'a' + 1));
53         }
54     }
55
56     Node *fix(Node *t)
57     {
58         t->sz = getSize(t->c[0]) + getSize(t->c[1]) + 1;
59         t->h = (((getHash(t->c[0]) * base + t->val) % mod) * pw[getSize(t->c[1])])
60              + getHash(t->c[1])) % mod;
61         return t;
62     }
63
64     array<Node *, 2> split(Node *t, int k)

```

```

64 {
65     if (!t)
66         return {0, 0};
67     if (getSize(t->c[0]) >= k)
68     {
69         auto ret = split(t->c[0], k);
70         t->c[0] = ret[1];
71         return {ret[0], fix(t)};
72     }
73     else
74     {
75         auto ret = split(t->c[1], k - getSize(t->c[0]) - 1);
76         t->c[1] = ret[0];
77         return {fix(t), ret[1]};
78     }
79 }
80
81 Node *merge(Node *u, Node *v)
82 {
83     if (!u || !v)
84         return u ? u : v;
85     if (u->pri > v->pri)
86     {
87         u->c[1] = merge(u->c[1], v);
88         return fix(u);
89     }
90     else
91     {
92         v->c[0] = merge(u, v->c[0]);
93         return fix(v);
94     }
95 }
96
97 void deleteRange(int l, int r)
98 {
99     auto a = split(root, l - 1);
100     auto b = split(a[1], r - l + 1);
101     root = merge(a[0], b[1]);
102 }
103
104 int getHash(int l, int r)
105 {
106     auto a = split(root, l - 1);
107     auto b = split(a[1], r - l + 1);
108     int ret = b[0]->h;
109     root = merge(a[0], merge(b[0], b[1]));
110     return ret;
111 }
112
113 void addChar(int pos, char c)
114 {
115     auto a = split(root, pos - 1);
116     root = merge(merge(a[0], new Node(c - 'a' + 1)), a[1]);
117 }
118
119 void print(Node *t)
120 {
121     if (!t)
122         return;
123     print(t->c[0]);
124     cout << t->h << '□';
125     print(t->c[1]);
126 }
127 };
128
129 int rev(int i, int n)
130 {
131     return n - i + 1;
132 }
133
134 signed main()
135 {
136     ios_base::sync_with_stdio(0);
137     cin.tie(0);
138     cout.tie(0);
139     init();
140     int n, q;
141     cin >> n >> q;
142     string s;
143     cin >> s;
144     string f = s;
145     reverse(f.begin(), f.end());
146     Treap a(s), b(f);
147     while (q--)
148     {
149         int op;
150         cin >> op;
151         if (op == 1)
152         {
153             int l, r;
154             cin >> l >> r;
155             int l2 = rev(l, n), r2 = rev(r, n);
156             n -= (r - l + 1);
157             a.deleteRange(l, r);
158             b.deleteRange(r2, l2);
159         }
160         else if (op == 2)
161         {
162             n++;
163             int pos;
164             char c;
165             cin >> c >> pos;
166             a.addChar(pos, c);
167             b.addChar(rev(pos, n), c);

```

```

168     }
169     else
170     {
171         int l, r;
172         cin >> l >> r;
173         int l2 = rev(l, n), r2 = rev(r, n);
174         if (a.getHash(l, r) == b.getHash(r2, l2))
175         {
176             cout << "yes\n";
177         }
178         else
179         {
180             cout << "no\n";
181         }
182     }
183 }
184 return 0;
185 }

```

## 11.7 INT 128

```

1  __int128 read() {
2      __int128 x = 0, f = 1;
3      char ch = getchar();
4      while (ch < '0' || ch > '9') {
5          if (ch == '-') f = -1;
6          ch = getchar();
7      }
8      while (ch >= '0' && ch <= '9') {
9          x = x * 10 + ch - '0';
10         ch = getchar();
11     }
12     return x * f;
13 }
14 void print(__int128 x) {
15     if (x < 0) {
16         putchar('-');
17         x = -x;
18     }
19     if (x > 9) print(x / 10);
20     putchar(x % 10 + '0');
21 }
22 bool cmp(__int128 x, __int128 y) { return x > y; }

```

## 11.8 MEX

```

1 int MEX(vector<int> &v)
2 {
3     int n = int(v.size());

```

```

4     vector<int> frq(n);
5     for (int i = 0; i < n; i++)
6     {
7         if (v[i] < n)
8             frq[v[i]]++;
9     }
10    for (int i = 0; i < n; i++)
11    {
12        if (frq[i] == 0)
13            return i;
14    }
15    return n;
16 }

```

## 11.9 Ordered set

```

1 #include <ext/pb_ds/assoc_container.hpp>
2 #include <ext/pb_ds/tree_policy.hpp>
3 using namespace __gnu_pbds;
4
5 typedef tree<int, null_type, less<>, rb_tree_tag,
6             tree_order_statistics_node_update> ordered_set;
7
8 void myerase(ordered_set &t, int v)
9 {
10     int rank = t.order_of_key(v);
11     auto it = t.find_by_order(rank);
12     t.erase(it);
13 }
14 /*
15  * t.order_of_key(v);           // Returns the NUMBER of elements strictly less
16  *                               than v (the rank)
17  * *t.find_by_order(k);         // Returns the value at index k (0-indexed)
18  * myerase(t, v);               // Safely erases exactly ONE instance of v
19 */

```

## 11.10 Power

```

1 ll power(ll base, ll pow)
2 {
3     base %= MOD;
4     ll res{1};
5     while (pow)
6     {
7         if (pow & 1)
8             res = res * base % MOD;
9         base = base * base % MOD;
10        pow >>= 1;
11    }

```

```

12     return res;
13 }

```

### 11.11 Query

```

1 struct Query
2 {
3     int x, a, id;
4
5     bool operator<(const Query &other) const
6     {
7         return a < other.a;
8     }
9 };

```

### 11.12 Solve Equations

```

1 pair<ll,ll> equ(ll a,ll b,ll c)
2 {
3     ll x1 = (-b + sqrt(b * b - 4 * a * c)) / (2 * a);
4     ll x2 = (-b - sqrt(b * b - 4 * a * c)) / (2 * a);
5     return {x1,x2};
6 }

```

### 11.13 Split mix

```

1 struct custom_hash
2 {
3     static uint64_t splitmix64(uint64_t x)
4     {
5         x += 0x9e3779b97f4a7c15;
6         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
7         x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
8         return x ^ (x >> 31);
9     }
10
11     size_t operator()(uint64_t x) const
12     {
13         static const uint64_t FIXED_RANDOM = chrono::steady_clock::now().
            time_since_epoch().count();
14         return splitmix64(x + FIXED_RANDOM);
15     }
16 };

```

### 11.14 debug

```

1 #define LOCAL ;
2
3 template <typename A, typename B>
4 string to_string(pair<A, B> p);
5
6 template <typename A, typename B, typename C>
7 string to_string(tuple<A, B, C> p);
8
9 template <typename A, typename B, typename C, typename D>
10 string to_string(tuple<A, B, C, D> p);
11
12 string to_string(const string &s)
13 {
14     return '"' + s + '"';
15 }
16
17 string to_string(const char *s)
18 {
19     return to_string((string)s);
20 }
21
22 string to_string(bool b)
23 {
24     return (b ? "true" : "false");
25 }
26
27 string to_string(vector<bool> v)
28 {
29     bool first = true;
30     string res = "{";
31     for (auto &i : v)
32     {
33         if (!first)
34         {
35             res += ", ";
36         }
37         first = false;
38         res += to_string(i);
39     }
40     res += "}";
41     return res;
42 }
43
44 template <size_t N>
45 string to_string(bitset<N> v)
46 {
47     string res = "";
48     for (size_t i = 0; i < N; i++)
49     {
50         res += static_cast<char>('0' + v[i]);

```

```

51     }
52     return res;
53 }
54
55 template <typename A>
56 string to_string(A v)
57 {
58     bool first = true;
59     string res = "{";
60     for (const auto &x : v)
61     {
62         if (!first)
63         {
64             res += ",_";
65         }
66         first = false;
67         res += to_string(x);
68     }
69     res += "}";
70     return res;
71 }
72
73 template <typename A, typename B>
74 string to_string(pair<A, B> p)
75 {
76     return "(" + to_string(p.first) + ",_" + to_string(p.second) + ")";
77 }
78
79 template <typename A, typename B, typename C>
80 string to_string(tuple<A, B, C> p)
81 {
82     return "(" + to_string(get<0>(p)) + ",_" + to_string(get<1>(p)) + ",_" +
83         to_string(get<2>(p)) + ")";
84 }
85
86 template <typename A, typename B, typename C, typename D>
87 string to_string(tuple<A, B, C, D> p)
88 {
89     return "(" + to_string(get<0>(p)) + ",_" + to_string(get<1>(p)) + ",_" +
90         to_string(get<2>(p)) + ",_" +
91         to_string(get<3>(p)) + ")";
92 }
93
94 template <typename T>
95 string to_string(queue<T> q)
96 {
97     string res = "{";
98     bool first = true;
99     while (!q.empty())
100     {
101         if (!first)
102             res += ",_";

```

```

101         first = false;
102         res += to_string(q.front());
103         q.pop();
104     }
105     res += "}";
106     return res;
107 }
108
109 template <typename T>
110 string to_string(stack<T> s)
111 {
112     string res = "{";
113     bool first = true;
114     while (!s.empty())
115     {
116         if (!first)
117             res += ",_";
118         first = false;
119         res += to_string(s.top());
120         s.pop();
121     }
122     res += "}";
123     return res;
124 }
125
126 template <typename T, typename Container, typename Compare>
127 string to_string(std::priority_queue<T, Container, Compare> pq)
128 {
129     string res = "{";
130     bool first = true;
131     while (!pq.empty())
132     {
133         if (!first)
134             res += ",_";
135         first = false;
136         res += to_string(pq.top());
137         pq.pop();
138     }
139     res += "}";
140     return res;
141 }
142
143 void debug_out() { cerr << endl; }
144
145 template <typename Head, typename... Tail>
146 void debug_out(Head H, Tail... T)
147 {
148     cerr << "_" << to_string(H);
149     debug_out(T...);
150 }
151
152 #ifdef LOCAL

```



```

153 #define debug(...) cerr << "[" << #__VA_ARGS__ << "]:", debug_out(__VA_ARGS__)
154 #else
155 #define debug(...) 42
156 #endif

```

### 11.15 dist

```

1 double dist(double x1, double y1, double x2, double y2)
2 {
3     return hypot(x2 - x1, y2 - y1);
4 }

```

### 11.16 div

```

1 vector<int> div(int n)
2 {
3     vector<int> divs;
4     for (int i = 1; i * i <= n; i++)
5     {
6         if (n % i == 0)
7         {
8             divs.push_back(i);
9             if (i * i != n)
10                divs.push_back(n / i);
11        }
12    }
13    sort(divs.begin(), divs.end());
14    return divs;
15 }

```

### 11.17 double compare

```

1 int dcomp(double a, double b)
2 {
3     return fabs(a - b) < EPS ? 0 : a > b ? 1
4         : -1;
5 }

```

### 11.18 edge

```

1 struct edge
2 {
3     ll from, to, w;
4
5     edge(ll from, ll to, ll w) : from(from), to(to), w(w)

```

```

6     {
7     }
8
9     bool operator<(const edge &e) const
10    {
11        return w < e.w;
12    }
13
14    bool operator==(const edge &e)
15    {
16        return from == e.from && to == e.to && w == e.w;
17    }
18 };

```

### 11.19 find last occurrence

```

1 find last occurrence for char to specific index
2 vector<vector<int>> nxt;
3 vector<int> last;
4 string s;
5 int n,k;
6 void pre() {
7     nxt.assign(n+5, vector<int>(k+5, 0));
8     last.assign(k+5, n);
9
10    for (int i = n-1; i >= 0; i--) {
11        for (int j = 0; j < k; j++) {
12            nxt[i][j] = last[j];
13        }
14        last[s[i] - 'a'] = i;
15    }
16
17 }

```

### 11.20 inv mod

```

1 ll power(ll base, ll pow)
2 {
3     base %= MOD;
4     ll res{1};
5     while (pow)
6     {
7         if (pow & 1)
8             res = res * base % MOD;
9         base = base * base % MOD;
10        pow >>= 1;
11    }
12    return res;
13 }

```

```

14
15 // MOD must be prime number
16 ll modInv(ll x)
17 {
18     return power(x, MOD - 2);
19 }

```

### 11.21 is prime

```

1 bool is_prime(int x)
2 {
3     if (x <= 1)
4         return false;
5     if (x % 2 == 0)
6         return x == 2;
7     for (int i = 3; i * i <= x; i += 2)
8     {
9         if (x % i == 0)
10            return false;
11     }
12     return true;
13 }

```

### 11.22 kadaness

```

1 int curr{a[0]};
2 int glob{a[0]};
3 for (int i = 1; i < n; i++)
4 {
5     curr = max(a[i], curr + a[i]);
6     glob = max(curr, glob);
7 }
8
9 //-----//
10
11 int mx{}, curr{};
12 pair<int, int> p{0, 0};
13 pair<int, int> best{};
14 for (int i = 0; i < n - 1; i++)
15 {
16     curr += a[i];
17     if (curr < 0)
18     {
19         curr = 0;
20         p.first = i + 1;
21     }
22     p.second = i;
23     if (curr > mx)
24     {

```

```

25         best = p;
26         mx = curr;
27     }
28     else if (mx == curr)
29     {
30         if (best.second - best.first < p.second - p.first)
31             best = p;
32     }
33 }
34
35 //-----//
36
37 int mx = -00;
38
39 for (int left = 0; left < n; left++)
40 {
41     vector<int> temp(n, 0);
42
43     for (int right = left; right < n; right++)
44     {
45         for (int i = 0; i < n; i++)
46         {
47             temp[i] += a[i][right];
48         }
49
50         int current_max = temp[0];
51         int current_sum = temp[0];
52
53         for (int i = 1; i < n; i++)
54         {
55             current_sum = max(temp[i], current_sum + temp[i]);
56             current_max = max(current_max, current_sum);
57         }
58
59         mx = max(mx, current_max);
60     }
61 }

```

### 11.23 nCr

```

1 int nCr(int n, int r)
2 {
3     if (r < 0 || n < 0 || r > n)
4         return 0;
5     int ret{1};
6     for (int i = 1; i <= min(r, n - r); i++)
7     {
8         ret *= n - i + 1;
9         ret /= i;
10    }
11    return ret;

```

```

12 }
13
14 int nPr(int n, int r)
15 {
16     if (r < 0 || n < 0 || r > n)
17         return 0;
18     int ret{1};
19     for (int i = 0; i < r; i++)
20         ret = mul(ret, n - i);
21     return ret;
22 }

```

## 11.24 next greater

```

1 vector<int> nextGreater(n, n);
2 for (int i = 0; i < n; i++)
3 {
4     while (!s.empty() && v[i] > v[s.top()])
5     {
6         nextGreater[s.top()] = i;
7         s.pop();
8     }
9     s.push(i);
10 }

```

## 11.25 permutation hashing

```

1 #include <bits/stdc++.h>
2 using namespace std;
3 #define ll long long
4 const int N = 2e5 + 5;
5 std::mt19937_64 rng(std::chrono::system_clock::now().time_since_epoch().count())
6 ll rand(ll l, ll r)
7 {
8     return uniform_int_distribution<ll>(l, r)(rng);
9 }
10
11 struct node
12 {
13     ll a, b, c, d, e;
14
15     node() : a(0), b(0), c(0), d(0), e(0)
16     {
17     }
18
19     node(ll x, ll xx, ll xxx, ll xxxx, ll xxxxx) : a(x), b(xx), c(xxx), d(xxxx),
20     {
21         e(xxxxx)
22     }
23 }

```

```

22
23     node operator+(const node &x)
24     {
25         node w;
26         w.a = a + x.a, w.b = b + x.b, w.c = c + x.c, w.d = d + x.d, w.e = e + x.e
27         ;
28         return w;
29     }
30
31     bool operator==(const node &x)
32     {
33         return a == x.a && b == x.b && c == x.c && d == x.d && e == x.e;
34     }
35
36     void operator+=(const node &x)
37     {
38         a += x.a, b += x.b, c += x.c, d += x.d, e += x.e;
39     }
40
41     node operator=(const node &x)
42     {
43         a = x.a, b = x.b, c = x.c, d = x.d, e = x.e;
44         return node(a, b, c, d, e);
45     }
46 };
47 node f[N], per[N];
48
49 void init()
50 {
51     for (int i = 1; i < N; i++)
52         f[i] = per[i] = node(rand(1, 1e9), rand(1, 1e9), rand(1, 1e9), rand(1, 1
53         e9),
54         rand(1, 1e9));
55     for (int i = 2; i < N; i++)
56         per[i] += per[i - 1];
57 }
58
59 signed main()
60 {
61     ios_base::sync_with_stdio(0);
62     cin.tie(0);
63     cout.tie(0);
64     init();
65     int t;
66     cin >> t;
67     while (t--)
68     {
69         int n;
70         cin >> n;
71         ll s[n + 1] = {};
72         for (int i = 1; i <= n; i++)

```

```
72         cin >> s[i];
73         node pre[n + 5], suf[n + 5];
74         for (int i = 1; i <= n; i++)
75             pre[i] = suf[i] = f[s[i]];
76         for (int i = 2; i <= n; i++)
77             pre[i] += pre[i - 1];
78         for (int i = n - 1; i >= 1; i--)
79             suf[i] += suf[i + 1];
80     }
81     return 0;
82 }
```

## 11.26 random

```
1 int getRandom(int min, int max)
2 {
3     static std::mt19937 gen(chrono::steady_clock::now().time_since_epoch().count
4         ());
5     std::uniform_int_distribution<int> distrib(min, max);
6     return distrib(gen);
7 }
```

## 11.27 valid

```
1 bool valid(int i, int j)
2 {
3     if (i >= r || i < 0 || j >= c || j < 0)
4         return false;
5     return true;
6 }
```